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PHYSICS 101

AN INTRODUCTION TO PHYSICS

This course of 45 video lectures, as well as accompanying notes, have been developed and presented by Dr. Pervez Amirali Hoodbhoy, professor of physics at Quaid-e-Azam University, Islamabad, for the Virtual University of Pakistan, Lahore.

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GENERAL INFORMATION

Purpose: This course aims at providing the student a good understanding of physics at the elementary level. Physics is essential for understanding the modern world, and is a definite part of its culture.

Background: It will be assumed that the student has taken physics and mathematics at the F.Sc level, i.e. the 12th year of schooling. However, B.Sc students are also likely to find the course useful. Calculus is not assumed and some essential concepts will be developed as the course progresses. Algebra and trigonometry are essential. However, for physics, the more mathematics one knows the better.

Scope and Duration: The course has 45 lectures, each of somewhat less than one hour duration. All main fields of physics will be covered, together with several applications in each.

Language: For ease of communication, all lectures are in Urdu. However, English or Latin technical terms have been used where necessary. The student must remember that further study and research in science is possible only if he or she has an adequate grasp of English.

Textbook: There is no prescribed textbook. However, you are strongly recommended to read a book at the level of “College Physics” by Halliday and Resnick (any edition). There are many other such books too, such as “University Physics” by Young and Freedman. Study any book that you are comfortable with, preferably by a well-established foreign author. Avoid local authors because they usually copy. After listening to a lecture, go read the relevant chapter. Please remember that these notes cover only some things that you should know and are not meant to be complete.

Assignments: There will be total Eight Assignment in this course and its schedules will be announced from time to time. The book you choose to consult will have many more. Those students who are seriously interested in the subject are advised to work out several of the questions posed there. In physics you cannot hope to gain mastery of the subject without extensive problem solving.

Examinations: Their schedules will be announced from time to time.

Tutors: Their duty is to help you, and they will respond to all genuine questions. However, please do not overload them as they have to deal with a large number of students. Happy studying!

Acknowledgements: I thank the Virtual University team and administration for excellent cooperation, as well as Mansoor Noori and Naeem Shahid, for valuable help.

Summary of Lecture 1 – INTRODUCTION TO PHYSICS

1. Physics is a science. Science works according to the *scientific method*. The *scientific method* accepts only reason, logic, and experimental evidence to tell between what is scientifically correct and what is not. Scientists do not simply believe – they test, and keep testing until satisfied. Just because some “big scientist” says something is right, that thing does not become a fact of science. Unless a discovery is repeatedly established in different laboratories at different times by different people, or the same theoretical result is derived by clear use of established rules, we do not accept it as a scientific discovery. The real strength of science lies in the fact that it continually keeps challenging itself.
2. It is thought that the laws of physics do not change from place to place. This is why experiments carried out in different countries by different scientists – of any religion or race – have always led to the same results if the experiments have been done honestly and correctly. We also think that the laws of physics today are the same as they were in the past. Evidence, contained in the light that left distant stars billions of years ago, strongly indicates that the laws operating at that time were no different than those today. The spectra of different elements then and now are impossible to tell apart, even though physicists have looked very carefully.
3. This course will cover the following broad categories:
 - a) *Classical Mechanics*, which deals with the motion of bodies under the action of forces. This is often called Newtonian mechanics as well.
 - b) *Electromagnetism*, whose objective is to study how charges behave under the influence of electric and magnetic fields as well as understand how charges can create these fields.
 - c) *Thermal Physics*, in which one studies the nature of heat and the changes that the addition of heat brings about in matter.
 - d) *Quantum Mechanics*, which primarily deals with the physics of small objects such as atoms, nuclei, quarks, etc. However, Quantum Mechanics will be treated only briefly for lack of time.
4. Every physical quantity can be expressed in terms of three fundamental dimensions: Mass (M), Length (L), Time (T). Some examples:

Speed	LT^{-1}
Acceleration	LT^{-2}
Force	MLT^{-2}
Energy	ML^2T^{-2}
Pressure	$ML^{-1}T^{-2}$

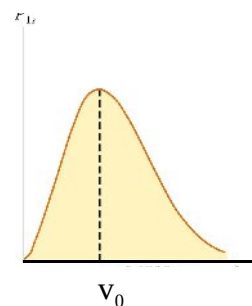
You cannot add quantities that have different dimensions. So force can be added to force, but force can never be added to energy, etc. A formula is definitely wrong if the dimensions on the left and right sides of the equal sign are different.

5. Remember that any function $f(x)$ takes as input a dimensionless *number* x and outputs a quantity f (which may, or may not have a dimension). Take, for example, the function $f(\theta) = \sin \theta$. You know the expansion: $\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$. If θ had a dimension then you would be adding up quantities of different dimensions, and that is not allowed.
6. Do not confuse units and dimensions. We can use different units to measure the same physical quantity. So, for example, you can measure the mass in units of kilograms, pounds, or even in *sair* and *chatak*! In this course we shall always use the MKS or **M**etre-**K**ilogram-**S**econd system. When you want to convert from one system to another, be methodical as in the example below:

$$1 \frac{mi}{hr} = 1 \frac{mi}{hr} \times 5280 \frac{ft}{mi} \times \frac{1}{3.28} \frac{m}{ft} \times \frac{1}{3600} \frac{hr}{s} = 0.447 \frac{m}{s}$$

When you write it out in this manner, note that various quantities cancel out cleanly in the numerator and denominator. So you never make a mistake!

7. A good scientist first thinks of the larger picture and then of the finer details. So, estimating *orders of magnitude* is extremely important. Students often make the mistake of trying to get the decimal points right instead of the first digit – which obviously matters the most! So if you are asked to calculate the height of some building using some data and you come up with 0.301219 metres or 4.01219×10^6 metres, then the answer is plain nonsense even though you may have miraculously got the last six digits right. Physics is commonsense first, so use your intelligence before submitting any answer.
8. Always check your equations to see if they have the same dimensions on the left side as on the right. So, for example, from this principle we can see the equation $v^2 = u^2 + 2at$ is clearly wrong, whereas $v^2 = u^2 + 13a^2t^2$ could possibly be a correct relation. (Here v and u are velocities, a is acceleration, and t is time.) Note here that I use the word *possibly* because the dimensions on both sides match up in this case.
9. Whenever you derive an equation that is a little complicated, see if you can find a special limit where it becomes simple and transparent. So, sometimes it is helpful to imagine that some quantity in it is very large or very small. Where possible, make a “mental graph” so that you can picture an equation. So, for example, a formula for the distribution of molecular speeds in a gas could look like $f(v) = ve^{-(v-v_0)^2/a^2}$. Even without knowing the value of a you can immediately see that
- $f(v)$ goes to zero for large values of v , and $v = 0$.
 - The maximum value of $f(v)$ occurs at v_0 and the function decreases on both side of this value.



Summary of Lecture 2 – KINEMATICS I

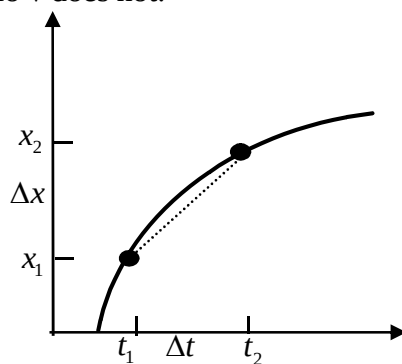
1. $x(t)$ is called displacement and it denotes the position of a body at time. If the displacement is positive then that body is to the right of the chosen origin and if negative, then it is to the left.
2. If a body is moving with average speed v then in time t it will cover a distance $d=vt$. But, in fact, the speed of a car changes from time to time and so one should limit the use of this formula to small time differences only. So, more accurately, one defines an average speed over the small time interval Δt as:

$$\text{average speed} = \frac{\text{distance travelled in time } \Delta t}{\Delta t}$$

3. We define *instantaneous velocity* at any time t as:

$$v = \frac{x(t_2) - x(t_1)}{t_2 - t_1} \equiv \frac{\Delta x}{\Delta t} .$$

Here Δx and Δt are both very small quantities that tend to zero but their ratio v does not.



4. Just as we have defined velocity as the rate of change of distance, similarly we can define *instantaneous acceleration* at any time t as:

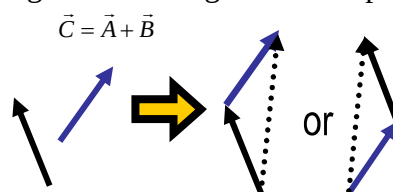
$$a = \frac{v(t_2) - v(t_1)}{t_2 - t_1} \equiv \frac{\Delta v}{\Delta t} .$$

Here Δv and Δt are both very small quantities that tend to zero but their ratio a is not zero, in general. Negative acceleration is called deceleration. The speed of a decelerating body decreases with time.

5. Some students are puzzled by the fact that a body can have a very large acceleration but can be standing still at a given time. In fact, it can be moving in the opposite direction to its acceleration. There is actually nothing strange here because position,

6. For constant acceleration and a body that starts from rest at $t = 0$, v increases linearly with time, $v \propto t$ (or $v = at$). If the body has speed v_0 at $t = 0$, then at time t , $v = at + v_0$.
7. We know in (6) above how far a body moving at constant speed moves in time t . But what if the body is changing its speed? If the speed is increasing linearly (i.e. constant acceleration), then the answer is particularly simple: just use the same formula as in (6) but use the average speed: $(v_0 + v_0 + at) / 2$. So we get

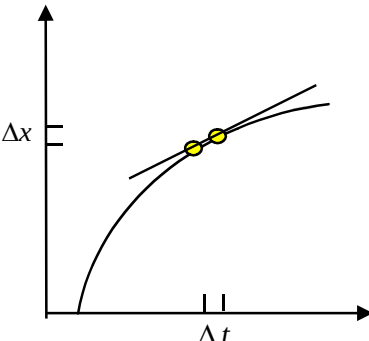
$$x = x_0 + (v_0 + v_0 + at)t / 2 = x_0 + v_0 t + \frac{1}{2} at^2$$
This formula tells you how far a body moves in time t if it moves with constant acceleration a , and if started at position x_0 at $t=0$ with speed v_0 .
8. We can eliminate the time using (7), and arrive at another useful formula that tells us what the final speed will be after the body has traveled a distance equal to $x - x_0$ after time t , $v^2 = v_0^2 + 2a(x - x_0)$.
9. Vectors: a quantity that has a size as well as direction is called a *vector*. So, for example, the wind blows with some speed and in some direction. So the *wind velocity* is a vector.
10. If we choose axes, then a vector is fixed by its components along those axes. In one dimension, a vector has only one component (call it the x-component). In two dimensions, a vector has both x and y components. In three dimensions, the components are along the x,y,z axes.
11. If we denote a vector $\vec{r} = (x, y)$ then, $r_x = x = r \cos \theta$, and $r_y = y = r \sin \theta$.
Note that $x^2 + y^2 = r^2$. Also, that $\tan \theta = y / x$.
13. Two vectors can be added together geometrically. We take any one vector, move it without changing its direction so that both vectors start from the same point, and then make a parallelogram. The diagonal of the parallelogram is the resultant.



14. Two vectors can also be added algebraically. In this case, we simply add the components of the two vectors along each axis separately. So, for example, Two vectors can be put together as $(1.5, 2.4) + (1, -1) = (2.5, 1.4)$.

Summary of Lecture 3 – KINEMATICS II

1. The concept of the derivative of a function is exceedingly important. The derivative shows how fast a function changes when its argument is changed. (Remember that for $f(x)$ we say that f is a function that depends upon the argument x . You should think of f as a machine that gives you the value f when you input x .)
2. Functions do not always have to be written as $f(x)$. $x(t)$ is also a function. It tells us where a body is at different times t .
3. The derivative of $x(t)$ at time t is defined as:

$$\begin{aligned}\frac{dx}{dt} &\equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t}.\end{aligned}$$


4. Let's see how to calculate the derivative of a simple function like $x(t) = t^2$. We must first calculate the difference in x at two slightly different values, t and $t + \Delta t$, while remembering that we choose Δt to be extremely small:

$$\begin{aligned}\Delta x &= (t + \Delta t)^2 - t^2 \\ &= t^2 + (\Delta t)^2 + 2t\Delta t - t^2 \\ \frac{\Delta x}{\Delta t} &= \Delta t + 2t \Rightarrow \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} = 2\end{aligned}$$

5. In exactly the same way you can show that if $x(t) = t^n$ then:

$$\frac{dx}{dt} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = nt^{n-1}$$

This is an extremely useful result.

6. Let us apply the above to the function $x(t)$ which represents the distance moved by a body with constant acceleration (see lecture 2):

$$x(t) = x_0 + v_0 t + \frac{1}{2} a t^2$$

$$\frac{dx}{dt} = 0 + v_0 + \frac{1}{2} a (2t) = v_0 + a t$$

This clearly shows that $\frac{dv}{dt} = 0 + a = a$ (acceleration is constant)

7. A stone dropped from rest increases its speed in the downward direction according to $\frac{dv}{dt} = g \approx 9.8 \text{ m/sec}$. This is true provided we are fairly close to the earth, otherwise the value of g decreases as we go further away from the earth. Also, note that if we measured distances from the ground up, then the acceleration would be negative.

8. A useful notation: write $\frac{dv}{dt} = \frac{d}{dt} \left(\frac{dx}{dt} \right) = \frac{d^2 x}{dt^2}$. We call $\frac{d^2 x}{dt^2}$ the second derivative of x with respect to t , or the rate of rate of change of x with respect to t .

9. It is easy to extend these ideas to a body moving in both the x and y directions. The position and velocity in 2 dimensions are:

$$\begin{aligned} \vec{r} &= x(t)\hat{i} + y(t)\hat{j} \\ \vec{v} &= \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} \\ &= v_x\hat{i} + v_y\hat{j} \end{aligned}$$

Here the unit vectors $\hat{i}$ and $\hat{j}$ are fixed, meaning that they do not depend upon time.

10. The scalar product of two vectors $\vec{A}$ and $\vec{B}$ is defined as:

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

You can think of:

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (A)(B \cos \theta) \\ &= (\text{length of } \vec{A}) \times (\text{projection of } \vec{B} \text{ on } \vec{A}) \end{aligned}$$

OR,

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (B)(A \cos \theta) \\ &= (\text{length of } \vec{B}) \times (\text{projection of } \vec{A} \text{ on } \vec{B}). \end{aligned}$$

Remember that for unit vectors $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = 1$ and $\hat{i} \cdot \hat{j} = 0$.

Summary of Lecture 4 – FORCE AND NEWTON'S LAWS

1. Ancient view: objects tend to stop if they are in motion; force is required to keep something moving. This was a natural thing to believe in because we see objects stop moving after some time; frictionless motion is possible to see only in rather special circumstances.
2. Modern view: objects tend to remain in their initial state; force is required to *change* motion. Resistance to changes in motion is called *inertia*. More inertia means it is harder to make a body accelerate or decelerate.
3. Newton's First Law: An object will remain at rest or move with constant velocity unless acted upon by a net external force. (A non-accelerating reference frame is called an inertial frame; Newton's First Law holds only in inertial frames.)
4. More force leads to more acceleration: $\Rightarrow a \propto F$
5. The greater the mass of a body, the harder it is to change its state of motion. More mass means more inertia. In other words, more mass leads to less acceleration:

$$\Rightarrow a \propto \frac{1}{m}$$

Combine both the above observations to conclude that:

$$a \propto \frac{F}{m}$$

6. Newton's Second Law: $a = \frac{F}{m}$ (or, if you prefer, write as $F = ma$).
7. $F = ma$ is one relation between three independent quantities (m, a, F). For it to be useful, we must have separate ways of measuring mass, acceleration, and force. Acceleration is measured from observing the rate of change of velocity; mass is a measure of the amount of matter in a body (e.g. two identical cars have twice the mass of a single one). Forces (due to gravity, a stretched spring, repulsion of two like charges, etc) will be discussed later.
8. Force has dimensions of $[\text{mass}] \times [\text{acceleration}] = M L T^{-2}$. In the MKS system the unit of force is the Newton. It has the symbol N where:
1 Newton = 1 kilogram.metre/second².
9. Forces can be internal or external. For example the mutual attraction of atoms within a block of wood are called internal forces. Something pushing the wood

is an external force. In the application of $F = ma$, remember that F stands for the total external force upon the body.

10. Forces are vectors, and so they must be added vectorially:

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots$$

This means that the components in the $\hat{x}$ direction must be added separately, those in the $\hat{y}$ direction separately, etc.

11. Gravity acts directly on the mass of a body - this is a very important experimental observation due to Newton and does not follow from $F = ma$. So a body of mass m_1 experiences a force $F_1 = m_1g$ while a body of mass m_2 experiences a force $F_2 = m_2g$, where g is the acceleration with which any body (big or small) falls under the influence of gravity. (Galileo had established this important fact when he dropped different masses from the famous leaning tower of Pisa!)
12. The weight of a body W is the force which gravity exerts upon it, $W = mg$. Mass and weight are two completely different quantities. So, for example, if you used a spring balance to weigh a kilo of grapes on earth, the same grapes would weigh only 1/7 kilo on the moon.
13. Newton's Third Law: for every action there is an equal and opposite reaction. More precisely, $F_{AB} = -F_{BA}$, where F_{AB} is the force exerted by body B upon A whereas F_{BA} is the force exerted by body A upon B . Ask yourself what would happen if this was not true. In that case, a system of two bodies, even if it is completely isolated from the surroundings, would have a net force acting upon it because the net force acting upon both bodies would be $F_{AB} + F_{BA} \neq 0$.
14. If action and reaction are always equal, then why does a body accelerate at all? Students are often confused by this. The answer: in considering the acceleration of a body you must consider only the (net) force acting upon that body. So, for example, the earth pulls a stone towards it and causes it to accelerate because there is a net force acting upon the stone. On the other hand, by the Third Law, the stone also pulls the earth towards it and this causes the earth to accelerate towards the stone. However, because the mass of the earth is so large, we are only able to see the acceleration of the stone and not that of the earth.

Summary of Lecture 5 – APPLICATIONS OF NEWTON'S LAWS – I

1. An obvious conclusion from $F = ma$ is that if $F = 0$ then $a = 0$! How simple, yet how powerful ! This says that for any body that is not accelerating the *sum of all the forces acting upon it* must vanish.
2. Examples of systems in equilibrium: a stone resting on the ground; a pencil balanced on your finger; a ladder placed against the wall, an aircraft flying at a constant speed and constant height.
3. Examples of systems out of equilibrium: a stone thrown upwards that is at its highest point; a plane diving downwards; a car at rest whose driver has just stepped on the car's accelerator.

4. If you know the acceleration of a body, it is easy to find the force that causes it to accelerate. Example: An aircraft of mass m has position vector,

$$\vec{r} = (at + bt^3)\hat{i} + (ct^2 + dt^4)\hat{j}$$

What force is acting upon it?

SOLUTION:

$$\begin{aligned}\vec{F} &= m \frac{d^2x}{dt^2} \hat{i} + m \frac{d^2y}{dt^2} \hat{j} \\ &= 6bmt\hat{i} + m(2c + 12dt^2)\hat{j}\end{aligned}$$

5. The other way around is not so simple: suppose that you know F and you want to find x . For this you must solve the equation,

$$\frac{d^2x}{dt^2} = \frac{F}{m}$$

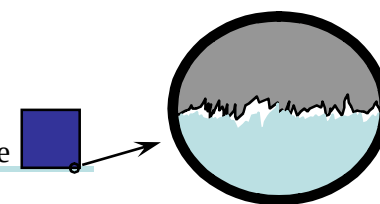
This may or may not be easy, depending upon F (which may depend upon both x as well as t if the force is not constant).

6. Ropes are useful because you can pull from a distance to change the direction of a force. The tension, often denoted by T , is the force you would feel if you cut the rope and grabbed the ends. For a massless rope (which may be a very good approximation in many situations) the tension is the same at every point along the rope. Why? Because if you take any small slice of the rope it weighs nothing (or very little). So if the force on one side of the slice was any different from the force on the other side, it would be accelerating hugely. All this was for the "ideal rope" which has no mass and never breaks. But this idealization is often good enough.

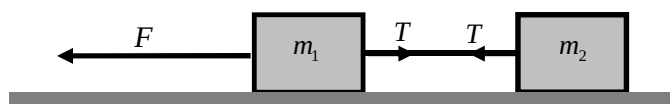
7. We are all familiar with frictional force. When two bodies rub against each other, the frictional force acts upon each body separately opposite to its direction of motion (i.e it acts to slow down the motion). The harder you press two bodies against each other, the greater the friction. Mathematically, $\vec{F} = \mu \vec{N}$, where $\vec{N}$ is the force with which you press the two bodies against each other (normal force). The quantity μ is called the coefficient of friction (obviously!). It is large for rough surfaces, and small for smooth ones. Remember that $\vec{F} = \mu \vec{N}$ is an empirical relation and holds only approximately. This is obviously true: if you put a large enough mass on a table, the table will start to bend and will eventually break.

8. Friction is caused by roughness at a microscopic level

- if you look at any surface with a powerful microscope you will see unevenness and jaggedness. If these big bumps are levelled somehow, friction will still not disappear because there will still be little bumps due to atoms. More precisely, atoms from the two bodies will interact each other because of the electrostatic interaction between their charges. Even if an atom is neutral, it can still exchange electrons and there will be a force because of surrounding atoms.



9. Consider the two blocks below on a frictionless surface:

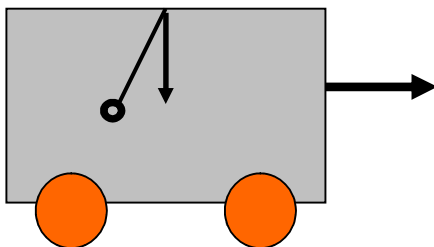


We want to find the tension and acceleration: The total force on the first mass is $F - T$ and so $F - T = m_1 a$. The force on the second mass is simply T and so $T = m_2 a$. Solving the above, we get: $T = \frac{m_2 F}{m_1 + m_2}$ and $a = \frac{F}{m_1 + m_2}$.

10. There is a general principle by which you solve equilibrium problems. For equilibrium, the sum of forces in every direction must vanish. So $F_x = F_y = F_z = 0$. You may always choose the x , y , z directions according to your convenience. So, for example, as in the lecture problem dealing with a body sliding down an inclined plane, you can choose the directions to be along and perpendicular to the surface of the plane.

Summary of Lecture 6 – APPLICATIONS OF NEWTON'S LAWS – II

1. As a body moves through a fluid it displaces the fluid. it has to exert a force on the fluid to push it out of the way. By Newton's third law, the fluid pushes back on the body with an equal and opposite force. The direction of the fluid resistance force on a body is always opposite to the direction of the body's velocity relative to the fluid.
2. The magnitude of the fluid resistance force usually increases with the speed of the body through the fluid. Typically, $f = kv$ (an empirical law!). Imagine that you drop a ball bearing into a deep container filled with oil. After a while the ball bearing will approach its maximum (terminal) speed when the forces of gravity and friction balance each other: $mg = kv$ from which $v_{\text{final}} = mg / k$.
3. The above was a simple example of equilibrium under two forces. In general, while solving problems you should a) draw a diagram, b) define an origin for a system of coordinates, c) identify all forces (tension, normal, friction, weight, etc) and their x and y components, d) Apply Newton's law separately along the x and y axes. e) find the accelerations, then velocities, then displacements. This sounds very cook-book, and in fact it will occur to you naturally how to do this while solving actual problems.
4. Your weight in a lift: suppose you are in a lift that is at rest or moving at constant velocity. In either case $a=0$ and the normal force N and the force due to gravity are exactly equal, $N - Mg = 0 \Rightarrow N = Mg$. But if the lift is accelerating downwards then $Mg - N = Ma$ or $N = M(g - a)$. So now the normal force (i.e. the force with which the floor of the lift is pushing on you) is decreased. Note that if the lift is accelerating downwards with acceleration a (which it will if the cable breaks!) then $N=0$ and you will experience weightlessness just like astronauts in space do. Finally, if the lift is accelerating upwards then a is negative and you will feel heavier.
5. Imagine that you are in a railway wagon and want to know how much you are accelerating. You are not able to look out of the windows. A mass is hung from the roof. Find the acceleration of the car from the angle made by the mass.

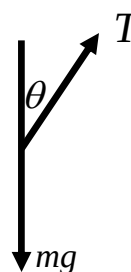


We first balance the forces vertically: $T \cos \theta = mg$

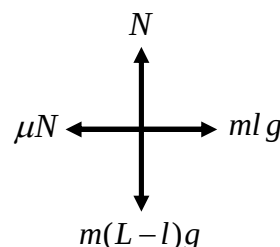
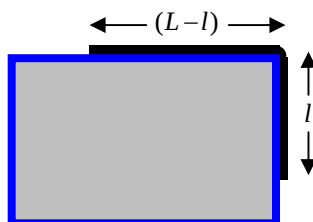
And then horizontally: $T \sin \theta = ma$

From these two equations we find that: $\tan \theta = \frac{a}{g}$

Note that the mass m doesn't matter - it cancels out!



6. Friction is a funny kind of force. It does not make up its mind which way to act until some other force compels it to decide. Imagine a block lying on the floor. If you push it forward, friction will act backward. And if you push it to the left, friction will act to the right. In other words, the direction of the frictional force is always in the opposite direction to the applied force.
7. Let us solve the following problem: a rope of total length L and mass per unit length m is put on a table with a length l hanging from one edge. What should be l such that the rope just begins to slip?



To solve this, look at the balance of forces in the diagram below: in the vertical direction, the normal force balances the weight of that part of the rope that lies on the table:

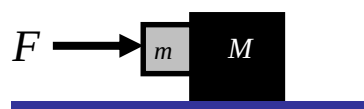
$N = m(L-l)g$. In the horizontal direction, the rope exerts a force mlg to the right, which is counteracted by the friction that acts to the left. Therefore $\mu N = mlg$. Substituting N

from the first equation we find that $l = \frac{\mu L}{\mu + 1}$. Note that if μ is very small then even a small piece of string that hangs over the edge will cause the entire string to slip down.

9. In this problem, we would like to calculate the minimum force

F such that the small block does not slip downwards. Clearly, since the 2 bodies move together, $F = (m+M)a$. This gives

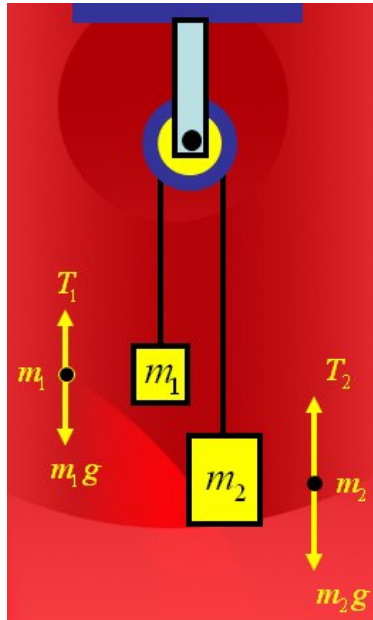
$a = \frac{F}{(m+M)}$. We want the friction μN to be at least as large as the downwards force, mg .



So, we put $N = ma = m \left(\frac{F}{(m+M)} \right)$ from which the minimum horizontal force needed to prevent slippage is $F = \frac{(m+M)g}{\mu}$.

There is an error in video lecture # 6 (video recording at 41:00) please consider the following problem with correct solution.

Consider two bodies of unequal masses m_1 and m_2 connected by the ends of a string, which passes over a frictionless pulley as shown in the diagram.



$$m_2 > m_1$$

$$T - m_1g = m_1a$$

$$m_2g - T = m_2a$$

$$m_2g - m_1g = (m_1 + m_2)a$$

$$a = \frac{(m_2 - m_1)}{(m_1 + m_2)} g$$

$$m_2 > m_1$$

$$T - m_1g = m_1a$$

$$a = \frac{T - m_1g}{m_1} \quad (i)$$

$$m_2g - T = m_2a$$

$$a = \frac{m_2g - T}{m_2} \quad (ii)$$

equating (i) and (ii)

$$\frac{T - m_1g}{m_1} = \frac{m_2g - T}{m_2}$$

$$m_2T - m_1m_2g = m_1m_2g - m_1T$$

$$m_2T + m_1T = m_1m_2g + m_1m_2g$$

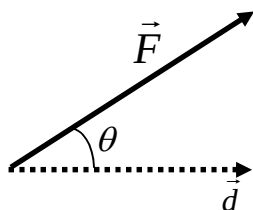
$$T(m_2 + m_1) = 2m_1m_2g$$

$$T = \frac{2m_1m_2g}{(m_2 + m_1)}$$

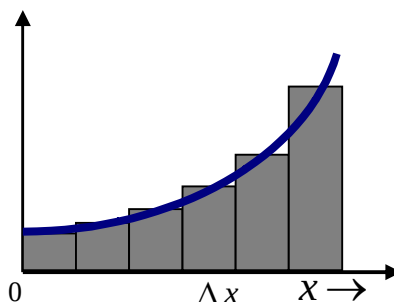
Summary of Lecture 7 – WORK AND ENERGY

1. Definition of work: force applied in direction of displacement $\times$ displacement. This means that if the force F acts at an angle θ with respect to the direction of motion, then

$$W = \vec{F} \cdot \vec{d} = Fd \cos \theta$$



2. a) Work is a scalar - it has magnitude but no direction.
b) Work has dimensions: $M \times (L T^{-2}) \times L = M L^2 T^{-2}$
c) Work has units: 1 Newton $\times$ 1 Metre $\equiv$ 1 Joule (J)
3. Suppose you lift a mass of 20 kg through a distance of 2 metres. Then the work you do is $20 \text{ kg} \times 9.8 \text{ Newtons} \times 2 \text{ metres} = 392 \text{ Joules}$. On the other hand, the force of gravity is directed opposite to the force you exert and the work done by gravity is -392 Joules .
4. What if the force varies with distance (say, a spring pulls harder as it becomes longer). In that case, we should break up the distance over which the force acts into small pieces so that the force is approximately constant over each bit. As we make the pieces smaller and smaller, we will approach the exact result:



Now add up all the little pieces of work:

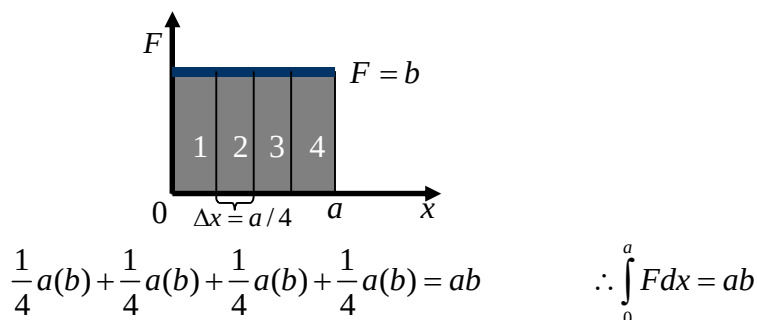
$$W = \Delta W_1 + \Delta W_2 + \cdots + \Delta W_N = F_1 \Delta x + F_2 \Delta x + \cdots + F_N \Delta x \equiv \sum_{n=1}^N F_n \Delta x$$

To get the exact result let $\Delta x \rightarrow 0$ and the number of intervals $N \rightarrow \infty$: $W = \lim_{\Delta x \rightarrow 0} \sum_{n=1}^{\infty} F_n \Delta x$

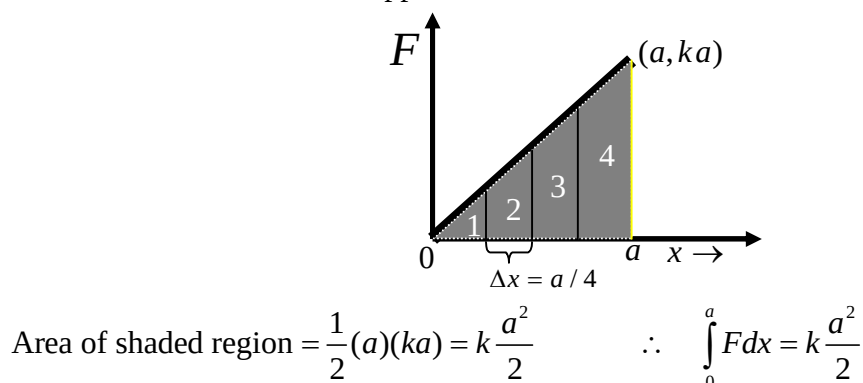
Definition: $W = \lim_{\Delta x \rightarrow 0} \sum_{n=1}^{\infty} F_n \Delta x \equiv \int_{x_i}^{x_f} F(x) dx$ is called the integral of F with respect to x from

x_i to x_f . This quantity is the work done by a force, constant or non-constant. So if the force is known as a function of position, we can always find the work done by calculating the definite integral.

5. Just to check what our result looks like for a constant force, let us calculate W if $F = b$,



6. Now for a less trivial case: suppose that $F=kx$, i.e. the force increases linearly with x .



7. Energy is the capacity of a physical system to do work:

- it comes in many forms – mechanical, electrical, chemical, nuclear, etc
- it can be stored
- it can be converted into different forms
- it can never be *created* or *destroyed*

8. Accepting the fact that energy is conserved, let us derive an expression for the kinetic energy of a body. Suppose a *constant force* accelerates a mass m from speed 0 to speed v over a *distance* d . What is the work done by the force? Obviously the answer is:

$$W = Fd. \text{ But } F = ma \text{ and } v^2 = 2ad. \text{ This gives } W = (ma)d = \frac{mv^2}{2d}d = \frac{1}{2}mv^2. \text{ So, we}$$

conclude that the work done by the force has gone into creating kinetic energy. and that the amount of kinetic energy possessed by a body moving with speed v is $\frac{1}{2}mv^2$.

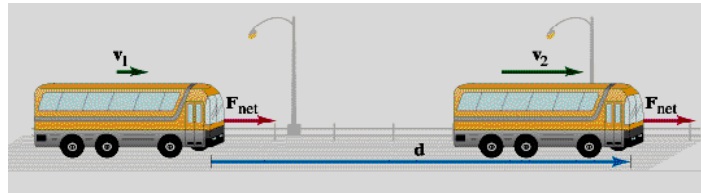
9. The work done by a force is just the force multiplied by the distance – it does not depend upon time. But suppose that the same amount of work is done in half the time. We then say that the *power* is twice as much.

We define:

$$\text{Power} = \frac{\text{Work done}}{\text{Time taken}}$$

If the force does not depend on time: $\frac{\text{Work}}{\text{Time}} = \frac{F \Delta x}{\Delta t} = F v$. Therefore, Power = $F v$.

10. Let's work out an example. A constant force accelerates a bus (mass m) from speed v_1 to speed v_2 over a distance d . What work is done by the engine?



Recall that for constant acceleration, $v_2^2 - v_1^2 = 2a(x_2 - x_1)$ where: v_2 = final velocity, x_2 = final position, v_1 = initial velocity, x_1 = initial position. Hence, $a = \frac{v_2^2 - v_1^2}{2d}$. Now

calculate the work done: $W = Fd = mad = m \frac{v_2^2 - v_1^2}{2d} d = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$. So the

work done has resulted in an increase in the quantity $\frac{1}{2}mv^2$, which is kinetic energy.

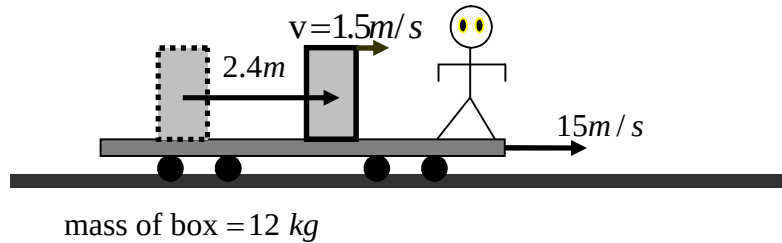
Summary of Lecture 8 – CONSERVATION OF ENERGY

1. Potential energy is, as the word suggests, the energy “locked up” up somewhere and which can do work. An object can store energy as the result of its position. Stored energy of position is referred to as potential energy. **Potential energy** is the stored energy of position possessed by an object. It has the ability or capacity to do work like other forms of energies. Potential energy can be converted into kinetic energy, $\frac{1}{2}mv^2$. As I showed you earlier, this follows directly from Newton’s Laws.
2. If you lift a stone of mass m from the ground up a distance x , you have to do work against gravity. The (constant) force is mg , and so $W = mgx$. By conservation of energy, the work done by you was transformed into gravitational potential energy whose values is exactly equal to mgx . Where is the energy stored? Answer: it is stored neither in the mass or in the earth - it is stored in the gravitational field of the combined system of stone+earth.
3. Suppose you pull on a spring and stretch it by an amount x away from its normal (equilibrium) position. How much energy is stored in the spring? Obviously, the spring gets harder and harder to pull as it becomes longer. When it is extended by length x and you pull it a further distance dx , the small amount of work done is $dW = Fdx = kx dx$. Adding up all the small bits of work gives the total work:

$$W = \int_0^x F dx = \int_0^x kx dx = \frac{1}{2} kx^2$$

This is the work you did. Maybe you got tired working so hard. What was the result of your working so hard? Answer: this work was transformed into energy stored in the spring. The spring contains energy exactly equal to $\frac{1}{2}kx^2$.

4. Kinetic energy obviously depends on the frame you choose to measure it in. If you are running with a ball, it has zero kinetic energy with respect to you. But someone who is standing will see that it has kinetic energy! Now consider the following situation: a box of mass 12kg is pushed with a constant force so that so that its speed goes from zero to 1.5m/sec (as measured by the person at rest on the cart) and it covers a distance of 2.4m. Assume there is no friction.



Let's first calculate the change in kinetic energy:

$$\Delta K = K_f - K_i = \frac{1}{2}(12\text{kg})(1.5\text{m/s})^2 - 0 = 13.5\text{J}$$

And then the (constant) acceleration:

$$a = \frac{v_f^2 - v_i^2}{2(x_f - x_i)} = \frac{(1.5\text{m/s})^2 - 0}{2(2.4\text{m})} = 0.469\text{m/s}^2$$

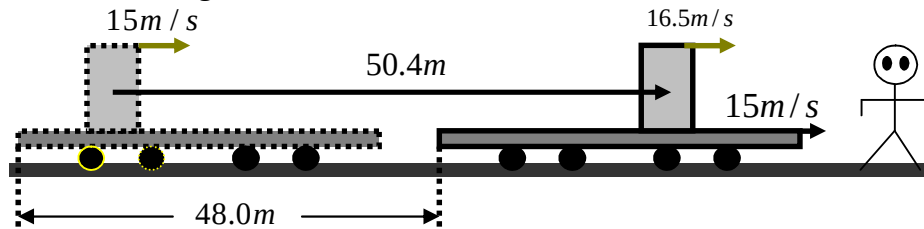
This acceleration results from a constant net force given by:

$$F = ma = (12\text{kg})(0.469\text{m/s}^2) = 5.63\text{N}$$

From this, the work done on the crate is:

$$W = F\Delta x = (5.63\text{N})(2.4\text{m}) = 13.5\text{J} \quad (\text{same as } \Delta K = 13.5\text{J} !)$$

5. Now suppose there is somebody standing on the ground, and that the trolley moves at 15 m/sec relative to the ground:



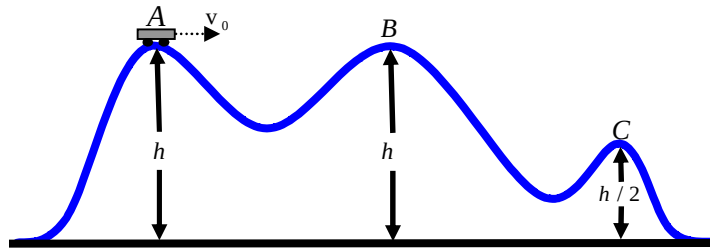
Let us repeat the same calculation:

$$\begin{aligned} \Delta K' &= K'_f - K'_i = \frac{1}{2}mv_f'^2 - \frac{1}{2}mv_i'^2 \\ &= \frac{1}{2}(12\text{kg})(16.5\text{m/s})^2 - \frac{1}{2}(12\text{kg})(15.0\text{m/s})^2 = 284\text{J} \end{aligned}$$

This example clearly shows that work and energy have different values in different frames.

6. The total mechanical energy is: $E_{\text{mech}} = KE + PE$. If there is no friction then E_{mech} is conserved. This means that the sum does not change with time. For example: a ball is thrown upwards at speed v_0 . How high will it go before it stops? The loss of potential energy is equal to the gain of potential energy. Hence, $\frac{1}{2}mv_0^2 = mgh \Rightarrow h = \frac{v_0^2}{2g}$.

Now look at the smooth, frictionless motion of a car over the hills below:



Even though the motion is complicated, we can use the fact that the total energy is a constant to get the speeds at the points B,C,D:

$$\text{At point A: } \frac{1}{2}mv_A^2 + mgh = \frac{1}{2}mv_B^2 + mgh \Rightarrow v_B = v_A$$

$$\text{At point C: } \frac{1}{2}mv_A^2 + mgh = \frac{1}{2}mv_C^2 + mg\frac{h}{2} \Rightarrow v_C = \sqrt{v_A^2 + gh}$$

$$\text{At point D: } \frac{1}{2}mv_A^2 + mgh = \frac{1}{2}mv_D^2 \Rightarrow v_D = \sqrt{v_A^2 + 2gh}$$

7. Remember that potential energy has meaning only for a force that is conservative. A conservative force is that for which the work done in going from point A to point B is independent of the path chosen. Friction is an example of a non-conservative force and a potential energy cannot be defined. For a conservative force, $F = -\frac{dV}{dx}$. So, for

a spring, $V = \frac{1}{2}kx^2$ and so $F = -kx$.

8. Derivation of $F = -\frac{dV}{dx}$: If the particle moves distance Δx in a potential V , then

change in PE is ΔV where, $\Delta V = -F \Delta x$. From this, $F = -\frac{\Delta V}{\Delta x}$. Now let $\Delta x \rightarrow 0$.

$$\text{Hence, } F = -\lim_{\Delta x \rightarrow 0} \frac{\Delta V}{\Delta x} = -\frac{dV}{dx}.$$

Summary of Lecture 9 – MOMENTUM

1. Momentum is the "quantity of motion" possessed by a body. More precisely, it is defined as:

Mass of the body $\times$ Velocity of the body.

The dimensions of momentum are MLT^{-1} and the units of momentum are kg-m/s.

2. Momentum is a vector quantity and has both magnitude and direction, $p = m\vec{v}$. We can easily see that Newton's Second Law can be reexpressed in terms of momentum. When

I wrote it down originally, it was in the form $m\vec{a} = \vec{F}$. But since $\frac{d\vec{v}}{dt} = \vec{a}$, this can also be

written as $\frac{d\vec{p}}{dt} = \vec{F}$ (new form). In words, the rate of change of momentum of a body equals the total force acting upon it. Of course, the old and new are exactly the same,

$$\frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} = m \frac{d\vec{v}}{dt} = m\vec{a} = \vec{F}.$$

3. When there are many particles, then the total momentum $\vec{P}$ is,

$$\begin{aligned}\vec{P} &= \vec{p}_1 + \vec{p}_2 + \cdots \vec{p}_N \\ \frac{d}{dt} \vec{P} &= \frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} + \cdots \frac{d\vec{p}_N}{dt} \\ &= \vec{F}_1 + \vec{F}_2 + \cdots \vec{F}_N = \vec{F}\end{aligned}$$

This shows that when there are several particles, the rate at which the total momentum changes is equal to the total force. It makes sense!

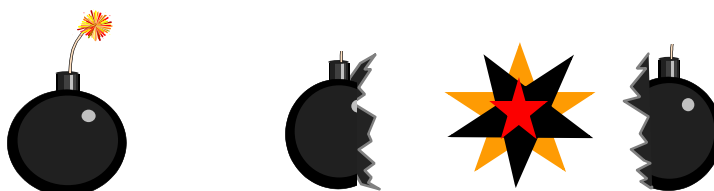
4. A very important conclusion of the above is that if the sum of the total external forces

vanishes, then the total momentum is conserved, $\sum \vec{F}_{ext} = 0 \Rightarrow \frac{d}{dt} \vec{P} = 0$. This is quite independent of what sort of forces act between the bodies - electric, gravitational, etc. - or how complicated these are. We shall see why this is so important from the following examples.

5. Two balls, which can only move along a straight line, collide with each other. The initial momentum is $P_i = m_1 u_1 + m_2 u_2$ and the final momentum is $P_f = m_1 v_1 + m_2 v_2$. Obviously one ball exerts a force on the other when they collide, so its momentum changes. But, from the fact that there is no external force acting on the balls, $P_i = P_f$, or $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$.

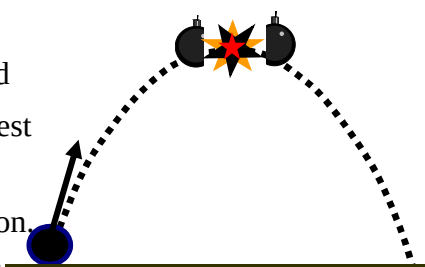
6. A bomb at rest explodes into two fragments. Before the explosion the total momentum is

zero. So obviously it is zero after the explosion as well, $\mathbf{P}_f = \mathbf{0}$. During the time that the explosion happens, the forces acting upon the pieces are very complicated and changing rapidly with time. But when all is said and done, there are two pieces flying away with a total zero final momentum $\mathbf{P}_f = m_1\mathbf{v}_1 + m_2\mathbf{v}_2$. Hence $m_1\mathbf{v}_1 = -m_2\mathbf{v}_2$. In other words, the fragments fly apart with equal momentum but in opposite directions. The centre-of-mass stays at rest. So, knowing the velocity of one fragment permits knowing the velocity of the other fragment.



7. If air resistance can be ignored, then we can do some interesting calculations with what we have learned.

So, suppose a shell is fired from a cannon with a speed 10 m/s at an angle 60° with the horizontal. At the highest point in its path it explodes into two pieces of equal masses. One of the pieces retraces its path to the cannon. Let us find the velocity of the other piece immediately after the explosion.



Solution: After the explosion: $P_{1x} = -5\frac{M}{2}$ (why?). But $P_{1x} + P_{2x} = P_x = M \times 10 \cos 60$

$$\Rightarrow P_{2x} = 5M + 5\frac{M}{2}. \text{ Now use: } P_{2x} = \frac{M}{2}v_{2x} \Rightarrow v_{2x} = 15 \text{ m/s.}$$

8. When you hit your thumb with a hammer it hurts, doesn't it? Why? Because a large amount of momentum has been destroyed in a short amount of time. If you wrap your thumb with foam, it will hurt less. To understand this better, remember that force is the rate of change of momentum: $F = \frac{dp}{dt} \Rightarrow dp = Fdt$. Now define the **impulse** I as:

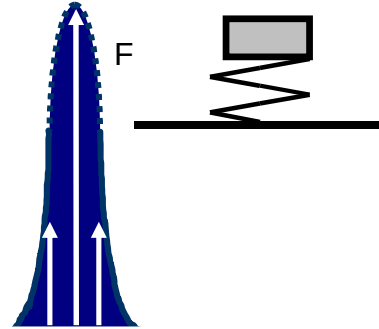
force $\times$ time over which the force acts.

If the force changes with time between the limits , then one should define I as,

$$I = \int_{t_1}^{t_2} Fdt. \text{ Since } \int_{t_1}^{t_2} Fdt = \int_{p_i}^{p_f} dp, \text{ therefore } I = p_f - p_i. \text{ In words, the change of momentum}$$

equals the impulse, which is equal to the area under the curve of force versus time. Even if you wrap your thumb in foam, the impulse is the same. But the force is definitely not!

9. Sometimes we only know the force numerically (i.e. there is no expression like $F=\text{something}$). But we still know what the integral means: it is the area under the curve of force versus time. The curve here is that of a hammer striking a table. Before the hammer strikes, the force is zero, reaches a peak, and goes back to zero.



Summary of Lecture 10 – COLLISIONS

1. Collisions are extremely important to understand because they happen all the time - electrons collide with atoms, a bat with a ball, cars with trucks, star galaxies with other galaxies,...In every case, the sum of the initial momenta equals the sum of the final momenta. This follows directly from Newton's Second Law, as we have already seen.
2. Take the simplest collision: two bodies of mass m_1 and m_2 moving with velocities u_1 and u_2 . After the collision they are moving with velocities v_1 and v_2 . From momentum conservation,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \Rightarrow m_1(u_1 - v_1) = m_2(v_2 - u_2)$$

This is as far as we can go. There are two unknowns but only one equation. However, if the collision is elastic then,

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \Rightarrow \frac{1}{2} m_1 (u_1^2 - v_1^2) = \frac{1}{2} m_2 (v_2^2 - u_2^2).$$

Combine the two equations above,

$$u_1 + v_1 = v_2 + u_2 \Rightarrow u_1 - u_2 = v_2 - v_1.$$

In words, this says that in an elastic collision the relative speed of the incoming particles equals the relative speed of the outgoing particles.

3. One can solve for v_1 and v_2 (please do it!) easily and find that:

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 + \left(\frac{2m_2}{m_1 + m_2} \right) u_2$$

$$v_2 = \left(\frac{2m_1}{m_1 + m_2} \right) u_1 + \left(\frac{m_2 - m_1}{m_1 + m_2} \right) u_2$$

Notice that if $m_1 = m_2$, then $v_1 = u_2$ and $v_2 = u_1$. So this says that after the collision, the bodies will just reverse their velocities and move on as before.

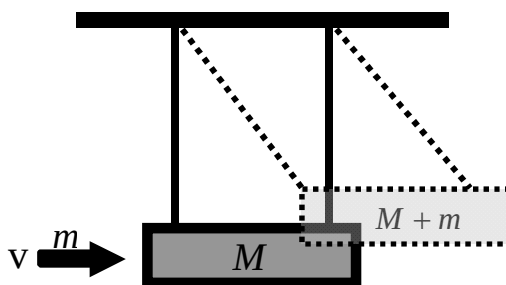
4. What if one of the bodies is much heavier than the other body, and the heavier body is at rest? In this case, $m_2 \gg m_1$ and $u_2 = 0$. We can immediately see that $v_1 = -u_1$ and $v_2 = 0$. This makes a lot of sense: the heavy body continues to stay at rest and the light body just bounces back with the same speed. In the lecture, you saw a demonstration of this!
5. And what if the lighter body (rickshaw) is at rest and is hit by the heavier body (truck)? In this case, $m_2 \ll m_1$ and $u_2 = 0$. From the above equation we see that $v_1 = u_1$ and $v_2 = 2u_1$. So the truck's speed is unaffected, but the poor rickshaw is thrust in the direction of the truck at twice the truck's speed!
6. Sometimes we wish to slow down particles by making them collide with other particles.

In a nuclear reactor, neutrons can be slowed down in this way. let's calculate the fraction by which the kinetic energy of a neutron of mass m_1 decreases in a head-on collision with an atomic nucleus of mass m_2 that is initially at rest:

Solution: $\frac{K_i - K_f}{K_i} = 1 - \frac{K_f}{K_i} = 1 - \frac{v_f^2}{v_i^2}$

For a target at rest: $v_f = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)v_i \quad \therefore \frac{K_i - K_f}{K_i} = \frac{4m_1m_2}{(m_1 + m_2)^2}$.

7. A bullet with mass m , is fired into a block of wood with mass M , suspended like a pendulum and makes a completely inelastic collision with it. After the impact, the block swings up to a maximum height y . What is the initial speed of the bullet?



Solution:

By conservation of momentum in the direction of the bullet,

$$mv = (m + M)V, \Rightarrow v = \frac{(m + M)}{m}V$$

The block goes up by distance y , and so gains potential energy. Now we can use the conservation of energy to give, $\frac{1}{2}(m + M)V^2 = (m + M)gy$, where V is the velocity acquired by the block+bullet in the upward direction just after the bullet strikes. Now use $V = \sqrt{2gy}$. So finally, the speed of the bullet is: $v = \frac{(m + M)}{m}\sqrt{2gy}$.

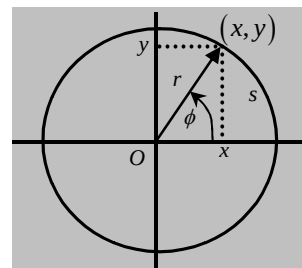
8. In 2 or 3 dimensions, you must apply conservation of momentum in each direction separately. The equation $\vec{P}_i = \vec{P}_f$ looks as if it is one equation, but it is actually 3 separate equations: $p_{ix} = p_{fx}, p_{iy} = p_{fy}, p_{iz} = p_{fz}$. On the other hand, suppose you had an elastic collision. In that case you would have only one extra equation coming from energy conservation, not three.

9. What happens to energy in an inelastic collision? Let's say that one body smashes into another body and breaks it into 20 pieces. To create 20 pieces requires doing work against the intermolecular forces, and the initial kinetic energy is used up for this.

Summary of Lecture 11 – ROTATIONAL KINEMATICS

1. Any rotation is specified by giving two pieces of information:

- The point about which the rotation occurs, i.e. the origin.
- The angle of rotation is denoted by ϕ in the diagram.



2. The arc length = radius $\times$ angular displacement, or $s = r\phi$.

Here ϕ is measured in radians. The maximum value of ϕ is 2π radians, which corresponds to 360 degrees or one full revolution. From this it follows that $1 \text{ radian} = 57.3^\circ$ or $1 \text{ radian} = 0.159 \text{ revolution}$. Obviously, if $\phi = 2\pi$, then $s = 2\pi r$, which is the total circumference.

3. Suppose that there is a particle located at the tip of the radius vector. Now we wish to describe the rotational *kinematics* of this particle, i.e. describe its motion as goes around the circle. So, suppose that the particle moves from angle ϕ_1 to ϕ_2 in time $t_2 - t_1$. Then, the

average angular speed $\bar{\omega}$ is defined as, $\bar{\omega} = \frac{\phi_2 - \phi_1}{t_2 - t_1} = \frac{\Delta\phi}{\Delta t}$. Suppose that we look at $\bar{\omega}$

over a very short time. Then, $\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\phi}{\Delta t} = \frac{d\phi}{dt}$ is called the instantaneous angular speed.

4. To familiarize ourselves with the notion of angular speed, let us compute ω for a clock second, minute and hour hands:

$$\omega_{\text{second}} = \frac{2\pi}{60} = 0.105 \text{ rad/s},$$

$$\omega_{\text{minute}} = \frac{2\pi}{60 \times 60} = 1.75 \times 10^{-3} \text{ rad/s},$$

$$\omega_{\text{hour}} = \frac{2\pi}{60 \times 60 \times 12} = 1.45 \times 10^{-4} \text{ rad/s}.$$

5. Just as we defined acceleration for linear motion, we also define acceleration for circular motion:

$$\bar{\alpha} \equiv \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\Delta\omega}{\Delta t} \quad (\text{average angular speed})$$

Hence, $\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t}$ becomes $\alpha = \frac{d\omega}{dt} = \frac{d}{dt} \frac{d\phi}{dt} = \frac{d^2\phi}{dt^2}$ (angular acceleration). Let us

see what this means for the speed with which a particle goes around. Now use $s = r\phi$.

Differentiate with respect to time t : $\frac{ds}{dt} = r \frac{d\phi}{dt}$. The rate of change of arc length s is clearly

what we should call the circular speed, v . So $v = r\omega$. Since r is held fixed, it follows that

$$\frac{dv}{dt} = r \frac{d\omega}{dt}. \text{ Now define } a_T = \frac{dv}{dt}. \text{ Obviously, } a_T = r\alpha. \text{ Here } T \text{ stands for tangential, i.e.}$$

in the direction of increasing s .

6. Compare the formulae for constant linear and angular accelerations:

LINEAR

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2} at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

ANGULAR

$$\omega = \omega_0 + \alpha t$$

$$\phi = \phi_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha(\phi - \phi_0)$$

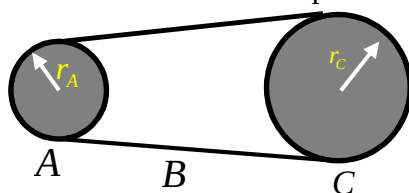
Why are they almost identical even though they describe two totally different physical situations. Answer: because the mathematics is identical!

7. The angular speed of a car engine is increased from 1170 rev/min to 2880 rev/min in 12.6 s. a) Find the average angular acceleration in rev/min^2 . (b) How many revolutions does the engine make during this time?

SOLUTION: this is a straightforward application of the formulae in point 5 above.

$$\alpha = \frac{\omega_f - \omega_i}{t} = 8140 \text{ rev/min}^2, \quad \phi = \omega_i t + \frac{1}{2} \alpha t^2 = 425 \text{ rev.}$$

8. Wheel A of radius $r_A = 10.0 \text{ cm}$ is coupled by a chain B to wheel C of radius $r_C = 25.0 \text{ cm}$. Wheel A increases its angular speed from rest at a uniform rate of 1.60 rad/s^2 . Determine the time for wheel C to reach a rotational speed of 100 rev/min.



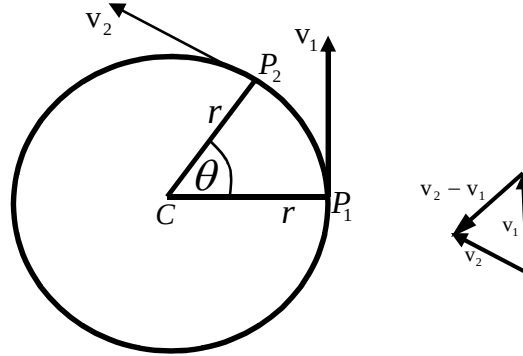
SOLUTION: Obviously every part of the chain moves with the same speed and so

$$v_A = v_C. \text{ Hence } r_A \omega_A = r_C \omega_C \Rightarrow \omega_A = \frac{r_C \omega_C}{r_A}. \text{ From the definition of acceleration,}$$

$$\alpha = \frac{\omega_A - 0}{t}. \text{ From this, } t = \frac{\omega_A}{\alpha} = \frac{r_C \omega_C}{r_A \alpha} = 16.4 \text{ s.}$$

9. Imagine a disc going around. All particles on the disc will have same ' ω ' and ' α ' but different ' v ' and ' a '. Clearly ' ω ' and ' α ' are simpler choices !!

10. Now consider a particle going around a circle at constant speed. You might think that constant speed means no acceleration. Bu this is wrong! It is changing its direction and accelerating. This is called "centripetal acceleration", meaning acceleration directed towards the centre of the circle. Look at the figure below:



Note that the distance between points P_1 and P_2 is $\Delta r = v\Delta t \approx r\theta$. Similarly,

$$\Delta v \approx v\theta \Rightarrow \bar{a} = \frac{\Delta v}{\Delta t} \approx \frac{v\theta}{r\theta/v} = \frac{v^2}{r}. \text{ More generally, } a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{v^2}{r}. \text{ In vector form,}$$

$$\vec{a}_R = -\frac{v^2}{r}\hat{r}. \text{ The negative sign indicates that the acceleration is towards the centre.}$$

12. Vector Cross Products: The vector crossproduct of two vectors is defined as:

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n} \text{ where } \hat{n} \text{ is a unit vector that is perpendicular to both } \vec{A} \text{ and } \vec{B}.$$

Apply this definition to unit vectors in 3-dimensions: $\hat{i} \times \hat{j} = \hat{k}$, $\hat{k} \times \hat{i} = \hat{j}$, $\hat{j} \times \hat{k} = \hat{i}$.

13. Some key properties of the crossproduct:

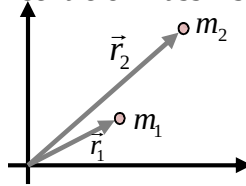
- $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$
- $\vec{A} \times \vec{A} = 0$
- $(\vec{A} + \vec{B}) \times \vec{C} = (\vec{A} \times \vec{B}) + (\vec{A} \times \vec{C})$
- $\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$

14. The cross product is only definable in 3 dimensions and has no meaning in 2-d. This is unlike the dot product which as a meaning in any number of dimensions.

Summary of Lecture 12 – PHYSICS OF MANY PARTICLES

1. A body is made of a collection of particles. We would like to think of this body having

a "centre". For two masses the "centre of mass" is defined as: $\vec{r}_{cm} \equiv \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$.



In 2 dimensions (i.e. a plane) this is actually two equations:

$$x_{cm} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2} \quad \text{and} \quad y_{cm} = \frac{m_1y_1 + m_2y_2}{m_1 + m_2}.$$

These give the coordinates of the centre of mass of the two-particle system.

2. Example: one mass is placed at $x = 2cm$ and a second mass, equal to the first, is placed at $x = 6cm$. The cm position lies halfway between the two as you can see from:

$$x_{cm} = \frac{mx_1 + mx_2}{m + m} = \frac{2m + 6m}{2m} = 4cm.$$

Note that there is no physical body that is actually located at $x_{cm} = 4cm$! So the centre of mass can actually be a point where there is no matter. Now suppose that the first mass is three times bigger than the first:

$$x_{cm} = \frac{(3m)x_1 + mx_2}{3m + m} = \frac{2(3m) + 6m}{4m} = 3cm$$

This shows that the cm lies closer to the heavier body. This is always true.

3. For N masses the obvious generalization of the centre of mass position is the following:

$$\vec{r}_{cm} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + \cdots + m_N\vec{r}_N}{m_1 + m_2 + \cdots + m_N} = \frac{1}{M} \left(\sum_{n=1}^{n=N} m_n \vec{r}_n \right).$$

In words, this says that the following: choose any origin and draw vectors $\vec{r}_1, \vec{r}_2, \cdots \vec{r}_N$ that connect to the masses $m_1, m_2, \cdots m_N$. Heavier masses get more importance in the sum.

So suppose that m_2 is much larger than any of the others. If so, $\vec{r}_{cm} \approx \frac{m_2\vec{r}_2}{m_2} = \vec{r}_2$. Hence, the cm is very close to the position vector of m_2 .

4. For symmetrical objects, it is easy to see where the cm position lies: for a sphere or circle it lies at the centre; for a cylinder it is on the axis halfway between the two faces, etc.

5. Our definition of the cm allows Newton's Second Law to be written for entire collection

of particles:

$$\vec{v}_{cm} = \frac{d\vec{r}_{cm}}{dt} = \frac{1}{M} \left(\sum m_n \vec{v}_n \right)$$

$$\vec{a}_{cm} = \frac{d\vec{v}_{cm}}{dt} = \frac{1}{M} \left(\sum m_n \vec{a}_n \right)$$

$$\therefore M \vec{a}_{cm} = \sum \vec{F}_n = \sum (\vec{F}_{ext} + \vec{F}_{int}) \quad \text{use } \sum \vec{F}_{int} = 0$$

$$\Rightarrow \sum \vec{F}_{ext} = M \vec{a}_{cm} \quad (\text{the sum of external forces is what causes acceleration})$$

In the above we have used Newton's Third Law as well: $\vec{F}_{12} + \vec{F}_{21} = 0$ etc.

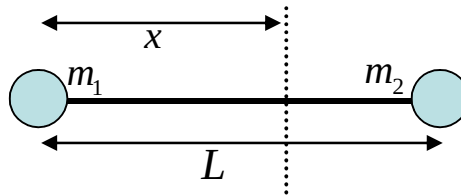
6. Consider rotational motion now for a rigid system of N particles. Rigid means that all particles have a fixed distance from the origin. The kinetic energy is,

$$\begin{aligned} K &= \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 + \frac{1}{2} m_3 v_3^2 + \dots \\ &= \frac{1}{2} m_1 r_1^2 \omega^2 + \frac{1}{2} m_2 r_2^2 \omega^2 + \frac{1}{2} m_3 r_3^2 \omega^2 + \dots \\ &= \frac{1}{2} \left(\sum m_i r_i^2 \right) \omega^2 \end{aligned}$$

Now suppose that we define the "moment of inertia" $I \equiv \sum m_i r_i^2$. Then clearly the

kinetic energy is $K = \frac{1}{2} I \omega^2$. How similar this is to $K = \frac{1}{2} M v^2$!

7. To familiarize ourselves with I, let us consider the following: Two particles m_1 and m_2 are connected by a light rigid rod of length L . Neglecting the mass of the rod, find the rotational inertia I of this system about an axis perpendicular to the rod and at a distance x from m_1 .



Answer: $I = m_1 x^2 + m_2 (L - x)^2$. Of course, this was quite trivial. Now we can ask a more interesting question: For what x is I the largest? Now, near a maximum, the slope of a function is zero. So calculate $\frac{dI}{dx}$ and then put it equal to zero:

$$\therefore \frac{dI}{dx} = 2m_1 x - 2m_2 (L - x) = 0 \quad \Rightarrow \quad x_{\max} = \frac{m_2 L}{m_1 + m_2}.$$

8. Although matter is made up of discrete atoms, even if one takes small pieces of any body, there are billions of atoms within it. So it is useful to think of matter as being continuously distributed. Since a sum $\sum$ becomes an integral $\int$, it is obvious that the new definitions of I and $\vec{R}_{cm}$ become:

$$I \equiv \int r^2 dm \quad \text{and} \quad \vec{R}_{cm} \equiv \frac{1}{M} \int \vec{r} dm.$$

9. A simple application: suppose there is a hoop with mass distributed uniformly over it.

The moment of inertia is: $I = \int r^2 dm = R^2 \int dm = MR^2$.

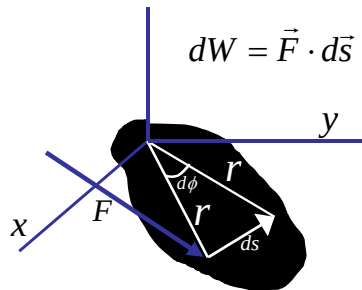
10. A less trivial application: instead of a hoop as above, now consider a solid plate:

$$I = \int r^2 dm \quad (dm = 2\pi r dr \rho_0)$$

$$= \int_0^R 2\pi r^3 dr \rho_0 = \frac{1}{2} (\pi R^2 \rho_0) R^2 = \frac{1}{2} M R^2$$

11. You have seen that it is easier to turn things (e.g. a nut, when changing a car's tyre after a puncture) when the applied force acts at a greater distance. This is because the *torque* τ is greater. We define $\vec{\tau} = \vec{r} \times \vec{F}$ from the magnitude is $\tau = rF \sin \theta$. Here θ is the angle between the radius vector and the force.

12. Remember that when a force $\vec{F}$ acts through a distance $d\vec{r}$ it does an amount of work equal to $\vec{F} \cdot d\vec{r}$. Now let us ask how much work is done when a torque acts through a certain angle as in the diagram below:



The small amount of work done is:

$$dW = \vec{F} \cdot d\vec{s} = F \cos \theta ds = (F \cos \theta)(r d\phi) = \tau d\phi$$

Add the contributions coming from from all particles,

$$dW_{net} = (F_1 \cos \theta_1) r_1 d\phi + (F_2 \cos \theta_2) r_2 d\phi + \dots + (F_n \cos \theta_n) r_n d\phi$$

$$= (\tau_1 + \tau_2 + \dots + \tau_n) d\phi$$

$$\therefore dW_{net} = \left(\sum \tau_{ext} \right) d\phi = \left(\sum \tau_{ext} \right) \omega dt \quad \dots(1)$$

Now consider the change in the kinetic energy K ,

$$dK = d \left(\frac{1}{2} I \omega^2 \right) = I \omega d\omega = (I \alpha) \omega dt \quad \dots(2)$$

By conservation of energy, the change in K must equal the work done, and so:

$$dW_{net} = dK \Rightarrow \sum \tau_{ext} = I \alpha$$

In words this says that the total torque equals the moment of inertia times the angular acceleration. This is just like Newton's second law, but for rotational motion !

13.A comparison between linear and rotational motion quantities and formulae:

LINEAR	ROTATIONAL
x, M	ϕ, I
$v = \frac{dx}{dt}$	$\omega = \frac{d\phi}{dt}$
$a = \frac{dv}{dt}$	$\alpha = \frac{d\omega}{dt}$
$F = Ma$	$\tau = I\alpha$
$K = \frac{1}{2} M v^2$	$K = \frac{1}{2} I \omega^2$
$W = \int F dx$	$W = \int \tau d\phi$

14. Rotational and translational motion can occur simultaneously. For example a car's wheel rotates and translates. In this case the total kinetic energy is clearly the sum of the energies of the two motions: $K = \frac{1}{2} M v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$.

15. It will take a little work to prove the following fact that I simply stated above: for a system of N particles, the total kinetic energy divides up neatly into the kinetic energy of rotation and translation. Start with the expression for kinetic energy and write $\vec{v}_{cm} + \vec{v}'_i$ where $\vec{v}'_i$ is the velocity of a particle with respect to the cm frame,

$$K = \sum \frac{1}{2} m_i v_i^2 = \sum \frac{1}{2} m_i \vec{v}_i \cdot \vec{v}_i$$

$$= \sum \frac{1}{2} m_i (\vec{v}_{cm} + \vec{v}'_i) \cdot (\vec{v}_{cm} + \vec{v}'_i)$$

$$= \sum \frac{1}{2} m_i (v_{cm}^2 + 2\vec{v}_{cm} \cdot \vec{v}'_i + v_i'^2)$$

Now, $\sum m_i \vec{v}_{\text{cm}} \cdot \vec{v}'_i = \vec{v}_{\text{cm}} \cdot \sum m_i \vec{v}'_i = \vec{v}_{\text{cm}} \cdot \sum \vec{p}'_i = 0$. Why? because the total momentum is zero in the cm frame! So this brings us to our result that,

$$\begin{aligned} K &= \sum \frac{1}{2} m_i v_{\text{cm}}^2 + \sum \frac{1}{2} m_i v_i'^2 \\ &= \frac{1}{2} M v_{\text{cm}}^2 + \sum \frac{1}{2} m_i r_i'^2 \omega^2. \end{aligned}$$

Summary of Lecture 13 – ANGULAR MOMENTUM

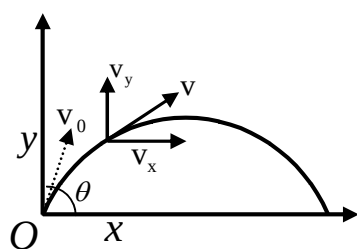
1. Recall the definition of angular momentum: $\vec{L} = \vec{r} \times \vec{p}$. The magnitude can be written in several different but equivalent ways,

$$(a) L = r p \sin \theta$$

$$(b) L = (r \sin \theta) p = r_{\perp} p$$

$$(c) L = r(p \sin \theta) = r p_{\perp}$$

2. Let us use this definition to calculate the angular momentum of a projectile thrown from the ground at an angle θ . Obviously, initial angular momentum is zero (why?).



We know what the projectile's coordinates will be at time t after launch,

$$x = (v_0 \cos \theta)t, \quad y = (v_0 \sin \theta)t - \frac{1}{2}gt^2$$

as well as the velocity components,

$$v_x = v_0 \cos \theta, \quad v_y = v_0 \sin \theta - gt.$$

$$\begin{aligned} \text{Hence, } \vec{L} = \vec{r} \times \vec{p} &= (x\hat{i} + y\hat{j}) \times (v_x\hat{i} + v_y\hat{j})m = m(xv_y - yv_x)\hat{k} \\ &= m\left(\frac{1}{2}gt^2v_0 \cos \theta - gt^2v_0 \cos \theta\right)\hat{k} = -\frac{m}{2}gt^2v_0 \cos \theta \hat{k}. \end{aligned}$$

In the above, $\hat{k} = \hat{i} \times \hat{j}$ is a unit vector perpendicular to the paper. You can see here that the angular momentum increases as t^2 .

2. Momentum changes because a force makes it change. What makes angular momentum change? Answer: torque. Here is the definition again: $\vec{\tau} = \vec{r} \times \vec{F}$. Now let us establish a very important relation between torque and rate of change of L.

Begin:

$$\begin{aligned} \vec{L} = \vec{r} \times \vec{p}. \text{ At a slightly later time, } \quad \vec{L} + \Delta\vec{L} &= (\vec{r} + \Delta\vec{r}) \times (\vec{p} + \Delta\vec{p}) \\ &= \vec{r} \times \vec{p} + \vec{r} \times \Delta\vec{p} + \Delta\vec{r} \times \vec{p} + \Delta\vec{r} \times \Delta\vec{p} \end{aligned}$$

$$\text{By subtracting, } \Delta\vec{L} = \vec{r} \times \Delta\vec{p} + \Delta\vec{r} \times \vec{p} \quad \frac{\Delta\vec{L}}{\Delta t} = \frac{\vec{r} \times \Delta\vec{p} + \Delta\vec{r} \times \vec{p}}{\Delta t} = \vec{r} \times \frac{\Delta\vec{p}}{\Delta t} + \frac{\Delta\vec{r}}{\Delta t} \times \vec{p}.$$

Now divide by the time difference and then take limit as $\Delta t \rightarrow 0$:

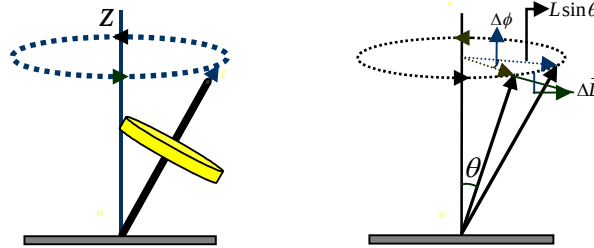
$$\therefore \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{L}}{\Delta t} = \frac{d\vec{L}}{dt} \quad \therefore \quad \frac{d\vec{L}}{dt} = \vec{r} \times \frac{d\vec{p}}{dt} + \frac{d\vec{r}}{dt} \times \vec{p}$$

But $\frac{d\vec{r}}{dt}$ is $\vec{v}$ and $\vec{p} = m\vec{v}$! Also, $\frac{d\vec{r}}{dt} \times \vec{p} = \vec{v} \times m\vec{v} = m(\vec{v} \times \vec{v}) = 0$. So we arrive

at $\frac{d\vec{L}}{dt} = \vec{r} \times \frac{d\vec{p}}{dt}$. Now use Newton's Law, $\vec{F} = \frac{d\vec{p}}{dt}$. Hence we get the fundamental

equation $\frac{d\vec{L}}{dt} = \vec{r} \times \vec{F}$, or $\frac{d\vec{L}}{dt} = \vec{\tau}$. So, just as a particle's momentum changes with time because of a force, a particle's *angular* momentum changes with time because of a torque.

3. As you saw in the lecture, the spinning top is an excellent application of $\frac{d\vec{L}}{dt} = \vec{\tau}$.



Start from $\vec{\tau} = \vec{r} \times \vec{F}$ where $\vec{F} = m\vec{g} \therefore \tau = Mgr \sin \theta$. But $\vec{\tau}$ is perpendicular to $\vec{L}$ and so it cannot change the magnitude of $\vec{L}$. Only the direction changes. Since $\Delta \vec{L} = \vec{\tau} \Delta t$,

you can see from the diagram that $\Delta \phi = \frac{\Delta L}{L \sin \theta} = \frac{\tau \Delta t}{L \sin \theta}$. So the precession speed ω_p

is: $\omega_p = \frac{\Delta \phi}{\Delta t} = \frac{\tau}{L \sin \theta} = \frac{Mgr \sin \theta}{L \sin \theta} = \frac{Mgr}{L}$. As the top slows down due to friction and L decreases, the top precesses faster and faster.

4. Now consider the case of many particles. Choose any origin with particles moving with respect to it. We want to write down the total angular momentum,

$$\vec{L} = \vec{L}_1 + \vec{L}_2 + \cdots + \vec{L}_N = \sum_{n=1}^N \vec{L}_n$$

$$\frac{d\vec{L}}{dt} = \frac{d\vec{L}_1}{dt} + \frac{d\vec{L}_2}{dt} + \cdots + \frac{d\vec{L}_N}{dt} = \sum_{n=1}^N \frac{d\vec{L}_n}{dt}$$

Since $\frac{d\vec{L}_n}{dt} = \vec{\tau}_n$, it follows that $\frac{d\vec{L}}{dt} = \sum_{n=1}^N \vec{\tau}_n$. Thus the time rate of change of the total angular momentum of a system of particles equals the net torque acting on the system. I showed earlier that internal forces cancel. So also do internal torques, as we shall see.

5. The torque on a system of particles can come both from external and internal forces. For example, there could be charged particles which attract/repel each other while they are all in an external gravitational field. Mathematically,

$\sum \vec{\tau} = \sum \vec{\tau}_{\text{int}} + \sum \vec{\tau}_{\text{ext}}$. Now, if the forces between two particles not only are equal and opposite but are also directed along the line joining the two particles, then can easily show that the total internal torque, $\sum \vec{\tau}_{\text{int}} = 0$. Take the case of two particles,

$$\sum \vec{\tau}_{\text{int}} = \vec{\tau}_1 + \vec{\tau}_2 = \vec{r}_1 \times \vec{F}_{12} + \vec{r}_2 \times \vec{F}_{21}$$

But $\vec{F}_{12} = -\vec{F}_{21} = F \hat{r}_{12}$, $\therefore \sum \vec{\tau}_{\text{int}} = (\vec{r}_1 - \vec{r}_2) \times \vec{F}_{12} = \vec{r}_{12} \times (F \hat{r}_{12}) = F (\vec{r}_{12} \times \hat{r}_{12}) = 0$

Thus net external torque acting on a system of particles is equal to the time rate of change of the of the total angular momentum of the system.

6. It follows from $\frac{d\vec{L}}{dt} = \vec{\tau}$ that if no net external torque acts on the system, then the angular

momentum of the system does not change with the time: $\frac{d\vec{L}}{dt} = 0 \Rightarrow \vec{L} = \text{a constant}$.

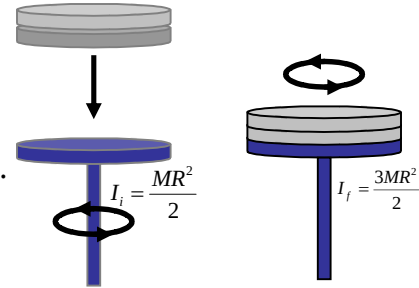
This is simple but extremely important. Let us apply this

to the system shown here. Two stationary discs, each with

$I = \frac{1}{2}MR^2$, fall on top of a rotating disc. The total angular

momentum is unchanged so, $I_i \omega_i = I_f \omega_f \Rightarrow \omega_f = \omega_i \left(\frac{I_i}{I_f} \right)$.

Hence, $\omega_f = \omega_i \left(\frac{MR^2}{2} \times \frac{2}{3MR^2} \right) = \frac{1}{3} \omega_i$.



7. You should be aware of the similarities and differences between the equations for linear

and rotational motion: $\vec{F} = \frac{d\vec{p}}{dt} \Leftrightarrow \vec{\tau} = \frac{d\vec{L}}{dt}$. One big difference is that for momentum,

$\vec{p} = m\vec{v}$, it does not matter where you pick the origin. But $\vec{L}$ definitely depends on the choice of the origin. So changing $\vec{r}$ to $\vec{c} + \vec{r}$ changes $\vec{L}$ to $\vec{L}'$:

$$\vec{L}' = \vec{r}' \times \vec{p} = (\vec{c} + \vec{r}) \times \vec{p} = \vec{c} \times \vec{p} + \vec{L}.$$

8. Linear and angular acceleration:

$$\begin{aligned} \vec{a} &= \frac{d\vec{v}}{dt} = \frac{d}{dt}(\vec{\omega} \times \vec{r}) = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt} \\ &= \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{v} \end{aligned}$$

So the acceleration has a tangential and radial part, $\vec{a} = \vec{a}_T + \vec{a}_R$.

Summary of Lecture 14 – EQUILIBRIUM OF RIGID BODIES

1. A rigid body is one where all parts of the body are fixed relative to each other (for example, a pencil). Fluids and gases are non-rigid.

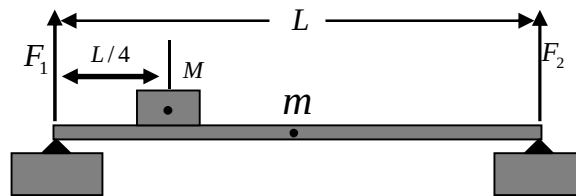
2. The translational motion of the centre of mass of a rigid body is governed by:

$$\frac{d\vec{P}}{dt} = \vec{F} \text{ where } \vec{F} = \sum \vec{F}_{\text{ext}} \text{ is the net external force.}$$

Similarly, for rotational motion, $\frac{d\vec{L}}{dt} = \vec{\tau}$ where $\vec{\tau} = \sum \vec{\tau}_{\text{ext}}$ is the net external torque.

3. A rigid body is in mechanical equilibrium if **both** the linear momentum $\vec{P}$ and angular momentum $\vec{L}$ have a constant value. i.e., $\frac{d\vec{P}}{dt} = 0$ and $\frac{d\vec{L}}{dt} = 0$. **Static equilibrium** refers to $\vec{P} = 0$ and $\vec{L} = 0$.

4. As an example of static equilibrium, consider a beam resting on supports:



We want to find the forces F_1 and F_2 with which the supports push on the rod in the upwards direction. First, balance forces in the vertical y direction:

$$\sum F_y = F_1 + F_2 - Mg - mg = 0$$

Now demand that the total torque vanishes:

$$\sum \tau_y = (F_1)(0) + (F_2)(L) - (Mg)(L/4) - (mg)(L/2) = 0$$

From these two conditions you can solve for F_1 and F_2 ,

$$F_1 = \frac{(3M + 2m)g}{4}, \quad \text{and} \quad F_2 = \frac{(M + 2m)g}{4}.$$

5. Angular momentum and torque depend on where you choose the origin of your coordinates. However, I shall now prove that for a body in equilibrium, the choice of origin does not matter. Let's start with the origin O and calculate the torque about O,

$$\vec{\tau}_O = \vec{\tau}_1 + \vec{\tau}_2 + \cdots + \vec{\tau}_N = \vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 + \cdots + \vec{r}_N \times \vec{F}_N$$

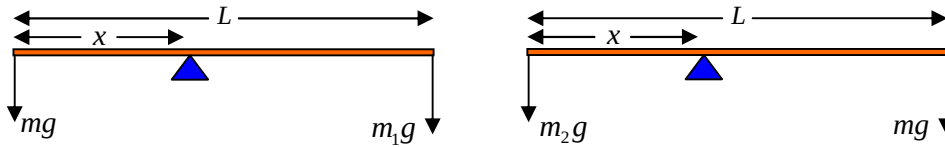
Now, if we take a second point P, then all distances will be measured from P and each

vector will be shifted by an amount $\vec{r}_p$. Hence the torque about P is,

$$\begin{aligned}\vec{\tau}_p &= (\vec{r}_1 - \vec{r}_p) \times \vec{F}_1 + (\vec{r}_2 - \vec{r}_p) \times \vec{F}_2 + \cdots + (\vec{r}_N - \vec{r}_p) \times \vec{F}_N \\ &= [\vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 + \cdots + \vec{r}_N \times \vec{F}_N] - [\vec{r}_p \times \vec{F}_1 + \vec{r}_p \times \vec{F}_2 + \cdots + \vec{r}_p \times \vec{F}_N] \\ &= \vec{\tau}_O - [\vec{r}_p \times (\vec{F}_1 + \vec{F}_2 + \cdots + \vec{F}_N)] \\ &= \vec{\tau}_O - [\vec{r}_p \times (\sum \vec{F}_{ext})]\end{aligned}$$

but $\sum \vec{F}_{ext} = 0$, for a body in translational equilibrium $\therefore \vec{\tau}_p = \vec{\tau}_O$.

6. Let us use the equilibrium conditions to do something of definite practical importance. Consider the balance below which is in equilibrium when two known weights are hung as shown. We want to know m in terms of m_1 and m_2 .

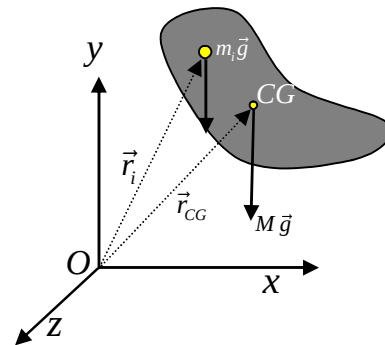


Taking the torques about the knife edge in the two cases, we have:

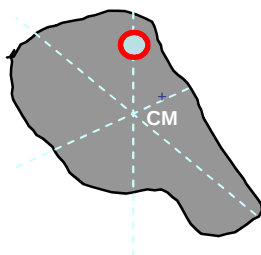
$$\begin{aligned}mgx &= m_1g(L-x) \text{ and } m_2gx = mg(L-x) \\ \Rightarrow \frac{m}{m_2} &= \frac{m_1}{m} \text{ or } m = \sqrt{m_1m_2}.\end{aligned}$$

Remarkably, we do not need the values of x or L .

7. **Centre of Gravity.** The centre of gravity is the average location of the weight of an object. This is not quite the same as the centre of mass of a body (see lecture 12) but suppose the gravitational acceleration $\vec{g}$ has the same value at all points of a body. Then: 1) The weight is equal to $M\vec{g}$, and 2) the centre of gravity coincides with the centre of mass. Remember that weight is force, so the CG is really the centre of gravitational force acting on the body. The net force on the whole body = sum of forces over all individual particles, $\sum \vec{F} = \sum m_i \vec{g}$. If $\vec{g}$ has the same value at all points of the body, then $\sum \vec{F} = \vec{g} \sum m_i = M\vec{g}$. So the net torque about the origin O is $\sum \vec{\tau} = \sum (\vec{r}_i \times m_i \vec{g}) = \sum (m_i \vec{r}_i \times \vec{g})$. Hence, $\sum \vec{\tau} = M \vec{r}_{cm} \times \vec{g} = \vec{r}_{cm} \times M \vec{g}$. So the torque due to gravity about the centre of mass of a body (i.e. at $\vec{r}_{cm} = 0$) is zero !!



8. In the demonstration I showed, you saw how to find the CG of an irregular object by simply suspending it on a pivot,

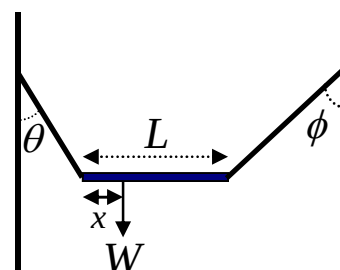


9. Let's solve a problem of static equilibrium: A non-uniform bar of weight W is suspended at rest in a horizontal position by two light cords. Find the distance x from the left-hand end to the center of gravity.

Solution : Call the tensions T_1 and T_2 . Put the forces in both directions equal to zero,

a) $T_2 \sin \phi - T_1 \sin \theta = 0$ (horizontal)

b) $T_2 \cos \phi + T_1 \cos \theta - W = 0$ (vertical) $\Rightarrow T_2 = \frac{W}{\sin(\theta + \phi)}$



The torque about any point must vanish. Let us choose that point to be one end of the bar,

$$-Wx + (T_2 \cos \phi)L = 0 \Rightarrow x = \frac{(T_2 \cos \phi)L}{W} = \frac{L \cos \phi}{\sin(\theta + \phi)}$$

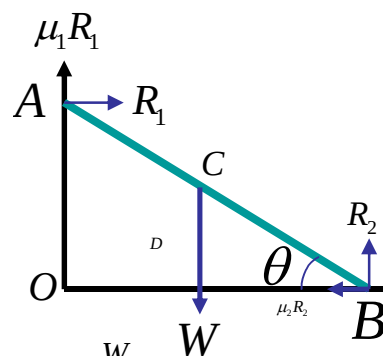
10. Here is another problem of the same kind: find the least angle θ at which the rod can lean to the horizontal without slipping.

Solution : Considering the translational equilibrium of the rod, $R_1 = \mu_2 R_2$ and $R_2 + \mu_1 R_1 = W$. This gives,

$$R_2 = \frac{W}{(1 + \mu_1 \mu_2)}.$$

Now consider rotational equilibrium about the point A: $R_2 \times OB = W \times OD + \mu_2 R_2 \times OA$

$$\text{or, } R_2 \times AB \cos \theta = W \times \frac{AB \cos \theta}{2} + \mu_2 R_2 \times AB \sin \theta.$$



$$\text{This gives } \cos \theta \left(R_2 - \frac{W}{2} \right) = \mu_2 R_2 \sin \theta \text{ from which } \tan \theta = \frac{R_2 - \frac{W}{2}}{\mu_2 R_2} \text{ with } R_2 = \frac{W}{(1 + \mu_1 \mu_2)}.$$

$$\text{Using this value of } R_2, \text{ we get } \tan \theta = \frac{1 - \mu_1 \mu_2}{2 \mu_2}.$$

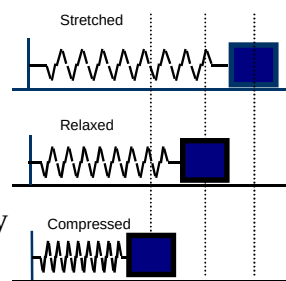
10. **Types of Equilibrium.** In the lecture you heard about:

- a) Stable equilibrium: object returns to its original position if displaced slightly.
- b) Unstable equilibrium: object moves farther away from its original position if displaced slightly.
- c) Neutral equilibrium: object stays in its new position if displaced slightly.

Summary of Lecture 15 – OSCILLATIONS: I

1. An oscillation is any self-repeating motion. This motion is characterized by:
 - a) The period T , which is the time for completing one full cycle.
 - b) The frequency $f = 1/T$, which is the number of cycles per second. (Another frequently used symbol is ν).
 - c) The amplitude A , which is the maximum displacement from equilibrium (or the size of the oscillation).
2. Why does a system oscillate? It does so because a force is always directed towards a central equilibrium position. In other words, the force always acts to return the object to its equilibrium position. So the object will oscillate around the equilibrium position. The restoring force depends on the displacement $F_{\text{restore}} = -k \Delta x$, where Δx is the distance away from the equilibrium point, the negative sign shows that the force acts towards the equilibrium point, and k is a constant that gives the strength of the restoring force.

3. Let us understand these matters in the context of a spring tied to a mass that can move freely over a frictionless surface. The force $F(x) = -kx$ (or $-k\Delta x$ because the extension Δx will be called x for short). In the first diagram x is positive, x is negative in the second, and zero in the middle one. The energy stored in the spring, $U(x) = \frac{1}{2}kx^2$, is positive in the first and



third diagrams and zero in the middle one. Now we will use Newton's second law to derive a differential equation that describes the motion of the mass: From $F(x) = -kx$ and $ma = F$ it follows that $m \frac{d^2x}{dt^2} = -kx$, or $\frac{d^2x}{dt^2} + \omega^2 x = 0$ where $\omega^2 \equiv \frac{k}{m}$. This is the equation of motion of a simple harmonic oscillator (SHO) and is seen widely in many different branches of physics. Although we have derived it for the case of a mass and spring, it occurs again and again. The only difference is that ω , which is called the oscillator frequency, is defined differently depending on the situation.

4. In order to solve the SHO equation, we shall first learn how to differentiate some elementary trigonometric functions. So let us first learn how to calculate $\frac{d}{dt} \cos \omega t$ starting from the basic definition of a derivative:

Start : $x(t) = \cos \omega t$ and $x(t + \Delta t) = \cos \omega(t + \Delta t)$. Take the difference:

$$\begin{aligned} x(t + \Delta t) - x(t) &= \cos \omega(t + \Delta t) - \cos \omega t \\ &= -\sin \omega \Delta t \sin(\omega t + \omega \Delta t / 2) \\ &\approx -\omega \Delta t \sin \omega t \quad \text{as } \Delta t \text{ becomes very small.} \end{aligned}$$

$$\therefore \frac{d}{dt} \cos \omega t = -\omega \sin \omega t.$$

(Here you should know that $\sin \theta \approx \theta$ for small θ , easily proved by drawing triangles.)

You should also derive and remember a second important result:

$$\frac{d}{dt} \sin \omega t = \omega \cos \omega t.$$

(Here you should know that $\cos \theta \approx 1$ for small θ .)

5. What happens if you differentiate twice?

$$\begin{aligned} \frac{d^2}{dt^2}(\sin \omega t) &= \omega \frac{d}{dt} \cos \omega t = -\omega^2 \sin \omega t \\ \frac{d^2}{dt^2}(\cos \omega t) &= -\omega \frac{d}{dt} \sin \omega t = -\omega^2 \cos \omega t. \end{aligned}$$

So twice differentiating either $\sin \omega t$ or $\cos \omega t$ gives the same function back!

6. Having done all the work above, now you can easily see that any function of the

form $x(t) = a \cos \omega t + b \sin \omega t$ satisfies $\frac{d^2 x}{dt^2} = -\omega^2 x$. But what do ω , a , b represent ?

a) The significance of ω becomes clear if you replace t by $t + \frac{2\pi}{\omega}$ in either $\sin \omega t$

or $\cos \omega t$. You can see that $\cos \omega \left(t + \frac{2\pi}{\omega} \right) = \cos(\omega t + 2\pi) = \cos \omega t$. That is, the

function merely repeats itself after a time $2\pi / \omega$. So $2\pi / \omega$ is really the period of the

motion T , $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$. The frequency ν of the oscillator is the number of

complete vibrations per unit time: $\nu = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$ so $\omega = 2\pi\nu = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$.

Sometimes ω is also called the angular frequency. Note that $\dim[\omega] = T^{-1}$, from it is clear that the unit of ω is radian/second.

b) To understand what a and b mean let us note that from $x(t) = a \cos \omega t + b \sin \omega t$ it

follows that $x(0) = a$ and that $\frac{d}{dt} x(t) = -\omega a \sin \omega t + \omega b \cos \omega t = \omega b$ (at $t = 0$). Thus,

a is the initial position, and b is the initial velocity divided by ω .

- c) To understand what a and b mean let us note that from $x(t) = a \cos \omega t + b \sin \omega t$ it follows that $x(0) = a$ and that $\frac{d}{dt} x(t) = -\omega a \sin \omega t + \omega b \cos \omega t = \omega b$ (at $t = 0$). Thus, a is the initial position, and b is the initial velocity divided by ω .
- d) The solution can also be written as: $x(t) = x_m \cos(\omega t + \phi)$. Since $\cos$ and $\sin$ never become bigger than 1, or less than -1, it follows that $-x_m \leq x \leq +x_m$. For obvious reason x_m is called the amplitude of the motion. The frequency of the simple harmonic motion is independent of the amplitude of the motion.
- e) The quantity $\theta = \omega t + \phi$ is called the phase of the motion. The constant ϕ is called the *phase constant*. A different value of ϕ just means that the origin of time has been chosen differently.

7. **Energy of simple harmonic motion.** Put $\phi = 0$ for convenience, and so imagine a mass whose position oscillates like $x = x_m \cos \omega t$. Let us first calculate the potential energy:

$$U = \frac{1}{2} kx^2 = \frac{1}{2} kx_m^2 \cos^2 \omega t.$$

Now calculate the kinetic energy:

$$K = \frac{1}{2} mv^2 = \frac{1}{2} m \left(\frac{dx}{dt} \right)^2 = \frac{1}{2} m \omega^2 x_m^2 \sin^2 \omega t = \frac{1}{2} kx_m^2 \sin^2 \omega t$$

The sum of potential + kinetic is:

$$\begin{aligned} E = K + U &= \frac{1}{2} kx_m^2 \cos^2 \omega t + \frac{1}{2} kx_m^2 \sin^2 \omega t \\ &= \frac{1}{2} kx_m^2 (\cos^2 \omega t + \sin^2 \omega t) = \frac{1}{2} kx_m^2. \end{aligned}$$

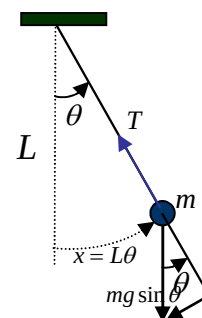
Note that this is independent of time and energy goes from kinetic to potential, then back to kinetic etc.

8. From the above, you can see that $v = \frac{dx}{dt} = \pm \sqrt{\frac{k}{m}(x_m^2 - x^2)}$. From this it is clear that the speed is maximum at $x = 0$ and that the speed is zero at $x = \pm x_m$.

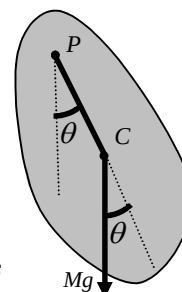
9. Putting two springs in parallel makes it harder to stretch them, and $k_{\text{eff}} = k_1 + k_2$. In series they are easier to stretch, and $k_{\text{eff}} = \left(\frac{k_1 k_2}{k_1 + k_2} \right)$. So a mass will oscillate faster in the first case as compared to the second.

Summary of Lecture 16 – OSCILLATIONS: II

1. In this chapter we shall continue with the concepts developed in the previous chapter that relate to simple harmonic motion and the simple harmonic oscillator (SHO). It is really very amazing that the SHO occurs again and again in physics, and in so many different branches.
2. As an example illustrating the above, consider a mass suspended a string. From the diagram, you can see that $F = -mg \sin \theta$. For small values of θ we know that $\sin \theta \approx \theta$. Using $x = L\theta$ (length of arc), we have $F = -mg\theta = -mg \frac{x}{L} = -\left(\frac{mg}{L}\right)x$. So now we have a restoring force that is proportional to the distance away from the equilibrium point. Hence we have a SHO with $\omega = \sqrt{g/L}$. What if we had not made the small θ approximation? We would still have an oscillator (i.e. the motion would be self repeating) but the solutions of the differential equation would be too complicated to discuss here.



3. If you take a common object (like a piece of cardboard) and pivot it at some point, it will oscillate when disturbed. But this is not the simple pendulum discussed above because all the mass is not concentrated at one point. So now let us use the ideas of torque and angular momentum discussed earlier for many particle systems. You can see that $\tau = -Mgd \sin \theta$. For small θ , $\sin \theta \approx \theta$ and so $\tau = -Mgd\theta$. But we also know that $\tau = I\alpha$ where I is the moment of



inertia and α is the angular acceleration, $\alpha = \frac{d^2\theta}{dt^2}$. Hence, we have

$$I \frac{d^2\theta}{dt^2} = -Mgd\theta, \text{ or, } \frac{d^2\theta}{dt^2} = -\left(\frac{Mgd}{I}\right)\theta. \text{ From this we immediately}$$

see that the oscillation frequency is $\omega = \sqrt{\frac{Mgd}{I}}$. Of course, we have used the small angle approximation over here again. Since all variables except I are known, we can use this formula to tell us what I is about any point. Note that we can choose to put the pivot at any point on the body. However, if you put the pivot exactly at the centre of mass then it will not oscillate. Why? Because there is no restoring force and the torque vanishes at the cm position, as we saw earlier.

4. Suppose you were to put the pivot at point P which is at a distance L from the centre of mass of the irregular object above. What should L be so that you get the same formula as for a simple pendulum?

$$\text{Answer: } T = 2\pi \sqrt{\frac{L}{g}} = 2\pi \sqrt{\frac{I}{Mgd}} \Rightarrow L = \frac{I}{Md}$$

P is then called the centre of gyration - when suspended from this point it appears as if all the mass is concentrated at the cm position.

5. Sum of two simple harmonic motions of the same period along the same line:

$$x_1 = A_1 \sin \omega t \text{ and } x_2 = A_2 \sin(\omega t + \phi)$$

Let us look at the sum of x_1 and x_2 ,

$$\begin{aligned} x &= x_1 + x_2 = A_1 \sin \omega t + A_2 \sin(\omega t + \phi) \\ &= A_1 \sin \omega t + A_2 \sin \omega t \cos \phi + A_2 \sin \phi \cos \omega t \\ &= \sin \omega t (A_1 + A_2 \cos \phi) + \cos \omega t (A_2 \sin \phi) \end{aligned}$$

Let $A_1 + A_2 \cos \phi = R \cos \theta$ and $A_2 \sin \phi = R \sin \theta$. Using some simple trigonometry, you can put x in the form, $x = R \sin(\omega t + \theta)$. It is easy to find R and θ :

$$R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi} \text{ and } \tan \theta = \frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi}.$$

Note that if $\phi = 0$ then $R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2} = \sqrt{(A_1 + A_2)^2} = A_1 + A_2$ and $\tan \theta = 0 \Rightarrow \theta = 0$. So we get $x = (A_1 + A_2) \sin \omega t$. This is an example of *constructive*

interference. If $\phi = \pi$ then $R = \sqrt{A_1^2 + A_2^2 - 2A_1A_2} = \sqrt{(A_1 - A_2)^2} = A_1 - A_2$ and $\tan \theta = 0 \Rightarrow \theta = 0$. Now we get $x = (A_1 - A_2) \sin \omega t$. This is *destructive interference*.

6. Composition of two simple harmonic motions of the same period but now at right angles to each other:

Suppose $x = A \sin \omega t$ and $y = B \sin(\omega t + \phi)$. These are two independent motions. We can write $\sin \omega t = \frac{x}{A}$ and $\cos \omega t = \sqrt{1 - x^2 / A^2}$.

From this, $\frac{y}{B} = \sin \omega t \cos \phi + \sin \phi \cos \omega t = \frac{x}{A} \cos \phi + \sin \phi \sqrt{1 - x^2 / A^2}$. Now square and rearrange terms to find:

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} - 2 \frac{xy}{AB} \cos \phi = \sin^2 \phi$$

This is the equation for an ellipse (see questions at the end of this section).

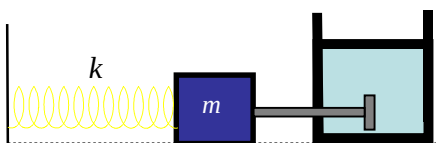
7. If two oscillations of different frequencies at right angles are combined, the resulting motion is more complicated. It is not even periodic unless the two frequencies are in the ratio of integers. This resulting curve are called Lissajous figures. Specifically, if

$$x = A \sin \omega_x t \text{ and } y = B \sin(\omega_y t + \phi), \text{ then periodic motion requires } \frac{\omega_x}{\omega_y} = \text{integers.}$$

You should look up a book for more details.

8. **Damped harmonic motion:** Typically the frictional force due to air resistance, or in a liquid, is proportional to the speed. So suppose that the damping force $= -b \frac{dx}{dt}$ (why negative sign?). Now apply Newton's law to a SHO that is damped: $-kx - b \frac{dx}{dt} = m \frac{d^2x}{dt^2}$

Rearrange slightly to get the equation for a damped SHO: $m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = 0.$



Its solution for $\frac{k}{m} \geq \left(\frac{b}{2m}\right)^2$ is $x = x_m e^{-bt/2m} \cos(\omega' t + \phi)$. The frequency is now

changed: $\omega' = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}$. The damping causes the amplitude to decrease with time and when $bt/2m = 1$, the amplitude is $1/e \approx 1/2.7$ of its initial value.

9. **Forced oscillation and resonance.** There is a characteristic value of the driving frequency ω at which the amplitude of oscillation is a maximum. This condition is called resonance. For negligible damping resonance occurs at $\omega = \omega_0$. Here ω_0 is

the natural frequency of the system and is given by $\omega_0 = \sqrt{\frac{k}{m}}$. The equation of

motion is: $m \frac{d^2x}{dt^2} + kx = F_0 \cos \omega t$. You should check that this is solved by putting

$$x = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos \omega t \text{ (just substitute into the equation and see!). Note that the}$$

amplitude "blows up" when $\omega \rightarrow \omega_0$. This is because we have no damping term here.

With damping, the amplitude is large when $\omega \rightarrow \omega_0$ but remains finite.

Summary of Lecture 17 – PHYSICS OF MATERIALS

1. **Elasticity** : the property by virtue of which a body tends to regain its original shape and size when external forces are removed. If a body completely recovers its original shape and size , it is called perfectly elastic. Quartz, steel and glass are very nearly elastic.
2. **Plasticity** : if a body has no tendency to regain its original shape and size , it is called perfectly plastic. Common plastics, kneaded dough, solid honey, etc are plastics.
3. **Stress** characterizes the strength of the forces causing the stretch, squeeze, or twist. It is defined usually as force/unit area but may have different definitions to suit different situations. We distinguish between three types of stresses:
 - a) If the deforming force is applied along some linear dimension of a body, the stress is called *longitudinal stress* or *tensile stress* or *compressive stress*.
 - b) If the force acts normally and uniformly from all sides of a body, the stress is called *volume stress*.
 - c) If the force is applied tangentially to one face of a rectangular body, keeping the other face fixed, the stress is called tangential or shearing stress.
4. **Strain**: When deforming forces are applied on a body, it undergoes a change in shape or size. The fractional (or relative) change in shape or size is called the strain.

$$\text{Strain} = \frac{\text{change in dimension}}{\text{original dimension}}$$

Strain is a ratio of similar quantities so it has no units. There are 3 different kinds of strain:

a) *Longitudinal (linear) strain* is the ratio of the change in length (ΔL) to original length (l), i.e., the linear strain $= \frac{\Delta l}{l}$.

b) *Volume strain* is the ratio of the change in volume (ΔV) to original volume (V)

$$\text{Volume strain} = \frac{\Delta V}{V}.$$

c) *Shearing strain* : The angular deformation (θ) in radians is called shearing stress.

$$\text{For small } \theta \text{ the shearing strain} \equiv \theta \approx \tan \theta = \frac{\Delta x}{l}.$$



5. Hooke's Law: for small deformations, stress is proportional to strain.

$$\text{Stress} = E \times \text{Strain}$$

The constant E is called the modulus of elasticity. E has the same units as stress because strain is dimensionless. There are three moduli of elasticity.

- (a) Young's modulus (Y) for linear strain:

$$Y \equiv \frac{\text{longitudinal stress}}{\text{longitudinal strain}} = \frac{F/A}{\Delta l/l}$$

- (b) Bulk Modulus (B) for volume strain: Let a body of volume V be subjected to a uniform pressure ΔP on its entire surface and let ΔV be the corresponding decrease in its volume. Then,

$$B \equiv \frac{\text{Volume Stress}}{\text{Volume Strain}} = -\frac{\Delta P}{\Delta V/V}.$$

$1/B$ is called the compressibility. A material having a small value of B can be compressed easily.

- (c) Shear Modulus (η) for shearing strain: Let a force F produce a strain θ as in the diagram in point 4 above. Then,

$$\eta \equiv \frac{\text{shearing stress}}{\text{shearing strain}} = \frac{F/A}{\theta} = \frac{F}{A \tan \theta} = \frac{Fl}{A \Delta x}.$$

6. When a wire is stretched, its length increases and radius decreases. The ratio of the lateral strain to the longitudinal strain is called Poisson's ratio, $\sigma = \frac{\Delta r/r}{\Delta l/l}$. Its value lies between 0 and 0.5.

7. We can calculate the work done in stretching a wire. Obviously, we must do work against a force. If x is the extension produced by the force F in a wire of length l ,

then $F = \frac{YA}{l}x$. The work done in extending the wire through Δl is given by,

$$\begin{aligned} W &= \int_0^{\Delta l} F dx = \frac{YA}{l} \int_0^{\Delta l} x dx = \frac{YA}{l} \frac{(\Delta l)^2}{2} \\ &= \frac{YA}{l} \frac{(\Delta l)^2}{2} = \frac{1}{2} (Al) \left(\frac{Y \Delta l}{l} \right) \left(\frac{\Delta l}{l} \right) = \frac{1}{2} \times \text{volume} \times \text{stress} \times \text{strain} \end{aligned}$$

Hence, Work / unit volume = $\frac{1}{2} \times \text{stress} \times \text{strain}$. We can also write this as,

$$W = \frac{1}{2} \left(\frac{YA \Delta l}{l} \right) \Delta l = \frac{1}{2} \times \text{load} \times \text{extension}.$$

8. A fluid is a substance that can flow and does not have a shape of its own. Thus all liquids and gases are fluids. Solids possess all the three moduli of elasticity whereas a fluid possess only the bulk modulus (B). A fluid at rest cannot sustain a tangential force. If such force is applied to a fluid, the different layers simply slide over one another. Therefore the forces acting on a fluid at rest have to be normal to the surface. This implies that the free surface of a liquid at rest, under gravity, in a container, is horizontal.

9. The normal force per unit area is called pressure, $P = \frac{\Delta F}{\Delta A}$. Pressure is a scalar quantity.

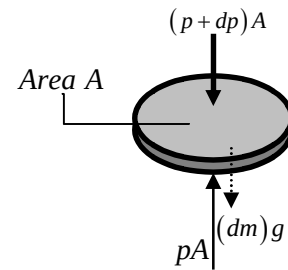
Its unit is Newtons/metre², or Pascal (Pa). Another scalar is density, $\rho = \frac{\Delta m}{\Delta V}$, where Δm is the mass of a small piece of the material and ΔV is the volume it occupies.

10. Let us calculate how the pressure in a fluid changes with depth.

So take a small element of fluid volume submerged within the body of the fluid: $dm = \rho dV = \rho A dy \quad \therefore (dm)g = \rho g A dy$

Now let us require that the sum of the forces on the fluid element

is zero: $pA - (p + dp)A - \rho g A dy = 0 \Rightarrow \frac{dp}{dy} = -\rho g.$



Note that we are taking the origin ($y = 0$) at the bottom of the liquid. Therefore as the elevation increases (dy positive), the pressure decreases (dp negative). The quantity ρg is the weight per unit volume of the fluid. For liquids, which are nearly incompressible, ρ is practically constant.

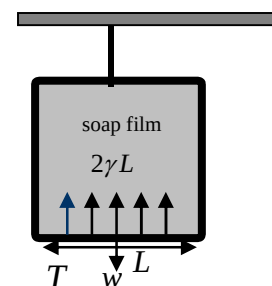
$$\therefore \rho g = \text{constant} \Rightarrow \frac{dp}{dy} = \frac{\Delta p}{\Delta y} = \frac{p_2 - p_1}{y_2 - y_1} = -\rho g \Rightarrow p_2 - p_1 = -\rho g(y_2 - y_1).$$

11. **Pascal's Principle:** Pressure applied to an enclosed fluid is transmitted to every portion of the fluid and to the walls of the containing vessel. This follows from the above: if h is the height below the liquid's surface, then $p = p_{ext} + \rho gh$. Here p_{ext} is the pressure at the surface of the liquid, and so the difference in pressure is $\Delta p = \Delta p_{ext} + \Delta(\rho gh)$. Therefore, $\Delta(\rho gh) = 0 \Rightarrow \Delta p = \Delta p_{ext}$. (I have used here the fact that liquids are incompressible). So the pressure is everywhere the same.

Summary of Lecture 18 – PHYSICS OF FLUIDS

1. A fluid is matter that has no definite shape and adjusts to the container that it is placed in. Gases and liquids are both fluids. All fluids are made of molecules. Every molecule attracts other molecules around it.
2. Liquids exhibit surface tension. A liquid has the property that its free surface tends to contract to minimum possible area and is therefore in a state of tension. The molecules of the liquid exert attractive forces on each other, which is called cohesive forces. Deep inside a liquid, a molecule is surrounded by other molecules in all directions. Therefore there is no net force on it. At the surface, a molecule is surrounded by only half as many molecules of the liquid, because there are no molecules above the surface.

3. The force of contraction is at right angles to an imaginary line of unit length, tangential to the surface of a liquid, is called its surface tension: $\gamma = \frac{F}{L}$. Here F is the force exerted by the "skin" of the liquid. The SI unit of the surface tension is N/m.

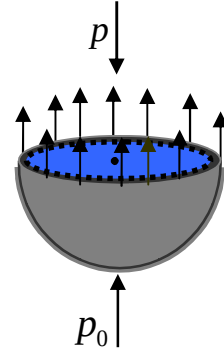


4. Quantitative measurement of surface tension: let w be the weight of the sliding wire, T = force with which you pull the wire downward. Obviously, $T + w = F$ = net downward force. Since film has both front and back surfaces, the force F acts along a total length of $2L$. The surface tension in the film is defined as, $\gamma = \frac{F}{2L} \Rightarrow F = 2\gamma L$.

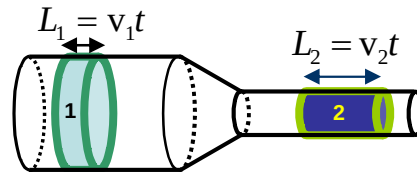
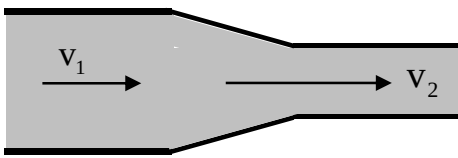
Hence, the surface tension is $\gamma = \frac{w + T}{2L}$.

5. Let's ask how much work is done when we stretch the skin of a liquid. If we move the sliding wire through a displacement Δx , the work done is $F\Delta x$. Now F is a conservative force, so there is potential energy $\Delta U = F\Delta x = \gamma L\Delta x$ where L is the length of the surface layer $L\Delta x = \Delta A$ = change in area of the surface. Thus $\gamma = \frac{\Delta U}{\Delta A}$. So we see that surface tension is the surface potential energy per unit area !

7. The surface tension causes a pressure difference between the inside and outside of a soap bubble or a liquid drop. A soap bubble consists of two spherical surface films with a thin layer of liquid between them. Let p = pressure exerted by the upper half, and p_0 = external pressure. $\therefore$ force exerted due to surface tension is $2(2\pi r\gamma)$ (the "2" is for two surfaces). In equilibrium the net forces must be equal: $\pi r^2 p = 2(2\pi r\gamma) + \pi r^2 p_0$. So the excess pressure is $p - p_0 = \frac{4\gamma}{r}$. For a liquid drop, the difference is that there is only surface and so, excess pressure = $p - p_0 = \frac{2\gamma}{r}$.



8. From the fact that liquids are incompressible, equal volumes of liquid must flow in both sections in time t , i.e. $V_1/t = V_2/t \Rightarrow V_1 = V_2$. But you can see that $V_1 = A_1 L_1 = A_1 v_1 t$ and similarly that $V_2 = A_2 L_2 = A_2 v_2 t$. Hence $A_1 v_1 = A_2 v_2$. This means that liquid will flow faster when the tube is narrower, and slower where it is wider.



9. **Bernoulli's Equation :** When a fluid flows steadily, it obeys the equation:

$$p + \frac{1}{2}\rho v^2 + \rho gy = \text{constant}$$

This famous equation, due to Daniel Bernoulli about 300 years ago, tells you how fast a fluid will flow when there is also a gravitational field acting upon. For a derivation, see any of the suggested references. In the following, I will only explain the meaning of the various terms in the formula and apply it to a couple of situations.

10. Let us apply this to water flowing in a pipe whose cross-section decreases along its length. (A_1 is area of the wide part, etc). There is no change in the height so $y_1 = y_2$ and the gravitational potential cancels,

$$p_1 + \frac{1}{2}\rho v_1^2 + \cancel{\rho g y_1} = p_2 + \frac{1}{2}\rho v_2^2 + \cancel{\rho g y_2} \Rightarrow p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2.$$

Now, since the liquid is incompressible, it flows faster in the narrower part:

$$A_1 v_1 = A_2 v_2 \Rightarrow v_2 = \left(\frac{A_1}{A_2}\right) v_1 \Rightarrow p_2 = p_1 - \frac{1}{2}\rho(v_2^2 - v_1^2).$$

This means that the pressure is smaller where the fluid is flowing faster! This is exactly why an aircraft flies: the wing shape is curved so that when the aircraft moves through the air, the air moves faster on the top part of the wing than on the lower part. Thus, the air pressure is lower on the top compared to the bottom and there is a net pressure upwards. This is called lift.

Summary of Lecture 19 – PHYSICS OF SOUND

1. Sound waves correspond to longitudinal oscillations of density. So if sound waves are moving from left to right, as you look along this direction you will find the density of air greater in some places and less in others. Sound waves carry energy. The minimum energy that humans can hear is about 10^{-12} watts per cm^3 (This is called I_0 , the threshold of hearing.)
2. To measure the intensity of sound, we use *decibels* as the unit. Decibels (db) are a relative measure to compare the intensity of different sounds with one another,

$$R \equiv \text{relative intensity of sound } I = \log_{10} \frac{I}{I_0} \text{ (decibels)}$$

Typically, on a street without traffic the sound level is about 30db, a pressure horn creates about 90db, and serious ear damage happens around 120db.

3. A sound wave moving in the x direction with speed v is described by

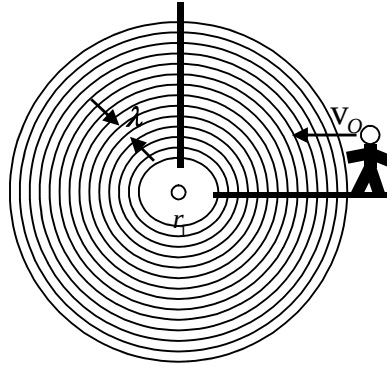
$$\rho(x, t) = \rho_m \sin \frac{2\pi}{\lambda} (x - vt)$$

where $\rho(x, t)$ is the density of air at a point x at time t . Let us understand various aspects of this formula.

- a) Suppose that as time t increases, we move in such a way as to keep $x - vt$ constant. So if at $t = 0$ the value of x is 0.23 (say), then at $t = 1$ the value of x would be $v + 0.23$, etc. In other words, to keep the density $\rho(x, t)$ constant, we would have to move with the speed of sound, i.e. v .
- b) In the expression for $\rho(x, t)$, replace x by $x + \lambda$. What happens? Answer: nothing, because $\sin \frac{2\pi}{\lambda} (x + \lambda - vt) = \sin \frac{2\pi}{\lambda} (x - vt)$. This is why we call λ the "wavelength", meaning that length after which a wave repeats itself.
- c) In the expression for $\rho(x, t)$, replace t by $t + T$ where $T = \lambda / v$. What happens? Again the answer: nothing. T is called the time period of the sound wave, meaning that time after which it repeats itself. The frequency is the number of cycles per second and is obviously related to T through $\nu = 1/T$.
- d) It is also common to introduce the *wavenumber* k and *angular frequency* ω :

$$k = \frac{2\pi}{\lambda} \quad \text{and} \quad \omega = \frac{2\pi}{T} = 2\pi\nu$$

4. **Doppler Effect:** The relative motion between source and observer causes the observer to receive a frequency that is different from that emitted by the source. One must distinguish between two cases:



Moving observer, source at rest. If the observer was at rest, the number of waves she would receive in time t would be t/T (or vt/λ). But if she is moving towards the source with speed v_o (as in the above figure), the additional number of waves received is obviously $v_o t/\lambda$. By definition,

$$\nu' = \text{frequency actually heard} = \frac{\text{number of waves received}}{\text{unit time}}$$

$$\therefore \nu' = \frac{\frac{vt}{\lambda} + \frac{v_o t}{\lambda}}{t} = \frac{v + v_o}{\lambda} = \frac{v + v_o}{v/\nu} = \nu \frac{v + v_o}{v}$$

We finally conclude that the frequency actually heard is $\nu' = \nu \left(1 + \frac{v_o}{v}\right)$. So as the observer runs towards the source, she hears a higher frequency (higher pitch).

Moving source, observer at rest : As the source runs towards the observer, more waves will have to be packed together. Each wavelength is reduced by $\frac{v_s}{v}$. So the wavelength seen by the observer is $\lambda' = \frac{v}{\nu} - \frac{v_s}{\nu}$. From this, the frequency that she hears is $\nu' = \frac{v}{\lambda} = \frac{v}{(v - v_s)/\nu} = \nu \frac{v}{(v - v_s)}$.

Moving source and moving observer : $\nu' = \nu \frac{v + v_o}{v - v_s}$. As you can easily see, the above two results are special cases of this.

Summary of Lecture 20 – WAVE MOTION

1. Wave motion is any kind of self-repeating (periodic, or oscillatory) motion that transports energy from one point to another. Waves are of two basic kinds:

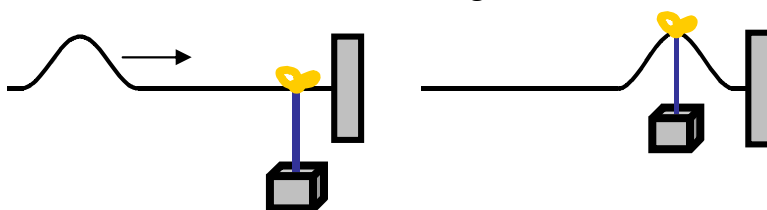
(a) **Longitudinal Waves:** the oscillation is parallel to the direction of wave travel.

Examples: sound, spring, "P-type" earthquake waves.

(b) **Transverse Waves:** the oscillation is perpendicular to the direction of wave travel.

Examples: radio or light waves, string, "S-type" earthquake waves.

2. Waves transport energy, not matter. Taking the vibration of a string as an example, each segment of the string stays in the same place, but the work done on the string at one end is transmitted to the other end. Work is done in lifting the mass at the other end below.



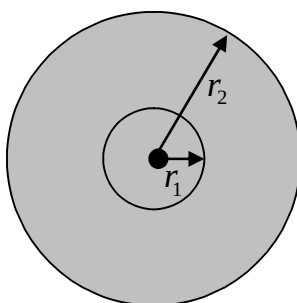
3. The height of a wave is called the amplitude. The average power (or intensity) in a wave is proportional to the square of the amplitude. So if $a(t) = a_0 \sin(\omega t - kx)$ is a wave of some kind, then a_0 is the amplitude and $I \propto a_0^2$.

4. A sound source placed at the origin will radiate sound waves in all directions equally. These are called spherical waves. For spherical waves the amplitude $\propto \frac{1}{r}$ and so the power $\propto \frac{1}{r^2}$.

We can easily see why this is so. Consider a source of sound and draw two spheres:

Let P_1 be the total radiated power and I_1 the intensity at r_1 , etc. All the power (and energy) that crosses r_1 also crosses r_2 since none is lost in between the two. We have that,

$$4\pi r_1^2 I_1 = P_1 \text{ and } 4\pi r_2^2 I_2 = P_2. \text{ But } P_1 = P_2 = P, \text{ and so } \frac{I_1}{I_2} = \frac{r_2^2}{r_1^2} \text{ or } I \propto \frac{1}{r^2}.$$



5. We have encountered waves of the kind $y(x, t) = y_0 \sin(kx - \omega t)$ in the previous lecture. Obviously $y(0, 0) = 0$. But what if the wave is not zero at $x = 0, t = 0$? Then it could be represented by $y(x, t) = y_m \sin(kx - \omega t - \phi)$, where $kx - \omega t - \phi$ is called the phase and ϕ is called the phase constant. Note that you can rewrite $y(x, t)$ either as,

$$\text{a) } y(x, t) = y_m \sin \left[k \left(x - \frac{\phi}{k} \right) - \omega t \right],$$

$$\text{or as, b) } y(x, t) = y_m \sin \left[kx - \omega \left(t + \frac{\phi}{\omega} \right) \right].$$

The two different ways of writing the same expression can be interpreted differently. In

(a) x has effectively been shifted to $x - \frac{\phi}{k}$ whereas in (b) t has been shifted to $t + \frac{\phi}{\omega}$. So the phase constant only moves the wave forward or backward in space or time.

6. When two sources are present the total amplitude at any point is the sum of the two separate amplitudes, $y(x, t) = y_1(x, t) + y_2(x, t)$. Now you remember that the power is proportional to the *square* of the amplitude, so $P \propto (y_1 + y_2)^2$. This is why *interference* happens. In the following we shall see why. Just to make things easier, suppose the two waves have equal amplitude. So let's take the two waves to be :

$$y_1(x, t) = y_m \sin(kx - \omega t - \phi_1) \quad \text{and} \quad y_2(x, t) = y_m \sin(kx - \omega t - \phi_2)$$

The total amplitudes is: $y(x, t) = y_1(x, t) + y_2(x, t)$

$$= y_m [\sin(kx - \omega t - \phi_1) + \sin(kx - \omega t - \phi_2)]$$

Now use the trigonometric formula, $\sin B + \sin C = 2 \sin \frac{1}{2}(B + C) \times \cos(B - C)$ to get,

$$\begin{aligned} y(x, t) &= y_m [\sin(kx - \omega t - \phi_1) + \sin(kx - \omega t - \phi_2)] \\ &= \left[2y_m \cos \left(\frac{\Delta\phi}{2} \right) \right] \times \sin(kx - \omega t - \phi'). \end{aligned}$$

Here $\Delta\phi = \phi_2 - \phi_1$ is the difference of phases, and $\phi' = \frac{(\phi_1 + \phi_2)}{2}$ is the sum. So what do

we learn from this? That if $\phi_2 = \phi_1$, then the two waves are in phase and the resultant amplitude is maximum (because $\cos 0 = 1$). And that if $\phi_2 = \phi_1 + \pi$, then the two waves are out of phase and the resultant amplitude is minimum (because $\cos \pi / 2 = 0$). The two waves have interfered with each other and have increased/decreased their amplitude in

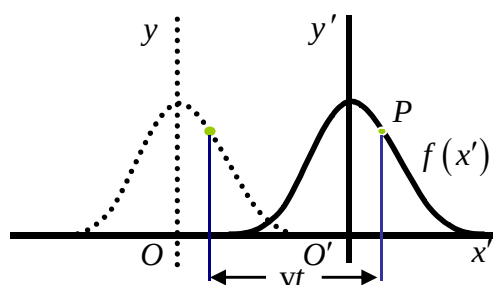
these two extreme cases. In general $\cos \left(\frac{\Delta\phi}{2} \right)$ will be some number that lies between

7. There was no time in the lecture to prove it, but you can look up any good book to

find a proof for the formula that the speed of sound in a medium is: $v = \sqrt{\frac{B}{\rho}}$ where

B is the bulk modulus and ρ is the density of the medium.

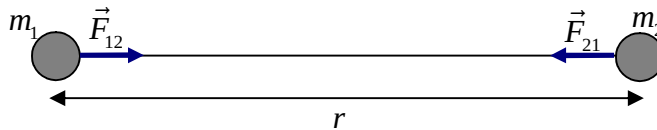
8. **The speed of a pulse.** A pulse is a burst of energy (sound, electromagnetic, heat,...) and could have any shape. Mathematically any pulse has the form $y(x,t) = f(x - vt)$. Here f is any function (e.g. sin, cos, exp,...). Note that at time $t = 0$, $y(x,0) = f(x)$ and the shape would look as on the left in the diagram below. At a late time t , it will look just the same, but shifted to the right. In other words at time t , $y(x,t) = f(x')$ where $x' = x - vt$. Fix your attention on any one point of the curve and follow it as the pulse moves to the right. From $x - vt = \text{constant}$ it follows that $\frac{dx}{dt} - v = 0$, or $v = \frac{dx}{dt}$. This is called the *phase velocity* because we derived it using the constancy of phase.



Summary of Lecture 21 – GRAVITY

1. Newton's law of universal gravitation states that the force of attraction between two masses m_1 and m_2 is $F \propto \frac{m_1 m_2}{r^2}$ and is directed along the line joining the two bodies.

Putting in a constant of proportionality, $F = G \frac{m_1 m_2}{r^2}$. Now let's be a bit careful of the direction of the force. Looking at the diagram below, $\vec{F}_{21}$ = Force on m_2 by m_1 , $\vec{F}_{12}$ = Force on m_1 by m_2 , $|\vec{F}_{12}| = |\vec{F}_{21}| = F$. By Newton's Third Law, $\vec{F}_{12} = -\vec{F}_{21}$.



2. The gravitational constant G is a very small quantity and needs a very sensitive experiment. An early experiment to find G involved suspending two masses and measuring the attractive force. From the figure you can see that the

gravitational torque is $2 \left(\frac{GmM}{r^2} \right) \frac{L}{2}$. A thread provides

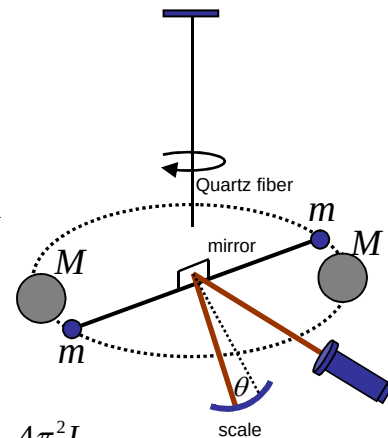
the restoring torque $\kappa\theta$. The deflection θ can be measured by observing the beam of the light reflected from the small

mirror. In equilibrium the torques balance, $\frac{GmML}{r^2} = \kappa\theta$.

Hence $G = \frac{\kappa\theta r^2}{GmML}$. How to find κ ? It can be found from

observing the period of free oscillations, $T = 2\pi \sqrt{\frac{I}{\kappa}} \Rightarrow \kappa = \frac{4\pi^2 I}{T^2}$

with $I = \frac{mL^2}{2}$. The modern value is $G = 6.67259 \times 10^{-11} \text{ N.m}^2 / \text{kg}^2$



3. The magnitude of the force with which the Earth attracts a body of mass m towards its centre is $F = \frac{GmM_E}{R_E^2}$, where $R_E = 6400 \text{ km}$ is the radius of the Earth and M_E is the mass.

The material does not matter - iron, wood, leather, etc. all feel the force in proportion to their masses. If the body can fall freely, then it will accelerate. So, $F = mg = \frac{GmM_E}{R_E^2}$.

We measure g , the acceleration due to gravity, as 9.8m/s^2 . From this we can immediately deduce the Earth's mass: $M_E = \frac{gR_E^2}{G} = 5.97 \times 10^{24} \text{ kg}$. What a remarkable achievement!

We can do still more: the volume of the Earth $= V_E = \frac{4}{3}\pi R_E^3 = 1.08 \times 10^{21} \text{ m}^3$. Hence the density of the Earth $= \rho_E = \frac{M_E}{V_E} = 5462 \text{ kg m}^{-3}$. So this is 5.462 times greater than the density of water and tells us that the earth must be quite dense inside.

4. The **gravitational potential** is an important quantity. It is the work done in moving a unit mass from infinity to a given point R , and equals $V(r) = -\frac{GM}{R}$.

Proof: Conservation of energy says, $dV = -Fdr \Rightarrow \int_{V(R)}^0 dV = -\int_R^\infty dr F(r)$

Integrate both sides: $0 - V(R) = GM \int_R^\infty \frac{dr}{r^2} = -GM \left[\frac{1}{r} \right]_R^\infty, \therefore V(R) = -\frac{GM}{R}$

5. Using the above formula, let us calculate the change in potential energy ΔU when we raise a body of mass m to a height h above the Earth's surface.

$$\Delta U = GMm \left(\frac{1}{R_E} - \frac{1}{R_E + h} \right) = GMm \left(1 - \frac{1}{1 + h/R_E} \right) = GMm \left(1 - (1 + h/R_E)^{-1} \right)$$

Now suppose that the distance h is much smaller than the Earth's radius. So, for $h \ll R_E$,

$$(1 + h/R_E)^{-1} = 1 - h/R_E. \text{ So we find } \Delta U = GMm \left(1 - (1 - h/R_E) \right) = m \left(\frac{GM}{R_E} \right) h = mgh.$$

6. We can use the expression for potential energy and the law of conservation of energy to find the minimum velocity needed for a body to escape the Earth' gravity. Far away from the Earth, the potential energy is zero, and the smallest value for the kinetic energy is

zero. Requiring that $(KE + PE)_{r=R} = (KE + PE)_{r=\infty}$ gives $\frac{1}{2}mv_e^2 - \frac{GMm}{R_E} = 0 + 0$. From

this, $v_e = \sqrt{\frac{2GM}{R_E}} = \sqrt{2gR_E}$. Putting in some numbers we find that for the Earth $v_e = 11.2 \text{ km/s}$

and for the Sun $v_e = 618 \text{ km/s}$. For a Black Hole, the escape velocity is so high that nothing can escape, even if it could move with the speed of light! (Nevertheless, Black Holes can be observed because when matter falls into them, a certain kind of radiation is emitted.)

7. **Satellite problems:** A satellite is in circular orbit over the Earth's surface. The condition

for equilibrium, $\frac{mv_o^2}{r} = \frac{GMm}{r^2} \Rightarrow v_o = \sqrt{\frac{GM}{r}}$. If R_E is the Earth's radius, and h is the

height of the satellite above the ground, then $r = R_E + h$. Hence, $v_o = \sqrt{\frac{GM}{R_E + h}}$. For

$h \ll R_E$, we can approximate $v_o = \sqrt{\frac{GM}{R_E}} = \sqrt{gR_E}$. We can easily calculate the time

for one complete revolution, $T = \frac{2\pi}{\omega} = \frac{2\pi r}{v_o} = 2\pi r \sqrt{\frac{r}{GM}} = \frac{2\pi}{\sqrt{GM}} r^{3/2}$. This gives the

important result, observed by Kepler nearly 3 centuries ago that $T^2 = \frac{4\pi^2}{GM} r^3$, or $T^2 \propto r^3$.

8. What is the total energy of a satellite moving in a circular orbit around the earth? Clearly, it has two parts, kinetic and potential. Remember that the potential energy is negative. So,

$$E = KE + PE = \frac{1}{2}mv_o^2 - \frac{GM_E m}{r}. \text{ But, } v_o^2 = \frac{GM}{r} \text{ as we saw earlier and therefore,}$$

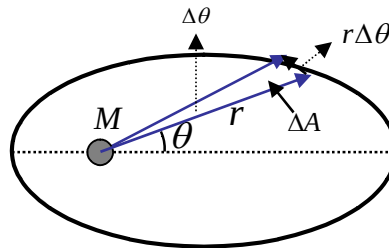
$$E = \frac{1}{2} \frac{GMm}{r} - \frac{GMm}{r} = -\frac{1}{2} \frac{GMm}{r}. \text{ Note that the magnitude of the potential energy is}$$

larger than the kinetic energy. If it wasn't, the satellite would not be bound to the Earth!

9. A famous discovery of the astronomer Johann Kepler some 300 years ago says that the line joining a planet to the Sun sweeps out equal areas in equal intervals of time. We can easily see this from the conservation of angular momentum. Call ΔA the area swept out in time Δt . Then from the diagram below you can see that $\Delta A = \frac{1}{2}r(r\Delta\theta)$. Divide this by Δt and then take the limit where it becomes very small,

$$\frac{dA}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta A}{\Delta t} = \frac{1}{2}r^2 \left(\lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} \right) = \frac{1}{2}r^2\omega = \frac{L}{2m}.$$

Since L is a constant, we have proved one of Kepler's laws (with so little effort)!



Summary of Lecture 22 – ELECTROSTATICS I

1. Like charges repel, unlike charges attract. But by how much? Coulomb's Law says that this depends both upon the strength of the two charges and the distance between them. In mathematical terms, $F \propto \frac{q_1 q_2}{r^2}$ which can be converted into an equality, $F = k \frac{q_1 q_2}{r^2}$. The constant of proportionality will take different values depending upon the units we choose. In the MKS system, charge is measured in Coulombs (C) and $k = \frac{1}{4\pi\epsilon_0}$ with $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 / \text{Nm}^2$ and hence $k = 8.99 \times 10^9 \text{ Nm}^2 / \text{C}^2$.
2. The situation is quite similar to that of gravity, except that electric charges and not masses are the source of force. In vector form, $\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$ is the force exerted by 2 on 1, where the unit vector is $\hat{r}_{12} = \frac{\vec{r}_{12}}{r_{12}}$. On the other hand, $\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{21}$ is the force exerted by 1 on 2. By Newton's Third Law, $\vec{F}_{12} = -\vec{F}_{21}$. For many charges, the force on charge 1 is given by, $\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \dots$
3. Charge is quantized. This means that charge comes in certain units only. So the size of a charge can only be 0, $\pm e, \pm 2e, \pm 3e, \dots$ where $e = 1.602 \times 10^{-19} \text{ C}$ is the value of the charge present on a proton. By definition we call the charge on a proton positive. This makes the charge on an electron negative.
4. Charge is conserved. This means that charge is never created or destroyed. Equivalently, in any possible situation, the total charge at an earlier time is equal to the charge at a later time. For example, in any of the reactions below the initial charge = final charge:

$$e^- + e^+ \rightarrow \gamma + \gamma \quad (\text{electron and positron annihilate into neutral photons})$$

$$\pi^0 \rightarrow \gamma + \gamma \quad (\text{neutral pion annihilates into neutral photons})$$

$$H^2 + H^2 \rightarrow H^3 + p \quad (\text{two deuterons turn into tritium and proton})$$
5. **Field**: this a quantity that has a definite value at any point in space and at any time. The simplest example is that of a scalar field, which is a *single number* for any value of x, y, z, t . Examples: temperature inside a room $T(x, y, z, t)$, density in a blowing wind $\rho(x, y, z, t)$, ... There are also *vector fields*, which comprise of three numbers at each value of x, y, z, t . Examples: the velocity of wind, the pressure inside a fluid, or even a sugarcane field. In

every case, there are 3 numbers: $\vec{V}(x, y, z, t) = \{ V_1(x, y, z, t), V_2(x, y, z, t), V_3(x, y, z, t) \}$.

6. The electric field is also an example of a vector field, and will be the most important for our purpose. It is defined as the force on a unit charge. Or, since we don't want the charge to disturb the field it is placed in, we should properly define it as the force on a "test"

charge q_0 , $E \equiv \frac{F}{q_0}$. Here q_0 is very very small. The electric field due to a point charge can

be calculated by considering two charges. The force between them is $F = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2}$ and so

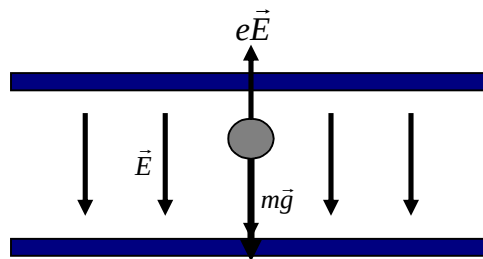
$E = \frac{F}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$. A way to visualize E fields is to think of lines starting on positive charges and ending on negative charges. The number of lines leaving/entering gives the amount of charge.

7. Typical values for the magnitude of the electric field E :

Inside an atom-	10^{11} N/C
Inside TV tube-	10^5 N/C
In atmosphere-	10^2 N/C
Inside a wire-	10^{-2} N/C

8. Measuring charge. One way to do this is to balance the gravitational force pulling a charged particle with mass m with the force exerted on it by a known electric field (see below). For equilibrium, the two forces must be equal and so $mg = qE$. The

unknown charge q can then be found from $q = \frac{mg}{E}$.



9. Given several charges, one can find the total electric field at any point as the sum of the fields produced by the charges at that point individually, $\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots$ or

$\vec{E} = \sum_i \vec{E}_i = k \sum_i \frac{q_i}{r_i^2} \hat{r}_i$ ($i = 1, 2, 3, \dots$). Here $\hat{r}_i$ is the unit vector pointing from the charge to the point of observation.

10. Let us apply the principle we have just learned to a system of two charges $+q$ and $-q$

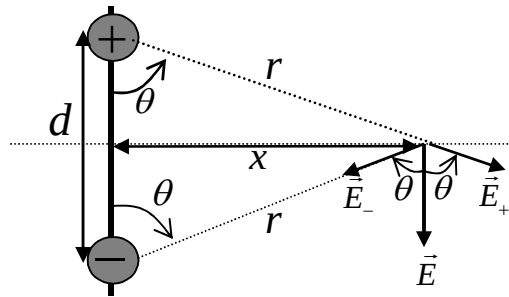
which are separated by a distance d (see diagram). Then, $\vec{E} = \vec{E}_+ + \vec{E}_-$. Just to make things easier (not necessary; one can do it for any point) I have taken a point that lies on the x-axis. The magnitudes of the electric field due to the two charges are equal;

$$E_+ = E_- = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{q}{x^2 + (d/2)^2}.$$

The vertical components cancel out, and the net electric field is directed downwards with magnitude, $E = E_+ \cos \theta + E_- \cos \theta = 2E_+ \cos \theta$.

From the diagram below you can see that $\cos \theta = \frac{d/2}{\sqrt{x^2 + (d/2)^2}}$. Substituting this, we

$$\text{find: } E = 2 \frac{1}{4\pi\epsilon_0} \frac{q}{x^2 + (d/2)^2} \frac{d/2}{\sqrt{x^2 + (d/2)^2}} = \frac{1}{4\pi\epsilon_0} \frac{qd}{[x^2 + (d/2)^2]^{3/2}}.$$



11. The result above is so important that we need to discuss it further. In particular, what happens if we are very far away from the dipole, meaning $x \gg d$? Let us first define the *dipole moment* as the product of the charge $\times$ the separation between them $p = qd$.

$$\text{Then, } E = \frac{1}{4\pi\epsilon_0} \frac{p}{x^3} \frac{1}{[1 + (d/2x)^2]^{3/2}} = \frac{1}{4\pi\epsilon_0} \frac{p}{x^3} \left[1 + \left(\frac{d}{2x} \right)^2 \right]^{-3/2} = \frac{1}{4\pi\epsilon_0} \frac{p}{x^3}.$$

$d/2x$ has been neglected in comparison to 1. So finally, we have found that for $x \gg d$,

$$E = \frac{1}{4\pi\epsilon_0} \frac{p}{x^3}.$$

12. It is easy to find the torque experienced by an electric dipole that is placed in a uniform

electric field: The magnitude is $\tau = F \frac{d}{2} \sin \theta + F \frac{d}{2} \sin \theta = Fd \sin \theta$, and the direction is perpendicular and into the plane. Here θ is the angle between the dipole and the electric field. So $\tau = (qE)d \sin \theta = pE \sin \theta$.

PHY101-PHYSICS MID TERM MCQS

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1. The ultimate strength of a sample is the stress at which the sample:

- A. Return to its original shape when the stress is removed
- B. Remains underwater
- C. Breaks
- D. Bends 180°

2. The value of k in Coulomb's law depends upon

- A. Medium between two charges
- B. Distance between charges
- C. Magnitudes of charges
- D. All of these

3. The law of inertia was first formulated by:

- A. Aristotle
- B. Galileo
- C. Newton
- D. Einstein

4. The ratio of circumference of a circle to its diameter is equal to;

- A. 2π
- B. π
- C. $\pi/2$
- D. one

5. A water bed that is 1.5 m wide and 2.5 m long weighs 1055 N. Assuming the entire lower surface of the bed is in contact with the floor, what is the pressure the bed exerts on the floor?

- i. 250 Pa
- ii. 260 Pa
- iii. 270 Pa
- iv. 280 Pa

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6. The center of gravity is the average location of the _____ of an object.

- A. Mass
- B. Weight
- C. Static equilibrium
- D. Dynamic

7. Acceleration in a body is always produced in the direction of:

- A. Velocity
- B. Weight
- C. Force
- D. Acceleration

8. As per Coulomb's law, the force of attraction or repulsion between two point charges is directly proportional to the.

- A. Cube of the distance
- B. Product of the magnitude of charges
- C. Sum of the magnitude of charges
- D. Square of the distance between them

9. One revolution is the same as:

- A. 2π rad
- B. π rad
- C. 57 rad
- D. 1 rad

10. A _____ vector is obtained by dividing the vector by its magnitude:

- A. Unit
- B. Position
- C. Normal
- D. Negative

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11. are the speed of sound to be 340 m/s. A thunder clap is heard about 3 s after the lightning is seen. The source of both light and sound is:

- A. moving overhead faster than the speed of sound
- B. emitting a much higher frequency than is heard
- C. emitting a much lower frequency than is heard
- D. about 1000 m away E. much more than 1000 m away

12. If the distance between all pairs of particles of the body do not change by applying a force then the body is said to be.

- A. Rigid
- B. Large
- C. Small
- D. Flexible

13. For an ideal fluid following through a horizontal pipe, Bernoulli's equation states that the sum of the pressure and energy per unit volume along the pipe des which of the following? (Assume measurement are taken along the pipe in the direction of fluid flow.)

- A. Increases as the pipe diameter increases
- B. Decreases as the pipe diameter increases
- C. Remains constant as the pipe diameter increases
- D. Increases, then decreases as the pipe diameter increases

14. Which of the following statement is true?

- A. Weight is a force, mass is a measure of inertia
- B. Mass depends on gravity, weight does not
- C. Gravity is necessary to measure both weight and mass
- D. Heaver objects weight more than light objects

15. A/an _____ is the basic reason to change in the motion of an object according to Newton's second law of motion

- A. Net force
- B. Decrease in inertia
- C. Change in velocity

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D. Acceleration

16. When a spring is compressed or stretched, The potential energy of the spring

A. Decreases

B. Stays constant

C. Increases

D. Becomes zero

17. The dimensional units of ratio of work and power is:

A. J

B. T

C. L

D. F

18. A particles oscillating in simple harmonic motion is:

A. In equilibrium at the end of its path because its velocity is zero there

B. In equilibrium at the center of its path because its acceleration is zero there

C. Never in equilibrium because it is in motion

D. Never in equilibrium because there is always a force

19. A Fire whistles emits a tone of 170 Hz Take the speed of sound in air to be 340m/s. The wavelength of this sound is about

A. 3.0m

B. 0.5m

C. 1.0m

D. 2.0m

20. How much pressure is exerted on a structure on a submarine at a depth of 8.50km in the Pacific Ocean? (The density of sea water= $1.025 \times 10^3 \text{ kg/m}^3$, and the atmospheric pressure at level= $1.01 \times 10^5 \text{ Pa}$).

A. $8.6 \times 10^5 \text{ Pa}$

B. $8.7 \times 10^6 \text{ Pa}$

C. $9.5 \times 10^6 \text{ Pa}$

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D. $8.6 \times 10^6 \text{ Pa}$

21. A force of 100N acts upon a body for five seconds. What will be the change in momentum?

A. 200Ns

B. 500Ns

C. 20 Ns

D. 50 Ns

22. One newton is a force that produces an acceleration of 0.5 m/s^2 in a body of mass:

A. 1 kg

B. 2 kg

C. 4kg

D. 8 kg

23. When breaks are applied to a fast moving car, the passenger will be throws:

A. Forward

B. Backward

C. Downward

D. Upward

24. In the formula $F = G \frac{m_1 m_2}{r^2}$, the quantity G:

A. Depends on the local value of g

B. Is used only when Earth is one of the two masses

C. Is greatest at the surface of Earth

D. Is a universal constant of nature

25. Per-second refers to the dimensions of physical quantity:

A. Angular displacement

B. Angular velocity

C. Angular acceleration

D. Angular momentum

26. Light years is a unit of:

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- A. Acceleration
- B. Velocity
- C. Time
- D. Distance

27. Young's modules can be used to calculate the strain for a stress that is:

- A. Just below the ultimate strength
- B. Just above the ultimate strength
- C. Well below the yield strength
- D. Well above the yield strength

28. A wheel of radius 50 cm having the angular speed of 5 rad/s will have linear speed in m/s?

- A. 0.5
- B. 1.5
- C. 2.5
- D. 3.5

29. Objects moves in a circle at constant speed. The work done by the centripetal force is zero because:

- A. The displacement for each revolution is zero
- B. The average force for each revolution is zero
- C. There is no friction
- D. The centripetal force is perpendicular to the velocity

30. A ball is thrown upward into the air with a speed that is greater than terminal speed. It lands at the place where it was thrown . During its flight the force of air resistance is the greatest.

- A. Just after it is thrown
- B. Half way up
- C. At the top its trajectory
- D. Halfway down

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31. A fire emits a tone of 170Hz Take the speed of sound in air to be 340m/s. The wavelength of this sound is about.

A. 2.0m

B. 3.0m

C. 0.5m

D. 1.0m

32. The angular momentum L of a rigid is given by:

A. $I\omega$

B. $R\omega$

C. mrl

D. mrv

33. A water bed that's is 1.5m wide and 2.5m long weighs 1055N assuming the entire lower surface of the bed is in contact with the floor. What is the pressure the bed exerts on the floor?

A. 270 Pa

B. 250 Pa

C. 260 Pa

D. 280 Pa

34. Young's modules can be used to calculate the strain for a stress that is:

A. Just below the ultimate strength

B. Just above the ultimate strength

C. Well below the yield strength

D. Well above the yield strength

35. A mosquito's buzz is often rated with a decibel rating of 40 dm. Normal conversation is often rated at 60 db. How many times more intense is normal conversation compared to a mosquito's buzz.

A. 100

B. 20

C. 400

D. 2

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36. Which of the following statement/s is/ are true?

- 1) The weight of a man on the Moon is smaller than on the Earth
- 2) The mass of a man is the same on both the Moon and the Earth.
- 3) We cannot determine our own mass in the outer space because there is no gravity.

A. 1 Only

B. (1) And (2) only

C. 2 Only

D. (2) And (3) only

37. The angular momentum vector of Earth about its rotation axis, Due to its daily rotation, is directed:

- A. Tangent to equator toward the east
- B. Tangent to equator toward the west
- C. North
- D. south

38. When body moves with constant acceleration the velocity time graph is:

- A. Hyperbola
- B. Straight line
- C. Curve
- D. Parabola

39. The persistence of audible sound due to the successive reflections from the surroundings objects even after the source has stopped to the produced that sound is called

- A. Reverberation
- B. Rarefaction
- C. Reflection
- D. Echo

40. A vector A is added to a vector B. The resultant vector $A+B$ have greatest magnitude when

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A. The magnitude of vector $A+B$ does not depend on the directions of A and B

B. When vectors A and B are perpendicular

C. When vectors A and B are parallel and in the same direction

D. When vectors A and B are parallel and in the opposite direction

41. The unit of intensity of sound is:

A. Candela

B. Decibel

C. Kelvin

D. Meter

42. When number of bodies are such that they can exert force upon one another and no external agency exerts a force on them, they are said to form:

A. Non-inertial

B. Non isolated system

C. An isolated system

D. An internal frames of reference

43. Because a buoyant force acts in the opposite direction of gravity:

A. Objects submerged in water have a net force larger than their weight

B. Objects submerged in water have a net force smaller than their weight

C. Objects submerged in water appear to weigh more than they do in air

D. Objects submerged in water have a net force equal to their weight

44. if the mass of moving objects is doubled then its K.E becomes:

A. 4 times

B. 5 times

C. 16 times

D. 2 times

45. acceleration of an object must be zero at a point where.

A. The average velocity is zero

B. The instantaneous velocity is constant

C. The instantaneous velocity is not zero but changing

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D. The instantaneous velocity is zero but changing

46. back and forth in a leftward and rightward direction. This type of wave is known as a ____.

- A. Mechanical
- B. **Longitudinal**
- C. Transverse
- D. Electromagnetic

47. Final take off velocity of an airplane is 67 m/s. The length of runway is 2km, The constant acceleration is:

- A. 2.24ms⁻¹
- B. **2.24 ms⁻²**
- C. 3.24 ms⁻²
- D. 3.24 ms⁻¹

48. The frequency which is note audible to the human ear is:

- A. **50000Hz**
- B. 500 Hz
- C. 50 Hz
- D. 5000 Hz

49. The speed of sound is medium depends upon

- A. **Properties of the medium**
- B. Amplitude
- C. Frequency
- D. Wavelength

50. If the body whose mass is much less than a body at rest collide with it elastically. Then it bounces back with:

- A. Half of the velocity
- B. Double velocity
- C. None of these
- D. **Same velocity**

51. SI Unit of time period is ____.

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- A. Nanosecond
- B. Minute
- C. Hour
- D. **Second**

52. Which of the following is not a characteristic of mechanical waves?

- A. **They travel in a direction which is at right angles to the direction of the particles of the medium.**
- B. They transport energy
- C. They are created by a vibrating source
- D. They consist of disturbances or oscillations of a medium

53. As the wavelength of a wave in a uniform medium increases, its frequency will ____.

- A. Remain the same
- B. None of these
- C. Increase
- D. **Decrease**

54. Which of the following has the smallest moment of inertia about the central axis if all have equal masses and radii?

- A. Ring
- B. Disc
- C. **Sphere**
- D. Spherical shell

55. Which of the following cases is/are NOT a uniformly accelerated motion?

- 1) 1. A feather falls from certain height inside a vacuum tube
- 2) 2. A ball rolls along a frictionless plane at uniform speed
- 3) 3. A coin falls from a certain height in air but air resistance is negligible

- A. 1 and 2 only
- B. **2 Only**
- C. 2 and 3 only

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D. 1 only

56. Before the density of an object can be found what two measurements are required?

A. Mass and volume

B. Temperature and volume

C. Mass and length

D. Volume and length

57. Distance covered during one vibration of an oscillating body in terms of amplitude A:

A. 4A

B. 2A

C. A/2

D. A

58. If the distance between all pairs of particles of the body do not change by applying a force then the body is said to be

A. Large

B. Flexible

C. Rigid

D. Small

59. Two seconds waves are travelling through a container of unknown gas. Wave A has a wavelength of 1.2m. Wave B has a wavelength of 3.6m. The frequency of wave B must be ___ the frequency of wave A

A. Three times larger than

B. The same as

C. One-third

D. One-ninth

60. The goal of all scientific inquiry (or scientific method) is:

A. To make everyone rich and happy

B. World dominion

C. Predicting natural events based on known patterns

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D. World peace

61. Pre-second refers to the dimension of physical quantity:

A. Angular velocity

B. Angular momentum

C. Angular displacement

D. Angular acceleration

62. The unit $\text{kg m}^2/\text{s}$ can be used for:

A. Rotational inertia

B. Rotational kinetic energy

C. Angular momentum

D. Torque

63. The center of mass of Earth's atmosphere is:

A. Near the surface of Earth

B. Near the center of Earth

C. A little less than half way between Earth's surface and the outer boundary of the atmosphere

D. Near the outer boundary of the atmosphere

64. Work may be done by _____

A. Only living organisms

B. Only non-living objects

C. Both living organisms and non-living organisms

D. Only vehicles

65. To determine if a rigid body is in equilibrium the vector sum of the gravitational forces acting on the particles of the body can be replaced by a single force acting at:

A. The center of gravity

B. A point on the boundary

C. The geometrical center

D. The center of mass

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66. Whenever an object strikes a stationary object of equal mass:

- A. The first object must stop
- B. The two objects can not stick together
- C. The collision must be elastic
- D. None of these**

67. A glass of water is placed on a sheet of paper. Quickly drag the sheet of paper from under the glass of water without spilling the water is due to:

- A. Newton's third law of motion
- B. Acceleration
- C. Lack of friction between paper and glass
- D. Inertia**

68. The dimension of Joule is:

- A. $[M L^2 T^{-2}]$**
- B. $[M L^2 T^{-1}]$
- C. $[M L^2 T^{-3}]$
- D. $[M L^2 T^{-2}]$

69. The slope of a velocity-time graph at any point may be identified with:

- A. Instantaneous velocity
- B. Instantaneous acceleration**
- C. Average acceleration
- D. Instantaneous acceleration

70. Ali wants to lift a mass of 7.5 kg with constant velocity by a rope that passes through a frictionless pulley which is attached to the ceiling of room. Calculate the tension in the rope by neglecting the mass of the rope.

- A. 0.75 N
- B. 7.5 kg
- C. 75 kg
- D. 75 N**

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71: If the magnitude of force applied is increased. The work done will

- A. Become zero
- B. Be decreased
- C. Have no change
- D. **Be increased**

72. A couple produces:

- A. No motion
- B. Linear and rotational
- C. **Purely rotational motion**
- D. Purely, linear motion

73. Which of the following statements are TRUE of sounds waves? Identify all that apply.

- A. A sound wave is a transverse waves
- B. To hear the sound of a tuning fork, the tines of the fork must move air from the fork to one's ear
- C. **A sound wave is a mechanical wave**
- D. Sound can travel through a vacuum

74. Add two vector of length 4m & 5m but their orientation is not known. The length after addition of these two vectors will be:

- A. 9m
- B. Less than 1 m
- C. Between 9m and 5
- D. **Between 9m and 1 m**

75. The application/s of dimensional analysis is/are:

- 1) To convert a physical quantity from one system of units to another
- 2) To check the dimensional correctness of a given equation
- 3) Establish a relationship between different physical quantities in an equation

- A. 1 & 3 only

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- B. 2 & 3 only
- C. 1 only
- D. 1,2 &3 only

76. People try to keep the ____ over their feet, in order to feel stable.

- A. Mass
- B. Center of gravity
- C. Weight
- D. Centre of mass

77. Which statement is not true for acceleration?

- A. Riding your bike straight down the street at a constant speed
- B. Riding your bike faster when you head down a hill
- C. Slowing your bike ride so you can make it up a hill
- D. Stopping your bike at an intersection

78. A force of 100N acts upon a body for five seconds. What will be the change in momentum?

- A. 50 NS
- B. 500 NS
- C. 200 NS
- D. 20 NS

79. In _____ objects return to its original position if displacement slightly

- A. Unstable equilibrium
- B. Dynamic equilibrium
- C. Rotational equilibrium
- D. Stable equilibrium

80. For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

- A. The center of gravity
- B. The same point
- C. The center of mass

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D. The geometrical center

81. The scalar product of two vectors is maximum when they are:

A. Parallel

B. Anti-parallel

C. Perpendicular

D. Null

82. Swimming becomes possible because of:

A. Third law of motion

B. Law of torque

C. Second law of motion

D. First law of motion

83. The projectile path is known as its:

A. Time of action

B. Range

C. Curve

D. Trajectory

84. A man, with his arms at his sides, is spinning on a light frictionless turntable. When he extends his arms:

A. His angular velocity

B. His angular velocity increases

C. His angular momentum remains the same

D. His rotational inertia decreases

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By Rj Afzal

1. Power is equal to the dot product of force and
 - a. Position vector
 - b. Displacement
 - c. **Velocity**
 - d. Acceleration
2. The frequency which is not audible for human ear is
 - a. **50000Hz**
 - b. 500Hz
 - c. 5000Hz
 - d. 50Hz
3. A Wheel of radius 50cm having the angular speed of 5rad/s will have linear speed in m/s?
 - a. 1.5
 - b. **2.5**
 - c. 3.5
 - d. 0.5
4. When breaks are applied to a fast moving car, the passenger will be throws:
 - a. Backward
 - b. **Forward**
 - c. Upward
 - d. downward
5. Stress can be measured
 - a. N.m
 - b. **N/m²**
 - c. Nm²
 - d. N/m
6. A heavy particle moving with 5m/s suffers an elastic collision with a light particle at rest. After collision, the velocity of light particle will be:
 - a. 10m/s
 - b. 2.5m/s
 - c. 20m/s
 - d. **5m/s**
- 7) Acceleration in a body is always produced in the direction of:

- a. Velocity
- b. Weight
- c. Acceleration
- d. Force

8) The fundamental dimensions of angular momentum are:

- a. $ML^{-2}T^{-2}$
- b. ML^2T^{-1}
- c. ML^2T^{-2}
- d. MLT^{-1}

9) Ali wants to lift a mass of 7.5kg with constant velocity by a rope that passes through a frictionless pulley which is attached to the ceiling of room. Calculate the tension in rope by neglecting the mass of the rope.

- a. 75N
- b. 7.5kg
- c. 75kg
- d. 0.75N

10) The first condition of equilibrium implies that:

- a. $\sum F_y = 0$
- b. $\sum F_x = \sum F_y$
- c. $\sum F = 0$
- d. $\sum F_x = 0$

11) Final take off velocity of an airplane is 67m/s. The length of runway 2km, the constant acceleration is :

- a. $3.24ms^{-1}$
- b. $3.24ms^{-2}$
- c. $2.24ms^{-2}$
- d. $2.24ms^{-1}$

12) The ultimate strength of a sample is the stress at which the sample :

- a. Remains under water
- b. Breaks
- c. Bends 180
- d. Returns to its original shape

13) The area under the velocity time graph is:

- a. Acceleration
- b. Torque
- c. Distance
- d. Force

14) If you travelled for 25hours with an average speed 48miles/hours, the distance travelled is:

- a. 19.2miles
- b. 120miles
- c. 300miles
- d. 48miles

15) A fire whistle emits a tone of 170Hz,Take the speed of sound in air to be 340m/s. The wavelength of this sound is about:

- a. 1.0m
- b. 2.0m
- c. 0.5m
- d. 3.0m

16) Motion defines by the pair of variables:

- a. Speed and passage of time
- b. Speed and distance
- c. Change of position and passage of time

17) The law of inertia was firstly formulated by:

- a. Newton
- b. Galileo
- c. Einstein
- d. Aristotle

18) When the velocity of an aeroplane is doubled, the momentum

- a. Is conserved
- b. Becomes Zero
- c. Increases uniformly
- d. Remains unchanged

19) As we move above, the body the surface of the earth the change in potential energy will be:

- a. Infinity
- b. Positive
- c. Zero
- d. Negative

20) The lowest tone produced by a certain organ comes from a 3.0-m pipe with both ends open. If the speed of sound is 340m/s, the frequency of this tone is approximately:

- A. 7Hz
- B. 14 Hz
- C. 28 Hz

D. 57 Hz page no 53

21) To raise the pitch of a certain piano string, the piano tuner

- A. loosens the string
- B. tightens the string
- C. shortens the string page no 54
- D. lengthens the string

22) A force of 5000N is applied outwardly to each end of a 5.0-m long rod with a radius of 34.0 cm and a Young's modulus of $125 \times 10^8 \text{ N/m}^2$. The elongation of the rod is:

- A. 0.0020mm
- B. 0.0040mm
- C. 0.14mm
- D. 0.55mm

23) A particle oscillating in simple harmonic motion is:

- A. never in equilibrium because it is in motion
- B. never in equilibrium because there is always a force
- C. in equilibrium at the ends of its path because its velocity is zero there
- D. in equilibrium at the center of its path because the acceleration is zero there

24) In simple harmonic motion, the restoring force must be proportional to the:

- A. amplitude
- B. frequency
- C. velocity
- D. displacement

page no:44 Reference: Now we have a restoring force that is proportional to the distance away from the equilibrium point

25) A 160-N child sits on a light swing and is pulled back and held with a horizontal force of 100 N. The magnitude of the tension force of each of the two supporting ropes is:

- A. 60N
- B. 94N
- C. 120N
- D. 190N

26) An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

- A. 12.6 rad/s
- B. 1.57 rad/s
- C. 2.0 rad/s
- D. 6.3 rad/s

Reference: Page no 42 $V=20$, $t=10$ $\Omega=2\pi v = 2 \times 3.14 \times 20 = 125.6(1/t) = 12.6 \text{ rad/sec}$

27) For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

- A. the center of mass
- B. the center of gravity
- C. the geometrical center
- D. the same point

28) Ten seconds after an electric fan is turned on, the fan rotates at 300 rev/min. Its average angular acceleration is:

C.30 rev/s²

Reference: Page no 28 Avg angular acceleration=(final angular velocity-intial angular velocity)/time
=(300-0)/10=30rev/s²

29) A 4.0-N puck is traveling at 3.0m/s. It strikes a 8.0-N puck, which is stationary. The two pucks stick together. Their common final speed is:

A.1.0m/s

B.1.5m/s

C.2.0m/s

D.2.3m/s

Reference: The case you're describing is called an inelastic collision.

30) Two objects collide, stick to each other and continue their motion as one body. Due to momentum conservation principle, sum of two bodies momenta before collision has to be equal to momentum of the one body after collision.

B before = p first + p second = $m_1v_1 + m_2v_2$ p after = $(m_1 + m_2)v$ common

Since p before = p after, $(m_1 + m_2)v$ common = $m_1v_1 + m_2v_2$

We can get v common from that: v common = $(m_1v_1 + m_2v_2) / (m_1 + m_2)$

31) An object moving in a circle at constant speed

D.has an acceleration of constant magnitude

Reference: Page no: 29

32) Now consider a particle going around a circle at constant speed. You might think that constant speed means no acceleration. Bu this is wrong! It is changing its direction and accelerating. This is called "centripetal acceleration", meaning acceleration directed towards the centre of the circle plane traveling north at 200m/s turns and then travels south at 200m/s. The change in its velocity is:

B.400m/s south

Reference:400 m/s south because it need 200 to overcome the 200 north then another 200 to get going 200 south.

33) At time $t = 0$ a car has a velocity of 16 m/s. It slows down with an acceleration given by $-0.50t$, in m/s² for t in seconds. It stops at $t =$

A. 64 s

B. 32 s

C. 16 s

D. 8.0 s

Reference: $V_f = v_i + at$ $V_f = 16 - 0.50t(0)$ $V_f = 16$ As we know $a = \text{final velocity}/t$ And $t = \text{final velocity}/a$ $16/.5 = t$ $32 = t$

34) 1 mi is equivalent to 1609 m so 55 mph is:

A. 15 m/s

B. 25 m/s

C. 66 m/s

D. 88 m/s

Reference: 1 mile = 1609 meters or 1.609 kilometers

so,

55 miles = $1609 \times 55 = 88495$ meters or 88.495 kilometers

35) The number of significant figures in 0.00150 is:

C. 3

37) One revolution is the same as:

A. 1 rad

B. 57 rad

C. $\pi/2$ rad

D. π rad

E. 2π rad

38) For a body to be in equilibrium under the combined action of several forces:

A. all the forces must be applied at the same point all the forces must be applied at the same point

B. all of the forces form pairs of equal and opposite forces

C. any two of these forces must be balanced by a third force

D.the sum of the torques about any point must equal zero

39) As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, a distance of 0.75m from the origin. The z component of the angular momentum around the origin is:

A. $6.0\text{kg} \cdot \text{m}^2/\text{s}$

B. $9.0\text{kg} \cdot \text{m}^2/\text{s}$

C. $11\text{ kg} \cdot \text{m}^2/\text{s}$

D. $14\text{ kg} \cdot \text{m}^2/\text{s}$

40) A net torque applied to a rigid object always tends to produce:

A. rotational equilibrium

B.linear acceleration

C.angular acceleration

D.rotational inertia

41) object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

A. 2.0 rad/s

B. 12.6 rad/s

C. 1.57 rad/s

D. 6.3 rad/s

42) In simple harmonic motion, the restoring force must be proportional to the:

A. displacement

B. amplitude

C. frequency

D. velocity

43) Mercury is a convenient liquid to use in a barometer because:

A. it has a high density

B. it is a metal

C. it has a high boiling point

D. it expands little with temperature

44) The units of the electric field are

A. J/m

B. J/(C·m)

C. J/C

D. J·C

45) A farad is the same as a

A. J/V

B. V/J

C. C/V

D. V/C

46) The wavelength of red light is 700 nm. Its frequency is _____.

A. 4.30×10^5 Hertz

47) Which of the following statements is NOT TRUE about electromagnetic waves?

A. The electromagnetic radiation from a burning candle is unpolarized.

48) Radio waves and light waves are _____.

A. Electromagnetic and transverse both

B. longitudinal waves

C. Transverse waves

D. Electromagnetic and transverse both

E. Electromagnetic and longitudinal both

49) Fahrenheit and Kelvin scales agree numerically at a reading of:

A. -40

50) According to the theory of relativity:

A. moving clocks run fast

51) Light from a stationary spaceship is observed, and then the spaceship moves directly away from the observer at high speed while still emitting the light. As a result, the light seen by the observer has:

D. lower frequency and a longer wavelength than before

52) How fast should you move away from a 6.0×10^{14} Hz light source to observe waves with a frequency of 4.0×10^{14} Hz?

A. 38c

53) The quantum number n is most closely associated with what property of the electron in a hydrogen atom?

A. Energy

B. Orbital angular momentum

C. Spin angular momentum

D. Magnetic moment

54) The quantum number m_s is most closely associated with what property of the electron in an atom?

A. Energy

B. Magnitude of the orbital angular momentum

C. z component of the spin angular momentum

D. z component of the orbital angular momentum

55) As the wavelength of a wave in a uniform medium increases, its speed will ____.

A. Remain the same

B. Decrease

C. Increase

D. Remain the same

E. None of these

56) Velocity is defined as:

A. rate of change of position with time

B. position divided by time

C. rate of change of acceleration with time

D. a speeding up or slowing down

57) Acceleration is defined as:

A. rate of change of position with time

B. speed divided by time

C. rate of change of velocity with time

D. a speeding up or slowing down

58) Which of the following is a scalar quantity?

A. Speed

B. Velocity

C. Displacement

D. Acceleration

59) Which of the following is a vector quantity?

A. Mass

B. Density

C. Speed

D. Temperature

E. None of these

60) Which of the following is NOT an example of accelerated motion?

A. Vertical component of projectile motion

B. Circular motion at constant speed

C. A swinging pendulum

D. Earth's motion about sun

E. Horizontal component of projectile motion

61) A particle goes from $x = -2\text{m}$, $y = 3\text{m}$, $z = 1\text{m}$ to $x = 3\text{m}$, $y = -1\text{m}$, $z = 4\text{m}$. Its displacement is:

- A. $(1\text{m})\hat{i} + (2\text{m})\hat{j} + (5\text{m})\hat{k}$
- B. $(5\text{m})\hat{i} - (4\text{m})\hat{j} + (3\text{m})\hat{k}$**
- C. $-(5\text{m})\hat{i} + (4\text{m})\hat{j} - (3\text{m})\hat{k}$
- D. $-(1\text{m})\hat{i} - (2\text{m})\hat{j} - (5\text{m})\hat{k}$

62) A jet plane in straight horizontal flight passes over your head. When it is directly above you, the sound seems to come from a point behind the plane in a direction 30° from the vertical. The speed of the plane is:

- A. the same as the speed of sound
- B. half the speed of sound**
- C. three-fifths the speed of sound
- D. 0.866 times the speed of sound

63) A plane traveling north at 200m/s turns and then travels south at 200m/s . The change in its velocity is:

- A. zero
- B. 200m/s north
- C. 200m/s south
- D. 400m/s north

E. 400m/s south

64) Two bodies are falling with negligible air resistance, side by side, above a horizontal plane. If one of the bodies is given an additional horizontal acceleration during its descent, it

- A. strikes the plane at the same time as the other body**
- B. strikes the plane earlier than the other body
- C. has the vertical component of its velocity altered
- D. has the vertical component of its acceleration altered

65) The velocity of a projectile equals its initial velocity added to: A. a constant horizontal velocity

- B. a constant vertical velocity
- C. a constantly increasing horizontal velocity
- D. a constantly increasing downward velocity

66) A stone thrown from the top of a tall building follows a path that is:

- A. circular
- B. made of two straight line segments
- C. hyperbolic
- D. parabolic

67) Identical guns fire identical bullets horizontally at the same speed from the same height above level planes, one on the Earth and one on the Moon. Which of the following three statements is/are true?

- I. The horizontal distance traveled by the bullet is greater for the Moon.
 - II. The flight time is less for the bullet on the Earth.
 - III. The velocity of the bullets at impact are the same.
- A. III only

B. I and II only

C. I and III only

D. II and III only

68) A stone is thrown horizontally and follows the path XYZ shown. The direction of the acceleration of the stone at point Y is:

A. ↓

B. →

C. 9

D. t

69) A bullet shot horizontally from a gun:

A. strikes the ground much later than one dropped vertically from the same point at the same instant

B. never strikes the ground

C. strikes the ground at approximately the same time as one dropped vertically from the same point at the same instant

D. travels in a straight line

E. strikes the ground much sooner than one dropped from the same point at the same instant

70) A bomber flying in level flight with constant velocity releases a bomb before it is over the target. Neglecting air resistance, which one of the following is NOT true?

A. The bomber is over the target when the bomb strikes

B. The acceleration of the bomb is constant

C. The horizontal velocity of the plane equals the vertical velocity of the bomb when it hits the target

D. The bomb travels in a curved path

E. The time of flight of the bomb is independent of the horizontal speed of the plane

71) An object is shot from the back of a railroad flatcar moving at 40km/h on a straight horizontal road. The launcher is aimed upward, perpendicular to the bed of the flatcar. The object falls:

A. in front of the flatcar

B. behind the flatcar

C. on the flatcar

D. either behind or in front of the flatcar, depending on the initial speed of the object

E. to the side of the flatcar

72) A ball is thrown horizontally from the top of a 20-m high hill. It strikes the ground at an angle of 45° . With what speed was it thrown?

A. 14m/s

B. 20m/s

C. 28m/s

73) The SI standard of time is based on:

- A. the daily rotation of the earth
- B. the frequency of light emitted by Kr86
- C. the yearly revolution of the earth about the sun
- D. a precision pendulum clock
- E. none of these

74) A nanosecond is:

- A. 109 s
- B. 10^{-9} s
- C. 10^{-10} s
- D. 10^{-10} s
- E. 10^{-12}

75) The SI standard of length is based on:

- A. the distance from the north pole to the equator along a meridian passing through Paris
- B. wavelength of light emitted by Hg198
- C. wavelength of light emitted by Kr86
- D. a precision meter stick in Paris
- E. the speed of light

76) In 1866, the U. S. Congress defined the U. S. yard as exactly 3600/3937 international meter. This was done primarily because:

- A. length can be measured more accurately in meters than in yards
- B. the meter is more stable than the yard
- C. this definition relates the common U. S. length units to a more widely used system
- D. there are more wavelengths in a yard than in a meter

77) Which of the following is closest to a yard in length?

- A. 0.01m

B. 0.1m

C. 1m

D. 100m

78) There is no SI base unit for area because:

A. an area has no thickness; hence no physical standard can be built

B. we live in a three (not a two) dimensional world

C. it is impossible to express square feet in terms of meters

D. area can be expressed in terms of square meters

79) The SI base unit for mass is:

A. gram

B. pound

C. kilogram

D. ounce

80) A gram is:

A. 10^{-6} kg

B. 10^{-3} kg

C. 1 kg

D. 103 kg

82) Which of the following weighs about a pound?

A. 0.05 kg

B. 0.5 kg

C. 5 kg

D. 50 kg

83) $(5.0 \times 10^4) \times (3.0 \times 10^6) =$

A. 1.5×10^9

B. 1.5×10^{10}

C. 1.5×10^{11}

D. 1.5×10^{12}

84) $(5.0 \times 10^4) \times (3.0 \times 10^{-6}) =$

A. 1.5×10^{-3}

B. 1.5×10^{-1}

C. 1.5×10^1

D. 1.5×10^3

85) $5.0 \times 10^5 + 3.0 \times 10^6 =$

A. 8.0×10^5

B. 8.0×10^6

C. 5.3×10^5

D. 3.5×10^5

E. 3.5×10^6

86) $(7.0 \times 10^6) / (2.0 \times 10^{-6}) =$

A. 3.5×10^{-12}

B. 3.5×10^{-6}

C. 3.5

D. 3.5×10^6

E. 3.5×10^{12}

87) The number of significant figures in 0.00150 is:

A. 2

B. 3

C. 4

D. 5

88) The number of significant figures in 15.0 is:

A. 1

B. 2

C. 3

D. 4

89) $3.2 \times 2.7 =$

A. 9

B. 8

C. 8.6

D. 8.64

90) $1.513 + 27.3 =$

A. 29

B. 28.8

C. 28.9

D. 28.81

91) 1 mi is equivalent to 1609 m so 55 mph is:

A. 15 m/s

B. 25 m/s

C. 66 m/s

D. 88 m/s

92) A sphere with a radius of 1.7 cm has a volume of:

A. $2.1 \times 10^{-5} \text{ m}^3$

B. $9.1 \times 10^{-4} \text{ m}^3$

C. $3.6 \times 10^{-3} \text{ m}^3$

D. 0.11 m^3

93) A sphere with a radius of 1.7 cm has a surface area of:

A. $2.1 \times 10^{-5} \text{ m}^2$

B. $9.1 \times 10^{-4} \text{ m}^2$

C. $3.6 \times 10^{-3} \text{ m}^2$

D. 0.11 m^2

94) A right circular cylinder with a radius of 2.3 cm and a height of 1.4 m has a volume of:

A. 0.20 m^3

B. 0.14 m^3

C. $9.3 \times 10^{-3} \text{ m}^3$

D. $2.3 \times 10^{-3} \text{ m}^3$

95) A right circular cylinder with a radius of 2.3 cm and a height of 1.4 cm has a total surface area of:

A. $1.7 \times 10^{-3} \text{ m}^2$

B. $3.2 \times 10^{-3} \text{ m}^2$

C. $2.0 \times 10^{-3} \text{ m}^3$

D. $5.3 \times 10^{-3} \text{ m}^2$

96) A cubic box with an edge of exactly 1 cm has a volume of:

A. 10^{-9} m^3

B. 10^{-6} m^3

C. 10^{-3} m^3

97) A square with an edge of exactly 1 cm has an area of:

A. 10^{-6} m^2

B. 10^{-4} m^2

C. 10^2 m^2

D. 10^4 m^2

98) 1 m is equivalent to 3.281 ft. A cube with an edge of 1.5 ft has a volume of:

A. $1.2 \times 10^2 \text{ m}^3$

B. $9.6 \times 10^{-2} \text{ m}^3$

C. 10.5 m^3

D. $9.5 \times 10^{-2} \text{ m}^3$

99) During a short interval of time the speed v in m/s of an automobile is given by $v = at^2 + bt^3$, where the time t is in seconds. The units of a and b are respectively:

A. $\text{m} \cdot \text{s}^2; \text{m} \cdot \text{s}^4$

B. $\text{s}^3/\text{m}; \text{s}^4/\text{m}$

C. $\text{m}/\text{s}^2; \text{m} / \text{s}^3$

D. $\text{m}/\text{s}^3; \text{m} / \text{s}^4$

100) Suppose $A = BC$, where A has the dimension L/M and C has the dimension L/T . Then B has the dimension:

A. T/M

B. L^2/TM

C. TM/L^2

D. L^2T/M

101) Suppose $A = B^n C^m$, where A has dimensions LT , B has dimensions L^2T^{-1} , and C has dimensions LT^2 . Then the exponents n and m have the values:

A. $2/3; 1/3$

B. $2; 3$

C. $4/5; -1/5$

D. $1/5; 3/5$

102) A particle moves along the x axis from x_i to x_f . Of the following values of the initial and final coordinates, which results in the displacement with the largest magnitude?

A. $x_i = 4\text{m}, x_f = 6\text{m}$

B. $x_i = -4\text{m}, x_f = -8\text{m}$

C. $x_i = -4\text{m}, x_f = 2\text{m}$

D. $x_i = 4\text{m}, x_f = -2\text{m}$

E. $x_i = -4\text{m}, x_f = 4\text{m}$

103) A particle moves along the x axis from x_i to x_f . Of the following values of the initial and final coordinates, which results in a negative displacement?

A. $x_i = 4\text{m}, x_f = 6\text{m}$

B. $x_i = -4\text{m}, x_f = -8\text{m}$

C. $x_i = -4\text{m}, x_f = 2\text{m}$

D. $x_i = -4\text{m}, x_f = -2\text{m}$

104) The average speed of a moving object during a given interval of time is always:

A. the magnitude of its average velocity over the interval

B. the distance covered during the time interval divided by the time interval

C. one-half its speed at the end of the interval

D. its acceleration multiplied by the time interval

105) Two automobiles are 150 kilometers apart and traveling toward each other. One automobile is moving at 60km/h and the other is moving at 40km/h mph. In how many hours will they meet?

A. 2.5

B. 2.0

C. 1.75

D. 1.5

106) A car travels 40 kilometers at an average speed of 80km/h and then travels 40 kilometers at an average speed of 40km/h. The average speed of the car for this 80-km trip is:

A. 40km/h

B. 45km/h

C. 48km/h

D. 53km/h

107) A car starts from Hither, goes 50 km in a straight line to Yon, immediately turns around, and returns to Hither. The time for this round trip is 2 hours. The magnitude of the average velocity of the car for this round trip is:

A. 0

B. 50 km/hr

C. 100 km/hr

D. 200 km/hr

108) A car starts from Hither, goes 50 km in a straight line to Yon, immediately turns around, and returns to Hither. The time for this round trip is 2 hours. The average speed of the car for this round trip is: A. 0

B. 50 km/h

C. 100 km/h

D. 200 km/h

110) The coordinate of a particle in meters is given by $x(t) = 16t - 3.0t^3$, where the time t is in seconds. The particle is momentarily at rest at $t =$

A. 0.75s

B. 1.3s

C. 5.3s

D. 7.3s

111) A drag racing car starts from rest at $t = 0$ and moves along a straight line with velocity given by $v = bt^2$, where b is a constant. The expression for the distance traveled by this car from its position at $t = 0$ is:

A. bt^3

B. $bt^3/3$

C. $4bt^2$

D. $3bt^2$

112) A ball rolls up a slope. At the end of three seconds its velocity is 20 cm/s; at the end of eight seconds its velocity is 0. What is the average acceleration from the third to the eighth second?

A. 2.5cm/s^2

B. 4.0cm/s^2

C. 5.0cm/s^2

D. 6.0cm/s^2

113) The coordinate of an object is given as a function of time by $x = 7t - 3t^2$, where x is in meters and t is in seconds. Its average velocity over the interval from $t = 0$ to $t = 4\text{s}$ is:

A. 5m/s

B. -5m/s

C. 11m/s

D. -11m/s

114) The velocity of an object is given as a function of time by $v = 4t - 3t^2$, where v is in m/s and t is in seconds. Its average velocity over the interval from $t = 0$ to $t = 2\text{s}$ is:

A. is 0

B. is -2m/s

C. is 2m/s

D. is -4m/s

115) The coordinate of an object is given as a function of time by $x = 4t^2 - 3t^3$, where x is in meters and t is in seconds. Its average acceleration over the interval from $t = 0$ to $t = 2\text{s}$ is:

A. -4m/s^2

B. 4m/s^2

C. -10m/s^2

D. 10m/s^2

116) Each of four particles move along an x axis. Their coordinates (in meters) as functions of time (in seconds) are given by particle 1: $x(t) = 3.5 - 2.7t^3$ particle 2: $x(t) = 3.5 + 2.7t^3$ particle 3: $x(t) = 3.5 + 2.7t^2$ particle 4: $x(t) = 3.5 - 3.4t - 2.7t^2$ Which of these particles have constant acceleration?

A. All four

B. Only 1 and 2

C. Only 2 and 3

D. Only 3 and 4

117) Each of four particles move along an x axis. Their coordinates (in meters) as functions of time (in seconds) are given by particle 1: $x(t) = 3.5 - 2.7t^3$ particle 2: $x(t) = 3.5 + 2.7t^3$ particle 3: $x(t) = 3.5 + 2.7t^2$ particle 4: $x(t) = 3.5 - 3.4t - 2.7t^2$ Which of these particles is speeding up for $t > 0$?

A. All four

B. Only 1

C. Only 2 and 3

D. Only 2, 3, and 4

118) An object starts from rest at the origin and moves along the x axis with a constant acceleration of 4 m/s^2 . Its average velocity as it goes from $x = 2\text{ m}$ to $x = 8\text{ m}$ is:

A. 1 m/s

B. 2 m/s

C. 3 m/s

D. 5 m/s

E. 6 m/s

119) Of the following situations, which one is impossible?

A. A body having velocity east and acceleration east

B. A body having velocity east and acceleration west

C. A body having zero velocity and non-zero acceleration

D. A body having constant acceleration and variable velocity

E. A body having constant velocity and variable acceleration

120) Throughout a time interval, while the speed of a particle increases as it moves along the x axis, its velocity and acceleration might be:

A. positive and negative, respectively

B. negative and positive, respectively

C. negative and negative, respectively

D. negative and zero, respectively

121) A particle moves on the x axis. When its acceleration is positive and increasing:

- A. its velocity must be positive
- B. its velocity must be negative
- C. it must be slowing down
- D. it must be speeding up

E. none of the above must be true

122) The position y of a particle moving along the y axis depends on the time t according to the equation $y = at - bt^2$. The dimensions of the quantities a and b are respectively:

- A. L^2/T , L^3/T^2
- B. L/T^2 , L^2/T
- C. L/T , L/T^2
- D. L^3/T , T^2/L

123) A particle moves along the x axis according to the equation $x = 6t^2$, where x is in meters and t is in seconds. Therefore:

- A. the acceleration of the particle is 6 m/s^2
- B. t cannot be negative
- C. the particle follows a parabolic path
- D. each second the velocity of the particle changes by 9.8 m/s

E. none of the above

124) Over a short interval near time $t = 0$ the coordinate of an automobile in meters is given by $x(t) = 27t - 4.0t^3$, where t is in seconds. At the end of 1.0 s the acceleration of the auto is:

- A. 27 m/s^2
- B. 4.0 m/s^2
- C. -4.0 m/s^2
- D. -12 m/s^2

E. -24 m/s^2

125) Over a short interval, starting at time $t = 0$, the coordinate of an automobile in meters is given by $x(t) = 27t - 4.0t^3$, where t is in seconds. The magnitudes of the initial (at $t = 0$) velocity and acceleration of the auto respectively are:

- A. 0; 12 m/s²
- B. 0; 24 m/s²
- C. 27 m/s; 0
- D. 27 m/s; 12 m/s²

126) At time $t = 0$ a car has a velocity of 16 m/s. It slows down with an acceleration given by $-0.50t$, in m/s² for t in seconds. It stops at $t =$

- A. 64 s
- B. 32 s
- C. 16 s
- D. 8.0s

127) At time $t = 0$ a car has a velocity of 16 m/s. It slows down with an acceleration given by $-0.50t$, in m/s² for t in seconds. At the end of 4.0 s it has traveled:

- A. 0
- B. 12 m
- C. 14 m
- D. 25 m
- E. 59 m

128) At time $t = 0$ a car has a velocity of 16 m/s. It slows down with an acceleration given by $-0.50t$, in m/s² for t in seconds. By the time it stops it has traveled:

- A. 15 m
- B. 31 m
- C. 62 m
- D. 85 m

129) Starting at time $t = 0$, an object moves along a straight line with velocity in m/s given by $v(t) = 98 - 2t^2$, where t is in seconds. When it momentarily stops its acceleration is:

- A. 0
- B. -4.0 m/s^2
- C. -9.8 m/s^2
- D. -28 m/s^2

130) Starting at time $t = 0$, an object moves along a straight line. Its coordinate in meters is given by $x(t) = 75t - 1.0t^3$, where t is in seconds. When it momentarily stops its acceleration is:

- A. 0
- B. -73 m/s^2
- C. -30 m/s^2
- D. -9.8 m/s^2

131) A car, initially at rest, travels 20 m in 4 s along a straight line with constant acceleration. The acceleration of the car is:

- A. 0.4 m/s^2
- B. 1.3 m/s^2
- C. 2.5 m/s^2
- D. 4.9 m/s^2

132) A racing car traveling with constant acceleration increases its speed from 10 m/s to 50 m/s over a distance of 60m. How long does this take?

- A. 2.0s
- B. 4.0s
- C. 5.0s
- D. 8.0s

133) A car starts from rest and goes down a slope with a constant acceleration of 5 m/s^2 . After 5 s the car reaches the bottom of the hill. Its speed at the bottom of the hill, in meters per second, is:

- A. 1
- B. 12.5
- C. 25

D. 50

134) A car moving with an initial velocity of 25 m/s north has a constant acceleration of 3 m/s² south. After 6 seconds its velocity will be:

A. 7 m/s north

B. 7 m/s south

C. 43 m/s north

D. 20 m/s north

135) An object with an initial velocity of 12 m/s west experiences a constant acceleration of 4 m/s² west for 3 seconds. During this time the object travels a distance of:

A. 12 m

B. 24 m

C. 36 m

D. 54 m

136) How far does a car travel in 6 s if its initial velocity is 2 m/s and its acceleration is 2 m/s² in the forward direction?

A. 12 m

B. 14 m

C. 24 m

D. 36 m

E. 48 m

137) At a stop light, a truck traveling at 15 m/s passes a car as it starts from rest. The truck travels at constant velocity and the car accelerates at 3 m/s². How much time does the car take to catch up to the truck?

A. 5s

B. 10s

C. 15s

D. 20s

138) A ball is in free fall. Its acceleration is:

- A. downward during both ascent and descent
- B. downward during ascent and upward during descent
- C. upward during ascent and downward during descent
- D. upward during both ascent and descent

139) A ball is in free fall. Upward is taken to be the positive direction. The displacement of the ball during a short time interval is:

- A. positive during both ascent and descent
- B. negative during both ascent and descent
- C. negative during ascent and positive during descent
- D. positive during ascent and negative during descent

140) Which one of the following statements is correct for an object released from rest?

- A. The average velocity during the first second of time is 4.9m/s
- B. During each second the object falls 9.8m
- C. The acceleration changes by 9.8m/s² every second
- D. The object falls 9.8m during the first second of time

141) A freely falling body has a constant acceleration of 9.8 m/s². This means that:

- A. the body falls 9.8 m during each second
- B. the body falls 9.8 m during the first second only
- C. the speed of the body increases by 9.8 m/s during each second
- D. the acceleration of the body increases by 9.8 m/s² during each second

142) An object is shot vertically upward. While it is rising:

- A. its velocity and acceleration are both upward
- B. its velocity is upward and its acceleration is downward
- C. its velocity and acceleration are both downward
- D. its velocity is downward and its acceleration is upward

143) An object is thrown straight up from ground level with a speed of 50 m/s. If $g = 10 \text{ m/s}^2$ its distance above ground level 1.0 s later is:

A. 40 m

B. 45 m

C. 50 m

D. 55 m

144) An object is thrown straight up from ground level with a speed of 50 m/s. If $g = 10 \text{ m/s}^2$ its distance above ground level 6.0 s later is:

A. 0.00 m

B. 270 m

C. 330 m

D. 480 m

E. none of these

145) At a location where $g = 9.80 \text{ m/s}^2$, an object is thrown vertically down with an initial speed of 1.00 m/s. After 5.00 s the object will have traveled:

A. 125 m

B. 127.5m

C. 245 m

D. 250 m

146) An object is thrown vertically upward at 35 m/s. Taking $g = 10 \text{ m/s}^2$, the velocity of the object 5 s later is:

A. 7.0 m/s up

B. 15 m/s down

C. 15 m/s up

D. 85 m/s down

147) A feather, initially at rest, is released in a vacuum 12 m above the surface of the earth. Which of the following statements is correct?

- A. The maximum velocity of the feather is 9.8 m/s
- B. The acceleration of the feather decreases until terminal velocity is reached
- C. The acceleration of the feather remains constant during the fall
- D. The acceleration of the feather increases during the fall

148) An object is released from rest. How far does it fall during the second second of its fall?

- A. 4.9m
- B. 9.8m
- C. 15m
- D. 20m

149) A heavy ball falls freely, starting from rest. Between the third and fourth second of time it travels a distance of:

- A. 4.9m
- B. 9.8m
- C. 29.4m
- D. 34.3m

150) As a rocket is accelerating vertically upward at 9.8 m/s^2 near Earth's surface, it releases a projectile. Immediately after release the acceleration (in m/s^2) of the projectile is:

- A. 9.8 down
- B. 0
- C. 9.8 up
- D. 19.6 up

151) A stone is released from a balloon that is descending at a constant speed of 10 m/s. Neglecting air resistance, after 20 s the speed of the stone is:

- A. 2160 m/s
- B. 1760 m/s
- C. 206 m/s
- D. 196 m/s

152) An object dropped from the window of a tall building hits the ground in 12.0 s. If its acceleration is 9.80 m/s^2 , the height of the window above the ground is:

- A. 29.4m
- B. 58.8m
- C. 118 m
- D. 353 m

E. 706 m

153) Neglecting the effect of air resistance a stone dropped off a 175-m high building lands on the ground in:

- A. 3 s
- B. 4 s

C. 6 s

D. 18 s

154) A stone is thrown vertically upward with an initial speed of 19.5 m/s . It will rise to a maximum height of:

- A. 4.9m
- B. 9.8m

C. 19.4m

D. 38.8m

155) A baseball is hit straight up and is caught by the catcher 2.0 s later. The maximum height of the ball during this interval is:

A. 4.9m

B. 7.4m

C. 9.8m

D. 12.6m

156) An object is thrown straight down with an initial speed of 4 m/s from a window which is 8 m above the ground. The time it takes the object to reach the ground is:

A. 0.80 s

B. 0.93 s

C. 1.3s

D. 1.7s

157) A stone is released from rest from the edge of a building roof 190 m above the ground. Neglecting air resistance, the speed of the stone, just before striking the ground, is:

A. 43 m/s

B. 61 m/s

C. 120 m/s

D. 190 m/s

158) An object is thrown vertically upward with a certain initial velocity in a world where the acceleration due to gravity is 19.6 m/s^2 . The height to which it rises is that to which the object would rise if thrown upward with the same initial velocity on the Earth. Neglect friction.

A. half

B. $\sqrt{2}$ times

C. twice

D. four times

159) A projectile is shot vertically upward with a given initial velocity. It reaches a maximum height of 100 m. If, on a second shot, the initial velocity is doubled then the projectile will reach a maximum height of:

A. 70.7m

B. 141.4m

C. 200 m

D. 241 m

E. 400 m

160) One object is thrown vertically upward with an initial velocity of 100 m/s and another object with an initial velocity of 10 m/s. The maximum height reached by the first object will be that of the other.

A. 10 times

B. 100 times

C. 1000 times

D. 10,000 times

161) The area under a velocity-time graph represents:

A. acceleration

B. change in acceleration

C. speed

D. change in velocity

E. displacement

162) Displacement can be obtained from:

A. the slope of an acceleration-time graph

B. the slope of a velocity-time graph

C. the area under an acceleration-time graph

D. the area under a velocity-time graph

163) An object has a constant acceleration of 3 m/s^2 . The coordinate versus time graph for this object has a slope:

A. that increases with time

B. that is constant

C. that decreases with time

D. of 3 m/s

164) The coordinate-time graph of an object is a straight line with a positive slope. The object has:

A. constant displacement

B. steadily increasing acceleration

C. steadily decreasing acceleration

D. constant velocity

165) We say that the displacement of a particle is a vector quantity. Our best justification for this assertion is:

A. displacement can be specified by a magnitude and a direction

B. operating with displacements according to the rules for manipulating vectors leads to results in agreement with experiments

C. a displacement is obviously not a scalar

D. displacement can be specified by three numbers

166) A vector of magnitude 3 CANNOT be added to a vector of magnitude 4 so that the magnitude of the resultant is:

A. zero

B. 1

C. 3

D. 5

167) A vector of magnitude 20 is added to a vector of magnitude 25. The magnitude of this sum might be:

A. zero

B. 3

C. 12

D. 47

168) A vector of magnitude 3 CANNOT be added to a vector of magnitude 4 so that the magnitude of the resultant is:

A. zero

B. 1

C. 3

D. 5

169) A vector of magnitude 20 is added to a vector of magnitude 25. The magnitude of this sum might be:

A. zero

B. 3

C. 12

D. 47

170) A vector of magnitude 3 CANNOT be added to a vector of magnitude 4 so that the magnitude of the resultant is:

A. zero

B. 1

C. 3

D. 5

171) A vector of magnitude 20 is added to a vector of magnitude 25. The magnitude of this sum might be:

A. zero

B. 3

C. 12

D. 47

172) Vectors A and B lie in the xy plane. We can deduce that $A = B$ if:

A. $A^2 x + A^2 y = B^2 x + B^2 y$

B. $A_x + A_y = B_x + B_y$

C. $A_x = B_x$ and $A_y = B_y$

D. $A_y/A_x = B_y/B_x$ E.

173) A vector has a magnitude of 12. When its tail is at the origin it lies between the positive x axis and the negative y axis and makes an angle of 30° with the x axis. Its y component is:

A. $6/\sqrt{3}$

B. $-6\sqrt{3}$

C. 6

D. -6

174) If the x component of a vector A, in the xy plane, is half as large as the magnitude of the vector, the tangent of the angle between the vector and the x axis is:

A. $\sqrt{3}$

B. $1/2$

C. $\sqrt{3}/2$

D. $3/2$

175) If $A = (6\text{m})\hat{i} - (8\text{m})\hat{j}$ then $4A$ has magnitude:

A. 10m

B. 20m

C. 30m

D. 40m

176) A vector has a component of 10m in the +x direction, a component of 10m in the +y direction, and a component of 5m in the +z direction. The magnitude of this vector is:

A. zero

B. 15m

C. 20m

D. 25m

177) Let $V = (2.00\text{m})\hat{i} + (6.00\text{m})\hat{j} - (3.00\text{m})\hat{k}$. The magnitude of V is:

A. 5.00m

B. 5.57m

C. 7.00m

D. 7.42m

178) A vector in the xy plane has a magnitude of 25m and an x component of 12m. The angle it makes with the positive x axis is:

A. 26°

B. 29°

C. 61°

D. 64°

179) The angle between $A = (25\text{m})\hat{i} + (45\text{m})\hat{j}$ and the positive x axis is:

A. 29°

B. 61°

C. 151°

D. 209°

180) The angle between $A = (-25\text{m})\hat{i} + (45\text{m})\hat{j}$ and the positive x axis is:

A. 29°

B. 61°

C. 119°

D. 151°

181) Let $A = (2\text{m})\hat{i} + (6\text{m})\hat{j} - (3\text{m})\hat{k}$ and $B = (4\text{m})\hat{i} + (2\text{m})\hat{j} + (1\text{m})\hat{k}$. The vector sum $S = A + B$ is:

A. $(6\text{m})\hat{i} + (8\text{m})\hat{j} - (2\text{m})\hat{k}$

B. $(-2\text{m})\hat{i} + (4\text{m})\hat{j} - (4\text{m})\hat{k}$

C. $(2\text{m})\hat{i} - (4\text{m})\hat{j} + (4\text{m})\hat{k}$

D. $(8\text{m})\hat{i} + (12\text{m})\hat{j} - (3\text{m})\hat{k}$

182) Let $A = (2\text{m})\hat{i} + (6\text{m})\hat{j} - (3\text{m})\hat{k}$ and $B = (4\text{m})\hat{i} + (2\text{m})\hat{j} + (1\text{m})\hat{k}$. The vector difference $D = A - B$ is:

A. $(6\text{m})\hat{i} + (8\text{m})\hat{j} - (2\text{m})\hat{k}$

B. $(-2\text{m})\hat{i} + (4\text{m})\hat{j} - (4\text{m})\hat{k}$

C. $(2\text{m})\hat{i} - (4\text{m})\hat{j} + (4\text{m})\hat{k}$

D. $(8\text{m})\hat{i} + (12\text{m})\hat{j} - (3\text{m})\hat{k}$

183) If $A = (2\text{m})\hat{i} - (3\text{m})\hat{j}$ and $B = (1\text{m})\hat{i} - (2\text{m})\hat{j}$, then $A - 2B =$

A. $(1\text{m})\hat{j}$

B. $(-1\text{m})\hat{j}$

C. $(4\text{m})\hat{i} - (7\text{m})\hat{j}$

D. $(4\text{m})\hat{i} + (1\text{m})\hat{j}$

184) A certain vector in the xy plane has an x component of 4m and a y component of 10m. It is then rotated in the xy plane so its x component is doubled. Its new y component is about:

A. 20m

B. 7.2m

C. 5.0m

D. 4.5m

185) Vectors A and B each have magnitude L. When drawn with their tails at the same point, the angle between them is 30° . The value of $A \cdot B$ is:

A. zero

B. L^2

C. $\sqrt{3}L^2/2$

D. $2L^2$

186) Let $A = (2\text{m})\hat{i} + (6\text{m})\hat{j} - (3\text{m})\hat{k}$ and $B = (4\text{m})\hat{i} + (2\text{m})\hat{j} + (1\text{m})\hat{k}$. Then $A \cdot B =$

A. $(8\text{m})\hat{i} + (12\text{m})\hat{j} - (3\text{m})\hat{k}$

B. $(12\text{m})\hat{i} - (14\text{m})\hat{j} - (20\text{m})\hat{k}$

C. 23m^2

D. 17m^2

187) Two vectors have magnitudes of 10m and 15m. The angle between them when they are drawn with their tails at the same point is 65° . The component of the longer vector along the line of the shorter is:

A. 0

B. 4.2m

C. 6.3m

D. 9.1m

188) Let $S = (1\text{m})\hat{i} + (2\text{m})\hat{j} + (2\text{m})\hat{k}$ and $T = (3\text{m})\hat{i} + (4\text{m})\hat{k}$. The angle between these two vectors is given by:

A. $\cos^{-1}(14/15)$

B. $\cos^{-1}(11/225)$

C. $\cos^{-1}(104/225)$

D. $\cos^{-1}(11/15)$

189) Two vectors lie with their tails at the same point. When the angle between them is increased by 20° their scalar product has the same magnitude but changes from positive to negative. The original angle between them was:

A. 0

B. 60°

C. 70°

D. 80°

190) If the magnitude of the sum of two vectors is less than the magnitude of either vector, then:

A. the scalar product of the vectors must be negative

B. the scalar product of the vectors must be positive

C. the vectors must be parallel and in opposite directions

D. the vectors must be parallel and in the same direction

191) If the magnitude of the sum of two vectors is greater than the magnitude of either vector, then:

A. the scalar product of the vectors must be negative

B. the scalar product of the vectors must be positive

C. the vectors must be parallel and in opposite directions

D. the vectors must be parallel and in the same direction

E. none of the above

192) Vectors A and B each have magnitude L. When drawn with their tails at the same point, the angle between them is 60° . The magnitude of the vector product $A \times B$ is:

A. $L^2/2$

B. L^2

C. $\sqrt{3}L^2/2$

D. $2L^2$

193) Two vectors lie with their tails at the same point. When the angle between them is increased by 20° the magnitude of their vector product doubles. The original angle between them was about:

A. 0

B. 18°

C. 25°

D. 45°

194) Two vectors have magnitudes of 10m and 15m. The angle between them when they are drawn with their tails at the same point is 65° . The component of the longer vector along the line perpendicular to the shorter vector, in the plane of the vectors, is:

A. 0

B. 4.2m

C. 6.3m

D. 9.1m

E. 14m

195) The two vectors $(3\text{m})^{\wedge}i - (2\text{m})^{\wedge}j$ and $(2\text{m})^{\wedge}i + (3\text{m})^{\wedge}j - (2\text{m})^{\wedge}k$ define a plane. It is the plane of the triangle with both tails at one vertex and each head at one of the other vertices. Which of the following vectors is perpendicular to the plane?

A. $(4\text{m})^{\wedge}i + (6\text{m})^{\wedge}j + (13\text{m})^{\wedge}k$

B. $(-4\text{m})^{\wedge}i + (6\text{m})^{\wedge}j + (13\text{m})^{\wedge}k$

C. $(4\text{m})^{\wedge}i - (6\text{m})^{\wedge}j + (13\text{m})^{\wedge}k$

D. $(4\text{m})^{\wedge}i + (6\text{m})^{\wedge}j - (13\text{m})^{\wedge}k$

196) Let $R = S \times T$ and $\theta = 90^\circ$, where θ is the angle between S and T when they are drawn with their tails at the same point. Which of the following is NOT true?

A. $|R| = |S| |T| \sin\theta$

B. $-R = T \times S$

C. $R \cdot S = 0$

D. $R \cdot T = 0$

E. $S \cdot T = 0$

197) The value of $\hat{i} \cdot (\hat{j} \times \hat{k})$ is:

A. zero

B. +1

C. -1

D. 3

198) The value of $\hat{k} \cdot (\hat{k} \times \hat{i})$ is:

A. zero

B. +1

C. -1

D. 3

199) Velocity is defined as:

A. rate of change of position with time

B. position divided by time

C. rate of change of acceleration with time

D. a speeding up or slowing down

200) Acceleration is defined as:

A. rate of change of position with time

B. speed divided by time

C. rate of change of velocity with time

D. a speeding up or slowing down

201) Which of the following is a scalar quantity?

A. Speed

B. Velocity

C. Displacement

D. Acceleration

202) Which of the following is a vector quantity?

A. Mass

B. Density

C. Speed

D. Temperature

E. None of these

203) Which of the following is NOT an example of accelerated motion?

A. Vertical component of projectile motion

B. Circular motion at constant speed

C. A swinging pendulum

D. Earth's motion about sun

E. Horizontal component of projectile motion

204) A particle goes from $x = -2\text{m}$, $y = 3\text{m}$, $z = 1\text{m}$ to $x = 3\text{m}$, $y = -1\text{m}$, $z = 4\text{m}$. Its displacement is:

A. $(1\text{m})^{\wedge}i + (2\text{m})^{\wedge}j + (5\text{m})^{\wedge}k$

B. $(5\text{m})^{\wedge}i - (4\text{m})^{\wedge}j + (3\text{m})^{\wedge}k$

C. $-(5\text{m})^{\wedge}i + (4\text{m})^{\wedge}j - (3\text{m})^{\wedge}k$

D. $-(1\text{m})^{\wedge}i - (2\text{m})^{\wedge}j - (5\text{m})^{\wedge}k$

205) A jet plane in straight horizontal flight passes over your head. When it is directly above you, the sound seems to come from a point behind the plane in a direction 30° from the vertical. The speed of the plane is:

A. the same as the speed of sound

B. half the speed of sound

C. three-fifths the speed of sound

D. 0.866 times the speed of sound

206) A plane traveling north at 200m/s turns and then travels south at 200m/s . The change in its velocity is:

A. zero

B. 200m/s north

C. 200m/s south

D. 400m/s north

E. 400m/s south

207) Two bodies are falling with negligible air resistance, side by side, above a horizontal plane. If one of the bodies is given an additional horizontal acceleration during its descent, it:

A. strikes the plane at the same time as the other body

B. strikes the plane earlier than the other body

- C. has the vertical component of its velocity altered
- D. has the vertical component of its acceleration altered

208) The velocity of a projectile equals its initial velocity added to:

- A. a constant horizontal velocity
- B. a constant vertical velocity
- C. a constantly increasing horizontal velocity
- D. a constantly increasing downward velocity

209) A stone thrown from the top of a tall building follows a path that is:

- A. circular
- B. made of two straight line segments
- C. hyperbolic
- D. parabolic

210) Identical guns fire identical bullets horizontally at the same speed from the same height above level planes, one on the Earth and one on the Moon. Which of the following three statements is/are true?

- I. The horizontal distance traveled by the bullet is greater for the Moon.
- II. The flight time is less for the bullet on the Earth.
- III. The velocity of the bullets at impact are the same.

A. III only

B. I and II only

C. I and III only

D. II and III only

211) A bullet shot horizontally from a gun:

A. strikes the ground much later than one dropped vertically from the same point at the same instant

B. never strikes the ground

C. strikes the ground at approximately the same time as one dropped vertically from the same point at the same instant

D. travels in a straight line

212) A bomber flying in level flight with constant velocity releases a bomb before it is over the target. Neglecting air resistance, which one of the following is NOT true?

A. The bomber is over the target when the bomb strikes

B. The acceleration of the bomb is constant

C. The horizontal velocity of the plane equals the vertical velocity of the bomb when it hits the target

D. The bomb travels in a curved path

213) An airplane makes a gradual 90° turn while flying at a constant speed of 200m/s . The process takes 20.0 seconds to complete. For this turn the magnitude of the average acceleration of the plane is:

- A. zero
- B. 40m/s^2
- C. 20m/s^2
- D. 14m/s^2

214) An airplane is flying north at 500 km/h . It makes a gradual 180° turn at constant speed, changing its direction of travel from north through east to south. The process takes 40s . The average acceleration of the plane for this turn (in $\text{km/h}\cdot\text{s}$) is:

- A. $12.5\text{km/h}\cdot\text{s}$, north
- B. $12.5\text{km/h}\cdot\text{s}$, east
- C. $12.5\text{km/h}\cdot\text{s}$, south
- D. $25\text{km/h}\cdot\text{s}$, north
- E. $25\text{km/h}\cdot\text{s}$, south

215) An object is moving on a circular path of radius π meters at a constant speed of 4.0m/s . The time required for one revolution is:

- A. $2/\pi^2\text{ s}$
- B. $\pi^2/2\text{s}$
- C. $\pi/2\text{s}$
- D. $\pi^2/4$

216) A particle moves at constant speed in a circular path. The instantaneous velocity and instantaneous acceleration vectors are:

- A. both tangent to the circular path
- B. both perpendicular to the circular path
- C. perpendicular to each other
- D. opposite to each other

217) A stone is tied to a string and whirled at constant speed in a horizontal circle. The speed is then doubled without changing the length of the string. Afterward the magnitude of the acceleration of the stone is:

- A. the same
- B. twice as great
- C. four times as great
- D. half as great

218) Two objects are traveling around different circular orbits with constant speed. They both have the same acceleration but object A is traveling twice as fast as object B. The orbit radius for object A is the orbit radius for object B.

- A. one-fourth
- B. one-half
- C. the same as
- D. twice
- E. four times

219) A stone is tied to a 0.50-m string and whirled at a constant speed of 4.0m/s in a vertical circle. Its acceleration at the top of the circle is:

- A. 9.8m/s^2 , up
- B. 9.8m/s^2 , down
- C. 8.0m/s^2 , down
- D. 32m/s^2 , up
- E. 32m/s^2 , down

220) A stone is tied to a 0.50-m string and whirled at a constant speed of 4.0m/s in a vertical circle. Its acceleration at the bottom of the circle is:

- A. 9.8m/s^2 , up
- B. 9.8m/s^2 , down
- C. 8.0m/s^2 , up
- D. 32m/s^2 , up

221) A car rounds a 20-m radius curve at 10m/s. The magnitude of its acceleration is:

- A. 0
- B. 0.20m/s^2
- C. 5.0m/s^2
- D. 40m/s^2

222) For a biological sample in a 1.0-m radius centrifuge to have a centripetal acceleration of 25g its speed must be:

- A. 11m/s
- B. 16m/s
- C. 50m/s
- D. 122m/s

223) A girl jogs around a horizontal circle with a constant speed. She travels one fourth of a revolution, a distance of 25m along the circumference of the circle, in 5.0s. The magnitude of her acceleration is

- A. 0.31m/s^2
- B. 1.3m/s^2
- C. 1.6m/s^2
- D. 3.9m/s^2
- E. 6.3m/s^2 ans: C

224) A stone is tied to the end of a string and is swung with constant speed around a horizontal circle with a radius of 1.5m. If it makes two complete revolutions each second, the magnitude of its acceleration is:

- A. 0.24m/s^2
- B. 2.4m/s^2
- C. 24m/s^2
- D. 240m/s^2

225) A Ferris wheel with a radius of 8.0m makes 1 revolution every 10s. When a passenger is at the top, essentially a diameter above the ground, he releases a ball. How far from the point on the ground directly under the release point does the ball land?

- A. 0
- B. 1.0m
- C. 8.0m
- D. 9.1m

226) A boat is able to move through still water at 20m/s. It makes a round trip to a town 3.0 km upstream. If the river flows at 5m/s, the time required for this round trip is:

- A. 120s
- B. 150s
- C. 200s
- D. 300s
- E. 320s

227) A boat is traveling upstream at 14km/h with respect to a river that is flowing at 6km/h (with respect to the ground). A man runs directly across the boat, from one side to the other, at 6km/h (with respect to the boat). The speed of the man with respect to the ground is:

- A. 10km/h
- B. 14km/h
- C. 18.5km/h
- D. 21km/h

228) A ferry boat is sailing at 12km/h 30° W of N with respect to a river that is flowing at 6.0km/h E. As observed from the shore, the ferry boat is sailing:

- A. 30° E of N
- B. due N
- C. 30° W of N
- D. 45° E of N
- E. none of these

229) An airplane makes a gradual 90° turn while flying at a constant speed of 200m/s. The process takes 20.0 seconds to complete. For this turn the magnitude of the average acceleration of the plane is:

- A. zero
- B. 40m/s²
- C. 20m/s²
- D. 14m/s²

230) An airplane is flying north at 500 km/h. It makes a gradual 180° turn at constant speed, changing its direction of travel from north through east to south. The process takes 40s. The average acceleration of the plane for this turn (in km/h·s) is:

- A. 12.5km/h·s, north
- B. 12.5km/h·s, east
- C. 12.5km/h·s, south
- D. 25km/h·s, north
- E. 25km/h·s, south

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B. $\pi^2/2$ s

C. $\pi/2$ s

D. $\pi^2/4$

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A. both tangent to the circular path

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A. one-fourth

B. one-half

C. the same as

D. twice

E. four times

235) A stone is tied to a 0.50-m string and whirled at a constant speed of 4.0 m/s in a vertical circle. Its acceleration at the top of the circle is:

A. 9.8 m/s^2 , up

B. 9.8 m/s^2 , down

C. 8.0 m/s^2 , down

D. 32 m/s^2 , up

E. 32 m/s^2 , down

236) A stone is tied to a 0.50-m string and whirled at a constant speed of 4.0 m/s in a vertical circle. Its acceleration at the bottom of the circle is:

A. 9.8 m/s^2 , up

B. 9.8 m/s^2 , down

C. 8.0 m/s^2 , up

D. 32 m/s^2 , up

237) A car rounds a 20-m radius curve at 10 m/s. The magnitude of its acceleration is:

A. 0

B. 0.20 m/s^2

C. 5.0 m/s^2

D. 40m/s^2

238) For a biological sample in a 1.0-m radius centrifuge to have a centripetal acceleration of $25g$ its speed must be:

A. 11m/s

B. 16m/s

C. 50m/s

D. 122m/s

239) A girl jogs around a horizontal circle with a constant speed. She travels one fourth of a revolution, a distance of 25m along the circumference of the circle, in 5.0s . The magnitude of her acceleration is:

A. 0.31m/s^2

B. 1.3m/s^2

C. 1.6m/s^2

D. 3.9m/s^2

240) A stone is tied to the end of a string and is swung with constant speed around a horizontal circle with a radius of 1.5m . If it makes two complete revolutions each second, the magnitude of its acceleration is:

A. 0.24m/s^2

B. 2.4m/s^2

C. 24m/s^2

D. 240m/s^2

241) A Ferris wheel with a radius of 8.0m makes 1 revolution every 10s . When a passenger is at the top, essentially a diameter above the ground, he releases a ball. How far from the point on the ground directly under the release point does the ball land?

A. 0

B. 1.0m

C. 8.0m

D. 9.1m

242) A boat is able to move through still water at 20m/s . It makes a round trip to a town 3.0 km upstream. If the river flows at 5m/s , the time required for this round trip is:

A. 120s

B. 150s

C. 200s

D. 300s

E. 320s

243) A boat is traveling upstream at 14km/h with respect to a river that is flowing at 6km/h (with respect to the ground). A man runs directly across the boat, from one side to the other, at 6km/h (with respect to the boat). The speed of the man with respect to the ground is:

A. 10km/h

B. 14km/h

C. 18.5km/h

D. 21km/h

244) A ferry boat is sailing at 12km/h 30° W of N with respect to a river that is flowing at 6.0km/h E. As observed from the shore, the ferry boat is sailing:

- A. 30° E of N
- B. due N
- C. 30° W of N
- D. 45° E of N
- E. none of these

245) An example of an inertial reference frame is:

- A. any reference frame that is not accelerating
- B. a frame attached to a particle on which there are no forces
- C. any reference frame that is at rest
- D. a reference frame attached to the center of the universe

246) An object moving at constant velocity in an inertial frame must:

- A. have a net force on it
- B. eventually stop due to gravity
- C. not have any force of gravity on it
- D. have zero net force on it

247) In SI units a force is numerically equal to the , when the force is applied to it.

- A. velocity of the standard kilogram
- B. speed of the standard kilogram
- C. velocity of any object
- D. acceleration of the standard kilogram

248) Which of the following quantities is NOT a vector?

- A. Mass
- B. Displacement
- C. Weight
- D. Acceleration

249) A newton is the force:

- A. of gravity on a 1kg body
- B. of gravity on a 1g body
- C. that gives a 1g body an acceleration of 1cm/s^2
- D. that gives a 1kg body an acceleration of 1m/s^2

250) The unit of force called the newton is:

- A. $9.8\text{kg}\cdot\text{m/s}^2$
- B. $1\text{kg}\cdot\text{m/s}^2$
- C. defined by means of Newton's third law
- D. 1kg of mass

251) A force of 1N is:

- A. 1kg/s
- B. $1\text{kg}\cdot\text{m/s}$
- C. $1\text{kg}\cdot\text{m/s}^2$
- D. $1\text{kg}\cdot\text{m}^2/\text{s}$

252) The standard 1-kg mass is attached to a compressed spring and the spring is released. If the mass initially has an acceleration of 5.6m/s^2 , the force of the spring has a magnitude of:

- A. 2.8N
- B. 5.6N
- C. 11.2N
- D. 0

253) Acceleration is always in the direction:

- A. of the displacement
- B. of the initial velocity
- C. of the final velocity
- D. of the net force

254) The term "mass" refers to the same physical concept as:

- A. weight
- B. inertia
- C. force
- D. acceleration

255) The inertia of a body tends to cause the body to:

- A. speed up
- B. slow down
- C. resist any change in its motion
- D. fall toward Earth

256) A heavy ball is suspended as shown. A quick jerk on the lower string will break that string but a slow pull on the upper string will break that string. The mass of a body:

- A. is slightly different at different places on Earth
- B. is a vector
- C. is independent of the free-fall acceleration
- D. is the same for all bodies of the same volume

257) The mass and weight of a body:

- A. differ by a factor of 9.8
- B. are identical
- C. are the same physical quantities expressed in different units
- D. are both a direct measure of the inertia of the body
- E. have the same ratio as that of any other body placed at that location

258) An object placed on an equal-arm balance requires 12kg to balance it. When placed on a spring scale, the scale reads 12kg. Everything (balance, scale, set of weights and object) is now transported to the Moon where the free-fall acceleration is one-sixth that on Earth. The new readings of the balance and spring scale (respectively) are:

- A. 12kg, 12kg
- B. 2kg, 2kg
- C. 12kg, 2kg
- D. 2kg, 12kg

259) Two objects, one having three times the mass of the other, are dropped from the same height in a vacuum. At the end of their fall, their velocities are equal because:

- A. anything falling in vacuum has constant velocity
- B. all objects reach the same terminal velocity
- C. the acceleration of the larger object is three times greater than that of the smaller object
- D. the force of gravity is the same for both objects
- E. none of the above

260) A feather and a lead ball are dropped from rest in vacuum on the Moon. The acceleration of the feather is:

- A. more than that of the lead ball
- B. the same as that of the lead ball
- C. less than that of the lead ball
- D. 9.8m/s^2

261) Equal forces F act on isolated bodies A and B. The mass of B is three times that of A. The magnitude of the acceleration of A is:

- A. three times that of B
- B. $1/3$ that of B
- C. the same as B
- D. nine times that of B

262) A car travels east at constant velocity. The net force on the car is:

- A. east
- B. west
- C. up
- D. down
- E. zero

263) A constant force of 8.0 N is exerted for 4.0 s on a 16-kg object initially at rest. The change in speed of this object will be:

- A. 0.5m/s
- B. 2m/s
- C. 4m/s
- D. 8m/s

264) A 6-kg object is moving south. A net force of 12N north on it results in the object having an acceleration of:

- A. 2m/s^2 , north
- B. 2m/s^2 , south
- C. 6m/s^2 , north
- D. 18m/s^2 , north

265) A 9000-N automobile is pushed along a level road by four students who apply a total forward force of 500N . Neglecting friction, the acceleration of the automobile is:

- A. 0.055m/s^2
- B. 0.54m/s^2
- C. 1.8m/s^2

D. 9.8m/s^2

266) An object rests on a horizontal frictionless surface. A horizontal force of magnitude F is applied. This force produces an acceleration:

A. only if F is larger than the weight of the object

B. only while the object suddenly changes from rest to motion

C. always

D. only if the inertia of the object decreases

267) A 25-kg crate is pushed across a frictionless horizontal floor with a force of 20N, directed 20° below the horizontal. The acceleration of the crate is:

A. 0.27m/s^2

B. 0.75m/s^2

C. 0.80m/s^2

D. 170m/s^2

268) A ball with a weight of 1.5N is thrown at an angle of 30° above the horizontal with an initial speed of 12m/s . At its highest point, the net force on the ball is:

A. 9.8N , 30° below horizontal

B. zero

C. 9.8N , up

D. 9.8N , down

E. 1.5N , down

269) Two forces are applied to a 5.0-kg crate; one is 6.0N to the north and the other is 8.0N to the west. The magnitude of the acceleration of the crate is:

A. 0.50m/s^2

B. 2.0m/s^2

C. 2.8m/s^2

D. 10m/s^2

270) A 400-N steel ball is suspended by a light rope from the ceiling. The tension in the rope is:

A. 400N

B. 800N

C. zero

D. 200N

271) A heavy steel ball B is suspended by a cord from a block of wood W. The entire system is dropped through the air. Neglecting air resistance, the tension in the cord is:

A. zero

B. the difference in the masses of B and W

C. the difference in the weights of B and W

D. the weight of B

272) A car moves horizontally with a constant acceleration of 3m/s^2 . A ball is suspended by a string from the ceiling of the car. The ball does not swing, being at rest with respect to the car. What angle does the string make with the vertical?

A. 17°

B. 35°

C. 52°

D. 73°

273) A man weighing 700N is in an elevator that is accelerating upward at 4m/s^2 . The force exerted on him by the elevator floor is:

A. 71N

B. 290N

C. 410N

D. 700N

E. 990N

274) You stand on a spring scale on the floor of an elevator. Of the following, the scale shows the highest reading when the elevator:

A. moves upward with increasing speed

B. moves upward with decreasing speed

C. remains stationary

D. moves downward with increasing speed

275) You stand on a spring scale on the floor of an elevator. Of the following, the scale shows the highest reading when the elevator:

A. moves downward with increasing speed

B. moves downward with decreasing speed

C. remains stationary

D. moves upward with decreasing speed

276) When a 25-kg crate is pushed across a frictionless horizontal floor with a force of 200N, directed 20° below the horizontal, the magnitude of the normal force of the floor on the crate is:

A. 25N

B. 68N

C. 180N

D. 250N

E. 310N

277) A block slides down a frictionless plane that makes an angle of 30° with the horizontal. The acceleration of the block is:

A. 980cm/s^2

B. 566cm/s^2

C. 849cm/s^2

D. zero

E. 490cm/s^2

278) A 25-N crate slides down a frictionless incline that is 25° above the horizontal. The magnitude of the normal force of the incline on the crate is:

A. 11N

B. 23N

C. 25N

D. 100N

279) A 25-N crate is held at rest on a frictionless incline by a force that is parallel to the incline. If the incline is 25° above the horizontal the magnitude of the applied force is:

A. 4.1N

B. 4.6N

C. 8.9N

D. 11N

E. 23N

280) A 25-N crate is held at rest on a frictionless incline by a force that is parallel to the incline. If the incline is 25° above the horizontal the magnitude of the normal force of the incline on the crate is:

A. 4.1N

B. 4.6N

C. 8.9N

D. 11N

E. 23N

281) A 32-N force, parallel to the incline, is required to push a certain crate at constant velocity up a frictionless incline that is 30° above the horizontal. The mass of the crate is:

A. 3.3kg

B. 3.8kg

C. 5.7kg

D. 6.5kg

E. 160kg

282) A sled is on an icy (frictionless) slope that is 30° above the horizontal. When a 40-N force, parallel to the incline and directed up the incline, is applied to the sled, the acceleration of the sled is 2.0m/s^2 , down the incline. The mass of the sled is:

A. 3.8kg

B. 4.1kg

C. 5.8kg

D. 6.2kg

E. 10kg

283) When a 40-N force, parallel to the incline and directed up the incline, is applied to a crate on a frictionless incline that is 30° above the horizontal, the acceleration of the crate is 2.0m/s^2 , up the incline. The mass of the crate is:

A. 3.8kg

B. 4.1kg

C. 5.8kg

D. 6.2kg

E. 10kg

284) The “reaction” force does not cancel the “action” force because:

- A. the action force is greater than the reaction force
- B. they are on different bodies**
- C. they are in the same direction
- D. the reaction force exists only after the action force is removed
- E. the reaction force is greater than the action force

285) A book rests on a table, exerting a downward force on the table. The reaction to this force is:

- A. the force of Earth on the book
- B. the force of the table on the book**
- C. the force of Earth on the table
- D. the force of the book on Earth
- E. the inertia of the book

286) A lead block is suspended from your hand by a string. The reaction to the force of gravity on the block is the force exerted by:

- A. the string on the block
- B. the block on the string
- C. the string on the hand
- D. the hand on the string
- E. the block on Earth**

287) A 5-kg concrete block is lowered with a downward acceleration of 2.8m/s^2 by means of a rope. The force of the block on the rope is:

- A. 14N, up
- B. 14N, down
- C. 35N, up
- D. 35N, down**
- E. 49N, up

288) A 90-kg man stands in an elevator that is moving up at a constant speed of 5.0m/s . The force exerted by him on the floor is about:

- A. zero
- B. 90N
- C. 880N**
- D. 450N
- E. 49N

289) A 90-kg man stands in an elevator that has a downward acceleration of 1.4m/s^2 . The force exerted by him on the floor is about:

- A. zero
- B. 90N
- C. 760N**
- D. 880N
- E. 1010N

290) A 5-kg concrete block is lowered with a downward acceleration of 2.8m/s^2 by means of a rope. The force of the block on Earth is:

- A. 14N, up
- B. 14N, down
- C. 35N, up
- D. 35N, down

E. 49N, up

291) A brick slides on a horizontal surface. Which of the following will increase the magnitude of the frictional force on it?

- A. Putting a second brick on top**
- B. Decreasing the surface area of contact
- C. Increasing the surface area of contact
- D. Decreasing the mass of the brick
- E. None of the above

292) The coefficient of kinetic friction:

- A. is in the direction of the frictional force
- B. is in the direction of the normal force
- C. is the ratio of force to area
- D. can have units of newtons

E. is none of the above

293) When the brakes of an automobile are applied, the road exerts the greatest retarding force:

- A. while the wheels are sliding
- B. just before the wheels start to slide**
- C. when the automobile is going fastest
- D. when the acceleration is least
- E. at the instant when the speed begins to change

295) A forward horizontal force of 12N is used to pull a 240-N crate at constant velocity across a horizontal floor. The coefficient of friction is:

- A. 0.5
- B. 0.05**
- C. 2
- D. 0.2
- E. 20

296) The speed of a 4.0-N hockey puck, sliding across a level ice surface, decreases at the rate of 0.61m/s^2 . The coefficient of kinetic friction between the puck and ice is:

- A. 0.062**
- B. 0.41
- C. 0.62
- D. 1.2
- E. 9.8

297) A 50-N force is applied to a crate on a horizontal rough floor, causing it to move horizontally. If the coefficient of kinetic friction is 0.50, in what direction should the force be applied to obtain the greatest acceleration?

- A. Horizontal
- B. 60° above the horizontal
- C. 30° above the horizontal
- D. 27° above the horizontal**
- E. 30° below the horizontal

298) A professor holds an eraser against a vertical chalkboard by pushing horizontally on it. He pushes with a force that is much greater than is required to hold the eraser. The force of friction exerted by the board on the eraser increases if he:

- A. pushes with slightly greater force
- B. pushes with slightly less force
- C. stops pushing
- D. pushes so his force is slightly downward but has the same magnitude**
- E. pushes so his force is slightly upward but has the same magnitude

299) A horizontal force of 12N pushes a 0.5-kg book against a vertical wall. The book is initially at rest. If the coefficients of friction are $\mu_s = 0.6$ and $\mu_k = 0.8$ which of the following is true?

- A. The magnitude of the frictional force is 4.9N**
- B. The magnitude of the frictional force is 7.2N
- C. The normal force is 4.9N
- D. The book will start moving and accelerate
- E. If started moving downward, the book will decelerate

300) A horizontal force of 5.0N pushes a 0.50-kg book against a vertical wall. The book is initially at rest. If the coefficients of friction are $\mu_s = 0.6$ and $\mu_k = 0.80$, the magnitude of the frictional force is:

- A. 0
- B. 4.9N
- C. 3.0N
- D. 5.0N
- E. 4.0N**

301) A horizontal force of 12N pushes a 0.50-kg book against a vertical wall. The book is initially at rest. If $\mu_s = 0.6$ and $\mu_k = 0.80$, the acceleration of the book in m/s^2 is:

- A. 0**
- B. $9.4m/s^2$
- C. $9.8m/s^2$
- D. $14.4m/s^2$
- E. $19.2m/s^2$

302) A horizontal force of 5.0N pushes a 0.50-kg block against a vertical wall. The block is initially at rest. If $\mu_s = 0.60$ and $\mu_k = 0.80$, the acceleration of the block in m/s^2 is:

- A. 0
- B. 1.8**
- C. 6.0
- D. 8.0

E. 9.8

303) A block is placed on a rough wooden plane. It is found that when the plane is tilted 30° to the horizontal, the block will slide down at constant speed. The coefficient of kinetic friction of the block with the plane is:

A. 0.500

B. 0.577

C. 1.73

D. 0.866

E. 4.90

304) A crate is sliding down an incline that is 35° above the horizontal. If the coefficient of kinetic friction is 0.40, the acceleration of the crate is:

A. 0

B. 2.4m/s^2

C. 5.8m/s^2

D. 8.8m/s^2

E. 10.3m/s^2

305) A 5.0-kg crate is resting on a horizontal plank. The coefficient of static friction is 0.50 and the coefficient of kinetic friction is 0.40. After one end of the plank is raised so the plank makes an angle of 25° with the horizontal, the force of friction is:

A. 0

B. 18N

C. 21N

D. 22N

E. 44N

306) A 5.0-kg crate is resting on a horizontal plank. The coefficient of static friction is 0.50 and the coefficient of kinetic friction is 0.40. After one end of the plank is raised so the plank makes an angle of 30° with the horizontal, the force of friction is:

A. 0

B. 18N

C. 21N

D. 22N

E. 44N

307) A 5.0-kg crate is on an incline that makes an angle of 30° with the horizontal. If the coefficient of static friction is 0.50, the minimum force that can be applied parallel to the plane to hold the crate at rest is:

A. 0

B. 3.3N

C. 30N

D. 46N

E. 55N

308) A 5.0-kg crate is on an incline that makes an angle of 30° with the horizontal. If the coefficient of static friction is 0.5, the maximum force that can be applied parallel to the plane without moving the crate is:

- A. 0
- B. 3.3N
- C. 30N
- D. 46N
- E. 55N

309) Block A, with mass m_A , is initially at rest on a horizontal floor. Block B, with mass m_B , is initially at rest on the horizontal top surface of A. The coefficient of static friction between the two blocks is μ_s . Block A is pulled with a horizontal force. It begins to slide out from under B if the force is greater than:

- A. $m_A g$
- B. $m_B g$
- C. $\mu_s m_A g$
- D. $\mu_s m_B g$
- E. $\mu_s (m_A + m_B) g$

310) When body moves with constant acceleration the velocity time graph is:

Straight line

311) A/an -----is basic reason to change in the motion of an object according to newton's second law of motion

Net force

312) when a spring is compressed or stretched, the potential energy of the spring

Increases

313) when the velocity of an aeroplane is doubled, the momentum-----

(Increases uniformly) not sure (is conserved)

314) the momentum of an object at a given instant is independent of its

Acceleration

315) the velocity time graph is parallel to the axis the acceleration of moving body is

Zero

316) final takeoff velocity of airplane is 67m/s. The length of runway is 2Km the constant acceleration is

2.24 ms^{-2}

317) The law of inertia was firstly formulated by

Einstein

318) According to the equation of work, when a body sits in one place and studies for the whole night, he does

No work

319) Acceleration of an object is must be zero at a point where

The average velocity is zero

320) The acceleration of projectile in X-direction

0m/s²

321) It is easier to walk on the concrete road than on ice because

The friction b/w the ice and feet is less than that b/w concrete and feet

322) An inelastic collision is one in which

Momentum is conserved but K.E is not conserved

323) Unit of distance is

Light year

324) The slope of velocity-time the graph at any point may be identified with

Average acceleration

325) The dot product of vector A with itself is equal to

A²

326) An elastic is one in which

Momentum is conserved but K.E is not conserved

327) if the magnitude of force applied is increased the work done will

Be increased

328) A particle, held by a string whose other end is attached to a fixed point C, moves in a circle on a horizontal frictionless surface

Does not change

329) If the distance b/w all pairs of particles of the body do not change by applying a force then the body is said to be

Rigid

330) A bullet shoot straight up returns to its starting point in 10sec the initial speed was

49m/s

331) One revolution per minute is about

0.0524rad/s

332) The goal of all scientific inquiry(or scientific method)is

Predicting natural events based on known patterns

333) The center of mass of a uniform disk of radius R is located

At the center

334) Force

Equals the time rate of change of momentum

335) A mosquito's buzz is often decibel rating of 40db normal conservation rated at 60db.How many times more intense is normal conservation compared to a mosquito's buzz?

100

336) When number of bodies they can exert force one another and no external agency exert a force on them

An isolated system

337) back and forth in a leftward direction. This type of wave is known

Longitudinal

338) The center of mass of system consisting of Earth,the Sun,and the planet Mars is

Closer to the Sun than to either of the other bodies

339) A plane produces a sonic boom only when

It flies faster than the speed of sound

340) Young's modulus can be stated to calculate strain for a stress that is

Well below the yield strength

341) As the wavelength of a wave in a uniform medium increases ,it speed will be

Remain the same

342) A sound wave has a wavelength of 3.0m.the adjacent rarefaction center is

1.5m

343) Which statement completely submerged object resting on the ocean bottom is correct

The apparent weight of the object depends upon the object's density

344) When the mass of colliding body is much larger than the mass of body at rest.its velocity at collision

Becomes zero

345) The rate of change of momentum of a body is equal

Applied force

346) Unit for gravitational constant G are

$\text{m}^3 /(\text{kg}.\text{s}^2)$

347) A sound wave has wavelength of 3.0m.the distance from a compression center of rarefaction

3.0

348) A net torque applied to a rigid object always tends to produce:

Rotational equilibrium

349) In simple harmonic motion, the restoring force must be proportional to the:

Displacement

350) The unit of Stress can be written as:

Nm

Nm²

Nm⁻²

N/m

351) A mass of 1kg lift vertically through a distance of 1m by a boy. What is the work done of boy? 10J

1J

0.1J

352) What is the basic unit of SI system?

Gram

Kilogram

Pressure

Watt

353) An example of SOP expression is

$A + B(C + D)$

$$A'B + AC' + AB'C$$

both (a) and (b)

354) 2's complement of 5 is

1011

355) How many data select lines are required for selecting eight inputs?

3

356) The boolean expression $X = AB + CD$ represents

two ANDs ORed together

357) which of the following rules states that if one input of an AND gate is always 1, the output is equal to the other input?

$A.1 = A$

358) Which one of the following is NOT a valid rule of Boolean algebra?

$A = A'$

359) The complement of a variable is always

the inverse of the variable

360) The boolean expression $(A + C)(AB' + AC)(A'C' + B')$ can be simplified to

$AB + BC$

361) The cell marked 6 in 4-variable K-Map represent minterm 6 or the maxterm 6 having the following binary value of variables A, B, C and D.

$A=0, B=1, C=1, D=0$

362) Adjacent 1s detector circuit will have active high output for the input

0011

363) A net torque applied to a rigid object always tends to produce:

rotational equilibrium

364) An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

2.0 rad/s

365) In simple harmonic motion, the restoring force must be proportional to the:

displacement

366) Mercury is a convenient liquid to use in a barometer because:

it has a high density

367) The units of the electric field are:

J/m

368) A farad is the same as a

J/V

369) We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5H. We also need a capacitance of:

100 μ F

370) The wavelength of red light is 700 nm. Its frequency is _____.

4.30×10^5 Hertz

371) Which of the following statements is NOT TRUE about electromagnetic waves?

The electromagnetic radiation from a burning candle is unpolarized

372) Radio waves and light waves are _____.

Electromagnetic and transverse both

373) Wien's Law states that, $\lambda_{\text{max}} =$ _____ K.

$2.90 \times 10^{-3} \text{ m}$

374) Interference of light is evidence that:

light is a wave phenomenon

375) Fahrenheit and Kelvin scales agree numerically at a reading of:

-40

376) According to the theory of relativity:

moving clocks run fast

377) Light from a stationary spaceship is observed, and then the spaceship moves directly away from the observer at high speed while still emitting the light. As a result, the light seen by the observer has:

lower frequency and a shorter wavelength than before

378) How fast should you move away from a $6.0 \times 10^{14} \text{ Hz}$ light source to observe waves with a frequency of $4.0 \times 10^{14} \text{ Hz}$?

38c

379) The quantum number n is most closely associated with what property of the electron in a hydrogen atom?

Energy

380) The quantum number m_s is most closely associated with what property of the electron in an atom?

Energy

381) As the wavelength of a wave in a uniform medium increases, its speed will _____.

Remain the same

382) The lowest tone produced by a certain organ comes from a 3.0-m pipe with both ends open. If the speed of sound is 340m/s, the frequency of this tone is approximately:

57 Hz

383) To raise the pitch of a certain piano string, the piano tuner:

loosens the string

384) An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

12.6 rad/s

385) For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

the same point

386) Ten seconds after an electric fan is turned on, the fan rotates at 300 rev/min. Its average angular acceleration is:

3.14 rad/s^2

387) A 4.0-N puck is traveling at 3.0m/s. It strikes a 8.0-N puck, which is stationary. The two pucks stick together. Their common final speed is:

1.0m/s

389) An object moving in a circle at constant speed:

has an acceleration of constant magnitude

390) A plane traveling north at 200m/s turns and then travels south at 200m/s. The change in its velocity is:

400m/s south

391) At time $t = 0$ a car has a velocity of 16 m/s. It slows down with an acceleration given by $-0.50t$, in m/s^2 for t in seconds. It stops at $t =$

8.0 s

392) 1 mi is equivalent to 1609 m so 55 mph is:

25 m/s

393) The number of significant figures in 0.00150 is:

3

394) One revolution is the same as: 2π rad

2π rad

395) For a body to be in equilibrium under the combined action of several forces:

any two of these forces must be balanced by a third force

396) A bucket of water is pushed from left to right with increasing speed across a horizontal surface. Consider the pressure at two points at the same level in the water.

It is higher at the point on the left

397) An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

600 Hz

398) Capacitors C_1 and C_2 are connected in series. The equivalent capacitance is given by

$C_1 C_2 / (C_1 + C_2)$

399) If the potential difference across a resistor is doubled:

only the current is doubled

400) By using only two resistors, R_1 and R_2 , a student is able to obtain resistances of 3Ω , 4Ω , 12Ω , and 16Ω . The values of R_1 and R_2 (in ohms) are:

4, 12

401) Faraday's law states that an induced emf is proportional to:

the rate of change of the magnetic flux

402) A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

1000V

403) Which of the following electromagnetic radiations has photons with the greatest energy?

x rays

404) A virtual image is one:

from which light rays diverge as they pass through

405) What is the unit of magnification factor?

no units

406) During an adiabatic process an object does 100 J of work and its temperature decreases by 5K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is.

15 J/K

407) An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

a change in entropy

408) The Stern-Gerlach experiment makes use of:

a strong non-uniform magnetic field

409) A large collection of nuclei are undergoing alpha decay. The rate of decay at any instant is proportional to:

the number of undecayed nuclei present at that instant

410) As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, a distance of 0.75m from the origin. The z component of the angular momentum around the origin is:

6.0kg · m²/s

RIZ MUGHAL

QUIZ MASTER

Grand Quiz(PHY101)

100% correct solution.

For more information you can visit my channel and for any type of help related to CS619 you can contact me.



YOUTUBE CHANNEL:

<https://www.youtube.com/channel/UCINsFwDiB62SValCcPDZbRQ/playlists>

FACEBOOK GROUP:

<https://www.facebook.com/groups/923887914750307>

PHY101:Grand Quiz

Question # 1 of 30 (Start time: 10:35:12 AM, 04 January 2021)

The ultimate strength of a sample is the stress at which the sample:

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | returns to its original shape when the stress is removed |
| ☐ | remains underwater |
| ☒ | breaks |
| ☐ | bends 180° |

PHY101:Grand Quiz

Question # 2 of 30 (Start time: 10:35:36 AM, 04 January 2021

The value of k in coulomb's law depends upon

Select the correct option

☒	medium between two charges
☐	distance between charges
☐	magnitude of charges
☐	all of these

PHY101:Grand Quiz

Question # 3 of 30 (Start time: 10:35:54 AM, 04 January 2020)

The law of inertia was firstly formulated by:

Select the correct option

☐	Aristotle
☐	Galileo
☒	Newton
☐	Einstein

RIZ MUGHAL

PHY101:Grand Quiz

Question # 4 of 30 (Start time: 10:36:10 AM, 04 January 2021)

The ratio of circumference of a circle to its diameter is equal to:

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | 2π |
| ☒ | π |
| ☐ | $\pi/2$ |
| ☐ | one steradian |

Question # 5 of 30 (Start time: 10:36:25 AM, 04 January 2021)

Total Marks: 1

A water bed that is 1.5 m wide and 2.5 m long weighs 1055 N. Assuming the entire lower surface of the bed is in contact with the floor, what is the pressure the bed exerts on the floor?

Select the correct option

- | | | |
|----------------------------------|--------|----|
| ☐ | 250 Pa | // |
| ☐ | 260 Pa | // |
| ☐ | 270 Pa | // |
| ☒ | 280 Pa | // |

PHY101:Grand Quiz

Question # 6 of 30 (Start time: 10:36:40 AM, 04 January 2021)

The centre of gravity is the average location of the of an object

Select the correct option

☐	mass
☒	weight
☐	static equilibrium
☐	dynamic

PHY101:Grand Quiz

Question # 7 of 30 (Start time: 10:36:54 AM, 04 January 2021)

Acceleration in a body is always produced in the direction of:

Select the correct option

☐	Velocity
☐	Weight
☒	Force
☐	Acceleration

Question # 8 of 30 (Start time: 10:37:10 AM, 04 January 2021)

Tot

As per Coulomb's law, the force of attraction or repulsion between two point charges directly proportional to the

Select the correct option

- | | |
|----------------------------------|-------------------------------------|
| ☐ | cube of the distance |
| ☒ | product of the magnitude of charges |
| ☐ | sum of the magnitude of charges |
| ☐ | square of the distance between them |

Question # 9 of 30 (Start time: 10:37:25 AM, 04 January 2021)

One revolution is the same as:

Select the correct option



2π rad



π rad



57 rad



1 rad

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Question # 10 of 30 (Start time: 10:37:39 AM, 04 January 2021)

A ----- vector is obtained by dividing the vector by its magnitude:

Select the correct option

☒	unit
☐	position
☐	normal
☐	negative

Question # 11 of 30 (Start time: 10:37:57 AM, 04 January 2021)

Total Marks:

Take the speed of sound to be 340m/s. A thunder clap is heard about 3 s after the lightning is seen. The source of both light and sound is:

Select the correct option



moving overhead faster than the speed of sound //



emitting a much higher frequency than is heard //



emitting a much lower frequency than is heard //



about 1000m away //

Question # 12 of 30 (Start time: 10:38:14 AM, 04 January 2021)

Total M

If the distance between all pairs of particles of the body do not change by applying a force then the body is said to be

Select the correct option



rigid



large



small



flexible

Question # 13 of 30 (Start time: 10:38:34 AM, 04 January 2021)

Total Marks: 1

For an ideal fluid flowing through a horizontal pipe, Bernoulli's equation states that the sum of the pressure and energy per unit volume along the pipe does which of the following? (Assume measurements are taken along the pipe in the direction of fluid flow.)

Select the correct option

- | | | |
|----------------------------------|--|----|
| ☐ | increases as the pipe diameter increases | // |
| ☐ | decreases as the pipe diameter increases | // |
| ☒ | remains constant as the pipe diameter increases | // |
| ☐ | increases, then decreases as the pipe diameter increases | // |

Question # 14 of 30 (Start time: 10:38:59 AM, 04 January 2021)

Which of the following statement is true?

Select the correct option



Weight is a force, mass is a measure of inertia



Mass depends on gravity, weight does not



Gravity is necessary to measure both weight and mass



Heavier objects weigh more than light objects

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Question # 15 of 30 (Start time: 10:39:14 AM, 04 January 2021)

Total M

A/an ----- is the basic reason to change in the motion of an object according to Newton's second law of motion.

Select the correct option

- | | |
|---|---------------------|
| ☒ | net force |
| ☐ | decrease in inertia |
| ☐ | change in velocity |
| ☐ | acceleration |

Question # 16 of 30 (Start time: 10:39:30 AM, 04 January 2021)

When a spring is compressed or stretched, the potential energy of the spring

Select the correct option

☐	Decreases
☐	Stays constant
☒	Increases
☐	Becomes zero

PHY101:Grand Quiz

Question # 17 of 30 (Start time: 10:39:44 AM, 04 January 2021)

The dimensional units of ratio of work and power is:

Select the correct option

☐	J
☒	T
☐	L
☐	F

Question # 18 of 30 (Start time: 10:39:58 AM, 04 January 2021)

A particle oscillating in simple harmonic motion is:

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | in equilibrium at the ends of its path because its velocity is zero there |
| ☒ | in equilibrium at the center of its path because the acceleration is zero there |
| ☐ | never in equilibrium because it is in motion |
| ☐ | never in equilibrium because there is always a force |

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Question # 19 of 30 (Start time: 10:40:13 AM, 04 January 2021)

Total Marks:

A fire whistle emits a tone of 170 Hz. Take the speed of sound in air to be 340m/s. The wavelength of this sound is about:

Select the correct option

☐	3.0m	//
☐	0.5m	//
☐	1.0m	//
☒	2.0m	//

Question # 20 of 30 (Start time: 10:40:28 AM, 04 January 2021)

Total Marks: 1

How much pressure is exerted on a submarine at a depth of 8.50 km in the Pacific Ocean? (The density of sea water = $1.025 \times 10^3 \text{ kg/m}^3$, and the atmospheric pressure at sea level = $1.01 \times 10^5 \text{ Pa}$.)

Select the correct option

- | | | |
|----------------------------------|------------------------------|----|
| ☐ | $8.6 \times 10^5 \text{ Pa}$ | // |
| ☐ | $8.7 \times 10^6 \text{ Pa}$ | // |
| ☐ | $9.5 \times 10^6 \text{ Pa}$ | // |
| ☒ | $8.6 \times 10^6 \text{ Pa}$ | // |

Question # 21 of 30 (Start time: 10:40:44 AM, 04 January 2021)

A force of 100 N acts upon a body for five seconds. What will be the change in momentum?

Select the correct option

- | | |
|----------------------------------|--------|
| ☐ | 200 NS |
| ☒ | 500 NS |
| ☐ | 20 NS |
| ☐ | 50 NS |

Question # 22 of 30 (Start time: 10:40:59 AM, 04 January 2021)

One newton is a force that produces an acceleration of 0.5 m/s^2 in a body of mass:

Select the correct option



1 kg



2 kg



4 kg



8 kg

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Question # 23 of 30 (Start time: 10:41:15 AM, 04 January 2021)

When breaks are applied to a fast moving car, the passenger will be throws:

Select the correct option

☒	Forward
☐	Backward
☐	Downward
☐	Upward

PHY101:Grand Quiz

Question # 24 of 30 (Start time: 10:41:30 AM, 04 January 2021)

In the formula $F = Gm_1m_2/r^2$, the quantity G:

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | depends on the local value of g |
| ☐ | is used only when Earth is one of the two masses |
| ☐ | is greatest at the surface of Earth |
| ☒ | is a universal constant of nature |

PHY101:Grand Quiz

Question # 25 of 30 (Start time: 10:41:45 AM, 04 January 2021)

Per-second refers to the dimensions of physical quantity:

Select the correct option

- | | |
|----------------------------------|----------------------|
| ☐ | Angular displacement |
| ☒ | Angular velocity |
| ☐ | Angular acceleration |
| ☐ | Angular momentum |

PHY101:Grand Quiz

Question # 26 of 30 (Start time: 10:41:59 AM, 04 January 2021)

Light year is a unit of:

Select the correct option

- | | |
|----------------------------------|--------------|
| ☐ | acceleration |
| ☐ | velociy |
| ☐ | time |
| ☒ | distance |

Question # 27 of 30 (Start time: 10:42:18 AM, 04 January 2021)

Young's modulus can be used to calculate the strain for a stress that is:

Select the correct option

- | | |
|----------------------------------|----------------------------------|
| ☐ | just below the ultimate strength |
| ☐ | just above the ultimate strength |
| ☒ | well below the yield strength |
| ☐ | well above the yield strength |

Question # 28 of 30 (Start time: 10:42:32 AM, 04 January 2021)

A wheel of radius 50 cm having the angular speed of 5 rad/s will have linear speed in m/s?

Select the correct option

- | | |
|----------------------------------|-----|
| ☐ | 0.5 |
| ☐ | 1.5 |
| ☒ | 2.5 |
| ☐ | 3.5 |

Question # 29 of 30 (Start time: 10:42:47 AM, 04 January 2021)

An object moves in a circle at constant speed. The work done by the centripetal force is zero because:

Select the correct option



the displacement for each revolution is zero



the average force for each revolution is zero



there is no friction



the centripetal force is perpendicular to the velocity

Question # 30 of 30 (Start time: 10:43:04 AM, 04 January 2021)

Total Marks: 1

A ball is thrown upward into the air with a speed that is greater than terminal speed. It lands at the place where it was thrown. During its flight the force of air resistance is the greatest:

Select the correct option

- | | | |
|----------------------------------|------------------------------|----|
| ☒ | just after it is thrown | // |
| ☐ | half way up | // |
| ☐ | at the top of its trajectory | // |
| ☐ | halfway down | // |

MID TERM EXAMINATION
Solved MCQS BY
MEHRAN ALI SHAH
PHY101- Physics
Mostly from MID TERM
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Question No: 2 (Marks: 1) - Please choose one

A net torque applied to a rigid object always tends to produce:

- ▶ **rotational equilibrium**
- ▶ linear acceleration
- ▶ angular acceleration
- ▶ rotational inertia

Question No: 3 (Marks: 1) - Please choose one

An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

- ▶ 12.6 rad/s
- ▶ 1.57 rad/s
- ▶ **2.0 rad/s**
- ▶ 6.3 rad/s

Question No: 4 (Marks: 1) - Please choose one

In simple harmonic motion, the restoring force must be proportional to the:

- ▶ amplitude
- ▶ frequency

- ▶ velocity
- ▶ **displacement**

Question No: 5 (Marks: 1) - Please choose one

Mercury is a convenient liquid to use in a barometer because:

- ▶ it is a metal
- ▶ it has a high boiling point
- ▶ it expands little with temperature
- ▶ **it has a high density**

Question No: 6 (Marks: 1) - Please choose one

The units of the electric field are:

- ▶ **J/m**
- ▶ J/(C·m)
- ▶ J/C
- ▶ J·C

Question No: 7 (Marks: 1) - Please choose one

A farad is the same as a

- ▶ **J/V**
- ▶ V/J
- ▶ C/V
- ▶ V/C

Question No: 8 (Marks: 1) - Please choose one

We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5H. We also need a capacitance of:

- ▶ 1 F
- ▶ 1mF
- ▶ 1 μ F
- ▶ **100 μ F**

Question No: 9 (Marks: 1) - Please choose one

The wavelength of red light is 700 nm. Its frequency is _____.

- ▶ $4.30 * 10^4$ Hertz
- ▶ $4.30 * 10^3$ Hertz
- ▶ **$4.30 * 10^5$ Hertz**
- ▶ $4.30 * 10^2$ Hertz

Question No: 10 (Marks: 1) - Please choose one

Which of the following statements is NOT TRUE about electromagnetic waves?

- ▶ Electromagnetic waves satisfy the Maswell's Equation.
- ▶ Electromagnetic waves can not travel through space.
- ▶ The receptions of electromagnetic waves require an antenna.
- ▶ **The electromagnetic radiation from a burning candle is unpolarized.**

Question No: 11 (Marks: 1) - Please choose one

Radio waves and light waves are _____.

- ▶ Longitudinal waves
- ▶ Transverse waves
- ▶ **Electromagnetic and transverse both**
- ▶ Electromagnetic and longitudinal both

Question No: 12 (Marks: 1) - Please choose one

Wien's Law states that, $\lambda_{max} =$ _____ K.

- ▶ $2.90 * 10^{-3}$ Hertz
- ▶ $2.90 * 10^{-3}$ s

- ▶ $2.90 \times 10^{-3} \text{ kg}$
- ▶ $2.90 \times 10^{-3} \text{ m}$

Question No: 13 (Marks: 1) - Please choose one

Interference of light is evidence that:

- ▶ the speed of light is very large
- ▶ light is a transverse wave
- ▶ **light is a wave phenomenon**
- ▶ light is electromagnetic in character

Question No: 14 (Marks: 1) - Please choose one

Fahrenheit and Kelvin scales agree numerically at a reading of:

- ▶ **-40**
- ▶ 0
- ▶ 273
- ▶ 574

Question No: 15 (Marks: 1) - Please choose one

According to the theory of relativity:

- ▶ **moving clocks run fast**
- ▶ energy is not conserved in high speed collisions
- ▶ the speed of light must be measured relative to the ether
- ▶ none of the above are true

Question No: 16 (Marks: 1) - Please choose one

Light from a stationary spaceship is observed, and then the spaceship moves directly away from the observer at high speed while still emitting the light. As a result, the light seen by the observer has:

- ▶ higher frequency and a longer wavelength than before
- ▶ **lower frequency and a shorter wavelength than before**
- ▶ higher frequency and a shorter wavelength than before
- ▶ lower frequency and a longer wavelength than before

Question No: 17 (Marks: 1) - Please choose one

How fast should you move away from a 6.0×10^{14} Hz light source to observe waves with a frequency of 4.0×10^{14} Hz?

- ▶ 20c
- ▶ **38c**
- ▶ 45c
- ▶ 51c

Question No: 18 (Marks: 1) - Please choose one

The quantum number n is most closely associated with what property of the electron in a hydrogen atom?

- ▶ **Energy**
- ▶ Orbital angular momentum
- ▶ Spin angular momentum
- ▶ Magnetic moment

Question No: 19 (Marks: 1) - Please choose one

The quantum number m_s is most closely associated with what property of the electron in an atom?

- ▶ Magnitude of the orbital angular momentum
- ▶ **Energy**
- ▶ z component of the spin angular momentum
- ▶ z component of the orbital angular momentum

Question No: 20 (Marks: 1) - Please choose one

As the wavelength of a wave in a uniform medium increases, its speed will

-
- ▶ Decrease
 - ▶ Increase
 - ▶ **Remain the same**
 - ▶ None of these

Question No: 1 (Marks: 1) - Please choose one

The lowest tone produced by a certain organ comes from a 3.0-m pipe with both ends open. If the speed of sound is 340m/s, the frequency of this tone is approximately:

- ▶ A. 7Hz
- ▶ B. 14 Hz
- ▶ C. 28 Hz
- ▶ **D. 57 Hz**

Question No: 2 (Marks: 1) - Please choose one

1. To raise the pitch of a certain piano string, the piano tuner:

- ▶ **A. loosens the string**
- ▶ B. tightens the string
- ▶ C. shortens the string
- ▶ D. lengthens the string

Question No: 7 (Marks: 1) - Please choose one

An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

- ▶ **12.6 rad/s**
- ▶ 1.57 rad/s
- ▶ 2.0 rad/s
- ▶ 6.3 rad/s

Question No: 8 (Marks: 1) - Please choose one

For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

- ▶ the center of mass

- ▶ the center of gravity
- ▶ the geometrical center
- ▶ **the same point**

Question No: 9 (Marks: 1) - Please choose one

Ten seconds after an electric fan is turned on, the fan rotates at 300 rev/min. Its average angular acceleration is:

- ▶ **3.14 rad/s²**
- ▶ 30 rad/s²
- ▶ 30 rev/s²
- ▶ 50 rev/min²
- ▶ 1800 rev/s²

Question No: 10 (Marks: 1) - Please choose one

A 4.0-N puck is traveling at 3.0m/s. It strikes a 8.0-N puck, which is stationary. The two pucks stick together. Their common final speed is:

- ▶ **1.0m/s**
- ▶ 1.5m/s
- ▶ 2.0m/s
- ▶ 2.3m/s

Question No: 11 (Marks: 1) - Please choose one

An object moving in a circle at constant speed:

- ▶ must have only one force acting on it
- ▶ is not accelerating
- ▶ is held to its path by centrifugal force
- ▶ **has an acceleration of constant magnitude**

Question No: 12 (Marks: 1) - Please choose one

A plane traveling north at 200m/s turns and then travels south at 200m/s.
The change in its velocity is:

- ▶ 400m/s north
- ▶ **400m/s south**
- ▶ zero
- ▶ 200m/s south

Question No: 13 (Marks: 1) - Please choose one

At time $t = 0$ a car has a velocity of 16 m/s. It slows down with an acceleration given by $-0.50t$, in m/s^2 for t in seconds. It stops at $t =$

- ▶ 64 s
- ▶ 32 s
- ▶ 16 s
- ▶ **8.0 s**

Question No: 14 (Marks: 1) - Please choose one

1 mi is equivalent to 1609 m so 55 mph is:

- ▶ 15 m/s
- ▶ **25 m/s**
- ▶ 66 m/s
- ▶ 88 m/s

The number of significant figures in 0.00150 is:

- ▶ 5
- ▶ 4
- ▶ **3**
- ▶ 2

Question No: 2 (Marks: 1)

One revolution is the same as:

2π rad

- ▶ 1 rad
- ▶ 57 rad
- ▶ $\pi/2$ rad
- ▶ π rad
- ▶ **2π rad**

Question No: 3 (M a r k s: 1)

For a body to be in equilibrium under the combined action of several forces:

- ▶ All the forces must be applied at the same point
- all the forces must be applied at the same point
- ▶ all of the forces form pairs of equal and opposite forces
- ▶ **any two of these forces must be balanced by a third force**
- ▶ the sum of the torques about any point must equal zero

PHY101 - Physics - Question No: 4 (M a r k s: 1)

A bucket of water is pushed from left to right with increasing speed across a horizontal surface.

Consider the pressure at two points at the same level in the water.

- ▶ It is the same
- ▶ **It is higher at the point on the left**
- ▶ It is higher at the point on the right
- ▶ At first it is higher at the point on the left but as the bucket speeds up it is lower there

PHY101 - Physics - Question No: 5 (M a r k s: 1)

An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

- ▶ A. 200 Hz
- ▶ B. 300 Hz
- ▶ C. 400 Hz
- ▶ **D. 600 Hz**

Question No: 6 (M a r k s: 1)

Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by

- ▶ $C_1 C_2 / (C_1 + C_2)$
- ▶ $(C_1 + C_2) / C_1 C_2$
- ▶ $1 / (C_1 + C_2)$
- ▶ C_1 / C_2

PHY101 - Physics - Question No: 7 (Marks: 1)

If the potential difference across a resistor is doubled:

- ▶ only the current is doubled
- ▶ only the current is halved
- ▶ only the resistance is doubled
- ▶ only the resistance is halved

PHY101 - Physics - Question No: 8 (Marks: 1)

By using only two resistors, R_1 and R_2 , a student is able to obtain resistances of 3Ω , 4Ω , 12Ω , and 16Ω . The values of R_1 and R_2 (in ohms) are:

- ▶ 3, 4
- ▶ 2, 12
- ▶ 3, 16
- ▶ 4, 12

PHY101 - Physics - Question No: 9 (Marks: 1)

Faraday's law states that an induced emf is proportional to:

- ▶ the rate of change of the electric field
- ▶ the rate of change of the magnetic flux
- ▶ the rate of change of the electric flux
- ▶ the rate of change of the magnetic field

PHY101 - Physics - Question No: 10 (Marks: 1)

A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

- ▶ 1000V
- ▶ 500V
- ▶ 250V
- ▶ 100V

PHY101 - Physics - Question No: 14 (Marks: 1)

Which of the following electromagnetic radiations has photons with the

greatest energy?

- ▶ blue light
- ▶ yellow light
- ▶ x rays
- ▶ radio waves

PHY101 - Physics - Question No: 15 (M a r k s: 1)

A virtual image is one:

- ▶ toward which light rays converge but do not pass through
- ▶ from which light rays diverge as they pass through
- ▶ toward which light rays converge and pass through
- ▶ from which light rays diverge but do not pass through

PHY101 - Physics - Question No: 16 (M a r k s: 1) vuzs

What is the unit of magnification factor?

- ▶ meter.Kelvin
- ▶ radian.Kelvin
- ▶ degree.Kelvin
- ▶ no units

Question No: 17 (M a r k s: 1)

During an adiabatic process an object does 100 J of work and its temperature decreases by 5K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is.

- ▶ 20 J/K
- ▶ 100 J/K
- ▶ 15 J/K
- ▶ 5 J/K

Question No: 18 (M a r k s: 1)

An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

- ▶ a change in entropy
- ▶ a decrease of internal energy
- ▶ an increase of pressure
- ▶ a change in temperature

Question No: 19 (Marks: 1)

The Stern-Gerlach experiment makes use of:

- ▶ a strong uniform magnetic field
- ▶ **a strong non-uniform magnetic field**
- ▶ a strong uniform electric field
- ▶ a strong non-uniform electric field

Question No: 20 (Marks: 1)

A large collection of nuclei are undergoing alpha decay. The rate of decay at any

instant is proportional to:

- ▶ **the number of undecayed nuclei present at that instant**
- ▶ the time since the decays started
- ▶ the time remaining before all have decayed
- ▶ the half-life of the decay

Question No: 1 (Marks: 1) - Please choose one

As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, a distance of 0.75m from the origin. The z component of the angular momentum around the origin is:

- ▶ **6.0kg · m²/s**
- ▶ 9.0kg · m²/s
- ▶ 11 kg · m²/s
- ▶ 14 kg · m²/s

Question No: 21 (Marks: 3)

Two people are carrying a uniform wooden board that is 3.00 m long and weighs 160 N. If one person applies an upward force equal to 60 N at one end, at what point does the other person lift? Begin with a free-body diagram of the board.

Forces in x direction = 0

Forces in Y = $F_1 + F_2 - W$

Given:

$L = 3.00 \text{ m}$ $F_1 = 60 \text{ N}$

$W = 160 \text{ N}$ $F_2 = ?$ and $x_2 = ?$

Sum of forces and torques = 0

Sum Force = $F_1 + F_2 - W = 0$

$60\text{N} + F_2 - 160 \text{ N} = 0$

$F_2 = 100 \text{ N}$

My pivot point is at F_2 .

Sum of torques = 0

Torque $F_1 = F_1(L - x_2)$

Torque $F_2 = 0$ b/c at pivot point

Torque $W = W(L/2 - x_2)$

$F_1L - F_1x_2 + (WL)/2 - Wx_2 = 0$

$(60)(3) - 60x_2 + (160 * 3)/2 - 160x_2 = 0$

$180 - 60x_2 + 240 - 160x_2 = 0$

$420 - 220x_2 = 0$

$x_2 = 1.9\text{m}$

Question No: 22 (Marks: 3)

If a charged particle moves in a straight line through some region of space, can you say that the magnetic field in that region is zero?

Question No: 26 (Marks: 5)

A bike accelerates uniformly from rest to a speed of 7.10 m/s over a distance of 35.4 m . Determine the acceleration of the bike.

$$2as = v_f^2 - v_i^2$$

$$2a(35.4) = (7.10)^2 - (0)^2$$

$$2a(35.4) = 50.41$$

$$A = .71 \text{ m/s}^2$$

Question No: 27 (Marks: 5)

A flat loop of wire consisting of a single turn of cross-sectional area 8.00 cm^2 is perpendicular to a magnetic field that increases uniformly in

magnitude from 0.500 T to 2.50 T in 1.00 s. What is the resulting induced current if the loop has a resistance of 2.00 W?

$$E = (B_f - B_i) \cdot A / t = (2.5 - 0.5) \cdot 8 \cdot 10^{-4} / 1 = 1.6 \cdot 10^{-3} \text{ V}$$
$$I = E / R = 1.23 \text{ mA}$$

MID TERM EXAMINATION

Solved MCQS BY

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Phy101 Searching file

by Orange Monkey Team

1. According to the equation of work, when a boy sits in one place and studies for the whole night, he does

No work

2. When the velocity of an aeroplane is doubled, the momentum_____

Increase uniformly

3. The momentum of an object at a given instant is independent of its.

Acceleration

4. What sort of energy does flying bird possess?

Both potential and kinetic energy

5. The acceleration of projectile in x-direction:

0ms^{-2}

6. The projectile path is also known as its

Trajectory

7. If force of friction is negligible, then acceleration of two freely falling objects of different masses is:

Smaller acceleration for smaller mass

8. Which of the following statement/s is/ are true?

- The weight of a man on the moon is smaller than that on the earth
- The mass of a man is the same on both the moon and the earth
- We cannot determine our own mass in the outer space because there is no gravity

1 only

9. When the momentum of a body is doubled, its kinetic energy_____

Becomes four times its initial energy

10. Which of the following statement is true

Gravity is necessary to measure both weight and mass

11. It is easier to walk on the concrete road than on ice because

the friction between the ice and the feet is less than that between the concrete and feet

12. Pre-second refers to the dimensions of physical quantity

_____ Velocity

13. If the mass of moving objects is doubled then its K.E becomes

4 times

14. The slope of a velocity-time the graph at any point may be identified with

Average acceleration

15. K.E can be defined as the dot product of

Mass and velocity

16. A wheel of radius 50cm having the angular speed of 5 rad/s will have linear speed in m/s

2.5

19. Lighter objects always reach the ground later than heavier objects

2 and 3 only

17. In _____ object returns to its original position if displaced slightly

Stable equilibrium

18. Newton first law is also called _____

Law of inertia

19. A mosquito's buzz is often rated with a decibel rating of 40 db _

20

20. The product of force and time is called change in _____

Momentum or impulse.... Both are correct

21. People try to keep the _____ over their feet in order to feel stable

Center of gravity

22. If the magnitude of force applied is incorrect the work done will _____ be increased unit of distance is

Light year

23. Power is equal to the dot product of force and Velocity

24. Product of vector $\mathbf{a}$ with itself is equal to

$$A^2$$

25. For a body to be in equilibrium under the combined action of several forces

The sum of the torques about any point must equal zero

26. The ratio of circumference of a circle to its diameter is equal to

π

27. A net torque applied to a rigid object always tends to produce

Angular acceleration

28. The work done by gravity during the descent of a projectile

Is positive

29. A man with his arms at his sides is spinning on a low friction stool

His angular momentum remains the same

30. Example of physical quantity is

Length

31. Swimming becomes possible because of

Third law of motion

32. The scalar product of two vectors is maximum when they are

Parallel

33. Which of the following statements are true of sound waves _____
identify all that apply

A sound wave is a mechanical wave

34. The angular momentum vector of earth about its rotation axis. Due to its daily rotation is directed

north

35. For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about

the same point

36. People try to keep the _____ over their feet in order to feel stable

Centre of gravity

37. Add two vectors of length 4m & 5m but their orientation is not known____the length after addition

Between 9m and 1 m

38. Which statement is not true for acceleration?

riding your bike straight down the street at a constant speed

39. Acceleration of an object must be zero at a point where

The instantaneous velocity is constant

40. Which of the following statements are true of sound waves____identify that apply

A sound wave is mechanical wave

41. If the distance between all pairs of particles of the body do not change by applying a force then the body is said to be

Rigid

42. A couple produces

Purely rotation motion

43. To determine if a rigid body is equilibrium the vector sum of the gravitational forces acting on the particles of the body can be replaced by a single force acting at

The center of gravity

44. Work may be done by

Only living organisms

45. The dimension of joule is

ML^2T^{-2}

46. The unit $kg\ m^2/s$ can be used for

Angular momentum

47. Unit of distance is

Meter

48. The law of inertia was firstly formulated by

Galileo

49. What is the angle of projection of projectile for which its maximum height and horizontal range are equal

90

50. A body will be in translational equilibrium if

$\Sigma F = 0$

51. Suitable units for the gravitational constant G are

$\text{m}^3/\text{kg s}^2$

54. As the wavelength of a wave in a uniform medium increases its frequency will

Remains the same

52. A sound wave is a mechanical wave not an electromagnetic wave this means that

A medium is required in order for sound waves to transport energy

53. The gravitational constant G has the derived units

Nm^2/kg^2

54. If the formula $F = Gm_1m_2/r^2$ the quantity G

Is a universal constant of nature

55. An object moving in a circle at constant speed

Has an acceleration of constant magnitude

56. The fundamental dimension of angular momentum are

$\text{ML}^2\text{T}^{-2} \text{Kg m}^2/\text{s}$

57. The area under the velocity time graph is

Distance

58. A particle oscillating in simple harmonic motion is

In equilibrium at the ends of its path because the acceleration is zero there

59. An elastic collision is one in which

K.E and momentum are both conserved

65. When brakes are applied to a fast moving car the passenger will be thrown

Forward

60. Work has dimension like power

Power

61. A 1kg block is lifted vertically 1m by a boy. The work done by the boy is about.

1J

62. 0.0003711 in scientific notation is

3.7×10^{-4}

68. A bullet shoot straight up returns to its starting point in 10 sec. the initial speed was

49m/s

63. The rate of change of momentum of a body is equal to

Applied force

64. If you travelled for 2.5 hours with an average speed 48 miles/hours. The distance travelled is

120 miles

65. A stone is thrown up from the surface of earth, then it reaches a maximum height, kinetic energy is equal to:

Zero

66. A force of 100N acts upon a body for 5 seconds. What will be the change in momentum?

500Ns

67. The relative speed of approach is always equal to relative speed of separation in?

Perfectly elastic collision

68. If the magnitude of force applied is increased, the work done will ?

Be increased

69. 6 joule of work done in 3 second then power is?

2 watt

70. An object moving in the circle at constant speed.

Must have only one force acting on it

71. A vector a is added to vector b. the resultant vector(a+b) have greatest magnitude when:

When vector A and B are perpendicular

72. A force of 120 newton is exerted on a 40 kg container which sits on a floor. If the friction force between floor and container is 80 Newton. What is the magnitude of acceleration of container?

2ms^{-2}

73. The angle between rectangular components of a vector is

45 degree

74. Newton second law of motion is also called ?

Law of acceleration

75. The goal of all scientific enquiry or scientific method is?

Predicting natural events based on known patterns

76. If the total momentum of a system is changing

A net external force must be acting on the system

77. Change in momentum in one second is called?

Impulse

78. A/An _____ is the basic reason to change the motion of an object according to Newton's second law of motion.

Net force

79. An object moves in a circle at constant speed. The work done by centripetal force is zero because_?

The centripetal force is perpendicular to the velocity

80. The law of conservation of momentum applies to a system of colliding object only if

The net external impulse is zero

81. Force is a

Universal quantity

82. When the mass of the colliding body is much larger than the mass of the body at rest, its velocity after collision

Become double

83. The slope of a velocity – time the graph at any point may be identified with

Average acceleration

84. If the force acting on a body is doubled then the acceleration becomes

Double

85. Force:

Equal the time rate of change of momentum

86. When a spring is compressed or stretched, the potential energy of the spring

Increases

87. Which of the following statement is true

Weight is a force, mass is a measure of inertia

88. A ball is allowed to fall freely from certain height. It covers a distance in 1st second equal to

$2g$

89. Acceleration of an object must be zero at a point where:

The average velocity is zero

90. A body is changing its direction but its speed is constant, the object is:

Accelerating

91. Per-second refers to the dimension of physical quantity

Velocity

92. The scalar product of two vectors is maximum when they are

Parallel

93. Physics is one of the branches of

Physical sciences

94. Suppose no reaction force exists then which of the following case\ s will occur?

- **We cannot swim**
- **We cannot jump**
- **We cannot walk**

1 2 and 3 all

95. Newton-Second is SI unit of:

Impulse

96. Work done in gravitational field

Negative

97. When body moves with constant acceleration the velocity time graph is:

Straight line

1. A sound wave has a wavelength of 3.0m. The distance from a compression center to the adjacent rarefaction center is:

1. 0.75m
2. 1.5m
3. 3.0m
4. need to know wave speed

2. The speed of sound in medium depends upon

1. amplitude
2. frequency
3. wavelength
4. properties of the medium

3. Which of the following will remain unchanged when a sound wave travels in air or in water?

1. Amplitude
2. Wavelength
3. Frequency
4. Speed

4. The vibrations or the pressure variations inside the inner ear are converted into electrical signals by the _____.

1. cochlea
2. tympanic membrane
3. pinna
4. anvil

5. Sound and light waves both

1. have similar wavelength
2. obey the laws of reflection
3. travel as longitudinal waves
4. travel through vacuum

6. A mosquito's buzz is often rated with a decibel rating of 40 dB. Normal conversation is often rated at 60 dB. How many times more intense is normal conversation compared to a mosquito's buzz?

1. 2
2. 20
3. 100
4. 400

7. Point where force causes system to move without rotation is

1. 2 times beyond centre of mass
2. edge of the body
3. mid point of body
4. center of mass

8. The frequency which is not audible to the human ear is

1. 50 Hz
2. 500 Hz
3. 5000 Hz
4. 50000 Hz

9. If the speed of sound is 340 m/s a plane flying at 400m/s creates a conical shock wave with an apex half angle of:

1. 0 (no shock wave)
2. 32°
3. 40°
4. 58°

10. A municipal water supply is provided by a tall water tower. Water from this tower flows to a building. How does the water flow out of a faucet on the ground floor of a building compare with the water flow out of an identical faucet on the second floor of the building?

1. Water flows more rapidly out of the ground-floor faucet.
2. Water flows more rapidly out of the second-floor faucet.
3. Water flows at the same speed out of both faucets.
4. The speed of water flow cannot be determined unless the height of the water tower is known.

11. A sound wave has a wavelength of 3.0m. The distance from a compression center to the adjacent to the rarefaction center is:

1. 0.75m
2. 1.5m
3. 3.0m
4. need to know frequency

12. A plane produces a sonic boom only when:

1. it emits sound waves of very long wavelength
2. it emits sound waves of high frequency it flies at high altitudes
3. it flies faster than the speed of sound
4. it flies faster than the speed of sound

13. acceleration of an object must be zero at a point where:

1. the instantaneous velocity is zero but changing
2. the instantaneous velocity is constant
3. the average velocity is zero
4. the instantaneous velocity is not zero but changing

14. Coulomb's law is only true for point charges whose sizes are

1. large
2. very small
3. very large
4. medium

15. A vector A is added to a vector B. The resultant vector A + B have greatest magnitude when:

1. The magnitude of vector A + B does not depend on the directions of A and B
2. When vectors A and B are parallel and in the opposite direction
3. When vectors A and B are parallel and in the same direction
4. When vectors A and B are perpendicular

16. Earth exerts a gravitational force on the Moon, keeping it in its orbit. The reaction to this force, in the sense of Newton's third law, is:

1. the centripetal force on the Moon
2. the nearly circular orbit on the Moon
3. the gravitational force on Earth by the Moon
4. the tides due to the Moon

17. The angular momentum vector of Earth about its rotation axis, due to its daily rotation, is directed:

1. south
2. north
3. tangent to the equator toward the west
4. tangent to the equator toward the east

18. Work may be done by _____.

1. only vehicles
2. only non-living objects
3. only living organisms
4. both living organisms and non-living objects

19. In simple harmonic motion, the restoring force must be proportional to the:

1. amplitude
2. frequency
3. velocity
4. displacement

20. The turning effect of a force is called the force's

1. momentum
2. torque/moment
3. distance
4. inertia

21. How much pressure is exerted on a submarine at a depth of 8.50 km in the Pacific Ocean? (The density of sea water = $1.025 \times 10^3 \text{ kg/m}^3$, and the atmospheric pressure at sea level = $1.01 \times 10^5 \text{ Pa}$.)

1. $8.6 \times 10^5 \text{ Pa}$
2. $8.7 \times 10^6 \text{ Pa}$
3. $9.5 \times 10^6 \text{ Pa}$

4. $8.6 \times 10^6 \text{ Pa}$

22. A 4.0-m long steel beam with a cross-sectional area of $1.0 \times 10^{-2} \text{ m}^2$ and a Young's modulus of $2.0 \times 10^{11} \text{ N/m}^2$ is wedged horizontally between two vertical walls. In order to wedge the beam, it is compressed by 0.020mm. If the coefficient of static friction between the beam and the walls is 0.70 the maximum mass (including its own) it can bear without slipping is:

1. 3.6 kg
2. 36 kg
3. 71 kg
4. 710 kg

23. The torque about any two points has the same value when the body is in

1. translational equilibrium
2. neutral equilibrium
3. dynamic equilibrium
4. static equilibrium

24. The work done by gravity during the descent of a projectile:

1. depends for its sign on the direction of the y axis
2. is positive
3. is negative
4. is zero

25. Take the speed of sound to be 340m/s. A thunder clap is heard about 3 s after the lightning is seen. The source of both light and sound is:

1. moving overhead faster than the speed of sound
2. emitting a much higher frequency than is heard
3. emitting a much lower frequency than is heard
4. about 1000m away

26. The magnitude of the acceleration of a planet in orbit around the Sun is proportional to:

1. the mass of the planet
2. the mass of the Sun
3. the distance between the planet and the Sun

4. the reciprocal of the distance between the planet and the Sun

27. The unit $\text{kg}\cdot\text{m}^2/\text{s}$ can be used for:

1. angular momentum
2. rational kinetic energy
3. rational inertia
4. torque

28. Which of the following statements about floating object is correct?

1. The object's density is greater than the density of the fluid on which it floats.
2. The object's density is equal to the density of the fluid on which it floats.
3. The displaced volume of fluid is greater than the volume of the object.
4. The buoyant force equals the object's weight.

29. When a spring is compressed or stretched, the potential energy of the spring

1. Decreases
2. stays constant
3. Increases
4. Becomes zero

30. An ultrasonic wave is sent from a ship towards the bottom of the sea. It is found that the time interval between the sending and receiving of the wave is 1.6 s. What is the depth of the sea, if the velocity of sound in the seawater is 1400 m/s?

1. 1120 m
2. 560 m
3. 1400 m
4. 112 m

31. To determine if a rigid body is in equilibrium the vector sum of the gravitational forces acting on the particles of the body can be replaced by a single force acting at:

1. a point on the boundary

2. the center of gravity
3. the geometrical center
4. the center of mass

32. A couple produces:

1. Purely linear motion
2. Purely rotational motion
3. Linear and rotational motion
4. No motion

33. In _____ object returns to its original position if displaced slightly

1. dynamic equilibrium
2. stable equilibrium
3. unstable equilibrium
4. rotational equilibrium

34. The momentum of an object at a given instant is independent of its:

1. acceleration
2. velocity
3. speed
4. mass

35. A vector is obtained by dividing the vector by its magnitude:

1. unit
2. position
3. normal
4. negative

36. According to the equation of work, when a boy sits in one place and studies for the whole night, he does

1. lot of work
2. no work
3. too much work
4. very little work

37. The application/s of dimensional analysis is/are:

- i. To convert a physical quantity from one system of units to another.

- ii. To check the dimensional correctness of a given equation.
iii. Establish a relationship between different physical quantities in an equation.

1. i only
2. ii & iii only
3. i & iii only
4. I, ii & iii

38. The angle between rectangular components of a vector is:

1. 0°
2. 45°
3. 60°
4. 90°

39. A qualitative definition of force is given by which law?

1. law of inertia

40. Which pair will always have the same magnitude to the rate of change of position?

1. Average speed and average velocity

41. Add two vectors of length 4 m & 5 m but their orientation is not known, The length after addition of these two vectors will be:

1. Less than 1 m
2. between 9 m and 1 m
3. between 9 m and 5 m
4. 9m

42. What is the angle of projection of projectile, for which its maximum height and horizontal range are equal?

1. 30°

2. 36*

3. 76*

4. 90*

43. A unit vector is obtained by dividing the vector by its magnitude:

1. unit

2. position

3. normal

4. negative

44. When the momentum of a body is doubled, its kinetic energy _____.

1. Becomes ten times its kinetic energy

2. Becomes four times its initial kinetic energy

45. A ball is thrown upward into air with a speed that is greater than terminal speed. It lands at the place where it was thrown. During its flight the force of air resistance is the greatest:

1. just after it is thrown

2. half way up

3. at the top of its trajectory

4. halfway down

46. An object moves in a circle at constant speed. The work done by the centripetal force is zero because:

1. the displacement for each revolution is zero

2. the average force for each revolution is zero

3. there is no friction

4. the centripetal force is perpendicular to the velocity

47. It is easier to walk on concrete road than on ice because:

1. the amount of firection is same for both (ice & concrete road)
2. there is more friction on the ice than on concrete
3. ice is soft and spongy where as concrete is hard
4. the friction between the ice and the feet is less than that between the concrete and feet

48. An inellastic collidion is one in which:

1. momentum is not conserved but kinetic energy is conserved
2. total mass is not conserved but momentum is conserved
3. neither kinetic energy nor momentum is conserved
4. momentum is conserved but kinetic energy is not conserved

49. When you step on the accelerator to increase the speed of your car, the force that accelerates the car is

1. the force of friction of the road on the tires

50. Light year is a unit of:

1. accelration
2. velocity
3. time
4. distance

51. A body is changing its direction but speed is constant, the object is:

1. accelerating
2. slowing down
3. speeding
4. Displacing

52. The ratio of circumference of a circle to its diameter is equal to:

1. 2π
2. π
3. $\pi/2$
4. one steradian

53. Which of the following statements are TRUE of sound waves? Identify all that apply.

1. Sound can travel through a vacuum
2. A sound wave is a transverse wave
3. A sound wave is a mechanical wave
4. To hear the sound of a tuning fork, the tines of the fork must move air from the fork to one's ear

54. A force of 120 N is exerted on a 40 kg container which sits on a floor. If the frictional force between floor and container is 80 N. What is the magnitude of the acceleration of the container?

1. 2 ms^{-2}
2. 5 ms^{-2}
3. 3 ms^{-2}
4. 1 ms^{-2}

55. If a simple pendulum oscillates with an amplitude 50 mm and time period 2s, then its maximum velocity is

1. 0.1 m/s
2. 0.15 m/s
3. 0.8 m/s
4. 0.16 m/s

56. Acceleration in a body is always produced in the direction of:

1. Velocity
2. Weight
3. Force
4. Acceleration

57. The center of mass of a uniform disk of radius R is located:

1. at the center
2. a distance $R/3$ from the center
3. a distance $R/2$ from the center

4. on the rim

58. For a body to be in equilibrium under the combined action of several forces:

1. any two of these forces must be balanced by a third force
2. the sum of components of all the forces in any direction must equal zero
3. all of the forces form pairs of equal and opposite forces
4. all the forces must be applied at the same point

59. A source of frequency of 500 Hz emits waves of wavelength 0.4 m, how long does the wave take to travel 600 m?

1. 3 s

60. Momentum may be expressed in:

1. kg/(m s)
2. N s
3. gram s
4. kg/m

61. A force of 5000N is applied outwardly to each end of a 5.0-m long rod with a radius of 34.0 cm and a Young's modulus of $125 \times 10^8 \text{ N/m}^2$. The elongation of the rod is:

1. 0.0020mm
2. 0.0040mm
3. 0.14mm
4. 0.55mm

62. The speed of a sound wave is determined by:

1. its amplitude

2. its intensity
3. the transmitting medium
4. number of harmonics present

63. Which of the following statement is TRUE of sound intensity and decibel levels? Identify all that apply.

1. The intensity of sound which corresponds to the threshold of pain is one trillion times more intense than the sound which corresponds to the threshold of hearing.
2. Two sounds which have a ratio of decibel ratings equal to 2.0. This means that the second sound is twice as intense as the first sound.
3. Sound A is 20 times more intense than the sound B. So if sound B is rated at 30 dB, then sound A is rated at 50 dB.
4. A machine produces a sound which is rated at 60 dB. If two of the machines were used at the same time, the decibel rating would be 120 dB.

64. Sound waves are

1. longitudinal

65. For an ideal fluid flowing through a horizontal pipe, Bernoulli's equation states that the sum of the pressure and energy per unit volume along the pipe does which of the following? (Assume measurements are taken along the pipe in the direction of fluid flow.)

1. increases as the pipe diameter increases
2. decreases as the pipe diameter increases
3. remains constant as the pipe diameter increases
4. increases, then decreases as the pipe diameter increases

66. A 2.0-kg block travels around a 0.5-m radius circle with an angular velocity of 12 rad/s. Its angular momentum about the center of the circle is:

1. 6.0 kg.m²/s
2. 12 kg.m²/s
3. 48 kg/m².s
4. 72 kg.m²/s²

67. The fundamental dimensions of angular momentum are

1. $\underline{ML^{-2}T^{-2}}$
2. $\underline{MLT^{-1}}$
3. $\underline{ML^2T^{-1}}$
4. $\underline{ML^2T^{-2}}$

12. Power is equal to the dot product of force and:

c) Velocity

2) The frequency which is not audible to the human ear is

Select the correct option

a) 50000 Hz

b) 500 Hz

c) 5000 Hz

d) 50 Hz

3) A wheel of radius 50 cm having the angular speed of 5 rad/s will have linear speed in m/s?

Select the correct option

a) 1.5

b) 2.5

c) 3.5

d) 0.5

4) When breaks are applied to a fast-moving car, the passenger will be thrown:

Select the correct option

a) Backward

b) Forward

c) Upward

d) Downward

5) Stress can be measured in:

Select the correct option

a) N.m

b) N/m²

c) N.m²

d) N/m

6) A heavy particle moving with 5 m/s suffers an elastic collision with a light particle at rest. After collision, velocity of light particle will be:

Select the correct option

a) 10 m/s

b) 2.5 m/s

c) 20 m/s

d) 5 m/s

6) Acceleration in a body is always produced on the direction of:

Select the correct option

a) Velocity

b) Weight

c) Acceleration

d) Force

7) The fundamental dimensions of angular momentum are

Select the correct option

a) $ML^{-2}T^{-2}$

b) ML^2T^{-1}

c) ML^2T^{-2}

d) MLT^{-1}

8) Ali wants to lift a mass of 7.5 kg with constant velocity by a rope that passes through a frictionless pulley which is attached to the ceiling of room. Calculate the tension in the rope by neglecting the mass of the rope.

Select the correct option

a) 75 N

b) 7.5 Kg

c) 75 Kg

d) 0.75 N

9) The first condition of equilibrium implies that:

Select the correct option

a) $\sum F_y = 0$

b) $\sum F_x = \sum F_y$

c) $\sum F = 0$

d) $\sum F_x = 0$

10) Final take off velocity of an airplane is 67 m/s. The length of runway is 2 km, the constant acceleration is:

Select the correct option

a) 3.24 ms^{-1}

b) 3.24 ms^{-2}

c) 2.24 ms^{-1}

d) 2.24 ms^{-2}

11) The ultimate strength of a sample in the stress at which the sample:

Select the correct option

a) remain underwater

b) breaks

c) bends 180°

d) returns to its original shape when the stress is moved

12) The area under the velocity-time graphs is:

Select the correct option

a) Acceleration

b) Torque

c) Distance

d) Force

13) if you traveled for 2.5 hours with an average speed 48 miles/hours, the distance traveled is:

Select the correct option

a) 19.2 miles

b) 120 miles

c) 300 miles

d) 48 miles

14) The law of inertia was firstly formulated by:

Select the correct option

a) Newton

b) Einstein

c) Galileo

d) Aristotle

15) When the velocity of an aeroplane is doubled, the momentum ____.

Select the correct option

a) Is conserved

b) Becomes zero

c) Increase uniformly

d) Remains unchanged

16) As we move above, the body above the surface of the earth, the change in potential energy will be:

Select the correct option

a) Infinity

b) Positive

c) Zero

d) Negative

17) The slope of a velocity – time the graph at any point may be identified with:

Select the correct option

a) average acceleration

18) A body of mass 2 kg moving with velocity 4 m/s has K.E equals to:

Select the correct option

a) 16 J

b) 8 J

c) 32 J

d) 2 J

19) A ____ velocity is obtained by dividing the vector by its magnitude:

Select the correct option

a) negative

b) unit

c) position

d) normal

20) A particle oscillating in simple harmonic motion is:

Select the correct option

a) in equilibrium at the center of its path because the acceleration is zero there

b) never in equilibrium because it is in motion

c) in equilibrium at the end of its path because the acceleration is zero there

d) never in equilibrium because there is always a force

22) Young's modulus is a proportionality constant that relates the force per unit area applied perpendicularly at the surface of an object to:

Select the correct option

a) the fractional change in volume

b) the fractional change in length

c) the shear

d) the pressure

23) A bullet shoot straight up returns to its sharing point in 10 sec. the initial speed was:

Select the correct option

a) 49 m/s

b) 9.8 m/s

c) 98 m/s

d) 24.5 m/s

24) An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

Select the correct option

a) 1.57 rad/s

b) 12.6 rad/s

c) 2.0 rad/s

d) 6.3 rad/s

25) The dimensional units of ratio of work and power is:

Select the correct option

a) F

b) J

c) T

d) L

26) The newton-second is a unit of:

Select the correct option

a) power

b) angular momentum

c) linear momentum

d) work

27) A sound wave has a wavelength of 3.0m. The distance from a compression center to the adjacent rarefaction center is:

Select the correct option

a) 3.0m

b) 1.5m

c) 0.75m

d) need to know wave speed

28) For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

Select the correct option

a) the center of gravity

b) the same point

c) the geometrical center

d) the center of mass

29) If a simple pendulum oscillates with an amplitude 50 mm and time period 2s, then its maximum velocity is

Select the correct option

a) 0.8 m/s

b) 0.16 m/s

c) 0.15 m/s

d) 0.1 m/s

30) bank and forth is a leftward and rightward direction. This type of wave is known as a ____.

Select the correct option

a) Mechanical

b) Electromagnetic

c) Transverse

d) Longitudinal

18) A/an ____ is the basic reason to change in the motion of an object according to Newton's second law of motion

Select the correct option

- a) acceleration
- b) change in velocity
- c) **net force**
- d) decrease in inertia

19) If the distance between all pairs of particles of the body do not change by applying a force, then a body is said to be

Select the correct option

- a) **rigid**
- b) small
- c) large
- d) flexible

20) The center of mass of Earth's atmosphere is:

Select the correct option

- a) near the outer boundary of the atmosphere
- b) **near the center of earth**
- c) a little less than halfway between earth's surface and the outer boundary of the atmosphere
- d) near the surface of earth

21) An object attached to one end of a spring makes 20 complete oscillations in 10 s. Its period is:

Select the correct option

- a) 0.5 s
- b) 0.5Hz
- c) **2 s**
- d) 2Hz

22) Whenever an object strikes a stationary object of equal mass:

Select the correct option

- a) the first object must stop
- b) the two objects cannot stick together
- c) **none of these**
- d) the collision must be elastic

23) If a wheel turns with constant angular speed then:

Select the correct option

- a) each point on its moves with constant velocity
- b) each point on its rim moves with constant acceleration
- c) the wheel turns through equal angles in equal times

d) the angle through which the wheel turns in each second increases as time goes on

24) A stone is thrown up from the surface of earth, the it reaches a maximum height, kinetic energy is equal to:

Select the correct option

- a) $\frac{1}{2} mv$
- b) mgh
- c) $2mgh$
- d) zero

25) In Simple harmonic motion, the magnitude of the acceleration is

Select the correct option

- a) greatest when the velocity is greatest
- b) proportional to the displacement
- c) inversely proportional to the displacement
- d) constant

The angle between rectangular components of a vector is:

Select the correct option

- a) 0°
- b) 45°
- c) 60°
- d) 90°

10) A ____ vector is obtained by dividing the vector by its magnitude

Select the correct option

- a) **unit**
- b) position
- c) normal
- d) negative

11) According to the equation of work, when a boy sits in one place and studies for the whole night, he does

Select the correct option

- a) lot of work
- b) **no work**
- c) too much work
- d) very little work

12) the projectile path is known as its:

Select the correct option

- a) range
- b) **trajectory**
- c) curve
- d) time of action

13) If force of friction is negligible, then acceleration of two freely falling objects of different masses is:

Select the correct option

- a) None of these
- b) **Smaller acceleration for smaller mass**
- c) Variable
- d) The same

14) Add two vectors of length 4 m & 5m but their orientation is not known. The length after addition of these two vectors will be:

Select the correct option

- a) **between 9 m and 1 m**
- b) 9 m
- c) between 9 m and 5
- d) Less than 1 m

15) The dimensional units of ratio of work and power is:

Select the correct option

- a) L
- b) F
- c) J
- d) **T**

16) A sphere of moment on inertia I rolls down an inclined plane without slipping. The ratio of the rotational kinetic energy to the transitional kinetic energy is nearly:

Select the correct option

- a) **5/7**
- b) 7/5
- c) 2/5

d) 5/2

17) Swimming becomes possible because of:

Select the correct option

- a) First law of motion
- b) Second law of motion
- c) **Third law of motion**
- d) Law of torque

An object moves in a circle at constant speed. The work done by the centripetal force is zero because:

- d) **the centripetal force is perpendicular to the velocity**

3) When the velocity of an aeroplane is doubled, the momentum ____.

Select the correct option

- a) **Increase uniformly**
- b) Becomes zero
- c) Is conserved
- d) Remains unchanged

4) The momentum of an object at a given instant is independent of its:

Select the correct option

- a) **acceleration**
- b) velocity
- c) speed

d) mass

5) What sort of energy does flying bird possess?

Select the correct option

- a) Potential energy
- b) Kinetic energy
- c) Elastic energy
- d) **Both potential and kinetic energy**

6) The acceleration of projectile in x-direction:

Select the correct option

- a) depends on initial velocity
- b) depends on how long it is in the air
- c) depends on y-acceleration
- d) **0 m/s²**

7) Light year is a unit of:

Select the correct option

- a) acceleration
- b) velocity
- c) time
- d) distance

8) The goal of all scientific inquiry (or scientific method) is:

Select the correct option

- a) world peace

b) predicting natural events based

c) to make everyone rich and happy

As the wavelength of a wave in a uniform medium increases, its frequency will _____.

Decrease

9

A bullet shot straight up returns to its starting point in 10 sec. the initial speed was:

49 m/s

11

Young's modulus is a proportionality constant that relates the force per unit area applied perpendicularly at the surface of an object to:

the fractional change in length

12

A ball is thrown upward into the air with a speed that is greater than terminal speed. It lands at the place where it was thrown. During its flight the force of air resistance is the greatest:

at the top of its trajectory

13

A sphere of the moment on inertia I roll down an inclined plane without slipping. The ratio of the rotational kinetic energy to the translational kinetic energy is near:

5/7

14

The persistence of audible sound due to the successive

Reverberation

	reflections from the surrounding objects even after the source has stopped to produce that sound is called	
15	A municipal water supply is provided by a tall water tower. Water from this tower flows to a building. How does the water flow out of a faucet on the ground floor of a building compare with the water flow out of an identical faucet on the second floor of the building?	Water flows more rapidly out of the ground-floor faucet.
16	An elastic collision is one in which:	kinetic energy and momentum are both conserved
17	The ultimate strength of a sample is the stress at which the sample	Breaks/ returns to its original shape when the stress is removed
18	The dot product of vector A with itself is equal to:	A^2
19	A heavy particle moving with 5 m/s suffers an elastic collision with a light particle at rest. After the collision, the velocity of light particle will be:	5 m/s
20	Young's modulus can be correctly given in:	N/m^2
21	A net torque applied to a rigid object always tends to produce:	angular acceleration
23	The velocity-time graph is parallel to the time axis the acceleration of moving body is:	zero
24	What is the angle of projection	76

	of projectile, for which its maximum height and horizontal range are equal?	
25	A particle	
26	The dimensional units of the ratio of work and power is:	T
27	The magnitude of the acceleration on the planet in orbit around the Sun is proportional to.	the mass of the Sun
28	A Prosonic wave I gent from @ ship towards the bottom of the sea. N is found that the time interval between the sending and receiving of the wave is 1.6 5. What is the depth of the sea. if the velocity of sound in the seawater is	1120m
29	Water flows into © house at e velocity of 15 m/s through a pipe that has 9 radii of 0.40 m. The water then flows through a smaller pipe at a velocity of 45 m/s. What is the radius of the smuttier pipe?	0.23m
30	A mosquito's buzz is often rated with a decibel rating of 40 dB. Normal conversation is often rated at 60 dB. How many times more intense is normal conversation compared to 6 mosquito's buzz?	20
31	People try to keep the... over their feet. in order to feel stable.	center of gravity
32	If force is applied at the center of mass then torque is.	Zero

33	As the wavelength of a wave in a uniform medium increases. its speed will	Decrease
34	For an ideal fluid flowing through a horizontal pipe, Bernoulli's equation state that the sum of the pressure and energy per unit volume along the pipe does which of the following?	Remains constant as the pipe diameter increases
35	Which of the following statement is true?	Weight is a force, mass is a measure of inertia
36	A/an ----- is the basic reason to change in the motion of an object according to Newton's second law of motion.	Net force
37	When spring is compressed or stretched, the potential energy of spring	Increase
38	The ultimate strength of a sample is the stress in which the sample:	Breaks
39	The value of k in coulomb's law depends upon:	Medium between two charges
40	The law of inertia was firstly formulated by	Newton
41	The ratio of the circumference of a circle to its diameter is equal to:	Pi
42	A water bed that is 1.5 m wide and 2.5 m long weighs 1055 N. assuming the entire lower surface of the bed in contact	280 pa

	with the floor, what is the pressure the bed exerts on the floor?	
43	The center of gravity is the average location of the of an object	Weight
44	Acceleration in a body is always produced in the direction of	Force
45	As per coulomb's law, the force of attraction or repulsion between two-point charge directly proportional to the	Product os the magnitude of charges
46	One revolution is the same as:	2pi rad
47	A ----- vector is obtained by dividing the vector by its magnitude:	Unit
48	Take the speed of the sound to be 340m/s. A thunderclap is heard about 3 s after the lightning is seen. The source of both light and sound is:	About 1000m away
49	The fundamental dimensions of angular momentum are :	ML ² T ⁻¹
50	The approximate value of get an altitude above Earth equal to one Earth diameter is:	4.9m/s ²
51	If the speed of sound is 340m/s a plane flying at 400m/s creates @ conical shock wave with an apex half-angle of:	Zero
52	If the distance between all pairs of particulars of the body do not change by applying a force that a body is said to be	Rigid

MID TERM EXAMINATION
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Question No: 2 (Marks: 1) - Please choose one

A net torque applied to a rigid object always tends to produce:

- ▶ **rotational equilibrium**
- ▶ linear acceleration
- ▶ angular acceleration
- ▶ rotational inertia

Question No: 3 (Marks: 1) - Please choose one

An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

- ▶ 12.6 rad/s
- ▶ 1.57 rad/s
- ▶ **2.0 rad/s**
- ▶ 6.3 rad/s

Question No: 4 (Marks: 1) - Please choose one

In simple harmonic motion, the restoring force must be proportional to the:

- ▶ amplitude
- ▶ frequency

- ▶ velocity
- ▶ **displacement**

Question No: 5 (Marks: 1) - Please choose one

Mercury is a convenient liquid to use in a barometer because:

- ▶ it is a metal
- ▶ it has a high boiling point
- ▶ it expands little with temperature
- ▶ **it has a high density**

Question No: 6 (Marks: 1) - Please choose one

The units of the electric field are:

- ▶ **J/m**
- ▶ J/(C·m)
- ▶ J/C
- ▶ J·C

Question No: 7 (Marks: 1) - Please choose one

A farad is the same as a

- ▶ **J/V**
- ▶ V/J
- ▶ C/V
- ▶ V/C

Question No: 8 (Marks: 1) - Please choose one

We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5H. We also need a capacitance of:

- ▶ 1 F
- ▶ 1mF
- ▶ 1 μ F
- ▶ **100 μ F**

Question No: 9 (Marks: 1) - Please choose one

The wavelength of red light is 700 nm. Its frequency is _____.

- ▶ $4.30 * 10^4$ Hertz
- ▶ $4.30 * 10^3$ Hertz
- ▶ **$4.30 * 10^5$ Hertz**
- ▶ $4.30 * 10^2$ Hertz

Question No: 10 (Marks: 1) - Please choose one

Which of the following statements is NOT TRUE about electromagnetic waves?

- ▶ Electromagnetic waves satisfy the Maswell's Equation.
- ▶ Electromagnetic waves can not travel through space.
- ▶ The receptions of electromagnetic waves require an antenna.
- ▶ **The electromagnetic radiation from a burning candle is unpolarized.**

Question No: 11 (Marks: 1) - Please choose one

Radio waves and light waves are _____.

- ▶ Longitudinal waves
- ▶ Transverse waves
- ▶ **Electromagnetic and transverse both**
- ▶ Electromagnetic and longitudinal both

Question No: 12 (Marks: 1) - Please choose one

Wien's Law states that, $\lambda_{max} =$ _____ K.

- ▶ $2.90 * 10^{-3}$ Hertz
- ▶ $2.90 * 10^{-3}$ s

- ▶ $2.90 \times 10^{-3} \text{ kg}$
- ▶ $2.90 \times 10^{-3} \text{ m}$

Question No: 13 (Marks: 1) - Please choose one

Interference of light is evidence that:

- ▶ the speed of light is very large
- ▶ light is a transverse wave
- ▶ **light is a wave phenomenon**
- ▶ light is electromagnetic in character

Question No: 14 (Marks: 1) - Please choose one

Fahrenheit and Kelvin scales agree numerically at a reading of:

- ▶ **-40**
- ▶ 0
- ▶ 273
- ▶ 574

Question No: 15 (Marks: 1) - Please choose one

According to the theory of relativity:

- ▶ **moving clocks run fast**
- ▶ energy is not conserved in high speed collisions
- ▶ the speed of light must be measured relative to the ether
- ▶ none of the above are true

Question No: 16 (Marks: 1) - Please choose one

Light from a stationary spaceship is observed, and then the spaceship moves directly away from the observer at high speed while still emitting the light. As a result, the light seen by the observer has:

- ▶ higher frequency and a longer wavelength than before
- ▶ **lower frequency and a shorter wavelength than before**
- ▶ higher frequency and a shorter wavelength than before
- ▶ lower frequency and a longer wavelength than before

Question No: 17 (Marks: 1) - Please choose one

How fast should you move away from a 6.0×10^{14} Hz light source to observe waves with a frequency of 4.0×10^{14} Hz?

- ▶ 20c
- ▶ **38c**
- ▶ 45c
- ▶ 51c

Question No: 18 (Marks: 1) - Please choose one

The quantum number n is most closely associated with what property of the electron in a hydrogen atom?

- ▶ **Energy**
- ▶ Orbital angular momentum
- ▶ Spin angular momentum
- ▶ Magnetic moment

Question No: 19 (Marks: 1) - Please choose one

The quantum number m_s is most closely associated with what property of the electron in an atom?

- ▶ Magnitude of the orbital angular momentum
- ▶ **Energy**
- ▶ z component of the spin angular momentum
- ▶ z component of the orbital angular momentum

Question No: 20 (Marks: 1) - Please choose one

As the wavelength of a wave in a uniform medium increases, its speed will

-
- ▶ Decrease
 - ▶ Increase
 - ▶ **Remain the same**
 - ▶ None of these

Question No: 1 (Marks: 1) - Please choose one

The lowest tone produced by a certain organ comes from a 3.0-m pipe with both ends open. If the speed of sound is 340m/s, the frequency of this tone is approximately:

- ▶ A. 7Hz
- ▶ B. 14 Hz
- ▶ C. 28 Hz
- ▶ **D. 57 Hz**

Question No: 2 (Marks: 1) - Please choose one

1. To raise the pitch of a certain piano string, the piano tuner:

- ▶ **A. loosens the string**
- ▶ B. tightens the string
- ▶ C. shortens the string
- ▶ D. lengthens the string

Question No: 7 (Marks: 1) - Please choose one

An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

- ▶ **12.6 rad/s**
- ▶ 1.57 rad/s
- ▶ 2.0 rad/s
- ▶ 6.3 rad/s

Question No: 8 (Marks: 1) - Please choose one

For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

- ▶ the center of mass

- ▶ the center of gravity
- ▶ the geometrical center
- ▶ **the same point**

Question No: 9 (Marks: 1) - Please choose one

Ten seconds after an electric fan is turned on, the fan rotates at 300 rev/min. Its average angular acceleration is:

- ▶ **3.14 rad/s²**
- ▶ 30 rad/s²
- ▶ 30 rev/s²
- ▶ 50 rev/min²
- ▶ 1800 rev/s²

Question No: 10 (Marks: 1) - Please choose one

A 4.0-N puck is traveling at 3.0m/s. It strikes a 8.0-N puck, which is stationary. The two pucks stick together. Their common final speed is:

- ▶ **1.0m/s**
- ▶ 1.5m/s
- ▶ 2.0m/s
- ▶ 2.3m/s

Question No: 11 (Marks: 1) - Please choose one

An object moving in a circle at constant speed:

- ▶ must have only one force acting on it
- ▶ is not accelerating
- ▶ is held to its path by centrifugal force
- ▶ **has an acceleration of constant magnitude**

Question No: 12 (Marks: 1) - Please choose one

A plane traveling north at 200m/s turns and then travels south at 200m/s.
The change in its velocity is:

- ▶ 400m/s north
- ▶ **400m/s south**
- ▶ zero
- ▶ 200m/s south

Question No: 13 (Marks: 1) - Please choose one

At time $t = 0$ a car has a velocity of 16 m/s. It slows down with an acceleration given by $-0.50t$, in m/s^2 for t in seconds. It stops at $t =$

- ▶ 64 s
- ▶ 32 s
- ▶ 16 s
- ▶ **8.0 s**

Question No: 14 (Marks: 1) - Please choose one

1 mi is equivalent to 1609 m so 55 mph is:

- ▶ 15 m/s
- ▶ **25 m/s**
- ▶ 66 m/s
- ▶ 88 m/s

The number of significant figures in 0.00150 is:

- ▶ 5
- ▶ 4
- ▶ **3**
- ▶ 2

Question No: 2 (Marks: 1)

One revolution is the same as:

2π rad

- ▶ 1 rad
- ▶ 57 rad
- ▶ $\pi/2$ rad
- ▶ π rad
- ▶ **2π rad**

Question No: 3 (M a r k s: 1)

For a body to be in equilibrium under the combined action of several forces:

- ▶ All the forces must be applied at the same point
- all the forces must be applied at the same point
- ▶ all of the forces form pairs of equal and opposite forces
- ▶ **any two of these forces must be balanced by a third force**
- ▶ the sum of the torques about any point must equal zero

PHY101 - Physics - Question No: 4 (M a r k s: 1)

A bucket of water is pushed from left to right with increasing speed across a horizontal surface.

Consider the pressure at two points at the same level in the water.

- ▶ It is the same
- ▶ **It is higher at the point on the left**
- ▶ It is higher at the point on the right
- ▶ At first it is higher at the point on the left but as the bucket speeds up it is lower there

PHY101 - Physics - Question No: 5 (M a r k s: 1)

An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

- ▶ A. 200 Hz
- ▶ B. 300 Hz
- ▶ C. 400 Hz
- ▶ **D. 600 Hz**

Question No: 6 (M a r k s: 1)

Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by

- ▶ $C_1 C_2 / (C_1 + C_2)$
- ▶ $(C_1 + C_2) / C_1 C_2$
- ▶ $1 / (C_1 + C_2)$
- ▶ C_1 / C_2

PHY101 - Physics - Question No: 7 (Marks: 1)

If the potential difference across a resistor is doubled:

- ▶ only the current is doubled
- ▶ only the current is halved
- ▶ only the resistance is doubled
- ▶ only the resistance is halved

PHY101 - Physics - Question No: 8 (Marks: 1)

By using only two resistors, R_1 and R_2 , a student is able to obtain resistances of 3Ω , 4Ω , 12Ω , and 16Ω . The values of R_1 and R_2 (in ohms) are:

- ▶ 3, 4
- ▶ 2, 12
- ▶ 3, 16
- ▶ 4, 12

PHY101 - Physics - Question No: 9 (Marks: 1)

Faraday's law states that an induced emf is proportional to:

- ▶ the rate of change of the electric field
- ▶ the rate of change of the magnetic flux
- ▶ the rate of change of the electric flux
- ▶ the rate of change of the magnetic field

PHY101 - Physics - Question No: 10 (Marks: 1)

A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

- ▶ 1000V
- ▶ 500V
- ▶ 250V
- ▶ 100V

PHY101 - Physics - Question No: 14 (Marks: 1)

Which of the following electromagnetic radiations has photons with the

greatest energy?

- ▶ blue light
- ▶ yellow light
- ▶ x rays
- ▶ radio waves

PHY101 - Physics - Question No: 15 (M a r k s: 1)

A virtual image is one:

- ▶ toward which light rays converge but do not pass through
- ▶ from which light rays diverge as they pass through
- ▶ toward which light rays converge and pass through
- ▶ from which light rays diverge but do not pass through

PHY101 - Physics - Question No: 16 (M a r k s: 1) vuzs

What is the unit of magnification factor?

- ▶ meter.Kelvin
- ▶ radian.Kelvin
- ▶ degree.Kelvin
- ▶ no units

Question No: 17 (M a r k s: 1)

During an adiabatic process an object does 100 J of work and its temperature decreases by 5K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is.

- ▶ 20 J/K
- ▶ 100 J/K
- ▶ 15 J/K
- ▶ 5 J/K

Question No: 18 (M a r k s: 1)

An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

- ▶ a change in entropy
- ▶ a decrease of internal energy
- ▶ an increase of pressure
- ▶ a change in temperature

Question No: 19 (Marks: 1)

The Stern-Gerlach experiment makes use of:

- ▶ a strong uniform magnetic field
- ▶ **a strong non-uniform magnetic field**
- ▶ a strong uniform electric field
- ▶ a strong non-uniform electric field

Question No: 20 (Marks: 1)

A large collection of nuclei are undergoing alpha decay. The rate of decay at any

instant is proportional to:

- ▶ **the number of undecayed nuclei present at that instant**
- ▶ the time since the decays started
- ▶ the time remaining before all have decayed
- ▶ the half-life of the decay

Question No: 1 (Marks: 1) - Please choose one

As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, a distance of 0.75m from the origin. The z component of the angular momentum around the origin is:

- ▶ **6.0kg · m²/s**
- ▶ 9.0kg · m²/s
- ▶ 11 kg · m²/s
- ▶ 14 kg · m²/s

Question No: 21 (Marks: 3)

Two people are carrying a uniform wooden board that is 3.00 m long and weighs 160 N. If one person applies an upward force equal to 60 N at one end, at what point does the other person lift? Begin with a free-body diagram of the board.

Forces in x direction = 0

Forces in Y = $F_1 + F_2 - W$

Given:

$L = 3.00 \text{ m}$ $F_1 = 60 \text{ N}$

$W = 160 \text{ N}$ $F_2 = ?$ and $x_2 = ?$

Sum of forces and torques = 0

Sum Force = $F_1 + F_2 - W = 0$

$60\text{N} + F_2 - 160 \text{ N} = 0$

$F_2 = 100 \text{ N}$

My pivot point is at F_2 .

Sum of torques = 0

Torque $F_1 = F_1(L - x_2)$

Torque $F_2 = 0$ b/c at pivot point

Torque $W = W(L/2 - x_2)$

$F_1L - F_1x_2 + (WL)/2 - Wx_2 = 0$

$(60)(3) - 60x_2 + (160 * 3)/2 - 160x_2 = 0$

$180 - 60x_2 + 240 - 160x_2 = 0$

$420 - 220x_2 = 0$

$x_2 = 1.9\text{m}$

Question No: 22 (Marks: 3)

If a charged particle moves in a straight line through some region of space, can you say that the magnetic field in that region is zero?

Question No: 26 (Marks: 5)

A bike accelerates uniformly from rest to a speed of 7.10 m/s over a distance of 35.4 m . Determine the acceleration of the bike.

$$2as = v_f^2 - v_i^2$$

$$2a(35.4) = (7.10)^2 - (0)^2$$

$$2a(35.4) = 50.41$$

$$A = .71 \text{ m/s}^2$$

Question No: 27 (Marks: 5)

A flat loop of wire consisting of a single turn of cross-sectional area 8.00 cm^2 is perpendicular to a magnetic field that increases uniformly in

magnitude from 0.500 T to 2.50 T in 1.00 s. What is the resulting induced current if the loop has a resistance of 2.00 W?

$$E = (B_f - B_i) \cdot A / t = (2.5 - 0.5) \cdot 8 \cdot 10^{-4} / 1 = 1.6 \cdot 10^{-3} \text{ V}$$
$$I = E / R = 1.23 \text{ mA}$$

MID TERM EXAMINATION

Solved MCQS BY

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PHY101- Physics

Mostly from MID TERM

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VU NET

Chapter 1: MEASUREMENT

1. The SI standard of time is based on:
 - A. the daily rotation of the earth
 - B. the frequency of light emitted by Kr^{86}
 - C. the yearly revolution of the earth about the sun
 - D. a precision pendulum clock
 - E. none of theseAns: E
2. A nanosecond is:
 - A. 10^9 s
 - B. 10^{-9} s
 - C. 10^{-10} s
 - D. 10^{-10} s
 - E. 10^{-12}Ans: B
3. The SI standard of length is based on:
 - A. the distance from the north pole to the equator along a meridian passing through Paris
 - B. wavelength of light emitted by Hg^{198}
 - C. wavelength of light emitted by Kr^{86}
 - D. a precision meter stick in Paris
 - E. the speed of lightAns: E
4. In 1866, the U. S. Congress defined the U. S. yard as exactly 3600/3937 international meter. This was done primarily because:
 - A. length can be measured more accurately in meters than in yards
 - B. the meter is more stable than the yard
 - C. this definition relates the common U. S. length units to a more widely used system
 - D. there are more wavelengths in a yard than in a meter
 - E. the members of this Congress were exceptionally intelligentAns: C
5. Which of the following is closest to a yard in length?
 - A. 0.01 m
 - B. 0.1 m
 - C. 1 m
 - D. 100 m
 - E. 1000 mAns: C

6. There is no SI base unit for area because:
- A. an area has no thickness; hence no physical standard can be built
 - B. we live in a three (not a two) dimensional world
 - C. it is impossible to express square feet in terms of meters
 - D. area can be expressed in terms of square meters
 - E. area is not an important physical quantity

Ans: D

7. The SI base unit for mass is:

- A. gram
- B. pound
- C. kilogram
- D. ounce
- E. kilopound

Ans: C

8. A gram is:

- A. 10^{-6} kg
- B. 10^{-3} kg
- C. 1 kg
- D. 10^3 kg
- E. 10^6 kg

Ans: B

9. Which of the following weighs about a pound?

- A. 0.05 kg
- B. 0.5 kg
- C. 5 kg
- D. 50 kg
- E. 500 kg

Ans: D

10. $(5.0 \times 10^4) \times (3.0 \times 10^6) =$

- A. 1.5×10^9
- B. 1.5×10^{10}
- C. 1.5×10^{11}
- D. 1.5×10^{12}
- E. 1.5×10^{13}

Ans: C

11. $(5.0 \times 10^4) \times (3.0 \times 10^{-6}) =$

- A. 1.5×10^{-3}
- B. 1.5×10^{-1}
- C. 1.5×10^1
- D. 1.5×10^3
- E. 1.5×10^5

Ans: B

12. $5.0 \times 10^5 + 3.0 \times 10^6 =$

- A. 8.0×10^5
- B. 8.0×10^6
- C. 5.3×10^5
- D. 3.5×10^5
- E. 3.5×10^6

Ans: E

13. $(7.0 \times 10^6)/(2.0 \times 10^{-6}) =$

- A. 3.5×10^{-12}
- B. 3.5×10^{-6}
- C. 3.5
- D. 3.5×10^6
- E. 3.5×10^{12}

Ans: E

14. The number of significant figures in 0.00150 is:

- A. 2
- B. 3
- C. 4
- D. 5
- E. 6

Ans: B

15. The number of significant figures in 15.0 is:

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Ans: C

16. $3.2 \times 2.7 =$

- A. 9
- B. 8
- C. 8.6
- D. 8.64
- E. 8.640

Ans: C

17. $1.513 + 27.3 =$

- A. 29
- B. 28.8
- C. 28.9
- D. 28.81
- E. 28.813

()Ans: B

18. 1 mi is equivalent to 1609 m so 55 mph is:

- A. 15 m/s
- B. 25 m/s
- C. 66 m/s
- D. 88 m/s
- E. 1500 m/s

Ans: B

19. A sphere with a radius of 1.7 cm has a volume of:

- A. $2.1 \times 10^{-5} \text{ m}^3$
- B. $9.1 \times 10^{-4} \text{ m}^3$
- C. $3.6 \times 10^{-3} \text{ m}^3$
- D. 0.11 m^3
- E. 21 m^3

Ans: A

20. A sphere with a radius of 1.7 cm has a surface area of:

- A. $2.1 \times 10^{-5} \text{ m}^2$
- B. $9.1 \times 10^{-4} \text{ m}^2$
- C. $3.6 \times 10^{-3} \text{ m}^2$
- D. 0.11 m^2
- E. 36 m^2

Ans: C

21. A right circular cylinder with a radius of 2.3 cm and a height of 1.4 m has a volume of:

- A. 0.20 m^3
- B. 0.14 m^3
- C. $9.3 \times 10^{-3} \text{ m}^3$
- D. $2.3 \times 10^{-3} \text{ m}^3$
- E. $7.4 \times 10^{-4} \text{ m}^3$

Ans: D

22. A right circular cylinder with a radius of 2.3 cm and a height of 1.4 cm has a total surface area of:

- A. $1.7 \times 10^{-3} \text{ m}^2$
- B. $3.2 \times 10^{-3} \text{ m}^2$
- C. $2.0 \times 10^{-3} \text{ m}^3$
- D. $5.3 \times 10^{-3} \text{ m}^2$
- E. $7.4 \times 10^{-3} \text{ m}^2$

Ans: D

23. A cubic box with an edge of exactly 1 cm has a volume of:

- A. 10^{-9} m^3
- B. 10^{-6} m^3
- C. 10^{-3} m^3
- D. 10^3 m^3
- E. 10^6 m^3

Ans: B

24. A square with an edge of exactly 1 cm has an area of:

- A. 10^{-6} m^2
- B. 10^{-4} m^2
- C. 10^2 m^2
- D. 10^4 m^2
- E. 10^6 m^2

Ans: B

25. 1 m is equivalent to 3.281 ft. A cube with an edge of 1.5 ft has a volume of:

- A. $1.2 \times 10^2 \text{ m}^3$
- B. $9.6 \times 10^{-2} \text{ m}^3$
- C. 10.5 m^3
- D. $9.5 \times 10^{-2} \text{ m}^3$
- E. 0.21 m^3

Ans: B

26. During a short interval of time the speed v in m/s of an automobile is given by $v = at^2 + bt^3$, where the time t is in seconds. The units of a and b are respectively:

- A. m s^2 ; m s^4
- B. s^3/m ; s^4/m
- C. m/s^2 ; m/s^3
- D. m/s^3 ; m/s^4
- E. m/s^4 ; m/s^5

Ans: D

27. Suppose $A = BC$, where A has the dimension L/M and C has the dimension L/T . Then B has the dimension:

- A. T/M
- B. L^2/TM
- C. TM/L^2
- D. L^2T/M
- E. M/L^2T

Ans: A

28. Suppose $A = B^n C^m$, where A has dimensions LT , B has dimensions L^2T^{-1} , and C has dimensions LT^2 . Then the exponents n and m have the values:
- A. $2/3$; $1/3$
 - B. 2 ; 3
 - C. $4/5$; $-1/5$
 - D. $1/5$; $3/5$
 - E. $1/2$; $1/2$
- Ans: D

Chapter 2: MOTION ALONG A STRAIGHT LINE

1. A particle moves along the x axis from x_i to x_f . Of the following values of the initial and final coordinates, which results in the displacement with the largest magnitude?
- A. $x_i = 4 \text{ m}, x_f = 6 \text{ m}$
 - B. $x_i = -4 \text{ m}, x_f = -8 \text{ m}$
 - C. $x_i = -4 \text{ m}, x_f = 2 \text{ m}$
 - D. $x_i = 4 \text{ m}, x_f = -2 \text{ m}$
 - E. $x_i = -4 \text{ m}, x_f = 4 \text{ m}$

ans: E

2. A particle moves along the x axis from x_i to x_f . Of the following values of the initial and final coordinates, which results in a negative displacement?
- A. $x_i = 4 \text{ m}, x_f = 6 \text{ m}$
 - B. $x_i = -4 \text{ m}, x_f = -8 \text{ m}$
 - C. $x_i = -4 \text{ m}, x_f = 2 \text{ m}$
 - D. $x_i = -4 \text{ m}, x_f = -2 \text{ m}$
 - E. $x_i = -4 \text{ m}, x_f = 4 \text{ m}$

ans: B

3. The average speed of a moving object during a given interval of time is always:
- A. the magnitude of its average velocity over the interval
 - B. the distance covered during the time interval divided by the time interval
 - C. one-half its speed at the end of the interval
 - D. its acceleration multiplied by the time interval
 - E. one-half its acceleration multiplied by the time interval.

ans: B

4. Two automobiles are 150 kilometers apart and traveling toward each other. One automobile is moving at 60 km/h and the other is moving at 40 km/h mph. In how many hours will they meet?
- A. 2.5
 - B. 2.0
 - C. 1.75
 - D. 1.5
 - E. 1.25

ans: D

5. A car travels 40 kilometers at an average speed of 80 km/h and then travels 40 kilometers at an average speed of 40 km/h. The average speed of the car for this 80-km trip is:
- A. 40 km/h
 - B. 45 km/h
 - C. 48 km/h
 - D. 53 km/h
 - E. 80 km/h

ans: D

6. A car starts from Hither, goes 50 km in a straight line to Yon, immediately turns around, and returns to Hither. The time for this round trip is 2 hours. The magnitude of the average velocity of the car for this round trip is:
- A. 0
 - B. 50 km/hr
 - C. 100 km/hr
 - D. 200 km/hr
 - E. cannot be calculated without knowing the acceleration
- ans: A

7. A car starts from Hither, goes 50 km in a straight line to Yon, immediately turns around, and returns to Hither. The time for this round trip is 2 hours. The average speed of the car for this round trip is:
- A. 0
 - B. 50 km/h
 - C. 100 km/h
 - D. 200 km/h
 - E. cannot be calculated without knowing the acceleration
- ans: B

8. The coordinate of a particle in meters is given by $x(t) = 16t - 3.0t^3$, where the time t is in seconds. The particle is momentarily at rest at $t =$
- A. 0.75 s
 - B. 1.3s
 - C. 5.3s
 - D. 7.3s
 - E. 9.3s
- ans: B

9. A drag racing car starts from rest at $t = 0$ and moves along a straight line with velocity given by $v = bt^2$, where b is a constant. The expression for the distance traveled by this car from its position at $t = 0$ is:
- A. bt^3
 - B. $bt^3/3$
 - C. $4bt^2$
 - D. $3bt^2$
 - E. $bt^{3/2}$
- ans: B

10. A ball rolls up a slope. At the end of three seconds its velocity is 20 cm/s; at the end of eight seconds its velocity is 0. What is the average acceleration from the third to the eighth second?
- A. 2.5cm/s²
 - B. 4.0cm/s²
 - C. 5.0cm/s²
 - D. 6.0cm/s²
 - E. 6.67 cm/s²
- ans: B

11. The coordinate of an object is given as a function of time by $x = 7t - 3t^2$, where x is in meters and t is in seconds. Its average velocity over the interval from $t = 0$ to $t = 4$ s is:
- A. 5 m/s
 - B. -5 m/s
 - C. 11 m/s
 - D. -11 m/s
 - E. -14.5 m/s

ans: B

12. The velocity of an object is given as a function of time by $v = 4t - 3t^2$, where v is in m/s and t is in seconds. Its average velocity over the interval from $t = 0$ to $t = 2$ s:
- A. is 0
 - B. is -2 m/s
 - C. is 2 m/s
 - D. is -4 m/s
 - E. cannot be calculated unless the initial position is given

ans: A

13. The coordinate of an object is given as a function of time by $x = 4t^2 - 3t^3$, where x is in meters and t is in seconds. Its average acceleration over the interval from $t = 0$ to $t = 2$ s is:
- A. -4 m/s²
 - B. 4 m/s²
 - C. -10 m/s²
 - D. 10 m/s²
 - E. -13 m/s²

ans: C

14. Each of four particles move along an x axis. Their coordinates (in meters) as functions of time (in seconds) are given by

particle 1: $x(t) = 3.5 - 2.7t^3$

particle 2: $x(t) = 3.5 + 2.7t^3$

particle 3: $x(t) = 3.5 + 2.7t^2$

particle 4: $x(t) = 3.5 - 3.4t - 2.7t^2$

Which of these particles have constant acceleration?

- A. All four
- B. Only 1 and 2
- C. Only 2 and 3
- D. Only 3 and 4
- E. None of them

ans: D

15. Each of four particles move along an x axis. Their coordinates (in meters) as functions of time (in seconds) are given by

particle 1: $x(t) = 3.5 - 2.7t^3$

particle 2: $x(t) = 3.5 + 2.7t^3$

particle 3: $x(t) = 3.5 + 2.7t^2$

particle 4: $x(t) = 3.5 - 3.4t - 2.7t^2$

Which of these particles is speeding up for $t > 0$?

- A. All four
- B. Only 1
- C. Only 2 and 3
- D. Only 2, 3, and 4
- E. None of them

ans: A

16. An object starts from rest at the origin and moves along the x axis with a constant acceleration of 4 m/s^2 . Its average velocity as it goes from $x = 2 \text{ m}$ to $x = 8 \text{ m}$ is:

- A. 1 m/s
- B. 2 m/s
- C. 3 m/s
- D. 5 m/s
- E. 6 m/s

ans: E

17. Of the following situations, which one is impossible?

- A. A body having velocity east and acceleration east
- B. A body having velocity east and acceleration west
- C. A body having zero velocity and non-zero acceleration
- D. A body having constant acceleration and variable velocity
- E. A body having constant velocity and variable acceleration

ans: E

18. Throughout a time interval, while the speed of a particle increases as it moves along the x axis, its velocity and acceleration might be:

- A. positive and negative, respectively
- B. negative and positive, respectively
- C. negative and negative, respectively
- D. negative and zero, respectively
- E. positive and zero, respectively

ans: C

19. A particle moves on the x axis. When its acceleration is positive and increasing:

- A. its velocity must be positive
- B. its velocity must be negative
- C. it must be slowing down
- D. it must be speeding up
- E. none of the above must be true

ans: E

20. The position y of a particle moving along the y axis depends on the time t according to the equation $y = at - bt^2$. The dimensions of the quantities a and b are respectively:
- A. $L^2/T, L^3/T^2$
 - B. $L/T^2, L^2/T$
 - C. $L/T, L/T^2$
 - D. $L^3/T, T^2/L$
 - E. none of these
- ans: C
21. A particle moves along the x axis according to the equation $x = 6t^2$, where x is in meters and t is in seconds. Therefore:
- A. the acceleration of the particle is 6 m/s^2
 - B. t cannot be negative
 - C. the particle follows a parabolic path
 - D. each second the velocity of the particle changes by 9.8 m/s
 - E. none of the above
- ans: E
22. Over a short interval near time $t = 0$ the coordinate of an automobile in meters is given by $x(t) = 27t - 4.0t^3$, where t is in seconds. At the end of 1.0 s the acceleration of the auto is:
- A. 27 m/s^2
 - B. 4.0 m/s^2
 - C. -4.0 m/s^2
 - D. -12 m/s^2
 - E. -24 m/s^2
- ans: E
23. Over a short interval, starting at time $t = 0$, the coordinate of an automobile in meters is given by $x(t) = 27t - 4.0t^3$, where t is in seconds. The magnitudes of the initial (at $t = 0$) velocity and acceleration of the auto respectively are:
- A. $0; 12 \text{ m/s}^2$
 - B. $0; 24 \text{ m/s}^2$
 - C. $27 \text{ m/s}; 0$
 - D. $27 \text{ m/s}; 12 \text{ m/s}^2$
 - E. $27 \text{ m/s}; 24 \text{ m/s}^2$
- ans: C
24. At time $t = 0$ a car has a velocity of 16 m/s . It slows down with an acceleration given by $-0.50t$, in m/s^2 for t in seconds. It stops at $t =$
- A. 64 s
 - B. 32 s
 - C. 16 s
 - D. 8.0 s
 - E. 4.0 s
- ans: D

25. At time $t = 0$ a car has a velocity of 16 m/s. It slows down with an acceleration given by $-0.50t$, in m/s² for t in seconds. At the end of 4.0 s it has traveled:
- A. 0
 - B. 12 m
 - C. 14 m
 - D. 25 m
 - E. 59 m
- ans: E
26. At time $t = 0$ a car has a velocity of 16 m/s. It slows down with an acceleration given by $-0.50t$, in m/s² for t in seconds. By the time it stops it has traveled:
- A. 15 m
 - B. 31 m
 - C. 62 m
 - D. 85 m
 - E. 100 m
- ans: D
27. Starting at time $t = 0$, an object moves along a straight line with velocity in m/s given by $v(t) = 98 - 2t^2$, where t is in seconds. When it momentarily stops its acceleration is:
- A. 0
 - B. -4.0 m/s^2
 - C. -9.8 m/s^2
 - D. -28 m/s^2
 - E. 49 m/s^2
- ans: D
28. Starting at time $t = 0$, an object moves along a straight line. Its coordinate in meters is given by $x(t) = 75t - 1.0t^3$, where t is in seconds. When it momentarily stops its acceleration is:
- A. 0
 - B. -73 m/s^2
 - C. -30 m/s^2
 - D. -9.8 m/s^2
 - E. $9.2 \times 10^3 \text{ m/s}^2$
- ans: C
29. A car, initially at rest, travels 20 m in 4 s along a straight line with constant acceleration. The acceleration of the car is:
- A. 0.4 m/s^2
 - B. 1.3 m/s^2
 - C. 2.5 m/s^2
 - D. 4.9 m/s^2
 - E. 9.8 m/s^2
- ans: C

30. A racing car traveling with constant acceleration increases its speed from 10 m/s to 50 m/s over a distance of 60 m. How long does this take?
- A. 2.0s
 - B. 4.0s
 - C. 5.0s
 - D. 8.0s
 - E. The time cannot be calculated since the speed is not constant
- ans: B
31. A car starts from rest and goes down a slope with a constant acceleration of 5 m/s^2 . After 5 s the car reaches the bottom of the hill. Its speed at the bottom of the hill, in meters per second, is:
- A. 1
 - B. 12.5
 - C. 25
 - D. 50
 - E. 160
- ans: C
32. A car moving with an initial velocity of 25 m/s north has a constant acceleration of 3 m/s^2 south. After 6 seconds its velocity will be:
- A. 7 m/s north
 - B. 7 m/s south
 - C. 43 m/s north
 - D. 20 m/s north
 - E. 20 m/s south
- ans: A
33. An object with an initial velocity of 12 m/s west experiences a constant acceleration of 4 m/s^2 west for 3 seconds. During this time the object travels a distance of:
- A. 12 m
 - B. 24 m
 - C. 36 m
 - D. 54 m
 - E. 144 m
- ans: D
34. How far does a car travel in 6 s if its initial velocity is 2 m/s and its acceleration is 2 m/s^2 in the forward direction?
- A. 12 m
 - B. 14 m
 - C. 24 m
 - D. 36 m
 - E. 48 m
- ans: E

35. At a stop light, a truck traveling at 15 m/s passes a car as it starts from rest. The truck travels at constant velocity and the car accelerates at 3 m/s^2 . How much time does the car take to catch up to the truck?

A. 5 s
B. 10 s
C. 15 s
D. 20 s
E. 25 s

ans: B

36. A ball is in free fall. Its acceleration is:

A. downward during both ascent and descent
B. downward during ascent and upward during descent
C. upward during ascent and downward during descent
D. upward during both ascent and descent
E. downward at all times except at the very top, when it is zero

ans: A

37. A ball is in free fall. Upward is taken to be the positive direction. The displacement of the ball during a short time interval is:

A. positive during both ascent and descent
B. negative during both ascent and descent
C. negative during ascent and positive during descent
D. positive during ascent and negative during descent
E. none of the above

ans: D

38. A baseball is thrown vertically into the air. The acceleration of the ball at its highest point is:

A. zero
B. g , down
C. g , up
D. $2g$, down
E. $2g$, up

ans: B

39. Which one of the following statements is correct for an object released from rest?

A. The average velocity during the first second of time is 4.9 m/s
B. During each second the object falls 9.8 m
C. The acceleration changes by 9.8 m/s^2 every second
D. The object falls 9.8 m during the first second of time
E. The acceleration of the object is proportional to its weight

ans: A

40. A freely falling body has a constant acceleration of 9.8 m/s^2 . This means that:
- A. the body falls 9.8 m during each second
 - B. the body falls 9.8 m during the first second only
 - C. the speed of the body increases by 9.8 m/s during each second
 - D. the acceleration of the body increases by 9.8 m/s^2 during each second
 - E. the acceleration of the body decreases by 9.8 m/s^2 during each second
- ans: C
41. An object is shot vertically upward. While it is rising:
- A. its velocity and acceleration are both upward
 - B. its velocity is upward and its acceleration is downward
 - C. its velocity and acceleration are both downward
 - D. its velocity is downward and its acceleration is upward
 - E. its velocity and acceleration are both decreasing
- ans: B
42. An object is thrown straight up from ground level with a speed of 50 m/s . If $g = 10 \text{ m/s}^2$ its distance above ground level 1.0 s later is:
- A. 40 m
 - B. 45 m
 - C. 50 m
 - D. 55 m
 - E. 60 m
- ans: B
43. An object is thrown straight up from ground level with a speed of 50 m/s . If $g = 10 \text{ m/s}^2$ its distance above ground level 6.0 s later is:
- A. 0.00 m
 - B. 270 m
 - C. 330 m
 - D. 480 m
 - E. none of these
- ans: E
44. At a location where $g = 9.80 \text{ m/s}^2$, an object is thrown vertically down with an initial speed of 1.00 m/s . After 5.00 s the object will have traveled:
- A. 125 m
 - B. 127.5 m
 - C. 245 m
 - D. 250 m
 - E. 255 m
- ans: B

45. An object is thrown vertically upward at 35 m/s. Taking $g = 10 \text{ m/s}^2$, the velocity of the object 5 s later is:
- A. 7.0 m/s up
 - B. 15 m/s down
 - C. 15 m/s up
 - D. 85 m/s down
 - E. 85 m/s up
- ans: B
46. A feather, initially at rest, is released in a vacuum 12 m above the surface of the earth. Which of the following statements is correct?
- A. The maximum velocity of the feather is 9.8 m/s
 - B. The acceleration of the feather decreases until terminal velocity is reached
 - C. The acceleration of the feather remains constant during the fall
 - D. The acceleration of the feather increases during the fall
 - E. The acceleration of the feather is zero
- ans: C
47. An object is released from rest. How far does it fall during the second second of its fall?
- A. 4.9m
 - B. 9.8m
 - C. 15 m
 - D. 20 m
 - E. 25 m
- ans: C
48. A heavy ball falls freely, starting from rest. Between the third and fourth second of time it travels a distance of:
- A. 4.9 m
 - B. 9.8 m
 - C. 29.4 m
 - D. 34.3 m
 - E. 39.8 m
- ans: D
49. As a rocket is accelerating vertically upward at 9.8 m/s^2 near Earth's surface, it releases a projectile. Immediately after release the acceleration (in m/s^2) of the projectile is:
- A. 9.8 down
 - B. 0
 - C. 9.8 up
 - D. 19.6 up
 - E. none of the above
- ans: A

50. A stone is released from a balloon that is descending at a constant speed of 10 m/s. Neglecting air resistance, after 20 s the speed of the stone is:
- A. 2160 m/s
 - B. 1760 m/s
 - C. 206 m/s
 - D. 196 m/s
 - E. 186 m/s
- ans: C
51. An object dropped from the window of a tall building hits the ground in 12.0 s. If its acceleration is 9.80 m/s^2 , the height of the window above the ground is:
- A. 29.4 m
 - B. 58.8 m
 - C. 118 m
 - D. 353 m
 - E. 706 m
- ans: E
52. Neglecting the effect of air resistance a stone dropped off a 175-m high building lands on the ground in:
- A. 3 s
 - B. 4 s
 - C. 6 s
 - D. 18 s
 - E. 36 s
- ans: C
53. A stone is thrown vertically upward with an initial speed of 19.5 m/s. It will rise to a maximum height of:
- A. 4.9 m
 - B. 9.8 m
 - C. 19.4 m
 - D. 38.8 m
 - E. none of these
- ans: C
54. A baseball is hit straight up and is caught by the catcher 2.0 s later. The maximum height of the ball during this interval is:
- A. 4.9 m
 - B. 7.4 m
 - C. 9.8 m
 - D. 12.6 m
 - E. 19.6 m
- ans: A

55. An object is thrown straight down with an initial speed of 4 m/s from a window which is 8 m above the ground. The time it takes the object to reach the ground is:

- A. 0.80 s
- B. 0.93 s
- C. 1.3 s
- D. 1.7 s
- E. 2.0 s

ans: B

56. A stone is released from rest from the edge of a building roof 190 m above the ground. Neglecting air resistance, the speed of the stone, just before striking the ground, is:

- A. 43 m/s
- B. 61 m/s
- C. 120 m/s
- D. 190 m/s
- E. 1400 m/s

ans: B

57. An object is thrown vertically upward with a certain initial velocity in a world where the acceleration due to gravity is 19.6 m/s^2 . The height to which it rises is _____ that to which the object would rise if thrown upward with the same initial velocity on the Earth. Neglect friction.

- A. half
- B. $\frac{1}{2}$ times
- C. twice
- D. four times
- E. cannot be calculated from the given data

ans: A

58. A projectile is shot vertically upward with a given initial velocity. It reaches a maximum height of 100 m. If, on a second shot, the initial velocity is doubled then the projectile will reach a maximum height of:

- A. 70.7 m
- B. 141.4 m
- C. 200 m
- D. 241 m
- E. 400 m

ans: E

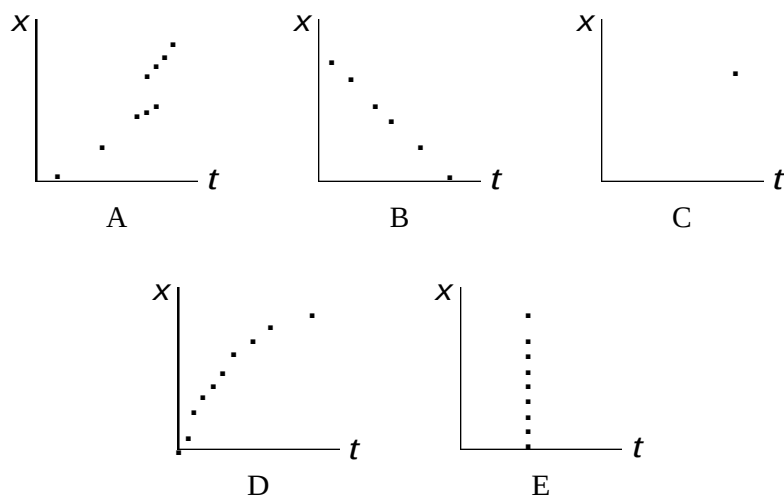
59. One object is thrown vertically upward with an initial velocity of 100 m/s and another object with an initial velocity of 10 m/s. The maximum height reached by the first object will be that of the other.

- A. 10 times
- B. 100 times
- C. 1000 times
- D. 10,000 times
- E. none of these

ans: B

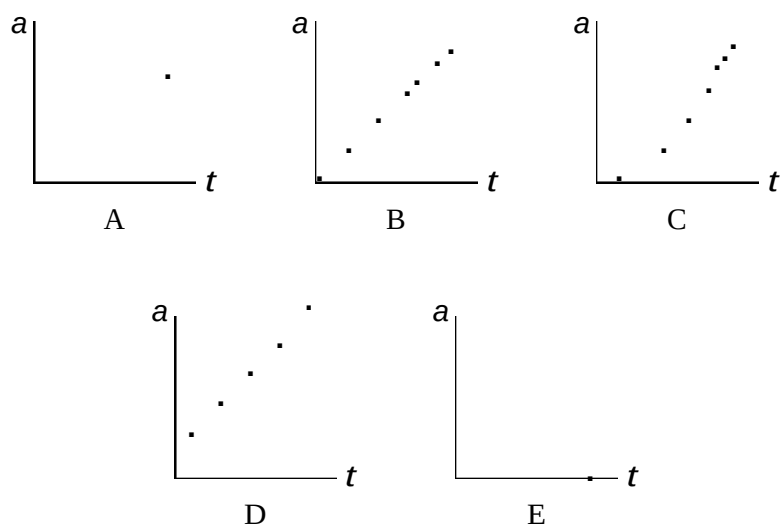
60. The area under a velocity-time graph represents:
- A. acceleration
 - B. change in acceleration
 - C. speed
 - D. change in velocity
 - E. displacement
- ans: E
61. Displacement can be obtained from:
- A. the slope of an acceleration-time graph
 - B. the slope of a velocity-time graph
 - C. the area under an acceleration-time graph
 - D. the area under a velocity-time graph
 - E. the slope of an acceleration-time graph
- ans: D
62. An object has a constant acceleration of 3 m/s^2 . The coordinate versus time graph for this object has a slope:
- A. that increases with time
 - B. that is constant
 - C. that decreases with time
 - D. of 3 m/s
 - E. of 3 m/s^2
- ans: A
63. The coordinate-time graph of an object is a straight line with a positive slope. The object has:
- A. constant displacement
 - B. steadily increasing acceleration
 - C. steadily decreasing acceleration
 - D. constant velocity
 - E. steadily increasing velocity
- ans: D

64. Which of the following five coordinate versus time graphs represents the motion of an object moving with a constant nonzero speed?



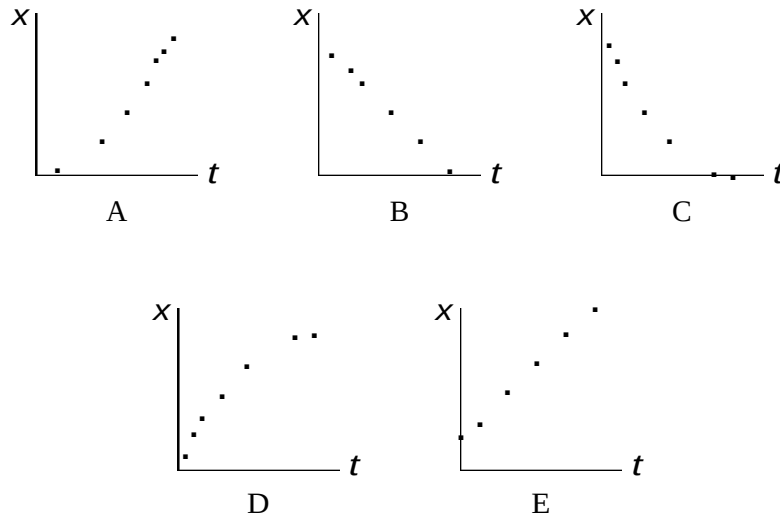
ans: B

65. Which of the following five acceleration versus time graphs is correct for an object moving in a straight line at a constant velocity of 20 m/s?



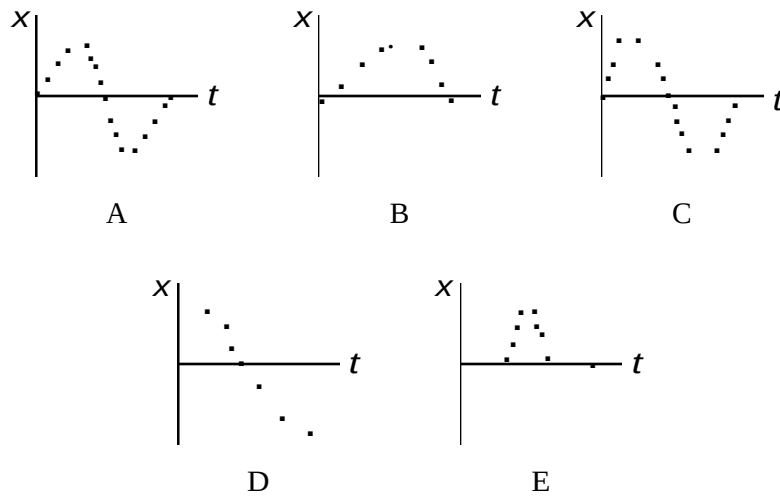
ans: E

66. Which of the following five coordinate versus time graphs represents the motion of an object whose speed is increasing?



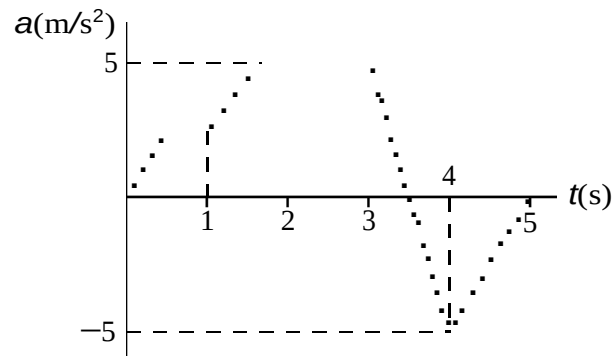
ans: A

67. A car accelerates from rest on a straight road. A short time later, the car decelerates to a stop and then returns to its original position in a similar manner, by speeding up and then slowing to a stop. Which of the following five coordinate versus time graphs best describes the motion?



ans: E

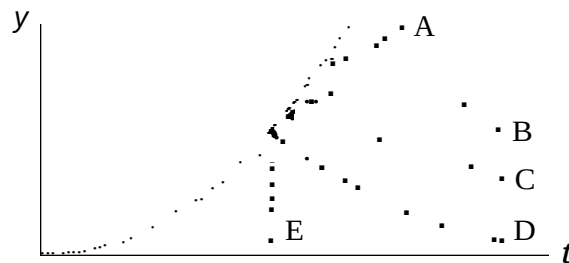
68. The acceleration of an object, starting from rest, is shown in the graph below. Other than at $t = 0$, when is the velocity of the object equal to zero?



- A. During the interval from 1.0 s to 3.0 s
- B. At $t = 3.5$ s
- C. At $t = 4.0$ s
- D. At $t = 5.0$ s
- E. At no other time less than or equal to 5 s

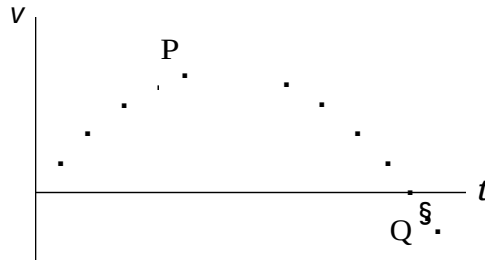
ans: E

69. An elevator is moving upward with constant acceleration. The dashed curve shows the position y of the ceiling of the elevator as a function of the time t . At the instant indicated by the dot, a bolt breaks loose and drops from the ceiling. Which curve best represents the position of the bolt as a function of time?



ans: B

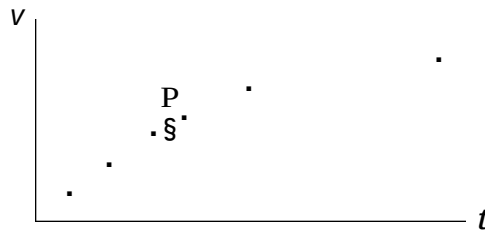
70. The diagram shows a velocity-time graph for a car moving in a straight line. At point Q the car must be:



- A. moving with zero acceleration
- B. traveling downhill
- C. traveling below ground-level
- D. reducing speed
- E. traveling in the reverse direction to that at point P

ans: E

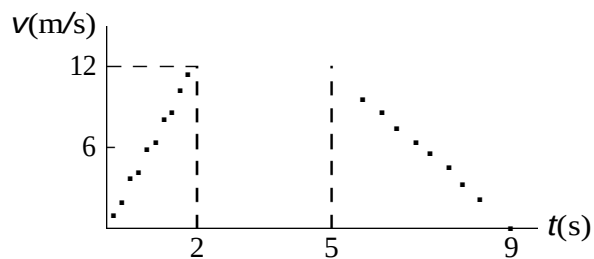
71. The diagram shows a velocity-time graph for a car moving in a straight line. At point P the car must be:



- A. moving with zero acceleration
- B. climbing the hill
- C. accelerating
- D. stationary
- E. moving at about 45° with respect to the x axis

ans: C

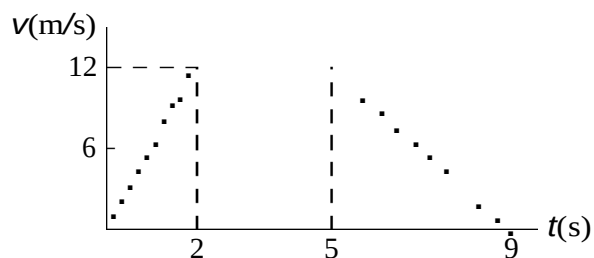
72. The graph represents the straight line motion of a car. How far does the car travel between $t = 2$ s and $t = 5$ s?



- A. 4 m
- B. 12 m
- C. 24 m
- D. 36 m
- E. 60 m

ans: D

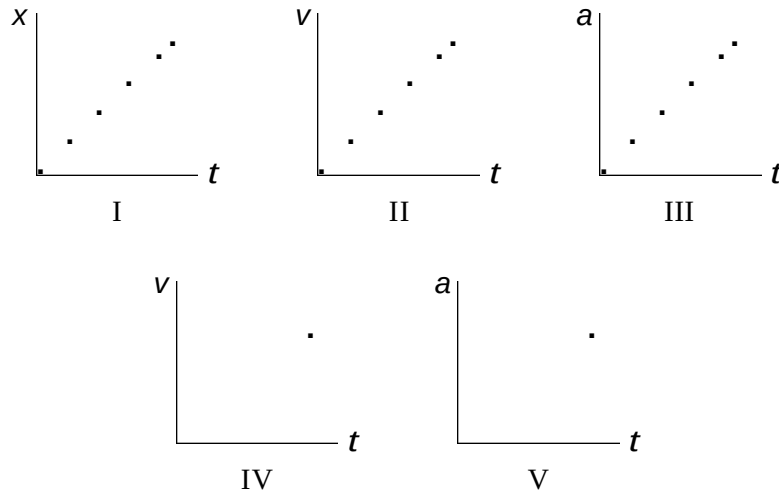
73. The diagram represents the straight line motion of a car. Which of the following statements is true?



- A. The car accelerates, stops, and reverses
- B. The car accelerates at 6 m/s^2 for the first 2 s
- C. The car is moving for a total time of 12 s
- D. The car decelerates at 12 m/s^2 for the last 4 s
- E. The car returns to its starting point when $t = 9$ s

ans: B

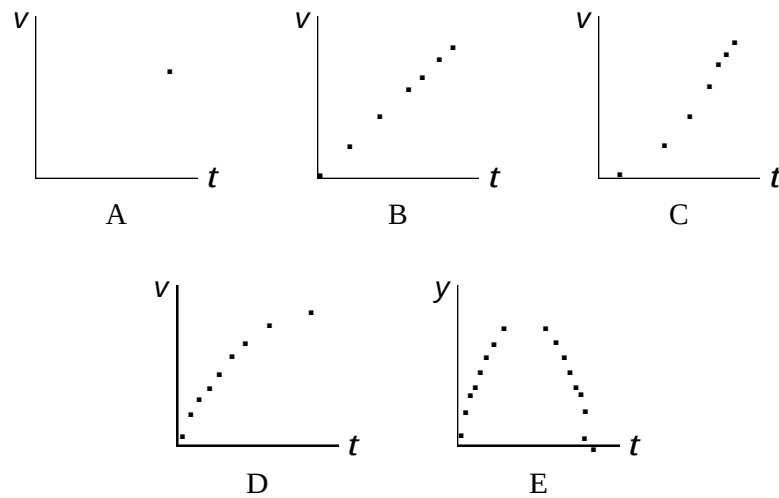
74. Consider the following five graphs (note the axes carefully). Which of these represents motion at constant speed?



- A. IV only
- B. IV and V only
- C. I, II, and III only
- D. I and II only
- E. I and IV only

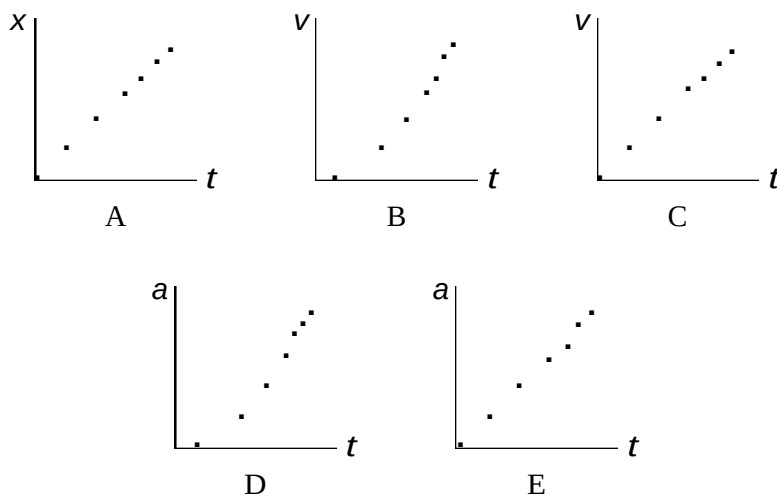
ans: E

75. An object is dropped from rest. Which of the following five graphs correctly represents its motion? The positive direction is taken to be downward.



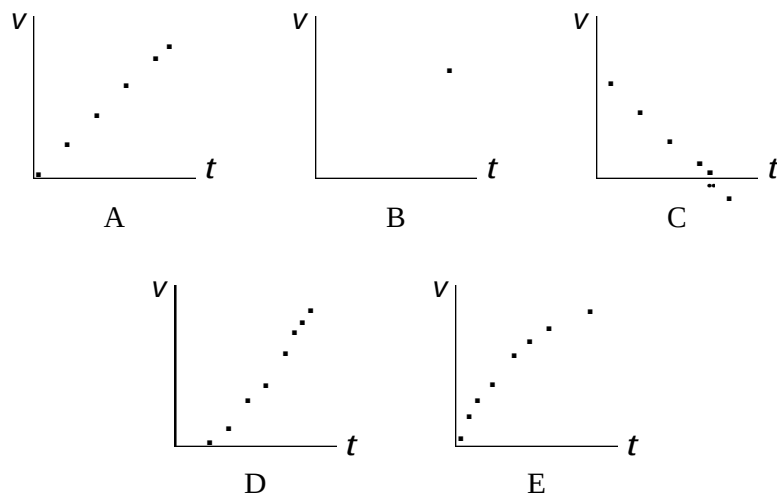
ans: B

76. A stone is dropped from a cliff. The graph (carefully note the axes) which best represents its motion while it falls is:



ans: C

77. An object is thrown vertically into the air. Which of the following five graphs represents the velocity (v) of the object as a function of the time (t)? The positive direction is taken to be upward.



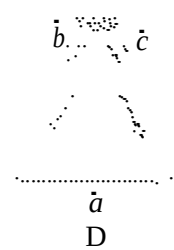
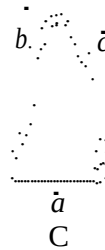
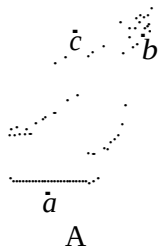
ans: C

Chapter 3: VECTORS

- We say that the displacement of a particle is a vector quantity. Our best justification for this assertion is:
 - displacement can be specified by a magnitude and a direction
 - operating with displacements according to the rules for manipulating vectors leads to results in agreement with experiments
 - a displacement is obviously not a scalar
 - displacement can be specified by three numbers
 - displacement is associated with motion

ans: B

- The vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ are related by $\vec{c} = \vec{b} - \vec{a}$. Which diagram below illustrates this relationship?



E. None of these

ans: D

- A vector of magnitude 3 CANNOT be added to a vector of magnitude 4 so that the magnitude of the resultant is:
 - zero
 - 1
 - 3
 - 5
 - 7

ans: A

- A vector of magnitude 20 is added to a vector of magnitude 25. The magnitude of this sum might be:
 - zero
 - 3
 - 12
 - 47
 - 50

ans: C

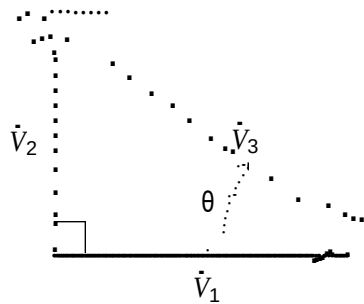
5. A vector $\vec{S}$ of magnitude 6 and another vector $\vec{T}$ have a sum of magnitude 12. The vector $\vec{T}$:
- must have a magnitude of at least 6 but no more than 18
 - may have a magnitude of 20
 - cannot have a magnitude greater than 12
 - must be perpendicular to $\vec{S}$
 - must be perpendicular to the vector sum

ans: A

6. The vector $-\vec{A}$ is:
- greater than $\vec{A}$ in magnitude
 - less than $\vec{A}$ in magnitude
 - in the same direction as $\vec{A}$
 - in the direction opposite to $\vec{A}$
 - perpendicular to $\vec{A}$

ans: D

7. The vector $\vec{V}_3$ in the diagram is equal to:



- $\vec{V}_1 - \vec{V}_2$
- $\vec{V}_1 + \vec{V}_2$
- $\vec{V}_2 - \vec{V}_1$
- $\vec{V}_1 \cos \theta$
- $\vec{V}_1 / (\cos \theta)$

ans: C

8. If $|\vec{A} + \vec{B}|^2 = A^2 + B^2$, then:
- $\vec{A}$ and $\vec{B}$ must be parallel and in the same direction
 - $\vec{A}$ and $\vec{B}$ must be parallel and in opposite directions
 - either $\vec{A}$ or $\vec{B}$ must be zero
 - the angle between $\vec{A}$ and $\vec{B}$ must be 60°
 - none of the above is true

ans: E

9. If $|\vec{A} + \vec{B}| = A + B$ and neither $\vec{A}$ nor $\vec{B}$ vanish, then:

- A. $\vec{A}$ and $\vec{B}$ are parallel and in the same direction
- B. $\vec{A}$ and $\vec{B}$ are parallel and in opposite directions
- C. the angle between $\vec{A}$ and $\vec{B}$ is 45°
- D. the angle between $\vec{A}$ and $\vec{B}$ is 60°
- E. $\vec{A}$ is perpendicular to $\vec{B}$

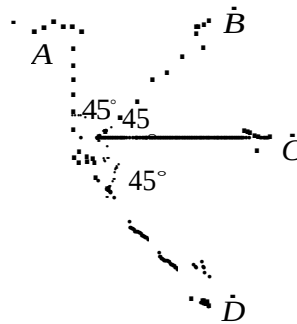
ans: A

10. If $|\vec{A} - \vec{B}| = A + B$ and neither $\vec{A}$ nor $\vec{B}$ vanish, then:

- A. $\vec{A}$ and $\vec{B}$ are parallel and in the same direction
- B. $\vec{A}$ and $\vec{B}$ are parallel and in opposite directions
- C. the angle between $\vec{A}$ and $\vec{B}$ is 45°
- D. the angle between $\vec{A}$ and $\vec{B}$ is 60°
- E. $\vec{A}$ is perpendicular to $\vec{B}$

ans: B

11. Four vectors ($\vec{A}$, $\vec{B}$, $\vec{C}$, $\vec{D}$) all have the same magnitude. The angle θ between adjacent vectors is 45° as shown. The correct vector equation is:



A. $\vec{A} - \vec{B} - \vec{C} + \vec{D} = 0$

B. $\vec{B} + \vec{D} - \sqrt{2}\vec{C} = 0$

C. $\vec{A} + \vec{B} + \vec{C} + \vec{D} = 0$

E. $(\vec{A} + \vec{C})/\sqrt{2} = -\vec{B}$

ans: B

12. Vectors $\vec{A}$ and $\vec{B}$ lie in the xy plane. We can deduce that $\vec{A} = \vec{B}$ if:

A. $A^2 + A^2 = B^2 + B^2$

B. $A_x + A_y = B_x + B_y$

C. $A_x = B_x$ and $A_y = B_y$

D. $A_y/A_x = B_y/B_x$

E. $A_x = A_y$ and $B_x = B_y$

ans: C

13. A vector has a magnitude of 12. When its tail is at the origin it lies between the positive x axis and the negative y axis and makes an angle of 30° with the x axis. Its y component is:

A. $6\sqrt{3}$
 B. $-6\sqrt{3}$
 C. 6
 D. -6
 E. 12

ans: D

14. If the x component of a vector $\vec{A}$, in the xy plane, is half as large as the magnitude of the vector, the tangent of the angle between the vector and the x axis is:

A. $\sqrt{3}$
 B. $1/2$
 C. $3/2$
 D. $3/2$
 E. 3

ans: D

15. If $\vec{A} = (6\text{ m})\hat{i} - (8\text{ m})\hat{j}$ then $4\vec{A}$ has magnitude:

A. 10 m
 B. 20 m
 C. 30 m
 D. 40 m
 E. 50 m

ans: D

16. A vector has a component of 10 m in the $+x$ direction, a component of 10 m in the $+y$ direction, and a component of 5 m in the $+z$ direction. The magnitude of this vector is:

A. zero
 B. 15 m
 C. 20 m
 D. 25 m
 E. 225 m

ans: B

17. Let $\vec{V} = (2.00\text{ m})\hat{i} + (6.00\text{ m})\hat{j} - (3.00\text{ m})\hat{k}$. The magnitude of $\vec{V}$ is:

A. 5.00 m
 B. 5.57 m
 C. 7.00 m
 D. 7.42 m
 E. 8.54 m

ans: C

18. A vector in the xy plane has a magnitude of 25 m and an x component of 12 m. The angle it makes with the positive x axis is:
- A. 26°
 - B. 29°
 - C. 61°
 - D. 64°
 - E. 241°

ans: C

19. The angle between $\vec{A} = (25 \text{ m})\hat{i} + (45 \text{ m})\hat{j}$ and the positive x axis is:
- A. 29°
 - B. 61°
 - C. 151°
 - D. 209°
 - E. 241°

ans: B

20. The angle between $\vec{A} = (-25 \text{ m})\hat{i} + (45 \text{ m})\hat{j}$ and the positive x axis is:
- A. 29°
 - B. 61°
 - C. 119°
 - D. 151°
 - E. 209°

ans: C

21. Let $\vec{A} = (2 \text{ m})\hat{i} + (6 \text{ m})\hat{j} - (3 \text{ m})\hat{k}$ and $\vec{B} = (4 \text{ m})\hat{i} + (2 \text{ m})\hat{j} + (1 \text{ m})\hat{k}$. The vector sum $\vec{S} = \vec{A} + \vec{B}$ is:
- A. $(6 \text{ m})\hat{i} + (8 \text{ m})\hat{j} - (2 \text{ m})\hat{k}$
 - B. $(-2 \text{ m})\hat{i} + (4 \text{ m})\hat{j} - (4 \text{ m})\hat{k}$
 - C. $(2 \text{ m})\hat{i} - (4 \text{ m})\hat{j} + (4 \text{ m})\hat{k}$
 - D. $(8 \text{ m})\hat{i} + (12 \text{ m})\hat{j} - (3 \text{ m})\hat{k}$
 - E. none of these

ans: A

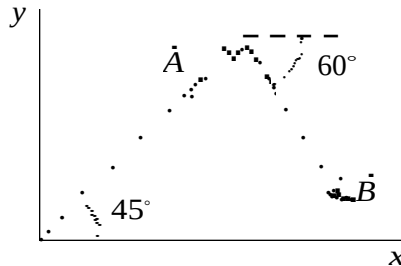
22. Let $\vec{A} = (2 \text{ m})\hat{i} + (6 \text{ m})\hat{j} - (3 \text{ m})\hat{k}$ and $\vec{B} = (4 \text{ m})\hat{i} + (2 \text{ m})\hat{j} + (1 \text{ m})\hat{k}$. The vector difference $\vec{D} = \vec{A} - \vec{B}$ is:
- A. $(6 \text{ m})\hat{i} + (8 \text{ m})\hat{j} - (2 \text{ m})\hat{k}$
 - B. $(-2 \text{ m})\hat{i} + (4 \text{ m})\hat{j} - (4 \text{ m})\hat{k}$
 - C. $(2 \text{ m})\hat{i} - (4 \text{ m})\hat{j} + (4 \text{ m})\hat{k}$
 - D. $(8 \text{ m})\hat{i} + (12 \text{ m})\hat{j} - (3 \text{ m})\hat{k}$
 - E. none of these

ans: B

23. If $\vec{A} = (2\text{ m})\hat{i} - (3\text{ m})\hat{j}$ and $\vec{B} = (1\text{ m})\hat{i} - (2\text{ m})\hat{j}$, then $\vec{A} - 2\vec{B} =$
- A. $(1\text{ m})\hat{j}$
 - B. $(-1\text{ m})\hat{j}$
 - C. $(4\text{ m})\hat{i} - (7\text{ m})\hat{j}$
 - D. $(4\text{ m})\hat{i} + (1\text{ m})\hat{j}$
 - E. $(-4\text{ m})\hat{i} + (7\text{ m})\hat{j}$

ans: A

24. In the diagram, $\vec{A}$ has magnitude 12 m and $\vec{B}$ has magnitude 8 m. The x component of $\vec{A} + \vec{B}$ is about:



- A. 5.5 m
- B. 7.6 m
- C. 12 m
- D. 14 m
- E. 15 m

ans: C

25. A certain vector in the xy plane has an x component of 4 m and a y component of 10 m. It is then rotated in the xy plane so its x component is doubled. Its new y component is about:
- A. 20 m
 - B. 7.2 m
 - C. 5.0 m
 - D. 4.5 m
 - E. 2.2 m

ans: B

26. Vectors $\vec{A}$ and $\vec{B}$ each have magnitude L . When drawn with their tails at the same point, the angle between them is 30° . The value of $\vec{A} \cdot \vec{B}$ is:
- A. zero
 - B. L^2
 - C. $\sqrt{3}L^2/2$
 - D. $2L^2$
 - E. none of these

ans: C

27. Let $\vec{A} = (2 \text{ m})\hat{i} + (6 \text{ m})\hat{j} - (3 \text{ m})\hat{k}$ and $\vec{B} = (4 \text{ m})\hat{i} + (2 \text{ m})\hat{j} + (1 \text{ m})\hat{k}$. Then $\vec{A} \cdot \vec{B} =$
- A. $(8 \text{ m})\hat{i} + (12 \text{ m})\hat{j} - (3 \text{ m})\hat{k}$
 - B. $(12 \text{ m})\hat{i} - (14 \text{ m})\hat{j} - (20 \text{ m})\hat{k}$
 - C. 23 m^2
 - D. 17 m^2
 - E. none of these
- ans: D
28. Two vectors have magnitudes of 10 m and 15 m. The angle between them when they are drawn with their tails at the same point is 65° . The component of the longer vector along the line of the shorter is:
- A. 0
 - B. 4.2 m
 - C. 6.3 m
 - D. 9.1 m
 - E. 14 m
- ans: C
29. Let $\vec{S} = (1 \text{ m})\hat{i} + (2 \text{ m})\hat{j} + (2 \text{ m})\hat{k}$ and $\vec{T} = (3 \text{ m})\hat{i} + (4 \text{ m})\hat{k}$. The angle between these two vectors is given by:
- A. $\cos^{-1}(14/15)$
 - B. $\cos^{-1}(11/225)$
 - C. $\cos^{-1}(104/225)$
 - D. $\cos^{-1}(11/15)$
 - E. cannot be found since $\vec{S}$ and $\vec{T}$ do not lie in the same plane
- ans: D
30. Two vectors lie with their tails at the same point. When the angle between them is increased by 20° their scalar product has the same magnitude but changes from positive to negative. The original angle between them was:
- A. 0
 - B. 60°
 - C. 70°
 - D. 80°
 - E. 90°
- ans: D
31. If the magnitude of the sum of two vectors is less than the magnitude of either vector, then:
- A. the scalar product of the vectors must be negative
 - B. the scalar product of the vectors must be positive
 - C. the vectors must be parallel and in opposite directions
 - D. the vectors must be parallel and in the same direction
 - E. none of the above
- ans: A

32. If the magnitude of the sum of two vectors is greater than the magnitude of either vector, then:
- A. the scalar product of the vectors must be negative
 - B. the scalar product of the vectors must be positive
 - C. the vectors must be parallel and in opposite directions
 - D. the vectors must be parallel and in the same direction
 - E. none of the above

ans: E

33. Vectors $\vec{A}$ and $\vec{B}$ each have magnitude L . When drawn with their tails at the same point, the angle between them is 60° . The magnitude of the vector product $\vec{A} \times \vec{B}$ is:

- A. $L^2/2$
- B. L^2
- C. $\sqrt{3}L^2/2$
- D. $2L^2$
- E. none of these

ans: C

34. Two vectors lie with their tails at the same point. When the angle between them is increased by 20° the magnitude of their vector product doubles. The original angle between them was about:

- A. 0
- B. 18°
- C. 25°
- D. 45°
- E. 90°

ans: B

35. Two vectors have magnitudes of 10 m and 15 m. The angle between them when they are drawn with their tails at the same point is 65° . The component of the longer vector along the line perpendicular to the shorter vector, in the plane of the vectors, is:

- A. 0
- B. 4.2 m
- C. 6.3 m
- D. 9.1 m
- E. 14 m

ans: E

36. The two vectors $(3\text{ m})\hat{i} - (2\text{ m})\hat{j}$ and $(2\text{ m})\hat{i} + (3\text{ m})\hat{j} - (2\text{ m})\hat{k}$ define a plane. It is the plane of the triangle with both tails at one vertex and each head at one of the other vertices. Which of the following vectors is perpendicular to the plane?

- A. $(4\text{ m})\hat{i} + (6\text{ m})\hat{j} + (13\text{ m})\hat{k}$
- B. $(-4\text{ m})\hat{i} + (6\text{ m})\hat{j} + (13\text{ m})\hat{k}$
- C. $(4\text{ m})\hat{i} - (6\text{ m})\hat{j} + (13\text{ m})\hat{k}$
- D. $(4\text{ m})\hat{i} + (6\text{ m})\hat{j} - (13\text{ m})\hat{k}$
- E. $(4\text{ m})\hat{i} + (6\text{ m})\hat{j}$

ans: A

37. Let $\vec{R} = \vec{S} \times \vec{T}$ and $\theta = 90^\circ$, where θ is the angle between $\vec{S}$ and $\vec{T}$ when they are drawn with their tails at the same point. Which of the following is NOT true?

A. $|\vec{R}| = |\vec{S}||\vec{T}|\sin\theta$

B. $-\vec{R} = \vec{T} \times \vec{S}$

C. $\vec{R} \cdot \vec{S} = 0$

D. $\vec{R} \cdot \vec{T} = 0$

E. $\vec{S} \cdot \vec{T} = 0$

ans: E

38. The value of $\hat{i} \cdot (\hat{j} \times \hat{k})$ is:

A. zero

B. +1

C. -1

D. 3

E. $\frac{1}{3}$

ans: B

39. The value of $\hat{k} \cdot (\hat{k} \times \hat{i})$ is:

A. zero

B. +1

C. -1

D. 3

E. $\frac{1}{3}$

ans: A

Chapter 4: MOTION IN TWO AND THREE DIMENSIONS

1. Velocity is defined as:
 - A. rate of change of position with time
 - B. position divided by time
 - C. rate of change of acceleration with time
 - D. a speeding up or slowing down
 - E. change of positionans: A
2. Acceleration is defined as:
 - A. rate of change of position with time
 - B. speed divided by time
 - C. rate of change of velocity with time
 - D. a speeding up or slowing down
 - E. change of velocityans: C
3. Which of the following is a scalar quantity?
 - A. Speed
 - B. Velocity
 - C. Displacement
 - D. Acceleration
 - E. None of theseans: A
4. Which of the following is a vector quantity?
 - A. Mass
 - B. Density
 - C. Speed
 - D. Temperature
 - E. None of theseans: E
5. Which of the following is NOT an example of accelerated motion?
 - A. Vertical component of projectile motion
 - B. Circular motion at constant speed
 - C. A swinging pendulum
 - D. Earth's motion about sun
 - E. Horizontal component of projectile motionans: E

6. A particle goes from $x = -2$ m, $y = 3$ m, $z = 1$ m to $x = 3$ m, $y = -1$ m, $z = 4$ m. Its displacement is:
- A. $(1 \text{ m})\hat{i} + (2 \text{ m})\hat{j} + (5 \text{ m})\hat{k}$
 - B. $(5 \text{ m})\hat{i} - (4 \text{ m})\hat{j} + (3 \text{ m})\hat{k}$
 - C. $-(5 \text{ m})\hat{i} + (4 \text{ m})\hat{j} - (3 \text{ m})\hat{k}$
 - D. $-(1 \text{ m})\hat{i} - (2 \text{ m})\hat{j} - (5 \text{ m})\hat{k}$
 - E. $-(5 \text{ m})\hat{i} - (2 \text{ m})\hat{j} + (3 \text{ m})\hat{k}$
- ans: B
7. A jet plane in straight horizontal flight passes over your head. When it is directly above you, the sound seems to come from a point behind the plane in a direction 30° from the vertical. The speed of the plane is:
- A. the same as the speed of sound
 - B. half the speed of sound
 - C. three-fifths the speed of sound
 - D. 0.866 times the speed of sound
 - E. twice the speed of sound
- ans: B
8. A plane traveling north at 200 m/s turns and then travels south at 200 m/s. The change in its velocity is:
- A. zero
 - B. 200 m/s north
 - C. 200 m/s south
 - D. 400 m/s north
 - E. 400 m/s south
- ans: E
9. Two bodies are falling with negligible air resistance, side by side, above a horizontal plane. If one of the bodies is given an additional horizontal acceleration during its descent, it:
- A. strikes the plane at the same time as the other body
 - B. strikes the plane earlier than the other body
 - C. has the vertical component of its velocity altered
 - D. has the vertical component of its acceleration altered
 - E. follows a straight line path along the resultant acceleration vector
- ans: A
10. The velocity of a projectile equals its initial velocity added to:
- A. a constant horizontal velocity
 - B. a constant vertical velocity
 - C. a constantly increasing horizontal velocity
 - D. a constantly increasing downward velocity
 - E. a constant velocity directed at the target
- ans: D

11. A stone thrown from the top of a tall building follows a path that is:

- A. circular
- B. made of two straight line segments
- C. hyperbolic
- D. parabolic
- E. a straight line

ans: D

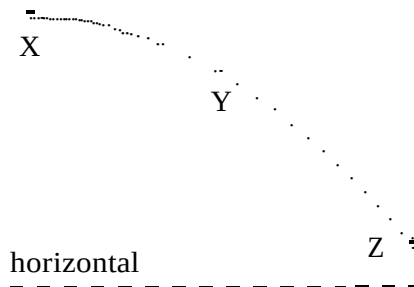
12. Identical guns fire identical bullets horizontally at the same speed from the same height above level planes, one on the Earth and one on the Moon. Which of the following three statements is/are true?

- I. The horizontal distance traveled by the bullet is greater for the Moon.
- II. The flight time is less for the bullet on the Earth.
- III. The velocity of the bullets at impact are the same.

- A. III only
- B. I and II only
- C. I and III only
- D. II and III only
- E. I, II, III

ans: B

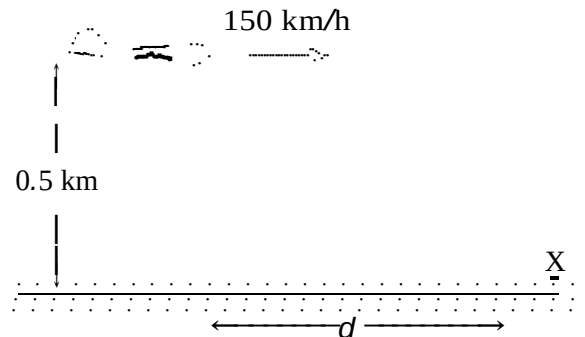
13. A stone is thrown horizontally and follows the path XYZ shown. The direction of the acceleration of the stone at point Y is:



- A. ↓
- B. →
- C. ≠
- D. <
- E. 3

ans: A

14. A bullet shot horizontally from a gun:
- A. strikes the ground much later than one dropped vertically from the same point at the same instant
 - B. never strikes the ground
 - C. strikes the ground at approximately the same time as one dropped vertically from the same point at the same instant
 - D. travels in a straight line
 - E. strikes the ground much sooner than one dropped from the same point at the same instant
- ans: C
15. A bomber flying in level flight with constant velocity releases a bomb before it is over the target. Neglecting air resistance, which one of the following is NOT true?
- A. The bomber is over the target when the bomb strikes
 - B. The acceleration of the bomb is constant
 - C. The horizontal velocity of the plane equals the vertical velocity of the bomb when it hits the target
 - D. The bomb travels in a curved path
 - E. The time of flight of the bomb is independent of the horizontal speed of the plane
- ans: C
16. The airplane shown is in level flight at an altitude of 0.50 km and a speed of 150 km/h. At what distance d should it release a heavy bomb to hit the target X? Take $g = 10 \text{ m/s}^2$.

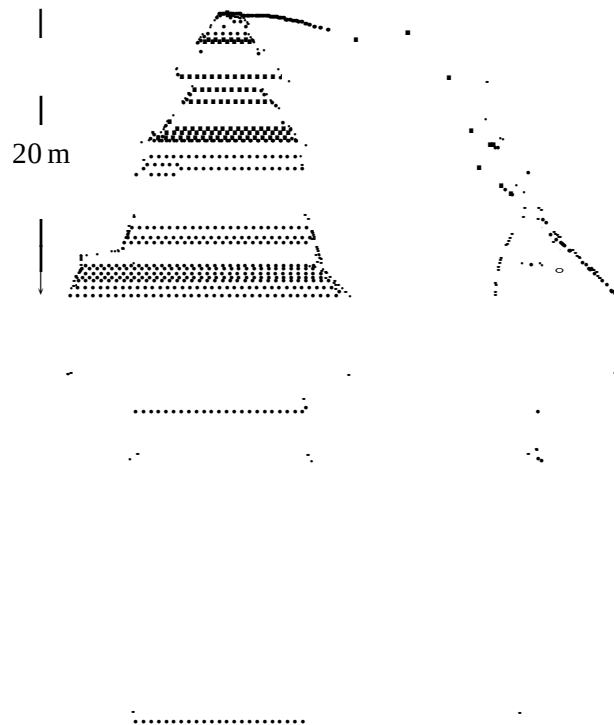


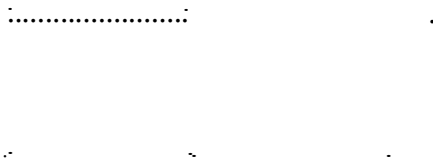
- A. 150 m
 - B. 295 m
 - C. 420 m
 - D. 2550 m
 - E. 15, 000 m
- ans: C

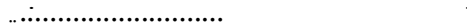
17. An object is shot from the back of a railroad flatcar moving at 40 km/h on a straight horizontal road. The launcher is aimed upward, perpendicular to the bed of the flatcar. The object falls:
- A. in front of the flatcar
 - B. behind the flatcar
 - C. on the flatcar
 - D. either behind or in front of the flatcar, depending on the initial speed of the object
 - E. to the side of the flatcar

ans: C

18. A ball is thrown horizontally from the top of a 20-m high hill. It strikes the ground at an angle of 45° . With what speed was it thrown?







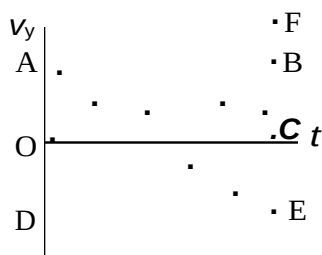
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21. A boy on the edge of a vertical cliff 20 m high throws a stone horizontally outward with a speed of 20 m/s. It strikes the ground at what horizontal distance from the foot of the cliff? Use $g = 10 \text{ m/s}^2$.
- A. 10 m
 - B. 40 m
 - C. $50\sqrt{2} \text{ m}$
 - D. 50.5 m
 - E. none of these

ans: B

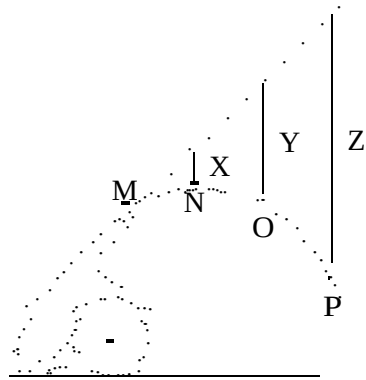
22. Which of the curves on the graph below best represents the vertical component v_y of the velocity versus the time t for a projectile fired at an angle of 45° above the horizontal?



- A. OC
- B. DE
- C. AB
- D. AE
- E. AF

ans: D

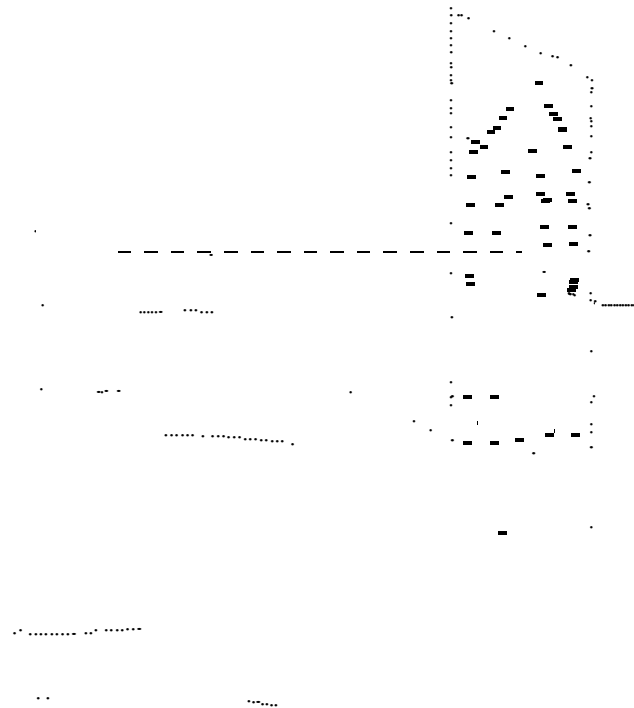
23. A cannon fires a projectile as shown. The dashed line shows the trajectory in the absence of gravity; points MNOP correspond to the position of the projectile at one second intervals. If $g = 10 \text{ m/s}^2$, the lengths X,Y,Z are:



- A. 5 m, 10 m, 15m
- B. 5 m, 20 m, 45m
- C. 10 m, 40 m, 90m
- D. 10 m, 20 m, 30 m
- E. 0.2 m, 0.8 m, 1.8m

ans: B

24. A dart is thrown horizontally toward X at 20 m/s as shown. It hits Y 0.1 s later. The distance XY is:



...

...

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...

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X



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A. 2 m

C. 0 5

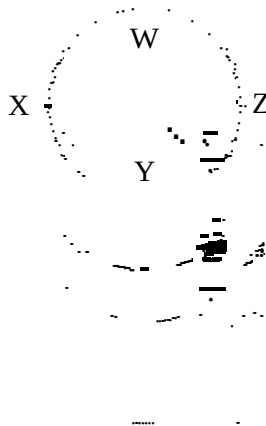
B. 1 m

^m
D. 0.1 m

E. 0.05m

ans: E

25. A projectile is fired from ground level over level ground with an initial velocity that has a vertical component of 20 m/s and a horizontal component of 30 m/s. Using $g = 10 \text{ m/s}^2$, the distance from launching to landing points is:
- A. 40 m
 - B. 60 m
 - C. 80 m
 - D. 120 m
 - E. 180 m
- ans: D
26. An object, tied to a string, moves in a circle at constant speed on a horizontal surface as shown. The direction of the displacement of this object, as it travels from W to X is:



A. ←

B. ↓

C. ↑

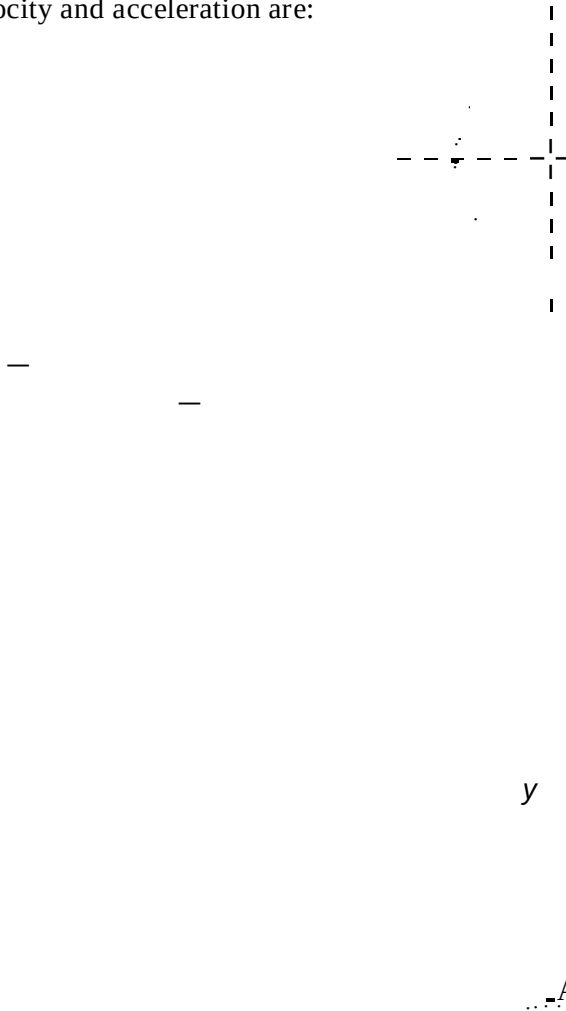
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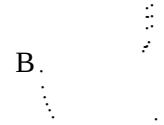
D. 3

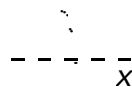
E. C

ans: E

27. A toy racing car moves with constant speed around the circle shown below. When it is at point A its coordinates are $x = 0$, $y = 3$ m and its velocity is $(6\text{ m/s})\mathbf{i}$. When it is at point B its velocity and acceleration are:







)

- A. $(6 \text{ m/s})\hat{j}$ and $(12 \text{ m/s}^2)\hat{i}$, respectively
 - B. $(6 \text{ m/s})\hat{i}$ and $(12 \text{ m/s}^2)\hat{i}$, respectively
 - C. $(6 \text{ m/s})\hat{j}$ and $(12 \text{ m/s}^2)\hat{i}$, respectively
 - D. $(6 \text{ m/s})\hat{i}$ and $(2 \text{ m/s}^2)\hat{j}$, respectively
 - E. $(6 \text{ m/s})\hat{j}$ and 0, respectively
- ans: C

28. An airplane makes a gradual 90° turn while flying at a constant speed of 200 m/s. The process takes 20.0 seconds to complete. For this turn the magnitude of the average acceleration of the plane is:
- A. zero
 - B. 40 m/s^2
 - C. 20 m/s^2
 - D. 14 m/s^2
 - E. 10 m/s^2
- ans: D
29. An airplane is flying north at 500 km/h. It makes a gradual 180° turn at constant speed, changing its direction of travel from north through east to south. The process takes 40 s. The average acceleration of the plane for this turn (in $\text{km/h} \cdot \text{s}$) is:
- A. $12.5 \text{ km/h} \cdot \text{s}$, north
 - B. $12.5 \text{ km/h} \cdot \text{s}$, east
 - C. $12.5 \text{ km/h} \cdot \text{s}$, south
 - D. $25 \text{ km/h} \cdot \text{s}$, north
 - E. $25 \text{ km/h} \cdot \text{s}$, south
- ans: E
30. An object is moving on a circular path of radius π meters at a constant speed of 4.0 m/s. The time required for one revolution is:
- A. $2/\pi^2 \text{ s}$
 - B. $\pi^2/2 \text{ s}$
 - C. $\pi/2 \text{ s}$
 - D. $\pi^2/4$
 - E. $2/\pi \text{ s}$
- ans: B
31. A particle moves at constant speed in a circular path. The instantaneous velocity and instantaneous acceleration vectors are:
- A. both tangent to the circular path
 - B. both perpendicular to the circular path
 - C. perpendicular to each other
 - D. opposite to each other
 - E. none of the above
- ans: C
32. A stone is tied to a string and whirled at constant speed in a horizontal circle. The speed is then doubled without changing the length of the string. Afterward the magnitude of the acceleration of the stone is:
- A. the same
 - B. twice as great
 - C. four times as great
 - D. half as great
 - E. one-fourth as great
- ans: C

33. Two objects are traveling around different circular orbits with constant speed. They both have the same acceleration but object A is traveling twice as fast as object B. The orbit radius for object A is _____ the orbit radius for object B.
- A. one-fourth
 - B. one-half
 - C. the same as
 - D. twice
 - E. four times
- ans: E
34. A stone is tied to a 0.50-m string and whirled at a constant speed of 4.0 m/s in a vertical circle. Its acceleration at the top of the circle is:
- A. 9.8 m/s^2 , up
 - B. 9.8 m/s^2 , down
 - C. 8.0 m/s^2 , down
 - D. 32 m/s^2 , up
 - E. 32 m/s^2 , down
- ans: E
35. A stone is tied to a 0.50-m string and whirled at a constant speed of 4.0 m/s in a vertical circle. Its acceleration at the bottom of the circle is:
- A. 9.8 m/s^2 , up
 - B. 9.8 m/s^2 , down
 - C. 8.0 m/s^2 , up
 - D. 32 m/s^2 , up
 - E. 32 m/s^2 , down
- ans: D
36. A car rounds a 20-m radius curve at 10 m/s. The magnitude of its acceleration is:
- A. 0
 - B. 0.20 m/s^2
 - C. 5.0 m/s^2
 - D. 40 m/s^2
 - E. 400 m/s^2
- ans: C
37. For a biological sample in a 1.0-m radius centrifuge to have a centripetal acceleration of $25g$ its speed must be:
- A. 11 m/s
 - B. 16 m/s
 - C. 50 m/s
 - D. 122 m/s
 - E. 245 m/s
- ans: B

38. A girl jogs around a horizontal circle with a constant speed. She travels one fourth of a revolution, a distance of 25 m along the circumference of the circle, in 5.0 s. The magnitude of her acceleration is:
- A. 0.31 m/s^2
 - B. 1.3 m/s^2
 - C. 1.6 m/s^2
 - D. 3.9 m/s^2
 - E. 6.3 m/s^2
- ans: C
39. A stone is tied to the end of a string and is swung with constant speed around a horizontal circle with a radius of 1.5 m. If it makes two complete revolutions each second, the magnitude of its acceleration is:
- A. 0.24 m/s^2
 - B. 2.4 m/s^2
 - C. 24 m/s^2
 - D. 240 m/s^2
 - E. 2400 m/s^2
- ans: D
40. A Ferris wheel with a radius of 8.0 m makes 1 revolution every 10 s. When a passenger is at the top, essentially a diameter above the ground, he releases a ball. How far from the point on the ground directly under the release point does the ball land?
- A. 0
 - B. 1.0m
 - C. 8.0m
 - D. 9.1m
 - E. 16 m
- ans: D
41. A boat is able to move through still water at 20 m/s. It makes a round trip to a town 3.0 km upstream. If the river flows at 5 m/s, the time required for this round trip is:
- A. 120 s
 - B. 150 s
 - C. 200 s
 - D. 300 s
 - E. 320 s
- ans: E

42. A boat is traveling upstream at 14 km/h with respect to a river that is flowing at 6 km/h (with respect to the ground). A man runs directly across the boat, from one side to the other, at 6 km/h (with respect to the boat). The speed of the man with respect to the ground is:

A. 10 km/h
B. 14 km/h
C. 18.5 km/h
D. 21 km/h
E. 26 km/h

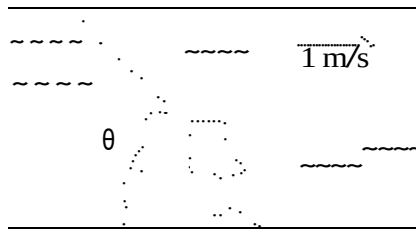
ans: A

43. A ferry boat is sailing at 12 km/h 30° W of N with respect to a river that is flowing at 6.0 km/h. As observed from the shore, the ferry boat is sailing:

A. 30° E of N
B. due N
C. 30° W of N
D. 45° E of N
E. none of these

ans: B

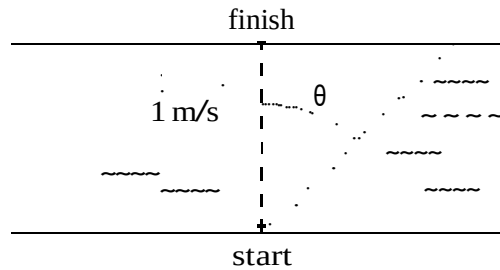
44. A boy wishes to row across a river in the shortest possible time. He can row at 2 m/s in still water and the river is flowing at 1 m/s. At what angle θ should he point the bow (front) of his boat?



A. 30°
B. 45°
C. 60°
D. 63°
E. 90°

ans: E

45. A girl wishes to swim across a river to a point directly opposite as shown. She can swim at 2 m/s in still water and the river is flowing at 1 m/s. At what angle θ with respect to the line joining the starting and finishing points should she swim?



- A. 30°
B. 45°
C. 60°
D. 63°
E. 90°
ans: A
46. A motor boat can travel at 10 km/h in still water. A river flows at 5 km/h west. A boater wishes to cross from the south bank to a point directly opposite on the north bank. At what angle must the boat be headed?
A. 27° E of N
B. 30° E of N
C. 45° E of N
D. 60° E of N
E. depends on the width of the river
ans: B
47. Two projectiles are in flight at the same time. The acceleration of one relative to the other:
A. is always 9.8 m/s^2
B. can be as large as 19.8 m/s^2
C. can be horizontal
D. is zero
E. none of these
ans: D

Chapter 5: FORCE AND MOTION — I

1. An example of an inertial reference frame is:
 - A. any reference frame that is not accelerating
 - B. a frame attached to a particle on which there are no forces
 - C. any reference frame that is at rest
 - D. a reference frame attached to the center of the universe
 - E. a reference frame attached to Earthans: B
2. An object moving at constant velocity in an inertial frame must:
 - A. have a net force on it
 - B. eventually stop due to gravity
 - C. not have any force of gravity on it
 - D. have zero net force on it
 - E. have no frictional force on itans: D
3. In SI units a force is numerically equal to the _____, when the force is applied to it.
 - A. velocity of the standard kilogram
 - B. speed of the standard kilogram
 - C. velocity of any object
 - D. acceleration of the standard kilogram
 - E. acceleration of any objectans: D
4. Which of the following quantities is NOT a vector?
 - A. Mass
 - B. Displacement
 - C. Weight
 - D. Acceleration
 - E. Forceans: A
5. A newton is the force:
 - A. of gravity on a 1 kg body
 - B. of gravity on a 1 g body
 - C. that gives a 1 g body an acceleration of 1 cm/s^2
 - D. that gives a 1 kg body an acceleration of 1 m/s^2
 - E. that gives a 1 kg body an acceleration of 9.8 m/s^2ans: D

6. The unit of force called the newton is:
- A. $9.8 \text{ kg} \cdot \text{m/s}^2$
 - B. $1 \text{ kg} \cdot \text{m/s}^2$
 - C. defined by means of Newton's third law
 - D. 1 kg of mass
 - E. 1 kg of force

ans: B

7. A force of 1 N is:

- A. 1 kg/s
- B. $1 \text{ kg} \cdot \text{m/s}$
- C. $1 \text{ kg} \cdot \text{m/s}^2$
- D. $1 \text{ kg} \cdot \text{m}^2/\text{s}$
- E. $1 \text{ kg} \cdot \text{m}^2/\text{s}^2$

ans: C

8. The standard 1-kg mass is attached to a compressed spring and the spring is released. If the mass initially has an acceleration of 5.6 m/s^2 , the force of the spring has a magnitude of:

- A. 2.8N
- B. 5.6N
- C. 11.2N
- D. 0
- E. an undetermined amount

ans: B

9. Acceleration is always in the direction:

- A. of the displacement
- B. of the initial velocity
- C. of the final velocity
- D. of the net force
- E. opposite to the frictional force

ans: D

10. The term "mass" refers to the same physical concept as:

- A. weight
- B. inertia
- C. force
- D. acceleration
- C. volume

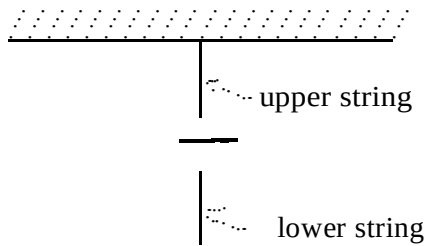
ans: B

11. The inertia of a body tends to cause the body to:

- A. speed up
- B. slow down
- C. resist any change in its motion
- D. fall toward Earth
- E. decelerate due to friction

ans: C

12. A heavy ball is suspended as shown. A quick jerk on the lower string will break that string but a slow pull on the lower string will break the upper string. The first result occurs because:



- A. the force is too small to move the ball
- B. action and reaction is operating
- C. the ball has inertia
- D. air friction holds the ball back
- E. the ball has too much energy

ans: C

13. When a certain force is applied to the standard kilogram its acceleration is 5.0 m/s^2 . When the same force is applied to another object its acceleration is one-fifth as much. The mass of the object is:

- A. 0.2 kg
- B. 0.5 kg
- C. 1.0 kg
- D. 5.0 kg
- E. 10 kg

ans: D

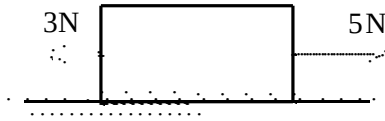
14. Mass differs from weight in that:

- A. all objects have weight but some lack mass
- B. weight is a force and mass is not
- C. the mass of an object is always more than its weight
- D. mass can be expressed only in the metric system
- E. there is no difference

ans: B

15. The mass of a body:
- A. is slightly different at different places on Earth
 - B. is a vector
 - C. is independent of the free-fall acceleration
 - D. is the same for all bodies of the same volume
 - E. can be measured most accurately on a spring scale
- ans: C
16. The mass and weight of a body:
- A. differ by a factor of 9.8
 - B. are identical
 - C. are the same physical quantities expressed in different units
 - D. are both a direct measure of the inertia of the body
 - E. have the same ratio as that of any other body placed at that location
- ans: E
17. An object placed on an equal-arm balance requires 12 kg to balance it. When placed on a spring scale, the scale reads 12 kg. Everything (balance, scale, set of weights and object) is now transported to the Moon where the free-fall acceleration is one-sixth that on Earth. The new readings of the balance and spring scale (respectively) are:
- A. 12 kg, 12 kg
 - B. 2 kg, 2 kg
 - C. 12 kg, 2 kg
 - D. 2 kg, 12 kg
 - E. 12 kg, 72 kg
- ans: C
18. Two objects, one having three times the mass of the other, are dropped from the same height in a vacuum. At the end of their fall, their velocities are equal because:
- A. anything falling in vacuum has constant velocity
 - B. all objects reach the same terminal velocity
 - C. the acceleration of the larger object is three times greater than that of the smaller object
 - D. the force of gravity is the same for both objects
 - E. none of the above
- ans: E
19. A feather and a lead ball are dropped from rest in vacuum on the Moon. The acceleration of the feather is:
- A. more than that of the lead ball
 - B. the same as that of the lead ball
 - C. less than that of the lead ball
 - D. 9.8 m/s^2
 - E. zero since it floats in a vacuum
- ans: B

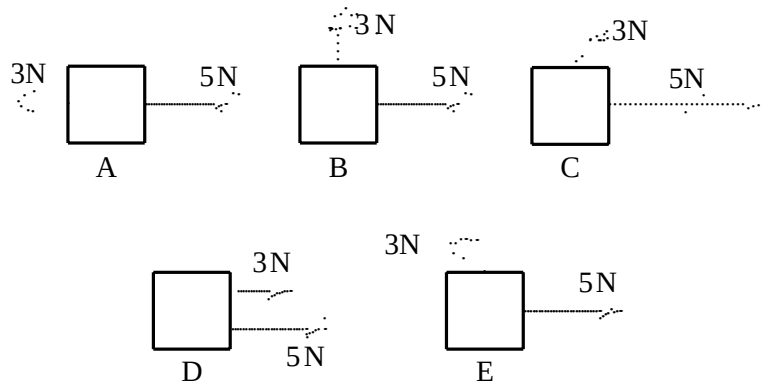
20. The block shown moves with constant velocity on a horizontal surface. Two of the forces on it are shown. A frictional force exerted by the surface is the only other horizontal force on the block. The frictional force is:



- A. 0
- B. 2 N, leftward
- C. 2 N, rightward
- D. slightly more than 2 N, leftward
- E. slightly less than 2 N, leftward

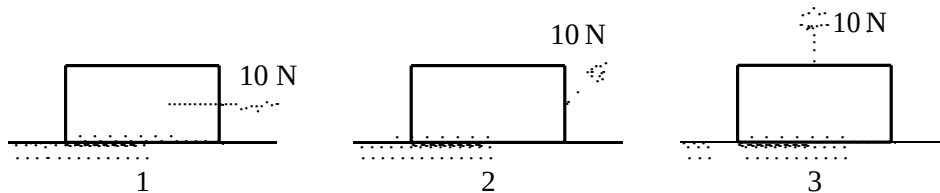
ans: B

21. Two forces, one with a magnitude of 3 N and the other with a magnitude of 5 N, are applied to an object. For which orientations of the forces shown in the diagrams is the magnitude of the acceleration of the object the least?



ans: A

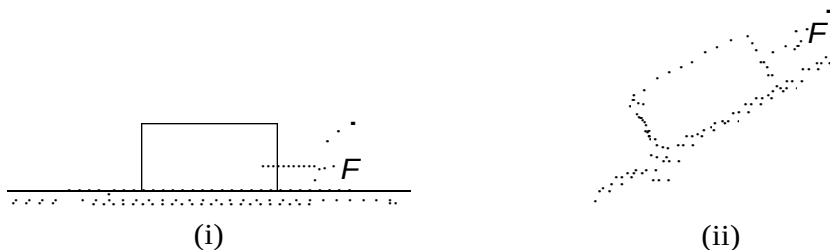
22. A crate rests on a horizontal surface and a woman pulls on it with a 10-N force. Rank the situations shown below according to the magnitude of the normal force exerted by the surface on the crate, least to greatest.



- A. 1, 2, 3
- B. 2, 1, 3
- C. 2, 3, 1
- D. 1, 3, 2
- E. 3, 2, 1

ans: E

23. A heavy wooden block is dragged by a force $\vec{F}$ along a rough steel plate, as shown in the diagrams for two cases. The magnitude of the applied force $\vec{F}$ is the same for both cases. The normal force in (ii), as compared with the normal force in (i) is:



- A. the same
- B. greater
- C. less
- D. less for some angles of the incline and greater for others
- E. less or greater, depending on the magnitude of the applied force $\vec{F}$.

ans: C

24. Equal forces $\vec{F}$ act on isolated bodies A and B. The mass of B is three times that of A. The magnitude of the acceleration of A is:

- A. three times that of B
- B. 1/3 that of B
- C. the same as B
- D. nine times that of B
- E. 1/9 that of B

ans: A

25. A car travels east at constant velocity. The net force on the car is:

- A. east
- B. west
- C. up
- D. down
- E. zero

ans: E

26. A constant force of 8.0 N is exerted for 4.0 s on a 16-kg object initially at rest. The change in speed of this object will be:

- A. 0.5 m/s
- B. 2 m/s
- C. 4 m/s
- D. 8 m/s
- E. 32 m/s

ans: B

27. A 6-kg object is moving south. A net force of 12 N north on it results in the object having an acceleration of:
- A. 2 m/s^2 , north
 - B. 2 m/s^2 , south
 - C. 6 m/s^2 , north
 - D. 18 m/s^2 , north
 - E. 18 m/s^2 , south
- ans: A

28. A 9000-N automobile is pushed along a level road by four students who apply a total forward force of 500 N. Neglecting friction, the acceleration of the automobile is:
- A. 0.055 m/s^2
 - B. 0.54 m/s^2
 - C. 1.8 m/s^2
 - D. 9.8 m/s^2
 - E. 18 m/s^2
- ans: B

29. An object rests on a horizontal frictionless surface. A horizontal force of magnitude F is applied. This force produces an acceleration:
- A. only if F is larger than the weight of the object
 - B. only while the object suddenly changes from rest to motion
 - C. always
 - D. only if the inertia of the object decreases
 - E. only if F is increasing
- ans: C

30. A 25-kg crate is pushed across a frictionless horizontal floor with a force of 20 N, directed 20° below the horizontal. The acceleration of the crate is:
- A. 0.27 m/s^2
 - B. 0.75 m/s^2
 - C. 0.80 m/s^2
 - D. 170 m/s^2
 - E. 470 m/s^2
- ans: B

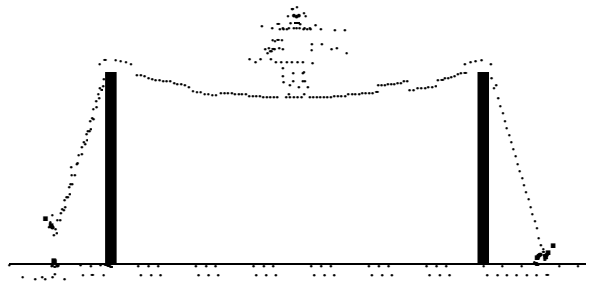
31. A ball with a weight of 1.5 N is thrown at an angle of 30° above the horizontal with an initial speed of 12 m/s. At its highest point, the net force on the ball is:
- A. 9.8 N, 30° below horizontal
 - B. zero
 - C. 9.8 N, up
 - D. 9.8 N, down
 - E. 1.5 N, down
- ans: E

32. Two forces are applied to a 5.0-kg crate; one is 6.0 N to the north and the other is 8.0 N to the west. The magnitude of the acceleration of the crate is:
- A. 0.50 m/s^2
 - B. 2.0 m/s^2
 - C. 2.8 m/s^2
 - D. 10 m/s^2
 - E. 50 m/s^2
- ans: B

33. A 400-N steel ball is suspended by a light rope from the ceiling. The tension in the rope is:
- A. 400 N
 - B. 800 N
 - C. zero
 - D. 200 N
 - E. 560 N
- ans: A

34. A heavy steel ball B is suspended by a cord from a block of wood W. The entire system is dropped through the air. Neglecting air resistance, the tension in the cord is:
- A. zero
 - B. the difference in the masses of B and W
 - C. the difference in the weights of B and W
 - D. the weight of B
 - E. none of these
- ans: A

35. A circus performer of weight W is walking along a “high wire” as shown. The tension in the wire:



- A. is approximately W
 - B. is approximately $W/2$
 - C. is much less than W
 - D. is much more than W
 - E. depends on whether he stands on one foot or two feet
- ans: D

36. A 1000-kg elevator is rising and its speed is increasing at 3 m/s^2 . The tension force of the cable on the elevator is:
- A. 6800 N
 - B. 1000 N
 - C. 3000 N
 - D. 9800 N
 - E. 12800 N

ans: E

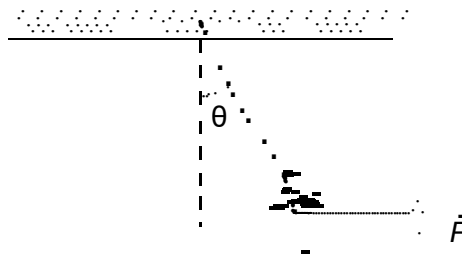
- *37. A 5-kg block is suspended by a rope from the ceiling of an elevator as the elevator accelerates downward at 3.0 m/s^2 . The tension force of the rope on the block is:
- A. 15 N, up
 - B. 34 N, up
 - C. 34 N, down
 - D. 64 N, up
 - E. 64 N, down

ans: B

38. A crane operator lowers a 16,000-N steel ball with a downward acceleration of 3 m/s^2 . The tension force of the cable is:
- A. 4900 N
 - B. 11,000 N
 - C. 16,000 N
 - D. 21,000 N
 - E. 48,000 N

ans: B

39. A 1-N pendulum bob is held at an angle θ from the vertical by a 2-N horizontal force F as shown. The tension in the string supporting the pendulum bob (in newtons) is:



- A. $\cos \theta$
- B. $2/\cos \theta$
- C. $\frac{1}{5}$
- D. 1
- E. none of these

ans: C

40. A car moves horizontally with a constant acceleration of 3 m/s^2 . A ball is suspended by a string from the ceiling of the car. The ball does not swing, being at rest with respect to the car. What angle does the string make with the vertical?
- A. 17°
 - B. 35°
 - C. 52°
 - D. 73°
 - E. Cannot be found without knowing the length of the string

ans: A

41. A man weighing 700 N is in an elevator that is accelerating upward at 4 m/s^2 . The force exerted on him by the elevator floor is:
- A. 71 N
 - B. 290 N
 - C. 410 N
 - D. 700 N
 - E. 990 N

ans: E

42. You stand on a spring scale on the floor of an elevator. Of the following, the scale shows the highest reading when the elevator:
- A. moves upward with increasing speed
 - B. moves upward with decreasing speed
 - C. remains stationary
 - D. moves downward with increasing speed
 - E. moves downward at constant speed

ans: A

43. You stand on a spring scale on the floor of an elevator. Of the following, the scale shows the highest reading when the elevator:
- A. moves downward with increasing speed
 - B. moves downward with decreasing speed
 - C. remains stationary
 - D. moves upward with decreasing speed
 - E. moves upward at constant speed

ans: B

44. When a 25-kg crate is pushed across a frictionless horizontal floor with a force of 200 N , directed 20° below the horizontal, the magnitude of the normal force of the floor on the crate is:
- A. 25 N
 - B. 68 N
 - C. 180 N
 - D. 250 N
 - E. 310 N

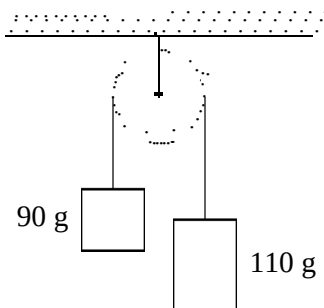
ans: E

45. A block slides down a frictionless plane that makes an angle of 30° with the horizontal. The acceleration of the block is:
- A. 980cm/s^2
 - B. 566cm/s^2
 - C. 849 cm/s^2
 - D. zero
 - E. 490cm/s^2
- ans: E
46. A 25-N crate slides down a frictionless incline that is 25° above the horizontal. The magnitude of the normal force of the incline on the crate is:
- A. 11 N
 - B. 23 N
 - C. 25 N
 - D. 100 N
 - E. 220 N
- ans: B
47. A 25-N crate is held at rest on a frictionless incline by a force that is parallel to the incline. If the incline is 25° above the horizontal the magnitude of the applied force is:
- A. 4.1N
 - B. 4.6N
 - C. 8.9N
 - D. 11 N
 - E. 23 N
- ans: D
48. A 25-N crate is held at rest on a frictionless incline by a force that is parallel to the incline. If the incline is 25° above the horizontal the magnitude of the normal force of the incline on the crate is:
- A. 4.1N
 - B. 4.6N
 - C. 8.9N
 - D. 11 N
 - E. 23 N
- ans: E
49. A 32-N force, parallel to the incline, is required to push a certain crate at constant velocity up a frictionless incline that is 30° above the horizontal. The mass of the crate is:
- A. 3.3 kg
 - B. 3.8 kg
 - C. 5.7 kg
 - D. 6.5 kg
 - E. 160 kg
- ans: D

50. A sled is on an icy (frictionless) slope that is 30° above the horizontal. When a 40-N force, parallel to the incline and directed up the incline, is applied to the sled, the acceleration of the sled is 2.0 m/s^2 , down the incline. The mass of the sled is:
- A. 3.8 kg
 - B. 4.1 kg
 - C. 5.8 kg
 - D. 6.2 kg
 - E. 10 kg
- ans: E
51. When a 40-N force, parallel to the incline and directed up the incline, is applied to a crate on a frictionless incline that is 30° above the horizontal, the acceleration of the crate is 2.0 m/s^2 , up the incline. The mass of the crate is:
- A. 3.8 kg
 - B. 4.1 kg
 - C. 5.8 kg
 - D. 6.2 kg
 - E. 10 kg
- ans: C
52. The “reaction” force does not cancel the “action” force because:
- A. the action force is greater than the reaction force
 - B. they are on different bodies
 - C. they are in the same direction
 - D. the reaction force exists only after the action force is removed
 - E. the reaction force is greater than the action force
- ans: B
53. A book rests on a table, exerting a downward force on the table. The reaction to this force is:
- A. the force of Earth on the book
 - B. the force of the table on the book
 - C. the force of Earth on the table
 - D. the force of the book on Earth
 - E. the inertia of the book
- ans: B
54. A lead block is suspended from your hand by a string. The reaction to the force of gravity on the block is the force exerted by:
- A. the string on the block
 - B. the block on the string
 - C. the string on the hand
 - D. the hand on the string
 - E. the block on Earth
- ans: E

55. A 5-kg concrete block is lowered with a downward acceleration of 2.8 m/s^2 by means of a rope. The force of the block on the rope is:
- A. 14 N, up
 - B. 14 N, down
 - C. 35 N, up
 - D. 35 N, down
 - E. 49 N, up
- ans: D
56. A 90-kg man stands in an elevator that is moving up at a constant speed of 5.0 m/s . The force exerted by him on the floor is about:
- A. zero
 - B. 90 N
 - C. 880 N
 - D. 450 N
 - E. 49 N
- ans: C
57. A 90-kg man stands in an elevator that has a downward acceleration of 1.4 m/s^2 . The force exerted by him on the floor is about:
- A. zero
 - B. 90 N
 - C. 760 N
 - D. 880 N
 - E. 1010 N
- ans: C
58. A 5-kg concrete block is lowered with a downward acceleration of 2.8 m/s^2 by means of a rope. The force of the block on Earth is:
- A. 14 N, up
 - B. 14 N, down
 - C. 35 N, up
 - D. 35 N, down
 - E. 49 N, up
- ans: E

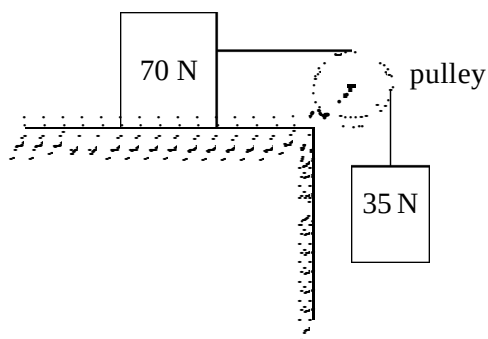
59. Two blocks are connected by a string and pulley as shown. Assuming that the string and pulley are massless, the magnitude of the acceleration of each block is:



- A. 0.049 m/s^2
- B. 0.020 m/s^2
- C. 0.0098 m/s^2
- D. 0.54 m/s^2
- E. 0.98 m/s^2

ans: E

60. A 70-N block and a 35-N block are connected by a string as shown. If the pulley is massless and the surface is frictionless, the magnitude of the acceleration of the 35-N block is:



- A. 1.6 m/s^2
- B. 3.3 m/s^2
- C. 4.9 m/s^2
- D. 6.7 m/s^2
- E. 9.8 m/s^2

ans: B

61. A 13-N weight and a 12-N weight are connected by a massless string over a massless, frictionless pulley. The 13-N weight has a downward acceleration with magnitude equal to that of a freely falling body times:

A. 1
B. $1/12$
C. $1/13$
D. $1/25$
E. $13/25$

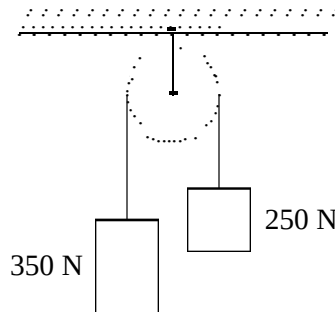
ans: D

62. A massless rope passes over a massless pulley suspended from the ceiling. A 4-kg block is attached to one end and a 5-kg block is attached to the other end. The acceleration of the 5-kg block is:

A. $g/4$
B. $5g/9$
C. $4g/9$
D. $g/5$
E. $g/9$

ans: E

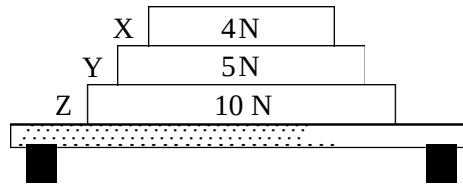
63. Two blocks, weighing 250 N and 350 N, respectively, are connected by a string that passes over a massless pulley as shown. The tension in the string is:



A. 210 N
B. 290 N
C. 410 N
D. 500 N
E. 4900 N

ans: B

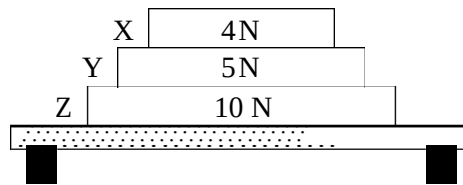
64. Three books (X, Y, and Z) rest on a table. The weight of each book is indicated. The net force acting on book Y is:



- A. 4 N down
- B. 5 N up
- C. 9 N down
- D. zero
- E. none of these

ans: D

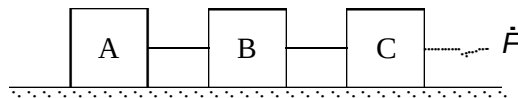
65. Three books (X, Y, and Z) rest on a table. The weight of each book is indicated. The force of book Z on book Y is:



- A. 0
- B. 5 N
- C. 9 N
- D. 14 N
- E. 19 N

ans: C

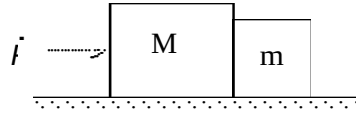
66. Three blocks (A,B,C), each having mass M , are connected by strings as shown. Block C is pulled to the right by a force $\vec{F}$ that causes the entire system to accelerate. Neglecting friction, the net force acting on block B is:



- A. zero
- B. $\vec{F}/3$
- C. $\vec{F}/2$
- D. $2\vec{F}/3$
- E. $\vec{F}$

ans: B

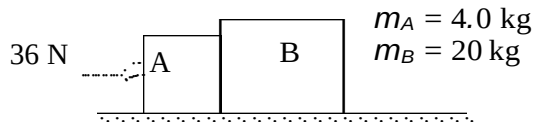
67. Two blocks with masses m and M are pushed along a horizontal frictionless surface by a horizontal applied force $\vec{F}$ as shown. The magnitude of the force of either of these blocks on the other is:



- A. $mF/(m+M)$
- B. mF/M
- C. $mF/(M-m)$
- D. $MF/(M+m)$
- E. MF/m

ans: A

68. Two blocks (A and B) are in contact on a horizontal frictionless surface. A 36-N constant force is applied to A as shown. The magnitude of the force of A on B is:



- A. 1.5N
- B. 6.0N
- C. 29 N
- D. 30 N
- E. 36 N

ans: D

69. A short 10-g string is used to pull a 50-g toy across a frictionless horizontal surface. If a 3.0×10^{-2} -N force is applied horizontally to the free end, the force of the string on the toy, at the other end, is:

- A. 0.15 N
- B. 6.0×10^{-3} N
- C. 2.5×10^{-2} N
- D. 3.0×10^{-2} N
- E. 3.5×10^{-2} N

ans: C

Chapter 6: FORCE AND MOTION — II

1. A brick slides on a horizontal surface. Which of the following will increase the magnitude of the frictional force on it?
 - A. Putting a second brick on top
 - B. Decreasing the surface area of contact
 - C. Increasing the surface area of contact
 - D. Decreasing the mass of the brick
 - E. None of the above

ans: A

2. The coefficient of kinetic friction:
 - A. is in the direction of the frictional force
 - B. is in the direction of the normal force
 - C. is the ratio of force to area
 - D. can have units of newtons
 - E. is none of the above

ans: E

3. When the brakes of an automobile are applied, the road exerts the greatest retarding force:
 - A. while the wheels are sliding
 - B. just before the wheels start to slide
 - C. when the automobile is going fastest
 - D. when the acceleration is least
 - E. at the instant when the speed begins to change

ans: B

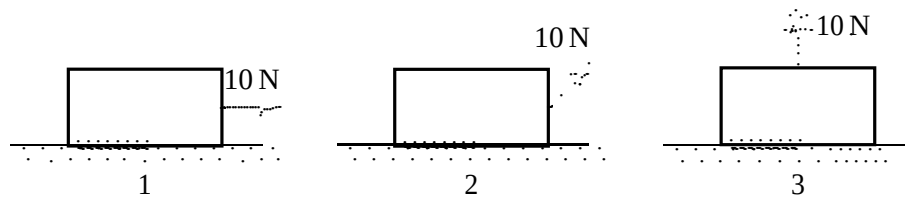
4. A forward horizontal force of 12 N is used to pull a 240-N crate at constant velocity across a horizontal floor. The coefficient of friction is:
 - A. 0.5
 - B. 0.05
 - C. 2
 - D. 0.2
 - E. 20

ans: B

5. The speed of a 4.0-N hockey puck, sliding across a level ice surface, decreases at the rate of 0.61 m/s^2 . The coefficient of kinetic friction between the puck and ice is:
 - A. 0.062
 - B. 0.41
 - C. 0.62
 - D. 1.2
 - E. 9.8

ans: A

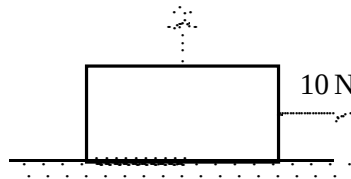
6. A crate rests on a horizontal surface and a woman pulls on it with a 10-N force. No matter what the orientation of the force, the crate does not move. Rank the situations shown below according to the magnitude of the frictional force of the surface on the crate, least to greatest.



- A. 1, 2, 3
B. 2, 1, 3
C. 2, 3, 1
D. 1, 3, 2
E. 3, 2, 1

ans: E

7. A crate with a weight of 50 N rests on a horizontal surface. A person pulls horizontally on it with a force of 10 N and it does not move. To start it moving, a second person pulls vertically upward on the crate. If the coefficient of static friction is 0.4, what is the smallest vertical force for which the crate moves?



- A. 4 N
B. 10 N
C. 14 N
D. 25 N
E. 35 N

ans: D

8. A 40-N crate rests on a rough horizontal floor. A 12-N horizontal force is then applied to it. If the coefficients of friction are $\mu_s = 0.5$ and $\mu_k = 0.4$, the magnitude of the frictional force on the crate is:

- A. 8 N
B. 12 N
C. 16 N
D. 20 N
E. 40 N

ans: B

9. A 24-N horizontal force is applied to a 40-N block initially at rest on a rough horizontal surface. If the coefficients of friction are $\mu_s = 0.5$ and $\mu_k = 0.4$, the magnitude of the frictional force on the block is:
- A. 8 N
 - B. 12 N
 - C. 16 N
 - D. 20 N
 - E. 400 N

ans: C

10. A horizontal shove of at least 200 N is required to start moving a 800-N crate initially at rest on a horizontal floor. The coefficient of static friction is:
- A. 0.25
 - B. 0.125
 - C. 0.50
 - D. 4.00
 - E. none of these

ans: A

11. A force $\vec{F}$ (larger than the largest possible force of static friction) is applied to the left to an object moving to the right on a horizontal surface. Then:
- A. the object must be moving at constant speed
 - B. $\vec{F}$ and the friction force act in opposite directions
 - C. the object must be slowing down
 - D. the object must be speeding up
 - E. the object must come to rest and remain at rest

ans: C

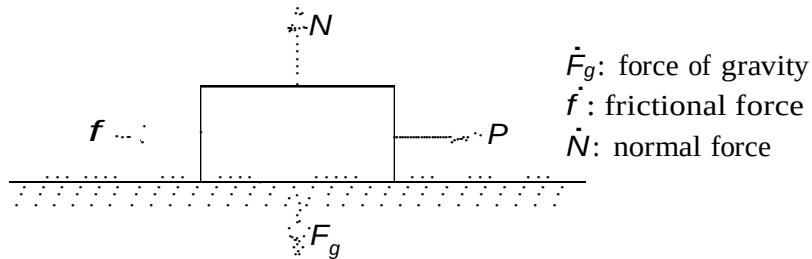
12. A bureau rests on a rough horizontal surface ($\mu_s = 0.50$, $\mu_k = 0.40$). A constant horizontal force, just sufficient to start the bureau in motion, is then applied. The acceleration of the bureau is:
- A. 0
 - B. 0.98 m/s^2
 - C. 3.3 m/s^2
 - D. 4.5 m/s^2
 - E. 8.9 m/s^2

ans: B

13. A car is traveling at 15 m/s on a horizontal road. The brakes are applied and the car skids to a stop in 4.0 s. The coefficient of kinetic friction between the tires and road is:
- A. 0.38
 - B. 0.69
 - C. 0.76
 - D. 0.92
 - E. 1.11

ans: A

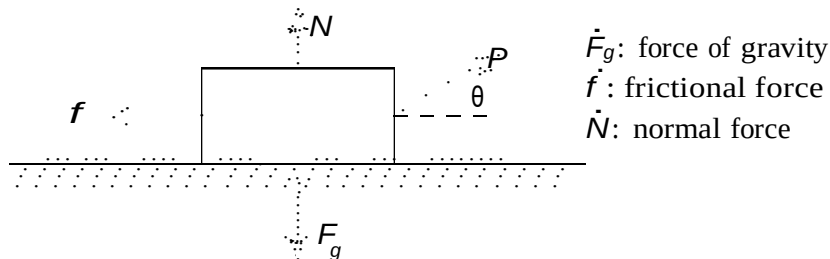
14. A boy pulls a wooden box along a rough horizontal floor at constant speed by means of a force $\vec{P}$ as shown. In the diagram f is the magnitude of the force of friction, N is the magnitude of the normal force, and F_g is the magnitude of the force of gravity. Which of the following must be true?



- A. $P = f$ and $N = F_g$
- B. $P = f$ and $N > F_g$
- C. $P > f$ and $N < F_g$
- D. $P > f$ and $N = F_g$
- E. none of these

ans: A

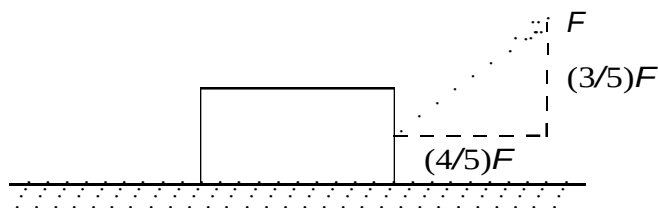
15. A boy pulls a wooden box along a rough horizontal floor at constant speed by means of a force $\vec{P}$ as shown. In the diagram f is the magnitude of the force of friction, N is the magnitude of the normal force, and F_g is the magnitude of the force of gravity. Which of the following must be true?



- A. $P = f$ and $N = F_g$
- B. $P = f$ and $N > F_g$
- C. $P > f$ and $N < F_g$
- D. $P > f$ and $N = F_g$
- E. none of these

ans: C

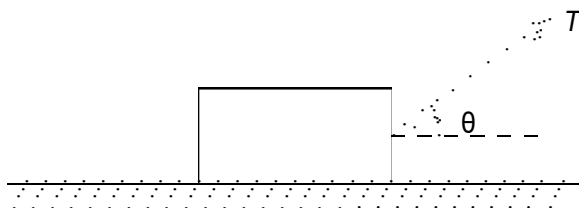
16. A 400-N block is dragged along a horizontal surface by an applied force $\vec{F}$ as shown. The coefficient of kinetic friction is $\mu_k = 0.4$ and the block moves at constant velocity. The magnitude of $\vec{F}$ is:



- A. 100 N
- B. 150 N
- C. 200 N
- D. 290 N
- E. 400 N

ans: B

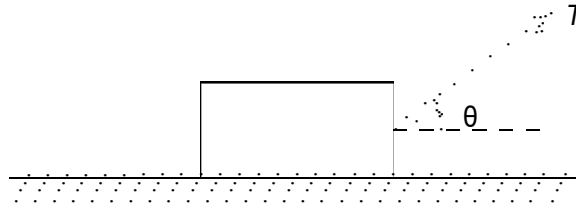
17. A block of mass m is pulled at constant velocity along a rough horizontal floor by an applied force $\vec{T}$ as shown. The magnitude of the frictional force is:



- A. $T \cos \theta$
- B. $T \sin \theta$
- C. zero
- D. mg
- E. $mg \cos \theta$

ans: A

18. A block of mass m is pulled along a rough horizontal floor by an applied force $\vec{T}$ as shown. The vertical component of the force exerted on the block by the floor is:



- A. mg
B. $mg - T \cos \theta$
C. $mg + T \cos \theta$
D. $mg - T \sin \theta$
E. $mg + T \sin \theta$
ans: D
19. A 12-kg crate rests on a horizontal surface and a boy pulls on it with a force that is 30° below the horizontal. If the coefficient of static friction is 0.40, the minimum magnitude force he needs to start the crate moving is:
A. 44 N
B. 47 N
C. 54 N
D. 56 N
E. 71 N
ans: E
20. A crate resting on a rough horizontal floor is to be moved horizontally. The coefficient of static friction is 0.40. To start the crate moving with the weakest possible applied force, in what direction should the force be applied?
A. Horizontal
B. 24° below the horizontal
C. 22° above the horizontal
D. 24° above the horizontal
E. 66° below the horizontal
ans: C
21. A 50-N force is applied to a crate on a horizontal rough floor, causing it to move horizontally. If the coefficient of kinetic friction is 0.50, in what direction should the force be applied to obtain the greatest acceleration?
A. Horizontal
B. 60° above the horizontal
C. 30° above the horizontal
D. 27° above the horizontal
E. 30° below the horizontal
ans: D

22. A professor holds an eraser against a vertical chalkboard by pushing horizontally on it. He pushes with a force that is much greater than is required to hold the eraser. The force of friction exerted by the board on the eraser increases if he:
- A. pushes with slightly greater force
 - B. pushes with slightly less force
 - C. stops pushing
 - D. pushes so his force is slightly downward but has the same magnitude
 - E. pushes so his force is slightly upward but has the same magnitude

ans: D

23. A horizontal force of 12 N pushes a 0.5-kg book against a vertical wall. The book is initially at rest. If the coefficients of friction are $\mu_s = 0.6$ and $\mu_k = 0.8$ which of the following is true?
- A. The magnitude of the frictional force is 4.9N
 - B. The magnitude of the frictional force is 7.2N
 - C. The normal force is 4.9N
 - D. The book will start moving and accelerate
 - E. If started moving downward, the book will decelerate

ans: A

24. A horizontal force of 5.0 N pushes a 0.50-kg book against a vertical wall. The book is initially at rest. If the coefficients of friction are $\mu_s = 0.6$ and $\mu_k = 0.80$, the magnitude of the frictional force is:
- A. 0
 - B. 4.9N
 - C. 3.0N
 - D. 5.0N
 - E. 4.0N

ans: E

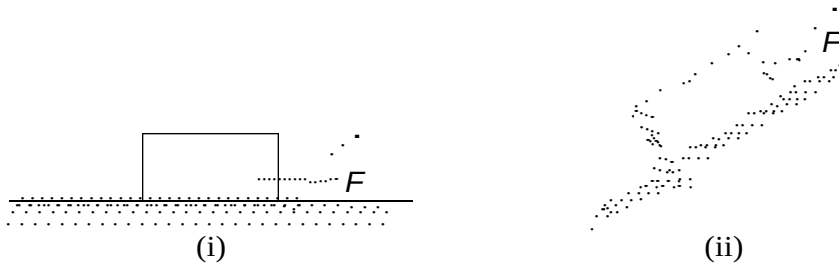
25. A horizontal force of 12 N pushes a 0.50-kg book against a vertical wall. The book is initially at rest. If $\mu_s = 0.6$ and $\mu_k = 0.80$, the acceleration of the book in m/s^2 is:
- A. 0
 - B. 9.4m/s^2
 - C. 9.8m/s^2
 - D. 14.4m/s^2
 - E. 19.2m/s^2

ans: A

26. A horizontal force of 5.0 N pushes a 0.50-kg block against a vertical wall. The block is initially at rest. If $\mu_s = 0.60$ and $\mu_k = 0.80$, the acceleration of the block in m/s^2 is:
- A. 0
 - B. 1.8
 - C. 6.0
 - D. 8.0
 - E. 9.8

ans: B

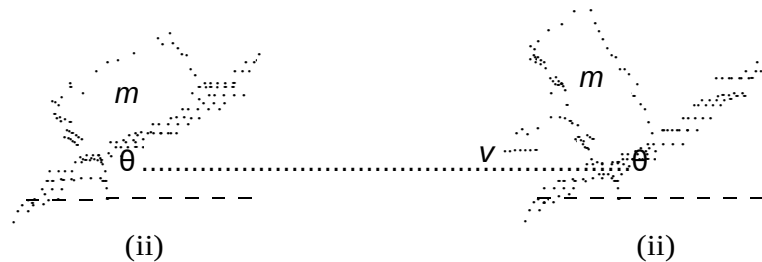
27. A heavy wooden block is dragged by a force $\vec{F}$ along a rough steel plate, as shown below for two possible situations. The magnitude of $\vec{F}$ is the same for the two situations. The magnitude of the frictional force in (ii), as compared with that in (i) is:



- A. the same
- B. greater
- C. less
- D. less for some angles and greater for others
- E. can be less or greater, depending on the magnitude of the applied force.

ans: C

28. A block is first placed on its long side and then on its short side on the same inclined plane, as shown. The block slides down the plane on its short side but remains at rest on its long side. A possible explanation is:



- A. the short side is smoother
- B. the frictional force is less because the contact area is less
- C. the center of gravity is higher in the second case
- D. the normal force is less in the second case
- E. the force of gravity is more nearly down the plane in the second case

ans: A

29. A box rests on a rough board 10 meters long. When one end of the board is slowly raised to a height of 6 meters above the other end, the box begins to slide. The coefficient of static friction is:

- A. 0.8
- B. 0.25
- C. 0.4
- D. 0.6
- E. 0.75

ans: E

30. A block is placed on a rough wooden plane. It is found that when the plane is tilted 30° to the horizontal, the block will slide down at constant speed. The coefficient of kinetic friction of the block with the plane is:
- A. 0.500
 - B. 0.577
 - C. 1.73
 - D. 0.866
 - E. 4.90

ans: B

31. A crate is sliding down an incline that is 35° above the horizontal. If the coefficient of kinetic friction is 0.40, the acceleration of the crate is:
- A. 0
 - B. 2.4m/s^2
 - C. 5.8m/s^2
 - D. 8.8m/s^2
 - E. 10.3 m/s^2

ans: B

32. A 5.0-kg crate is resting on a horizontal plank. The coefficient of static friction is 0.50 and the coefficient of kinetic friction is 0.40. After one end of the plank is raised so the plank makes an angle of 25° with the horizontal, the force of friction is:
- A. 0
 - B. 18 N
 - C. 21 N
 - D. 22 N
 - E. 44 N

ans: C

33. A 5.0-kg crate is resting on a horizontal plank. The coefficient of static friction is 0.50 and the coefficient of kinetic friction is 0.40. After one end of the plank is raised so the plank makes an angle of 30° with the horizontal, the force of friction is:
- A. 0
 - B. 18 N
 - C. 21 N
 - D. 22 N
 - E. 44 N

ans: B

34. A 5.0-kg crate is on an incline that makes an angle of 30° with the horizontal. If the coefficient of static friction is 0.50, the minimum force that can be applied parallel to the plane to hold the crate at rest is:
- 0
 - 3.3 N
 - 30 N
 - 46 N
 - 55 N

ans: B

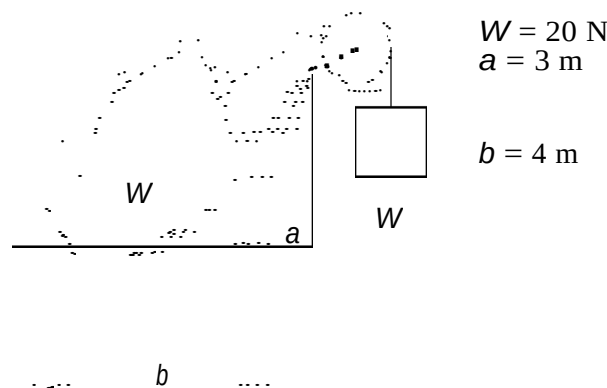
35. A 5.0-kg crate is on an incline that makes an angle of 30° with the horizontal. If the coefficient of static friction is 0.5, the maximum force that can be applied parallel to the plane without moving the crate is:
- 0
 - 3.3 N
 - 30 N
 - 46 N
 - 55 N

ans: D

36. Block A, with mass m_A , is initially at rest on a horizontal floor. Block B, with mass m_B , is initially at rest on the horizontal top surface of A. The coefficient of static friction between the two blocks is μ_s . Block A is pulled with a horizontal force. It begins to slide out from under B if the force is greater than:
- $m_A g$
 - $m_B g$
 - $\mu_s m_A g$
 - $\mu_s m_B g$
 - $\mu_s (m_A + m_B) g$

ans: E

37. The system shown remains at rest. Each block weighs 20 N. The force of friction on the upper block is:



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A. 4 N

B. 8 N

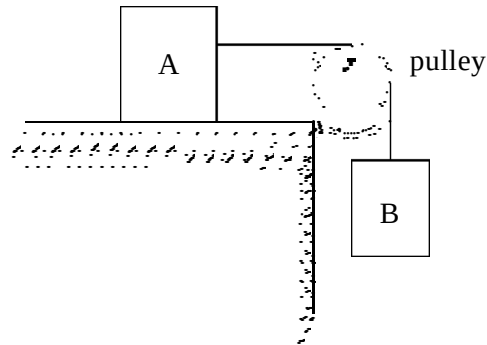
C. 12 N

D. 16 N

E. 20 N

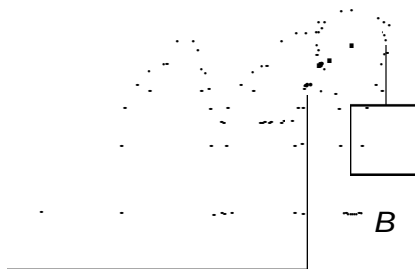
ans: B

38. Block A, with a mass of 50 kg, rests on a horizontal table top. The coefficient of static friction is 0.40. A horizontal string is attached to A and passes over a massless, frictionless pulley as shown. The smallest mass m_B of block B, attached to the dangling end, that will start A moving when it is attached to the other end of the string is:



- A. 20 kg
 - B. 30 kg
 - C. 40 kg
 - D. 50 kg
 - E. 70 kg
- ans: A

39. Block A, with a mass of 10 kg, rests on a 35° incline. The coefficient of static friction is 0.40. An attached string is parallel to the incline and passes over a massless, frictionless pulley at the top. The largest mass m_B of block B, attached to the dangling end, for which A begins to slide down the incline is:





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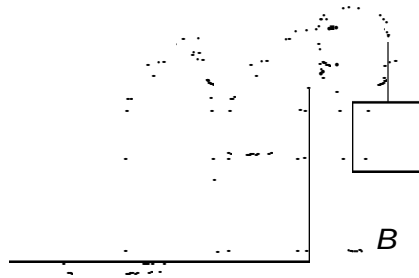
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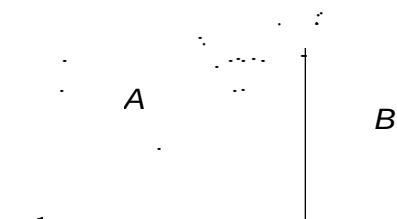
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- A. 2.5 kg
 - B. 3.5 kg
 - C. 5.9 kg
 - D. 9.0 kg
 - E. 10.5 kg
- ans: A

40. Block A, with a mass of 10 kg, rests on a 35° incline. The coefficient of static friction is 0.40. An attached string is parallel to the incline and passes over a massless, frictionless pulley at the top. The largest mass m_B , attached to the dangling end, for which A remains at rest is:





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A. 2.5 kg

B. 3.5 kg

C. 5.9 kg

D. 9.0 kg

E. 10.5 kg

ans: D



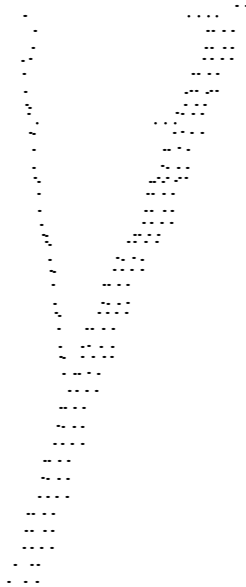
41. Block A, with a mass of 10 kg, rests on a 30° incline. The coefficient of kinetic friction is 0.20.

The attached string is parallel to the incline and passes over a massless, frictionless pulley at

the top. Block B, with a mass of 8.0 kg, is attached to the dangling end of the string. The

acceleration of B is:

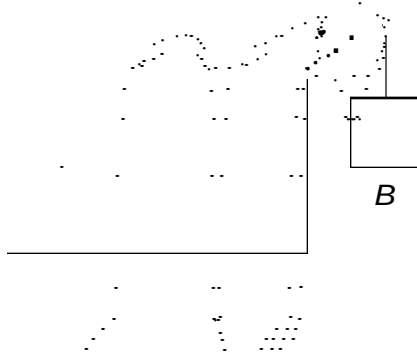




- A. 0.69 m/s^2 , up the plane
- B. 0.69 m/s^2 , down the plane
- C. 2.6 m/s^2 , up the plane
- D. 2.6 m/s^2 , down the plane
- E. 0

ans: B

42. Block A, with a mass of 10 kg, rests on a 30° incline. The coefficient of kinetic friction is 0.20. The attached string is parallel to the incline and passes over a massless, frictionless pulley at the top. Block B, with a mass of 3.0 kg, is attached to the dangling end of the string. The acceleration of B is:



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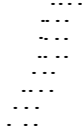
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- A. 0.20 m/s^2 , up
- B. 0.20 m/s^2 , down
- C. 2.8 m/s^2 , up
- D. 2.8 m/s^2 , down
- E. 0

ans: A

43. A 1000-kg airplane moves in straight flight at constant speed. The force of air friction is 1800 N. The net force on the plane is:
- A. zero
 - B. 11800 N
 - C. 1800N
 - D. 9800 N
 - E. none of these

ans: A

44. Why do raindrops fall with constant speed during the later stages of their descent?
- A. The gravitational force is the same for all drops
 - B. Air resistance just balances the force of gravity
 - C. The drops all fall from the same height
 - D. The force of gravity is negligible for objects as small as raindrops
 - E. Gravity cannot increase the speed of a falling object to more than 9.8 m/s

ans: B

45. A ball is thrown downward from the edge of a cliff with an initial speed that is three times the terminal speed. Initially its acceleration is
- A. upward and greater than g
 - B. upward and less than g
 - C. downward and greater than g
 - D. downward and less than g
 - E. downward and equal to g

ans: A

46. A ball is thrown upward into the air with a speed that is greater than terminal speed. On the way up it slows down and, after its speed equals the terminal speed but before it gets to the top of its trajectory:
- A. its speed is constant
 - B. it continues to slow down
 - C. it speeds up
 - D. its motion becomes jerky
 - E. none of the above
- ans: B
47. A ball is thrown upward into the air with a speed that is greater than terminal speed. It lands at the place where it was thrown. During its flight the force of air resistance is the greatest:
- A. just after it is thrown
 - B. halfway up
 - C. at the top of its trajectory
 - D. halfway down
 - E. just before it lands.
- ans: A
48. Uniform circular motion is the direct consequence of:
- A. Newton's third law
 - B. a force that is always tangent to the path
 - C. an acceleration tangent to the path
 - D. a force of constant magnitude that is always directed away from the same fixed point
 - E. a force of constant magnitude that is always directed toward the same fixed point
- ans: E
49. An object moving in a circle at constant speed:
- A. must have only one force acting on it
 - B. is not accelerating
 - C. is held to its path by centrifugal force
 - D. has an acceleration of constant magnitude
 - E. has an acceleration that is tangent to the circle
- ans: D
50. An object of mass m and another object of mass $2m$ are each forced to move along a circle of radius 1.0 m at a constant speed of 1.0 m/s. The magnitudes of their accelerations are:
- A. equal
 - B. in the ratio of $\sqrt{2} : 1$
 - C. in the ratio of 2 : 1
 - D. in the ratio of 4 : 1
 - E. zero
- ans: A

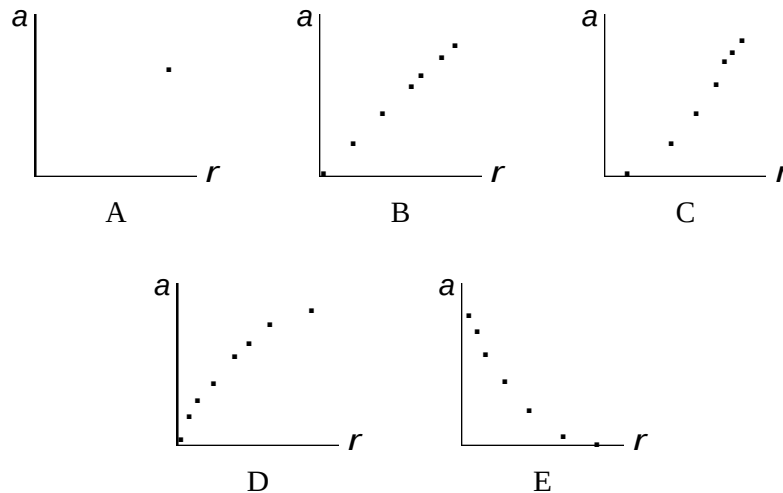
51. The magnitude of the force required to cause a 0.04-kg object to move at 0.6 m/s in a circle of radius 1.0 m is:
- A. 2.4×10^{-2} N
 - B. 1.4×10^{-2} N
 - C. $1.4\pi \times 10^{-2}$ N
 - D. $2.4\pi^2 \times 10^{-2}$ N
 - E. 3.13 N

ans: B

52. A 0.2-kg stone is attached to a string and swung in a circle of radius 0.6 m on a horizontal and frictionless surface. If the stone makes 150 revolutions per minute, the tension force of the string on the stone is:
- A. 0.03 N
 - B. 0.2 N
 - C. 0.9 N
 - D. 1.96 N
 - E. 30 N

ans: E

53. Which of the following five graphs is correct for a particle moving in a circle of radius r at a constant speed of 10 m/s?



ans: E

54. An object moves around a circle. If the radius is doubled keeping the speed the same then the magnitude of the centripetal force must be:
- A. twice as great
 - B. half as great
 - C. four times as great
 - D. one-fourth as great
 - E. the same

ans: B

55. An object moves in a circle. If the mass is tripled, the speed halved, and the radius unchanged, then the magnitude of the centripetal force must be multiplied by a factor of:

A. $3/2$
B. $3/4$
C. $9/4$
D. 6
E. 12

ans: B

56. If a satellite moves above Earth's atmosphere in a circular orbit with constant speed, then:

A. its acceleration and velocity are always in the same direction
B. the net force on it is zero
C. its velocity is constant
D. it will fall back to Earth when its fuel is used up
E. its acceleration is toward the Earth

ans: E

57. A 800-N passenger in a car presses against the car door with a 200 N force when the car makes a left turn at 13 m/s. The (faulty) door will pop open under a force of 800 N. Of the following, the least speed for which the passenger is thrown out of the car is:

A. 14m/s
B. 19m/s
C. 20m/s
D. 26m/s
E. 54m/s

ans: D

58. If a certain car, going with speed v_1 , rounds a level curve with a radius R_1 , it is just on the verge of skidding. If its speed is now doubled, the radius of the tightest curve on the same road that it can round without skidding is:

A. $2R_1$
B. $4R_1$
C. $R_1/2$
D. $R_1/4$
E. R_1

ans: B

59. An automobile moves on a level horizontal road in a circle of radius 30 m. The coefficient of friction between tires and road is 0.50. The maximum speed with which this car can round this curve is:

A. 3.0 m/s
B. 4.9 m/s
C. 9.8 m/s
D. 12m/s
E. 13m/s

ans: D

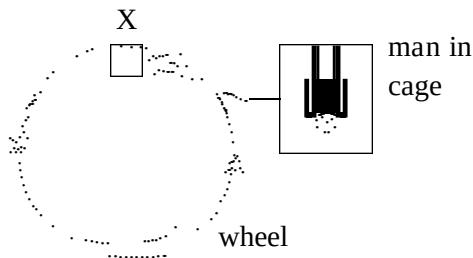
60. The driver of a 1000-kg car tries to turn through a circle of radius 100 m on an unbanked curve at a speed of 10 m/s. The actual frictional force between the tires and slippery road has a magnitude of 900 N. The car:
- A. slides into the inside of the curve
 - B. makes the turn
 - C. slows down due to the frictional force
 - D. makes the turn only if it goes faster
 - E. slides off to the outside of the curve

ans: E

61. A car rounds a 75-m radius curve at a constant speed of 18 m/s. A ball is suspended by a string from the ceiling the car and moves with the car. The angle between the string and the vertical is:
- A. 0
 - B. 1.4°
 - C. 24°
 - D. 90°
 - E. cannot be found without knowing the mass of the ball

ans: C

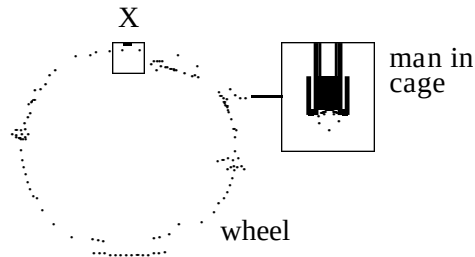
62. A giant wheel, having a diameter of 40 m, is fitted with a cage and platform on which a man of mass m stands. The wheel is rotated in a vertical plane at such a speed that the force exerted by the man on the platform is equal to his weight when the cage is at X, as shown. The net force on the man at point X is:



- A. zero
- B. mg , down
- C. mg , up
- D. $2mg$, down
- E. $2mg$, up

ans: D

63. A giant wheel, 40 m in diameter, is fitted with a cage and platform on which a man can stand. The wheel rotates at such a speed that when the cage is at X (as shown) the force exerted by the man on the platform is equal to his weight. The speed of the man is:



- A. 14m/s
- B. 20m/s
- C. 28m/s
- D. 80m/s
- E. 120 m/s

ans: B

64. A person riding a Ferris wheel is strapped into her seat by a seat belt. The wheel is spun so that the centripetal acceleration is g . Select the correct combination of forces that act on her when she is at the top. In the table F_g = force of gravity, down; F_b = seat belt force, down; and F_s = seat force, up.

	F_g	F_b	F_s
A.	0	mg	0
B.	mg	0	0
C.	0	0	mg
D.	mg	mg	0
E.	mg	0	mg

ans: B

65. One end of a 1.0-m long string is fixed, the other end is attached to a 2.0-kg stone. The stone swings in a vertical circle, passing the bottom point at 4.0 m/s. The tension force of the string at this point is about:

- A. 0
- B. 12 N
- C. 20 N
- D. 32 N
- E. 52 N

ans: E

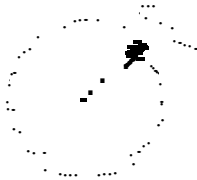
66. One end of a 1.0-m string is fixed, the other end is attached to a 2.0-kg stone. The stone swings in a vertical circle, passing the top point at 4.0 m/s. The tension force of the string (in newtons) at this point is about:
- A. 0
 - B. 12
 - C. 20
 - D. 32
 - E. 52

ans: B

67. A coin is placed on a horizontal phonograph turntable. Let N be the magnitude of the normal force exerted by the turntable on the coin, f be the magnitude of the frictional force exerted by the turntable on the coin, and $f_{s, \max}$ be the maximum possible force of static friction. The speed of the turntable is increased in small steps. If the coin does not slide, then
- A. N increases, f increases, and $f_{s, \max}$ stays the same
 - B. N increases, f increases, and $f_{s, \max}$ increases
 - C. f increases and both N and $f_{s, \max}$ stay the same
 - D. N , f , and $f_{s, \max}$ all stay the same
 - E. N , f , and $f_{s, \max}$ all increase

ans: C

68. The iron ball shown is being swung in a vertical circle at the end of a 0.7-m long string. How slowly can the ball go through its top position without having the string go slack?



- A. 1.3 m/s
- B. 2.6 m/s
- C. 3.9 m/s
- D. 6.9 m/s
- E. 9.8 m/s

ans: B

69. A block is suspended by a rope from the ceiling of a car. When the car rounds a 45-m radius horizontal curve at 22 m/s (about 50 mph), what angle does the rope make with the vertical?
- A. 0
 - B. 25°
 - C. 48°
 - D. 65°
 - E. 90°

ans: C

70. Circular freeway entrance and exit ramps are commonly banked to handle a car moving at 13 m/s. To design a similar ramp for 26 m/s one should:

- A. increase radius by factor of 2
- B. decrease radius by factor of 2
- C. increase radius by factor of 4
- D. decrease radius by factor of $\sqrt{2}$
- E. increase radius by factor of $\sqrt{2}$

ans: C

71. At what angle should the roadway on a curve with a 50 m radius be banked to allow cars to negotiate the curve at 12 m/s even if the roadway is icy (and the frictional force is zero)?

- A. 0
- B. 16°
- C. 18°
- D. 35°
- E. 73°

ans: B

Chapter 7: KINETIC ENERGY AND WORK

1. Which of the following is NOT a correct unit for work?

A. erg
B. ft·lb
C. watt
D. newton·meter
E. joule

ans: C

2. Which of the following groups does NOT contain a scalar quantity?

A. velocity, force, power
B. displacement, acceleration, force
C. acceleration, speed, work
D. energy, work, distance
E. pressure, weight, time

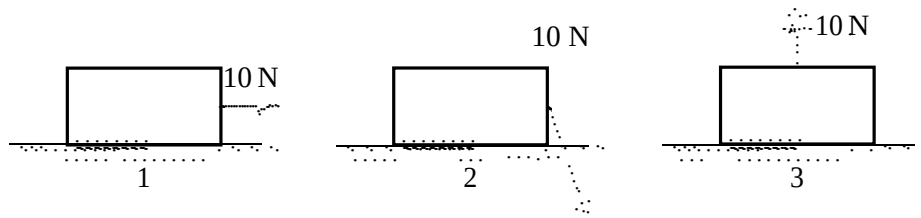
ans: B

3. A boy holds a 40-N weight at arm's length for 10 s. His arm is 1.5 m above the ground. The work done by the force of the boy on the weight while he is holding it is:

A. 0
B. 6.1J
C. 40J
D. 60J
E. 90 J

ans: A

4. A crate moves 10 m to the right on a horizontal surface as a woman pulls on it with a 10-N force. Rank the situations shown below according to the work done by her force, least to greatest.



A. 1, 2, 3
B. 2, 1, 3
C. 2, 3, 1
D. 1, 3, 2
E. 3, 2, 1

ans: E

5. An object moves in a circle at constant speed. The work done by the centripetal force is zero because:
- A. the displacement for each revolution is zero
 - B. the average force for each revolution is zero
 - C. there is no friction
 - D. the magnitude of the acceleration is zero
 - E. the centripetal force is perpendicular to the velocity
- ans: E
6. An object of mass 1 g is whirled in a horizontal circle of radius 0.5 m at a constant speed of 2 m/s. The work done on the object during one revolution is:
- A. 0
 - B. 1 J
 - C. 2 J
 - D. 4 J
 - E. 16 J
- ans: A
7. The work done by gravity during the descent of a projectile:
- A. is positive
 - B. is negative
 - C. is zero
 - D. depends for its sign on the direction of the y axis
 - E. depends for its sign on the direction of both the x and y axes
- ans: A
8. A baseball is hit high into the upper bleachers of left field. Over its entire flight the work done by gravity and the work done by air resistance, respectively, are:
- A. positive; positive
 - B. positive; negative
 - C. negative; positive
 - D. negative; negative
 - E. unknown since vital information is lacking
- ans: D
9. A line drive to the shortstop is caught at the same height as it was originally hit. Over its entire flight the work done by gravity and the work done by air resistance, respectively, are:
- A. zero; positive
 - B. zero; negative
 - C. positive; negative
 - D. negative; positive
 - E. negative; negative
- ans: B

10. A 2-kg object is moving at 3 m/s. A 4-N force is applied in the direction of motion and then removed after the object has traveled an additional 5 m. The work done by this force is:
- A. 12J
 - B. 15J
 - C. 18J
 - D. 20J
 - E. 38 J

ans: D

11. A sledge (including load) weighs 5000 N. It is pulled on level snow by a dog team exerting a horizontal force on it. The coefficient of kinetic friction between sledge and snow is 0.05. How much work is done by the dog team pulling the sledge 1000 m at constant speed?
- A. 2.5×10^4 J
 - B. 2.5×10^5 J
 - C. 5.0×10^5 J
 - D. 2.5×10^6 J
 - E. 5.0×10^6 J

ans: B

12. Camping equipment weighing 6000 N is pulled across a frozen lake by means of a horizontal rope. The coefficient of kinetic friction is 0.05. The work done by the campers in pulling the equipment 1000 m at constant velocity is:
- A. 3.1×10^4 J
 - B. 1.5×10^5 J
 - C. 3.0×10^5 J
 - D. 2.9×10^6 J
 - E. 6.0×10^6 J

ans: C

13. Camping equipment weighing 6000 N is pulled across a frozen lake by means of a horizontal rope. The coefficient of kinetic friction is 0.05. How much work is done by the campers in pulling the equipment 1000 m if its speed is increasing at the constant rate of 0.20 m/s^2 ?
- A. -1.2×10^6 J
 - B. 1.8×10^5 J
 - C. 3.0×10^5 J
 - D. 4.2×10^5 J
 - E. 1.2×10^6 J

ans: D

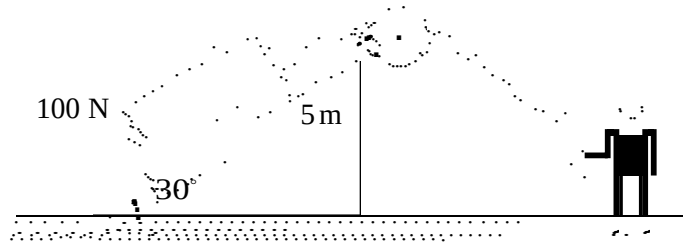
14. A 1-kg block is lifted vertically 1 m by a boy. The work done by the boy is about:
- A. 1 ft lb
 - B. 1 J
 - C. 10 J
 - D. 0.1J
 - E. zero

ans: C

15. A 0.50-kg object moves in a horizontal circular track with a radius of 2.5 m. An external force of 3.0 N, always tangent to the track, causes the object to speed up as it goes around. The work done by the external force as the mass makes one revolution is:
- A. 24J
 - B. 47J
 - C. 59J
 - D. 94J
 - E. 120 J

ans: B

16. A man pulls a 100-N crate up a frictionless 30° slope 5 m high, as shown. Assuming that the crate moves at constant speed, the work done by the man is:



- A. -500 J
- B. -250 J
- C. 0
- D. 250 J
- E. 500 J

ans: E

17. A man pushes an 80-N crate a distance of 5.0m upward along a frictionless slope that makes an angle of 30° with the horizontal. His force is parallel to the slope. If the speed of the crate decreases at a rate of 1.5 m/s^2 , then the work done by the man is:
- A. -200 J
 - B. 61 J
 - C. 140J
 - D. 200J
 - E. 260 J

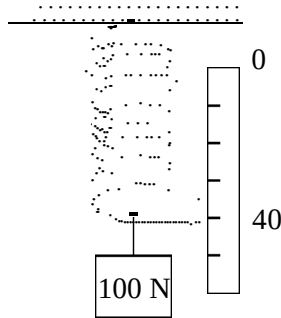
ans: C

18. A man pushes an 80-N crate a distance of 5.0m upward along a frictionless slope that makes an angle of 30° with the horizontal. The force he exerts is parallel to the slope. If the speed of the crate is constant, then the work done by the man is:
- A. -200 J
 - B. 61 J
 - C. 140J
 - D. 200J
 - E. 260 J

ans: D

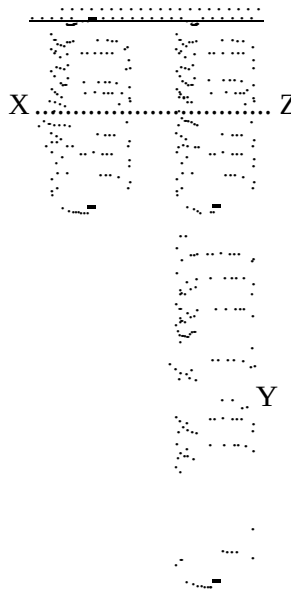
19. An 80-N crate slides with constant speed a distance of 5.0 m downward along a rough slope that makes an angle of 30° with the horizontal. The work done by the force of gravity is:
- 400 J
 - 200 J
 - 69 J
 - 200 J
 - 400 J
- ans: D
20. A man pulls a sled along a rough horizontal surface by applying a constant force $\vec{F}$ at an angle θ above the horizontal. In pulling the sled a horizontal distance d , the work done by the man is:
- Fd
 - $Fd \cos \theta$
 - $Fd \sin \theta$
 - $Fd/\cos \theta$
 - $Fd/\sin \theta$
- ans: B
21. A man wishes to pull a crate 15 m across a rough floor by exerting a force of 100 N. The coefficient of kinetic friction is 0.25. For the man to do the least work, the angle between the force and the horizontal should be:
- 0
 - 14°
 - 43°
 - 66°
 - 76°
- ans: A
22. A particle moves 5 m in the positive x direction while being acted upon by a constant force $\vec{F} = (4 \text{ N})\hat{i} + (2 \text{ N})\hat{j} - (4 \text{ N})\hat{k}$. The work done on the particle by this force is:
- 20 J
 - 10 J
 - 20 J
 - 30 J
 - is impossible to calculate without knowing other forces
- ans: A
23. A block is attached to the end of an ideal spring and moved from coordinate x_i to coordinate x_f . The relaxed position is at $x = 0$. The work done by spring is positive if:
- $x_i = 2 \text{ cm}$ and $x_f = 4 \text{ cm}$
 - $x_i = -2 \text{ cm}$ and $x_f = 4 \text{ cm}$
 - $x_i = -2 \text{ cm}$ and $x_f = -4 \text{ cm}$
 - $x_i = 2 \text{ cm}$ and $x_f = -4 \text{ cm}$
 - $x_i = -4 \text{ cm}$ and $x_f = -2 \text{ cm}$
- ans: E

24. An ideal spring, with a pointer attached to its end, hangs next to a scale. With a 100-N weight attached, the pointer indicates “40” on the scale as shown. Using a 200-N weight instead results in “60” on the scale. Using an unknown weight X instead results in “30” on the scale. The weight of X is:



- A. 10 N
 - B. 20 N
 - C. 30 N
 - D. 40 N
 - E. 50 N
- ans: E

25. Three identical ideal springs (X,Y,Z) are arranged as shown. When a 4.0-kg mass is hung on X, the mass descends 3.0 cm. When a 6.0-kg mass is hung on Y, the mass descends:



- A. 2.0 cm
 - B. 4.0 cm
 - C. 4.5 cm
 - D. 6.0 cm
 - E. 9.0 cm
- ans: E

26. When a certain rubber band is stretched a distance x , it exerts a restoring force of magnitude $F = Ax$, where A is a constant. The work done by a person in stretching this rubber band from $x = 0$ to $x = L$, beginning and ending at rest, is:

A. AL^2
B. $A + 2L$
C. $A + 2L^2$
D. A/L
E. $AL^2/2$

ans: E

27. When a certain rubber band is stretched a distance x , it exerts a restoring force of magnitude $F = ax + bx^2$, where a and b are constants. The work done in stretching this rubber band from $x = 0$ to $x = L$ is:

A. $aL^2 + bLx^3$
B. $aL + 2bL^2$
C. $a + 2bL$
D. bL
E. $aL^2/2 + bL^3/3$

ans: E

28. An ideal spring is hung vertically from the ceiling. When a 2.0-kg mass hangs at rest from it the spring is extended 6.0 cm from its relaxed length. A downward external force is now applied to the mass to extend the spring an additional 10 cm. While the spring is being extended by the force, the work done by the spring is:

A. -3.6J
B. -3.3J
C. $-3.4 \times 10^{-5}\text{J}$
D. 3.3J
E. 3.6J

ans: A

29. An ideal spring is hung vertically from the ceiling. When a 2.0-kg block hangs at rest from it the spring is extended 6.0 cm from its relaxed length. A upward external force is then applied to the block to move it upward a distance of 16 cm. While the block is moving upward the work done by the spring is:

A. -1.0J
B. -0.52J
C. -0.26J
D. 0.52J
E. 1.0J

ans: A

30. Which of the following bodies has the largest kinetic energy?
- A. Mass $3M$ and speed V
 - B. Mass $3M$ and speed $2V$
 - C. Mass $2M$ and speed $3V$
 - D. Mass M and speed $4V$
 - E. All four of the above have the same kinetic energy
- ans: C
31. Two trailers, X with mass 500 kg and Y with mass 2000 kg, are being pulled at the same speed. The ratio of the kinetic energy of Y to that of X is:
- A. 1:1
 - B. 2:1
 - C. 4:1
 - D. 9:1
 - E. 1500:1
- ans: C
32. A 8000-N car is traveling at 12 m/s along a horizontal road when the brakes are applied. The car skids to a stop in 4.0 s. How much kinetic energy does the car lose in this time?
- A. 4.8×10^4 J
 - B. 5.9×10^4 J
 - C. 1.2×10^5 J
 - D. 5.8×10^5 J
 - E. 4.8×10^6 J
- ans: B
33. The velocity of a particle moving along the x axis changes from v_i to v_f . For which values of v_i and v_f is the total work done on the particle positive?
- A. $v_i = 5 \text{ m/s}$, $v_f = 2 \text{ m/s}$
 - B. $v_i = 5 \text{ m/s}$, $v_f = -2 \text{ m/s}$
 - C. $v_i = -5 \text{ m/s}$, $v_f = -2 \text{ m/s}$
 - D. $v_i = -5 \text{ m/s}$, $v_f = 2 \text{ m/s}$
 - E. $v_i = 2 \text{ m/s}$, $v_f = -5 \text{ m/s}$
- ans: E
34. An object is constrained by a cord to move in a circular path of radius 0.5 m on a horizontal frictionless surface. The cord will break if its tension exceeds 16 N. The maximum kinetic energy the object can have is:
- A. 4 J
 - B. 8 J
 - C. 16 J
 - D. 32 J
 - E. 64 J
- ans: A

35. The weight of an object on the moon is one-sixth of its weight on Earth. The ratio of the kinetic energy of a body on Earth moving with speed V to that of the same body moving with speed V on the moon is:

A. 6:1
B. 36:1
C. 1:1
D. 1:6
E. 1:36

ans: C

36. Which of the following is the correct combination of dimensions for energy?

A. MLT
B. LT^2/m
C. ML^2/T^2
D. M^2L^3T
E. ML/T^2

ans: C

37. The amount of work required to stop a moving object is equal to:

A. the velocity of the object
B. the kinetic energy of the object
C. the mass of the object times its acceleration
D. the mass of the object times its velocity
E. the square of the velocity of the object

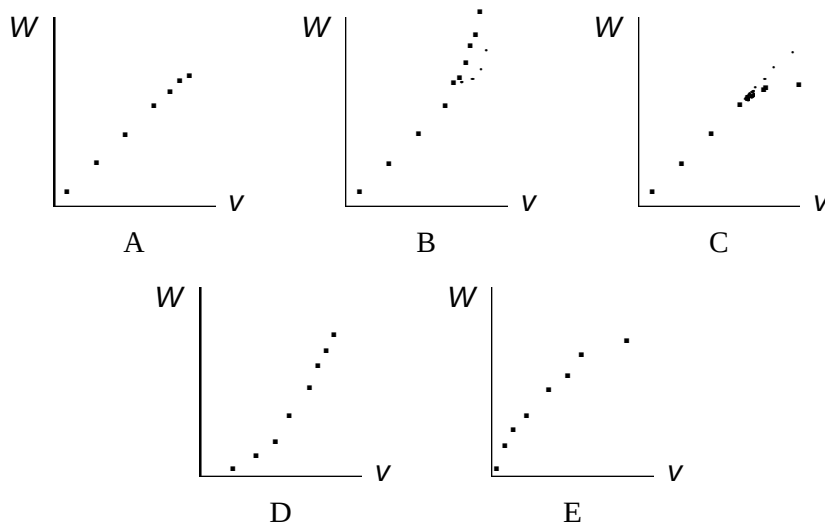
ans: B

38. A 5.0-kg cart is moving horizontally at 6.0 m/s. In order to change its speed to 10.0 m/s, the net work done on the cart must be:

A. 40J
B. 90J
C. 160J
D. 400J
E. 550 J

ans: C

39. A crate is initially at rest on a horizontal frictionless table. A constant horizontal force F is applied. Which of the following five graphs is a correct plot of work W as a function of the crate's speed v ?



ans: D

40. An 8-N block slides down an incline. It has an initial speed of 7 m/s. The work done by the resultant force on this block is:
- 3 J
 - 6 J
 - 56 J
 - impossible to calculate without more information
 - none of these

ans: D

41. A 4-kg cart starts up an incline with a speed of 3 m/s and comes to rest 2 m up the incline. The total work done on the car is:
- 6 J
 - 8 J
 - 12 J
 - 18 J
 - impossible to calculate without more information

ans: D

42. Two objects with masses of m_1 and m_2 have the same kinetic energy and are both moving to the right. The same constant force $\vec{F}$ is applied to the left to both masses. If $m_1 = 4m_2$, the ratio of the stopping distance of m_1 to that of m_2 is:
- 1:4
 - 4:1
 - 1:2
 - 2:1
 - 1:1

ans: E

43. A Boston Red Sox baseball player catches a ball of mass m that is moving toward him with speed v . While bringing the ball to rest, his hand moves back a distance d . Assuming constant deceleration, the horizontal force exerted on the ball by his hand is:
- A. mv/d
 - B. mvd
 - C. mv^2/d
 - D. $2mv/d$
 - E. $mv^2/(2d)$
- ans: E
44. A 0.50-kg object moves on a horizontal circular track with a radius of 2.5 m. An external force of 3.0 N, always tangent to the track, causes the object to speed up as it goes around. If it starts from rest its speed at the end of one revolution is:
- A. 9.8 m/s
 - B. 14 m/s
 - C. 15 m/s
 - D. 19 m/s
 - E. 21 m/s
- ans: B
45. A 0.50-kg object moves on a horizontal frictionless circular track with a radius of 2.5 m. An external force of 3.0 N, always tangent to the track, causes the object to speed up as it goes around. If it starts from rest, then at the end of one revolution the radial component of the force of the track on it is:
- A. 19 N
 - B. 38 N
 - C. 47 N
 - D. 75 N
 - E. 96 N
- ans: B
46. A 2-kg block is attached to a horizontal ideal spring with a spring constant of 200 N/m. When the spring has its equilibrium length the block is given a speed of 5 m/s. What is the maximum elongation of the spring?
- A. 0
 - B. 0.05 m
 - C. 5 m
 - D. 10 m
 - E. 100 m
- ans: C

47. At time $t = 0$ a particle starts moving along the x axis. If its kinetic energy increases uniformly with t the net force acting on it must be:
- constant
 - proportional to t
 - inversely proportional to t
 - proportional to $\sqrt{t}$
 - proportional to $1/\sqrt{t}$
- ans: E
48. At time $t = 0$ a 2-kg particle has a velocity of $(4 \text{ m/s})\hat{i} - (3 \text{ m/s})\hat{j}$. At $t = 3 \text{ s}$ its velocity is $(2 \text{ m/s})\hat{i} + (3 \text{ m/s})\hat{j}$. During this time the work done on it was:
- 4 J
 - 4 J
 - 12 J
 - 40 J
 - $(4 \text{ J})\hat{i} + (36 \text{ J})\hat{j}$
- ans: C
49. A particle starts from rest at time $t = 0$ and moves along the x axis. If the net force on it is proportional to t , its kinetic energy is proportional to:
- t
 - t^2
 - t^4
 - $1/t^2$
 - none of the above
- ans: C
50. A 1.5-kg crate falls from a height of 2.0 m onto an industrial spring scale with a spring constant of $1.5 \times 10^5 \text{ N/m}$. At its greatest compression the reading on the scale is:
- 15 N
 - 30 N
 - $1.5 \times 10^3 \text{ N}$
 - $2.1 \times 10^3 \text{ N}$
 - $3.0 \times 10^3 \text{ N}$
- ans: E
51. A particle moving along the x axis is acted upon by a single force $F = F_0 e^{-kx}$, where F_0 and k are constants. The particle is released from rest at $x = 0$. It will attain a maximum kinetic energy of:
- F_0/k
 - F_0/e^k
 - kF_0
 - $1/2(kF_0)^2$
 - $ke^k F_0$
- ans: A

52. The mechanical advantage of any machine is:
- A. the efficiency of the machine
 - B. the work done by the machine
 - C. the ratio of the work done by the machine to the work expended on it
 - D. the ratio of the force exerted by the machine to the force applied to it
 - E. the ratio of the force applied to the machine to the force exerted by it
- ans: D
53. In raising an object to a given height by means of an inclined plane, as compared with raising the object vertically, there is a reduction in:
- A. work required
 - B. distance pushed
 - C. friction
 - D. force required
 - E. value of the acceleration due to gravity
- ans: D
54. A watt is:
- A. $\text{kg} \cdot \text{m}/\text{s}^3$
 - B. $\text{kg} \cdot \text{m}^2/\text{s}$
 - C. $\text{kg} \cdot \text{m}^2/\text{s}^3$
 - D. $\text{kg} \cdot \text{m}/\text{s}$
 - E. $\text{kg} \cdot \text{m}^2/\text{s}^2$
- ans: C
55. Power has the dimensions of:
- A. ML^2/T^2
 - B. MT/L^2
 - C. ML/T^2
 - D. ML^2/T^3
 - E. none of these
- ans: D
56. Which of the following five units represents a quantity that is NOT the same as the other four?
- A. joule
 - B. erg
 - C. watt
 - D. foot·pound
 - E. newton·meter
- ans: C

57. Which of the following five quantities is NOT an expression for energy? Here m is a mass, g is the acceleration due to gravity, h and d are distances, F is a force, v is a speed, a is an acceleration, P is power, and t is time.

- A. mgh
- B. Fd
- C. $\frac{1}{2}mv^2$
- D. ma
- E. Pt

ans: D

58. A watt-second is a unit of:

- A. force
- B. power
- C. displacement
- D. speed
- E. energy

ans: E

59. A watt per hour is a unit of:

- A. energy
- B. power
- C. force
- D. acceleration
- E. none of these

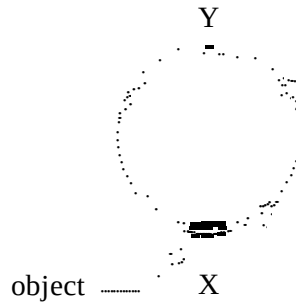
ans: E

60. A kilowatt-hour is a unit of:

- A. power
- B. energy/time
- C. work
- D. power/time
- E. force/distance

ans: C

61. A man moves the 10-g object shown in a vertical plane from position X to position Y along a circular track of radius 20 m. The process takes 0.75 min. The work done by the man is about:



- A. 1 J
- B. 2 J
- C. 4 J
- D. 6 J
- E. 12 J

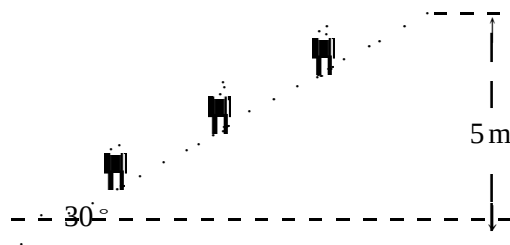
ans: C

62. A woman lifts a barbell 2.0 m in 5.0 s. If she lifts it the same distance in 10 s, the work done by her is:

- A. four times as great
- B. two times as great
- C. the same
- D. half as great
- E. one-fourth as great

ans: C

63. An escalator is used to move 20 people (60 kg each) per minute from the first floor of a department store to the second floor, 5 m above. Neglecting friction, the power required is approximately:



- A. 100 W
- B. 200 W
- C. 1000 W
- D. 2000 W
- E. 60,000 W

ans: C

64. A person holds an 80-N weight 2 m above the floor for 30 seconds. The power required to do this is:
- A. 80 W
 - B. 40 W
 - C. 20 W
 - D. 10 W
 - E. none of these
- ans: E
65. A 50-N force is the only force on a 2-kg object that starts from rest. When the force has been acting for 2 s the rate at which it is doing work is:
- A. 75 W
 - B. 100 W
 - C. 1000 W
 - D. 2500 W
 - E. 5000 W
- ans: D
66. A 50-N force is the only force a 2-kg crate that starts from rest. At the instant the object has gone 2 m the rate at which the force is doing work is:
- A. 2.5W
 - B. 25 W
 - C. 75 W
 - D. 100 W
 - E. 500 W
- ans: E
67. A particle starts from rest and is acted on by a net force that does work at a rate that is proportional to the time t . The speed of the particle is proportional to:
- A. $\sqrt{t}$
 - B. t
 - C. t^2
 - D. $1/t$
 - E. $1/t$
- ans: A

Chapter 8: POTENTIAL ENERGY AND CONSERVATION OF ENERGY

1. Only if a force on a particle is conservative:
 - A. is its work zero when the particle moves exactly once around any closed path
 - B. is its work always equal to the change in the kinetic energy of the particle
 - C. does it obey Newton's second law
 - D. does it obey Newton's third law
 - E. is it not a frictional forceans: A
2. A nonconservative force:
 - A. violates Newton's second law
 - B. violates Newton's third law
 - C. cannot do any work
 - D. must be perpendicular to the velocity of the particle on which it acts
 - E. none of the aboveans: E
3. The sum of the kinetic and potential energies of a system of objects is conserved:
 - A. only when no external force acts on the objects
 - B. only when the objects move along closed paths
 - C. only when the work done by the resultant external force is zero
 - D. always
 - E. none of the aboveans: E
4. A force on a particle is conservative if:
 - A. its work equals the change in the kinetic energy of the particle
 - B. it obeys Newton's second law
 - C. it obeys Newton's third law
 - D. its work depends on the end points of every motion, not on the path between
 - E. it is not a frictional forceans: D
5. Two particles interact by conservative forces. In addition, an external force acts on each particle. They complete round trips, ending at the points where they started. Which of the following must have the same values at the beginning and end of this trip?
 - A. the total kinetic energy of the two-particle system
 - B. the potential energy of the two-particle system
 - C. the mechanical energy of the two-particle system
 - D. the total linear momentum of the two-particle system
 - E. none of the aboveans: B

6. Two objects interact with each other and with no other objects. Initially object A has a speed of 5 m/s and object B has a speed of 10 m/s. In the course of their motion they return to their initial positions. Then A has a speed of 4 m/s and B has a speed of 7 m/s. We can conclude:
- A. the potential energy changed from the beginning to the end of the trip
 - B. mechanical energy was increased by nonconservative forces
 - C. mechanical energy was decreased by nonconservative forces
 - D. mechanical energy was increased by conservative forces
 - E. mechanical energy was decreased by conservative forces

ans: C

7. A good example of kinetic energy is provided by:
- A. a wound clock spring
 - B. the raised weights of a grandfather's clock
 - C. a tornado
 - D. a gallon of gasoline
 - E. an automobile storage battery

ans: C

8. No kinetic energy is possessed by:
- A. a shooting star
 - B. a rotating propeller on a moving airplane
 - C. a pendulum at the bottom of its swing
 - D. an elevator standing at the fifth floor
 - E. a cyclone

ans: D

9. The wound spring of a clock possesses:
- A. kinetic but no potential energy
 - B. potential but no kinetic energy
 - C. both potential and kinetic energy in equal amounts
 - D. neither potential nor kinetic energy
 - E. both potential and kinetic energy, but more kinetic energy than potential energy

ans: B

10. A body at rest in a system is capable of doing work if:
- A. the potential energy of the system is positive
 - B. the potential energy of the system is negative
 - C. it is free to move in such a way as to decrease its kinetic energy
 - D. it is free to move in such a way as to decrease the potential energy of the system
 - E. it is free to move in such a way as to increase the potential energy of the system

ans: D

11. Which one of the following five quantities CANNOT be used as a unit of potential energy?

- A. watt second
- B. $\text{gram}\cdot\text{cm}/\text{s}^2$
- C. joule
- D. $\text{kg}\cdot\text{m}^2/\text{s}^2$
- E. $\text{ft}\cdot\text{lb}$

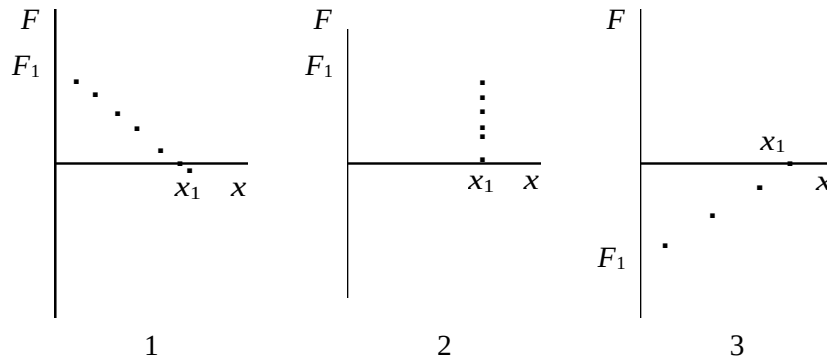
ans: B

12. Suppose that the fundamental dimensions are taken to be: force (F), velocity (V) and time (T). The dimensions of potential energy are then:

- A. F/T
- B. FVT
- C. FV/T
- D. F/T^2
- E. FV^2/T^2

ans: B

13. The graphs below show the magnitude of the force on a particle as the particle moves along the positive x axis from the origin to $x = x_1$. The force is parallel to the x axis and is conservative. The maximum magnitude F_1 has the same value for all graphs. Rank the situations according to the change in the potential energy associated with the force, least (or most negative) to greatest (or most positive).



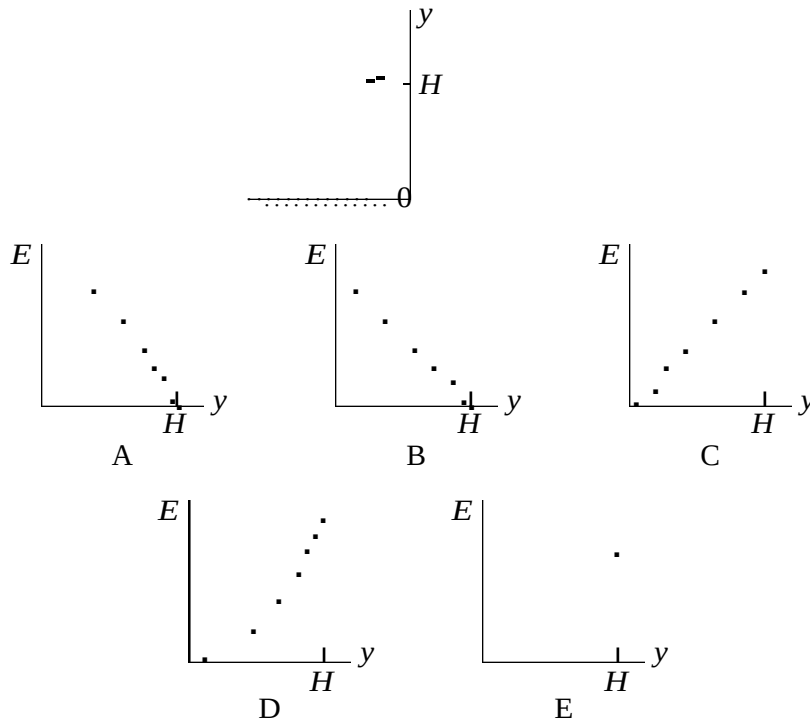
- A. 1, 2, 3
- B. 1, 3, 2
- C. 2, 3, 1
- D. 3, 2, 1
- E. 2, 1, 3

ans: E

14. A golf ball is struck by a golf club and falls on a green three meters above the tee. The potential energy of the Earth-ball system is greatest:
- just before the ball is struck
 - just after the ball is struck
 - just after the ball lands on the green
 - when the ball comes to rest on the green
 - when the ball reaches the highest point in its flight

ans: E

15. A ball is held at a height H above a floor. It is then released and falls to the floor. If air resistance can be ignored, which of the five graphs below correctly gives the mechanical energy E of the Earth-ball system as a function of the altitude y of the ball?



ans: E

16. A 6.0-kg block is released from rest 80 m above the ground. When it has fallen 60 m its kinetic energy is approximately:
- 4800 J
 - 3500 J
 - 1200 J
 - 120 J
 - 60 J

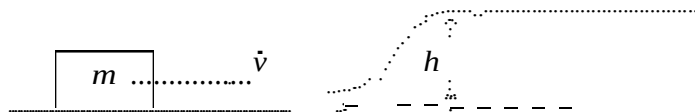
ans: B

17. A 2-kg block is thrown upward from a point 20 m above Earth's surface. At what height above Earth's surface will the gravitational potential energy of the Earth-block system have increased by 500 J?
- 5 m
 - 25 m
 - 46 m
 - 70 m
 - 270 m
- ans: C

18. An elevator is rising at constant speed. Consider the following statements:
- the upward cable force is constant
 - the kinetic energy of the elevator is constant
 - the gravitational potential energy of the Earth-elevator system is constant
 - the acceleration of the elevator is zero
 - the mechanical energy of the Earth-elevator system is constant
- all five are true
 - only II and V are true
 - only IV and V are true
 - only I, II, and III are true
 - only I, II, and IV are true
- ans: E

19. A projectile of mass 0.50 kg is fired with an initial speed of 10 m/s at an angle of 60° above the horizontal. The potential energy of the projectile-Earth system (relative potential energy when the projectile is at ground level) is:
- 25 J
 - 18.75 J
 - 12.5 J
 - 6.25 J
 - none of these
- ans: B

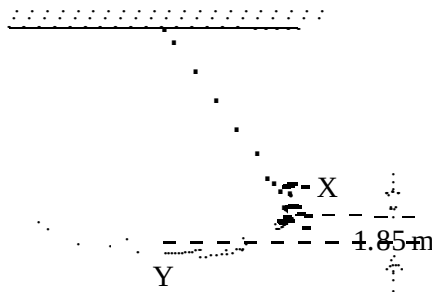
20. For a block of mass m to slide without friction up the rise of height h shown, it must have a minimum initial kinetic energy of:



- gh
 - mgh
 - $gh/2$
 - $mgh/2$
 - $2mgh$
- ans: B

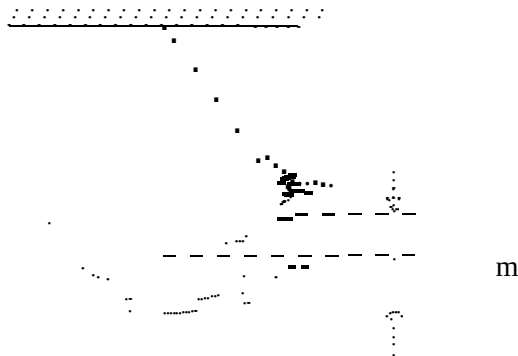
21. A 2.2-kg block starts from rest on a rough inclined plane that makes an angle of 25° with the horizontal. The coefficient of kinetic friction is 0.25. As the block goes 2.0 m down the plane, the mechanical energy of the Earth-block system changes by:
- 0
 - 9.8 J
 - 9.8 J
 - 18 J
 - 18 J
- ans: B

22. A simple pendulum consists of a 2.0-kg mass attached to a string. It is released from rest at X as shown. Its speed at the lowest point Y is about:



- 0.90 m/s
 - 3.6 m/s
 - 3.6 m/s
 - 6.0 m/s
 - 36 m/s
- ans: D

23. The long pendulum shown is drawn aside until the ball has risen 0.50 m. It is then given an initial speed of 3.0 m/s. The speed of the ball at its lowest position is:



- zero
 - 0.89 m/s
 - 3.1 m/s
 - 3.7 m/s
 - 4.3 m/s
- ans: E

24. A particle moves along the x axis under the influence of a stationary object. The net force on the particle is given by $F = (8 \text{ N/m}^3)x^3$. If the potential energy is taken to be zero for $x = 0$ then the potential energy is given by:

A. $(2 \text{ J/m}^4)x^4$
 B. $(-2 \text{ J/m}^4)x^4$
 C. $(24 \text{ J/m}^2)x^2$
 D. $(-24 \text{ J/m}^2)x^2$
 E. $5 \text{ J} - (2 \text{ J/m}^4)x^4$

ans: B

25. A 0.20-kg particle moves along the x axis under the influence of a stationary object. The potential energy is given by

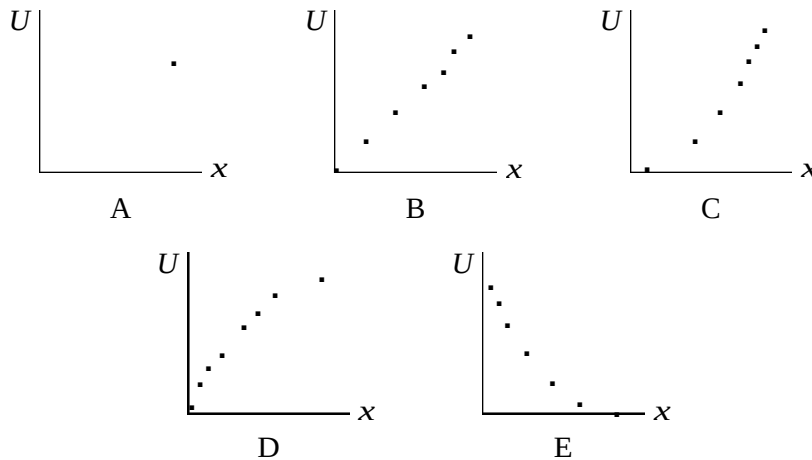
$$U(x) = (8.0 \text{ J/m}^2)x^2 + (2.0 \text{ J/m}^4)x^4,$$

where x is in coordinate of the particle. If the particle has a speed of 5.0 m/s when it is at $x = 1.0 \text{ m}$, its speed when it is at the origin is:

A. 0
 B. 2.5 m/s
 C. 5.7 m/s
 D. 7.9 m/s
 E. 11 m/s

ans: E

26. Which of the five graphs correctly shows the potential energy of a spring as a function of its elongation x ?



ans: C

27. A force of 10 N holds an ideal spring with a 20-N/m spring constant in compression. The potential energy stored in the spring is:
- A. 0.5J
 - B. 2.5J
 - C. 5 J
 - D. 10 J
 - E. 200 J
- ans: B

28. An ideal spring is used to fire a 15.0-g pellet horizontally. The spring has a spring constant of 20 N/m and is initially compressed by 7.0 cm. The kinetic energy of the pellet as it leaves the spring is:
- A. zero
 - B. 2.5×10^{-2} J
 - C. 4.9×10^{-2} J
 - D. 9.8×10^{-2} J
 - E. 1.4J
- ans: C

29. A 0.50-kg block attached to an ideal spring with a spring constant of 80 N/m oscillates on a horizontal frictionless surface. The total mechanical energy is 0.12 J. The greatest extension of the spring from its equilibrium length is:
- A. 1.5×10^{-3} m
 - B. 3.0×10^{-3} m
 - C. 0.039 m
 - D. 0.054 m
 - E. 18 m
- ans: D

30. A 0.50-kg block attached to an ideal spring with a spring constant of 80 N/m oscillates on a horizontal frictionless surface. The total mechanical energy is 0.12 J. The greatest speed of the block is:
- A. 0.15m/s
 - B. 0.24m/s
 - C. 0.49m/s
 - D. 0.69m/s
 - E. 1.46 m/s
- ans: D

31. A 0.50-kg block attached to an ideal spring with a spring constant of 80 N/m oscillates on a horizontal frictionless surface. When the spring is 4.0 cm longer than its equilibrium length, the speed of the block is 0.50 m/s. The greatest speed of the block is:
- A. 0.23m/s
 - B. 0.32m/s
 - C. 0.55m/s
 - D. 0.71m/s
 - E. 0.93 m/s
- ans: D

32. A 0.5-kg block slides along a horizontal frictionless surface at 2 m/s. It is brought to rest by compressing a very long spring of spring constant 800 N/m. The maximum spring compression is:

A. 0
 B. 3 cm
 C. 5 cm
 D. 80 cm
 E. 80 m

ans: C

33. A block of mass m is initially moving to the right on a horizontal frictionless surface at a speed v . It then compresses a spring of spring constant k . At the instant when the kinetic energy of the block is equal to the potential energy of the spring, the spring is compressed a distance of:

A. $v \sqrt{m/2k}$
 B. $(1/2)mv^2$
 C. $(1/4)mv^2$
 D. $mv^2/4k \sqrt{\quad}$
 E. $(1/4) \sqrt{mv/k}$

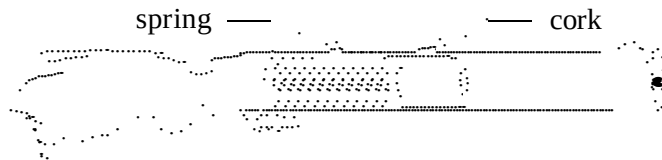
ans: A

34. A 700-N man jumps out of a window into a fire net 10 m below. The net stretches 2 m before bringing the man to rest and tossing him back into the air. The maximum potential energy of the net, compared to its unstretched potential energy, is:

A. 300 J
 B. 710 J
 C. 850 J
 D. 7000 J
 E. 8400 J

ans: E

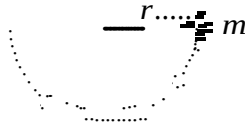
35. A toy cork gun contains a spring whose spring constant is 10.0 N/m. The spring is compressed 5.00 cm and then used to propel a 6.00-g cork. The cork, however, sticks to the spring for 1.00 cm beyond its unstretched length before separation occurs. The muzzle velocity of this cork is:



A. 1.02 m/s
 B. 1.41 m/s
 C. 2.00 m/s
 D. 2.04 m/s
 E. 4.00 m/s

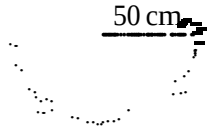
ans: C

36. A small object of mass m , on the end of a light cord, is held horizontally at a distance r from a fixed support as shown. The object is then released. What is the tension force of the cord when the object is at the lowest point of its swing?



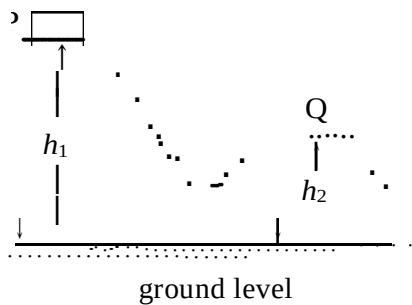
- A. $mg/2$
 - B. mg
 - C. $2mg$
 - D. $3mg$
 - E. mgr
- ans: D

37. The string in the figure is 50 cm long. When the ball is released from rest, it swings along the dotted arc. How fast is it going at the lowest point in its swing?



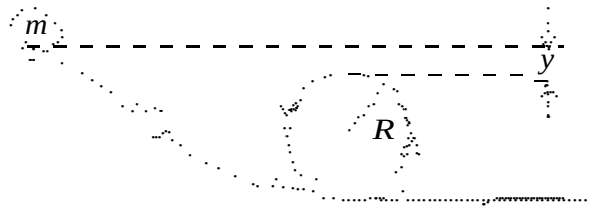
- A. 2.0m/s
 - B. 2.2m/s
 - C. 3.1m/s
 - D. 4.4m/s
 - E. 6.0m/s
- ans: C

38. A block is released from rest at point P and slides along the frictionless track shown. At point Q, its speed is:



- A. $2g\sqrt{h_1 - h_2}$
 - B. $2g(h_1 - h_2)$
 - C. $(h_1 - h_2)/2g$
 - D. $\sqrt{2g(h_1 - h_2)}$
 - E. $(h_1 - h_2)^2/2g$
- ans: D

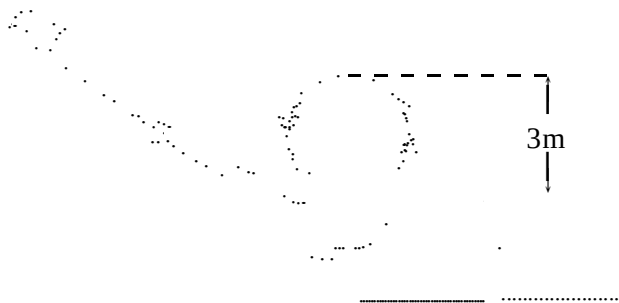
39. A small object of mass m starts from rest at the position shown and slides along the frictionless loop-the-loop track of radius R . What is the smallest value of y such that the object will slide without losing contact with the track?



- A. $R/4$
- B. $R/2$
- C. R
- D. $2R$
- E. zero

ans: B

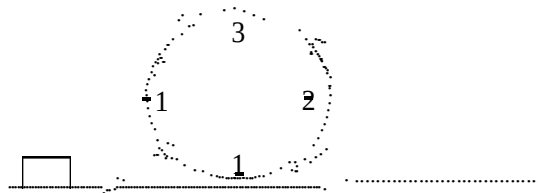
40. A small object slides along the frictionless loop-the-loop with a diameter of 3 m. What minimum speed must it have at the top of the loop?



- A. 1.9m/s
- B. 3.8m/s
- C. 5.4m/s
- D. 15m/s
- E. 29m/s

ans: B

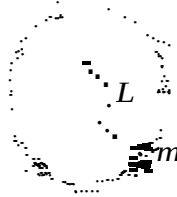
41. A rectangular block is moving along a frictionless path when it encounters the circular loop as shown. The block passes points 1, 2, 3, 4, 1 before returning to the horizontal track. At point 3:



- A. its mechanical energy is a minimum
- B. the forces on it are balanced
- C. it is not accelerating
- D. its speed is a minimum
- E. it experiences a net upward force

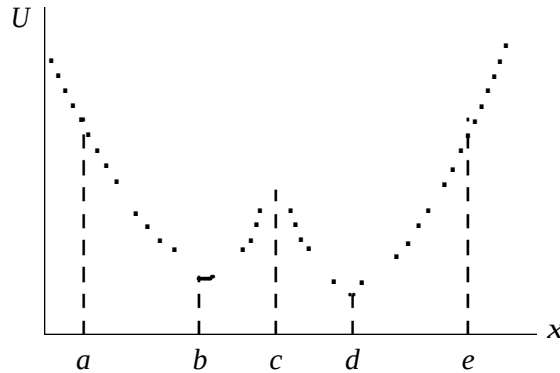
ans: D

42. A ball of mass m , at one end of a string of length L , rotates in a vertical circle just fast enough to prevent the string from going slack at the top of the circle. The speed of the ball at the bottom of the circle is:



- A. $\sqrt{2gL}$
 - B. $\sqrt{3gL}$
 - C. $\sqrt{4gL}$
 - D. $\sqrt{5gL}$
 - E. $\sqrt{7gL}$
- ans: D

43. A particle is released from rest at the point $x = a$ and moves along the x axis subject to the potential energy function $U(x)$ shown. The particle:



- A. moves to a point to the left of $x = e$, stops, and remains at rest
 - B. moves to a point to $x = e$, then moves to the left
 - C. moves to infinity at varying speed
 - D. moves to $x = b$, where it remains at rest
 - E. moves to $x = e$ and then to $x = d$, where it remains at rest
- ans: B

44. The potential energy of a particle moving along the x axis is given by

$$U(x) = (8.0 \text{ J/m}^2)x^2 + (2.0 \text{ J/m}^4)x^4.$$

If the total mechanical energy is 9.0 J, the limits of motion are:

- A. $-0.96 \text{ m}; +0.96 \text{ m}$
- B. $-2.2 \text{ m}; +2.2 \text{ m}$
- C. $-1.6 \text{ m}; +1.6 \text{ m}$
- D. $-0.96 \text{ m}; +2.2 \text{ m}$
- E. $-0.96 \text{ m}; +1.6 \text{ m}$

ans: A

45. The potential energy of a 0.20-kg particle moving along the x axis is given by

$$U(x) = (8.0 \text{ J/m}^2)x^2 + (2.0 \text{ J/m}^4)x^4.$$

When the particle is at $x = 1.0 \text{ m}$ it is traveling in the positive x direction with a speed of 5.0 m/s . It next stops momentarily to turn around at $x =$

- A. 0
- B. -1.1 m
- C. 1.1 m
- D. -2.3 m
- E. 2.3 m

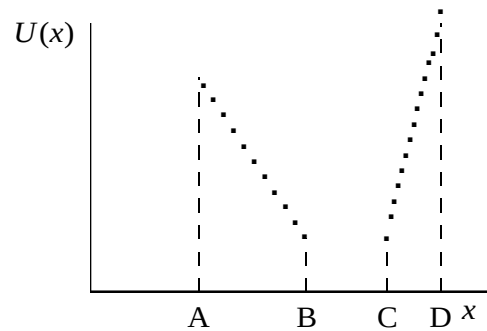
ans: C

46. Given a potential energy function $U(x)$, the corresponding force $\vec{F}$ is in the positive x direction if:

- A. U is positive
- B. U is negative
- C. U is an increasing function of x
- D. U is a decreasing function of x
- E. it is impossible to obtain the direction of $\vec{F}$ from U

ans: D

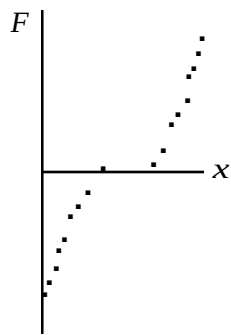
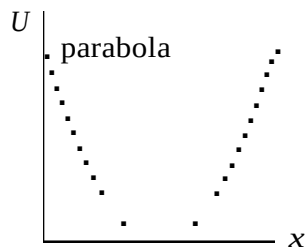
47. As a particle moves along the x axis it is acted upon by a conservative force. The potential energy is shown below as a function of the coordinate x of the particle. Rank the labeled regions according to the magnitude of the force, least to greatest.



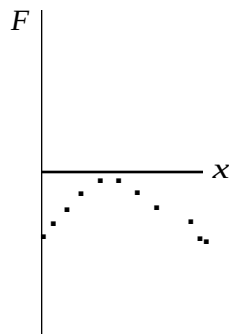
- A. AB, BC, CD
- B. AB, CD, BC
- C. BC, CD, AB
- D. BC, AB, CD
- E. CD, BC, AB

ans: D

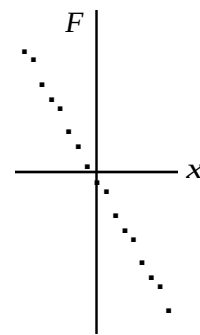
48. The first graph shows the potential energy $U(x)$ for a particle moving on the x axis. Which of the other five graphs correctly gives the force F exerted on the particle?



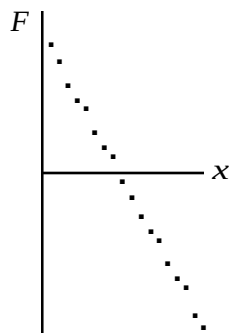
A



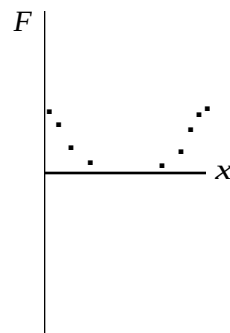
B



C



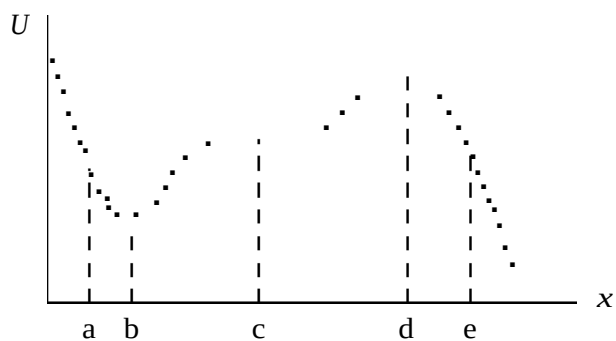
D



E

ans: D

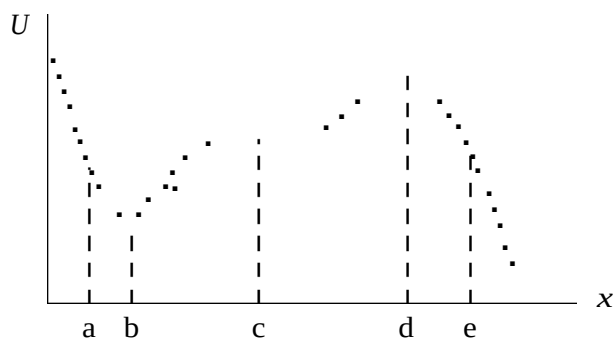
49. The diagram shows a plot of the potential energy as a function of x for a particle moving along the x axis. The points of stable equilibrium are:



- A. only a
- B. only b
- C. only c
- D. only d
- E. b and d

ans: B

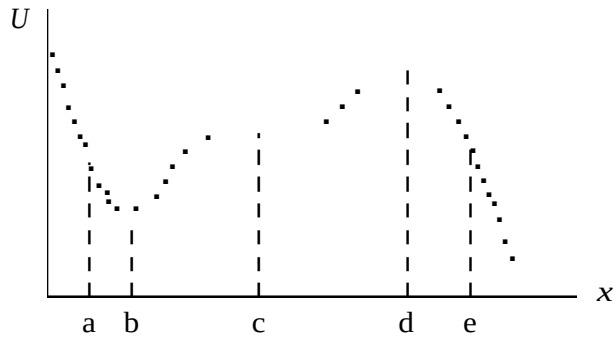
50. The diagram shows a plot of the potential energy as a function of x for a particle moving along the x axis. The points of unstable equilibrium are:



- A. only a
- B. only b
- C. only c
- D. only d
- E. b and d

ans: D

51. The diagram shows a plot of the potential energy as a function of x for a particle moving along the x axis. Of the labeled points, the points of neutral equilibrium are:



- A. only a
- B. only b
- C. only c
- D. only d
- E. b and d

ans: C

52. The potential energy of a body of mass m is given by $U = -mgx + \frac{1}{2}kx^2$. The corresponding force is:

- A. $-mgx^2/2 + kx^3/6$
- B. $mgx^2/2 - kx^3/6$
- C. $-mg + kx/2$
- D. $-mg + kx$
- E. $mg - kx$

ans: E

53. The potential energy of a 0.20-kg particle moving along the x axis is given by

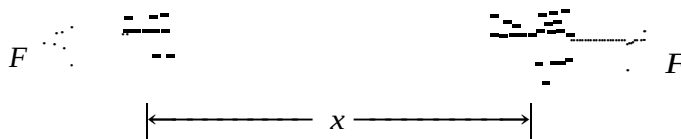
$$U(x) = (8.0 \text{ J/m}^2)x^2 + (2.0 \text{ J/m}^4)x^4.$$

When the particle is at $x = 1.0$ m the magnitude of its acceleration is:

- A. 0
- B. -8 m/s^2
- C. 8 m/s^2
- D. -40 m/s^2
- E. 40 m/s^2

ans: D

54. The potential energy for the interaction between the two atoms in a diatomic molecule is $U = A/x^{12} - B/x^6$, where A and B are constants and x is the interatomic distance. The magnitude of the force of one atom on the other is:



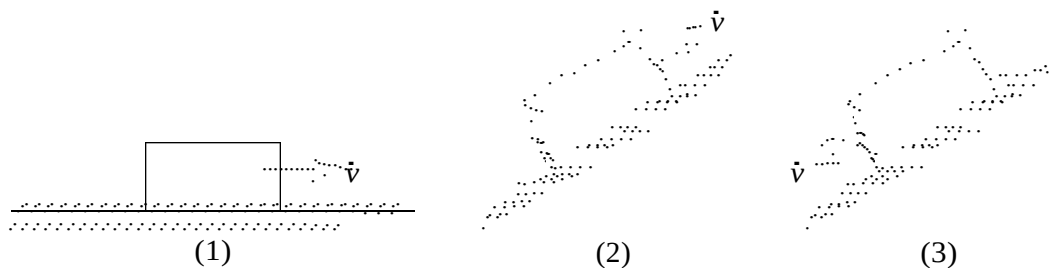
- A. $12A/x^{13} - 6B/x^7$
- B. $-13A/x^{13} + 7B/x^7$
- C. $-11A/x^{11} + 5B/x^5$
- D. $72A/x^{12} - 72B/x^6$
- E. $A/x^{13} - B/x^7$

ans: A

55. The thermal energy of a system consisting of a thrown ball, Earth, and the air is most closely associated with:
- A. the gravitational interaction of Earth and the ball
 - B. the kinetic energy of the ball as a whole
 - C. motions of the individual particles within the ball
 - D. motions of individual particles within the ball and the air
 - E. the kinetic energy of Earth as a whole

ans: D

56. Three identical blocks move either on a horizontal surface, up a plane, or down a plane, as shown below. They start with different speeds and continue to move until brought to rest by friction. They all move the same distance. Rank the three situations according to the initial speeds, least to greatest.



- A. The same for all cases
- B. 1, 2, 3
- C. 1, then 2 and 3 tie
- D. 3, 1, 2
- E. 2, 1, 3

ans: D

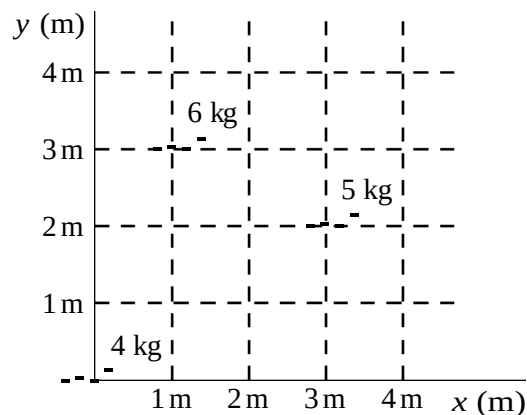
57. Objects A and B interact with each other via both conservative and nonconservative forces. Let K_A and K_B be the kinetic energies, U be the potential energy, and E_{int} be the thermal energy. If no external agent does work on the objects then:
- A. $K_A + U$ is conserved
 - B. $K_A + U + E_{\text{int}}$ is conserved
 - C. $K_A + K_B + E_{\text{int}}$ is conserved
 - D. $K_A + K_B + U$ is conserved
 - E. $K_A + K_B + U + E_{\text{int}}$ is conserved
- ans: E
58. A block slides across a rough horizontal table top. The work done by friction changes:
- A. only the kinetic energy
 - B. only the potential energy
 - C. only the internal energy
 - D. only the kinetic and potential energies
 - E. only the kinetic and internal energies
- ans: E
59. A 25-g ball is released from rest 80 m above the surface of Earth. During the fall the total internal energy of the ball and air increases by 15 J. Just before it hits the surface its speed is
- A. 19m/s
 - B. 36m/s
 - C. 40m/s
 - D. 45m/s
 - E. 53m/s
- ans: A
60. A 5-kg projectile is fired over level ground with a velocity of 200 m/s at an angle of 25° above the horizontal. Just before it hits the ground its speed is 150 m/s. Over the entire trip the change in the internal energy of the projectile and air is:
- A. +19, 000 J
 - B. -19,000J
 - C. +44, 000 J
 - D. -44,000J
 - E. 0
- ans: C
61. A 0.75-kg block slides on a rough horizontal table top. Just before it hits a horizontal ideal spring its speed is 3.5 m/s. It compresses the spring 5.7 cm before coming to rest. If the spring constant is 1200 N/m, the internal energy of the block and the table top must have:
- A. not changed
 - B. decreased by 1.9J
 - C. decreased by 2.6J
 - D. increased by 1.9J
 - E. increased by 2.6J
- ans: C

Chapter 9: CENTER OF MASS AND LINEAR MOMENTUM

1. Which one of the following statements is true?
- A. the center of mass of an object must lie within the object
 - B. all the mass of an object is actually concentrated at its center of mass
 - C. the center of mass of an object cannot move if there is zero net force on the object
 - D. the center of mass of a cylinder must lie on its axis
 - E. none of the above

ans: E

2. The x and y coordinates of the center of mass of the three-particle system shown below are:



- A. 0, 0
- B. 1.3 m, 1.7m
- C. 1.4 m, 1.9m
- D. 1.9 m, 2.5m
- E. 1.4 m, 2.5m

ans: C

3. The center of mass of a uniform disk of radius R is located:
- A. on the rim
 - B. a distance $R/2$ from the center
 - C. a distance $R/3$ from the center
 - D. a distance $2R/3$ from the center
 - E. at the center

ans: E

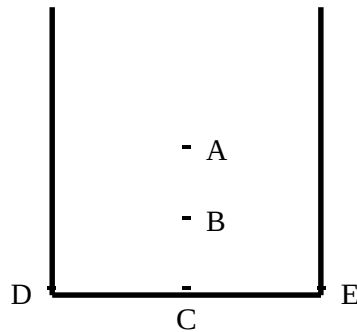
4. The center of mass of the system consisting of Earth, the Sun, and the planet Mars is:
- A. closer to Earth than to either of the other bodies
 - B. closer to the Sun than to either of the other bodies
 - C. closer to Mars than to either of the other bodies
 - D. at the geometric center of the triangle formed by the three bodies
 - E. at the center of the line joining Earth and Mars

ans: B

5. The center of mass of Earth's atmosphere is:
- a little less than halfway between Earth's surface and the outer boundary of the atmosphere
 - near the surface of Earth
 - near the outer boundary of the atmosphere
 - near the center of Earth
 - none of the above

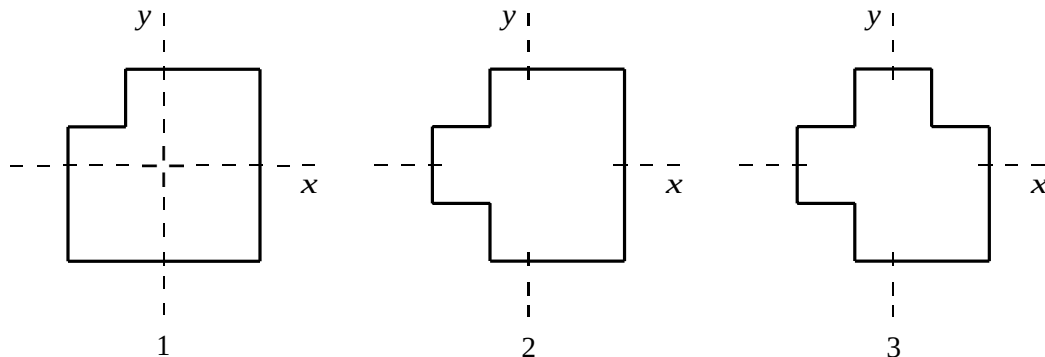
ans: D

6. A thick uniform wire is bent into the shape of the letter "U" as shown. Which point indicates the location of the center of mass of this wire?



ans: B

7. A machinist starts with three identical square plates but cuts one corner from one of them, two corners from the second, and three corners from the third. Rank the three plates according to the x coordinate of their centers of mass, from smallest to largest.



- 1, 2, 3
- 1 and 2 tie, then 3
- 1, then 2 and 3 tie
- 3, 2, 1
- 1 and 3 tie, then 2

ans: E

8. Block A, with a mass of 4 kg, is moving with a speed of 2.0 m/s while block B, with a mass of 8 kg, is moving in the opposite direction with a speed of 3 m/s. The center of mass of the two block-system is moving with a velocity of:
- A. 1.3 m/s in the same direction as A
 - B. 1.3 m/s in the same direction as B
 - C. 2.7 m/s in the same direction as A
 - D. 1.0 m/s in the same direction as B
 - E. 5.0 m/s in the same direction as A

ans: B

9. At the same instant that a 0.50-kg ball is dropped from 25 m above Earth, a second ball, with a mass of 0.25 kg, is thrown straight upward from Earth's surface with an initial speed of 15 m/s. They move along nearby lines and pass each other without colliding. At the end of 2.0 s the height above Earth's surface of the center of mass of the two-ball system is:

- A. 2.9m
- B. 4.0m
- C. 5.0m
- D. 7.1m
- E. 10.4m

ans: D

10. At the same instant that a 0.50-kg ball is dropped from 25 m above Earth, a second ball, with a mass of 0.25 kg, is thrown straight upward from Earth's surface with an initial speed of 15 m/s. They move along nearby lines and pass without colliding. At the end of 2.0 s the velocity of the center of mass of the two-ball system is:

- A. 11 m/s, down
- B. 11 m/s, up
- C. 15 m/s, down
- D. 15 m/s, up
- E. 20 m/s, down

ans: C

11. At the same instant that a 0.50-kg ball is dropped from 25 m above Earth, a second ball, with a mass of 0.25 kg, is thrown straight upward from Earth's surface with an initial speed of 15 m/s. They move along nearby lines and pass without colliding. At the end of 2.0 s the magnitude of the acceleration of the center of mass of the two-ball system is:

- A. $0.25g$
- B. $0.50g$
- C. $0.75g$
- D. g
- E. $g/0.75$

ans: D

12. A light rope passes over a light frictionless pulley attached to the ceiling. An object with a large mass is tied to one end and an object with a smaller mass is tied to the other end. Starting from rest the heavier object moves downward and the lighter object moves upward with the same magnitude acceleration. Which of the following statements is true for the system consisting of the two masses?
- A. The center of mass remains at rest.
 - B. The net external force is zero.
 - C. The velocity of the center of mass is a constant.
 - D. The acceleration of the center of mass is g , downward.
 - E. None of the above statements are true.

ans: E

13. Two 4.0-kg blocks are tied together with a compressed spring between them. They are thrown from the ground with an initial velocity of 35 m/s, 45° above the horizontal. At the highest point of the trajectory they become untied and spring apart. About how far below the highest point is the center of mass of the two-block system 2.0 s later, before either fragment has hit the ground?
- A. 12 m
 - B. 20 m
 - C. 31 m
 - D. Can't tell because the velocities of the fragments are not given.
 - E. Can't tell because the coordinates of the highest point are not given.

ans: B

14. The center of mass of a system of particles has a constant velocity if:
- A. the forces exerted by the particles on each other sum to zero
 - B. the external forces acting on particles of the system sum to zero
 - C. the velocity of the center of mass is initially zero
 - D. the particles are distributed symmetrically around the center of mass
 - E. the center of mass is at the geometric center of the system

ans: B

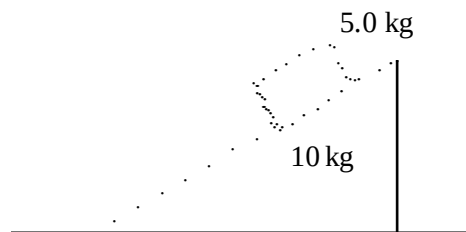
15. The center of mass of a system of particles remains at the same place if:
- A. it is initially at rest and the external forces sum to zero
 - B. it is initially at rest and the internal forces sum to zero
 - C. the sum of the external forces is less than the maximum force of static friction
 - D. no friction acts internally
 - E. none of the above

ans: A

16. A man sits in the back of a canoe in still water. He then moves to the front of the canoe and sits there. Afterwards the canoe:
- A. is forward of its original position and moving forward
 - B. is forward of its original position and moving backward
 - C. is rearward of its original position and moving forward
 - D. is rearward of its original position and moving backward
 - E. is rearward of its original position and not moving

ans: E

17. A 640-N hunter gets a rope around a 3200-N polar bear. They are stationary, 20 m apart, on frictionless level ice. When the hunter pulls the polar bear to him, the polar bear will move:
- 1.0m
 - 3.3m
 - 10m
 - 12m
 - 17m
- ans: B
18. Two boys, with masses of 40 kg and 60 kg, respectively, stand on a horizontal frictionless surface holding the ends of a light 10-m long rod. The boys pull themselves together along the rod. When they meet the 60-kg boy will have moved what distance?
- 4m
 - 5m
 - 6m
 - 10m
 - need to know the forces they exert
- ans: A
19. The center of mass of a system of particles obeys an equation similar to Newton's second law $\vec{F} = m\vec{a}_{\text{com}}$, where:
- $\vec{F}$ is the net internal force and m is the total mass of the system
 - $\vec{F}$ is the net internal force and m is the mass acting on the system
 - $\vec{F}$ is the net external force and m is the total mass of the system
 - $\vec{F}$ is the force of gravity and m is the mass of Earth
 - $\vec{F}$ is the force of gravity and m is the total mass of the system
- ans: C
20. A large wedge with a mass of 10 kg rests on a horizontal frictionless surface, as shown. A block with a mass of 5.0 kg starts from rest and slides down the inclined surface of the wedge, which is rough. At one instant the vertical component of the block's velocity is 3.0 m/s and the horizontal component is 6.0 m/s. At that instant the velocity of the wedge is:



- 3.0 m/s to the left
- 3.0m/s to the right
- 6.0m/s to the right
- 6.0 m/s to the left
- 17 m/s to the right

ans: B

21. A 2.0-kg block is attached to one end of a spring with a spring constant of 100 N/m and a 4.0-kg block is attached to the other end. The blocks are placed on a horizontal frictionless surface and set into motion. At one instant the 2.0-kg block is observed to be traveling to the right with a speed of 0.50 m/s and the 4.0-kg block is observed to be traveling to the left with a speed of 0.30 m/s. Since the only forces on the blocks are the force of gravity, the normal force of the surface, and the force of the spring, we conclude that:
- A. the spring is compressed at the time of the observation
 - B. the spring is not compressed at the time of observation
 - C. the motion was started with the masses at rest
 - D. the motion was started with at least one of masses moving
 - E. the motion was started by compressing the spring

ans: D

22. A 2.0-kg mass is attached to one end of a spring with a spring constant of 100 N/m and a 4.0-kg mass is attached to the other end. The masses are placed on a horizontal frictionless surface and the spring is compressed 10 cm. The spring is then released with the masses at rest and the masses oscillate. When the spring has its equilibrium length for the first time the 2.0-kg mass has a speed of 0.36 m/s. The mechanical energy that has been lost to the instant is:
- A. zero
 - B. 0.31J
 - C. 0.61J
 - D. 0.81 J
 - E. 1.2J

ans: B

23. Momentum may be expressed in:
- A. kg/m
 - B. gram·s
 - C. N·s
 - D. kg/(m·s)
 - E. N/s

ans: C

24. The momentum of an object at a given instant is independent of its:
- A. inertia
 - B. mass
 - C. speed
 - D. velocity
 - E. acceleration

ans: E

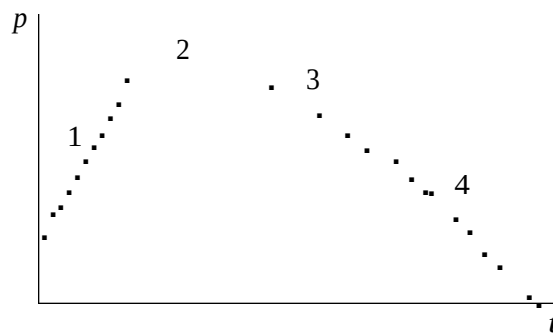
25. Two bodies, A and B, have equal kinetic energies. The mass of A is nine times that of B. The ratio of the momentum of A to that of B is:
- 1:9
 - 1:3
 - 1:1
 - 3:1
 - 9:1

ans: D

26. Two objects, P and Q, have the same momentum. Q has more kinetic energy than P if it:
- weighs more than P
 - is moving faster than P
 - weighs the same as P
 - is moving slower than P
 - is moving at the same speed as P

ans: B

27. A particle moves along the x axis. Its momentum is graphed below as a function of time. Rank the numbered regions according to the magnitude of the force acting on the particle, least to greatest.



- 1, 2, 3, 4
- 2, 3, 4, 1
- 1, 4, 3, 2
- 1, 3, 4, 2
- 2, 4, 3, 1

ans: B

28. A 1.0-kg ball moving at 2.0 m/s perpendicular to a wall rebounds from the wall at 1.5 m/s. The change in the momentum of the ball is:
- zero
 - $0.5\text{N} \cdot \text{s}$ away from wall
 - $0.5\text{N} \cdot \text{s}$ toward wall
 - $3.5\text{N} \cdot \text{s}$ away from wall
 - $3.5\text{N} \cdot \text{s}$ toward wall

ans: D

29. If the total momentum of a system is changing:
- A. particles of the system must be exerting forces on each other
 - B. the system must be under the influence of gravity
 - C. the center of mass must have constant velocity
 - D. a net external force must be acting on the system
 - E. none of the above
- ans: D
30. When you step on the accelerator to increase the speed of your car, the force that accelerates the car is:
- A. the force of your foot on the accelerator
 - B. the force of friction of the road on the tires
 - C. the force of the engine on the drive shaft
 - D. the normal force of the road on the tires
 - E. none of the above
- ans: B
31. A 2.5-kg stone is released from rest and falls toward Earth. After 4.0 s, the magnitude of its momentum is:
- A. 98 kg· m/s
 - B. 78 kg· m/s
 - C. 39 kg· m/s
 - D. 24 kg· m/s
 - E. zero
- ans: A
32. A 64-kg woman stands on frictionless level ice with a 0.10-kg stone at her feet. She kicks the stone with her foot so that she acquires a velocity of 0.0017 m/s in the forward direction. The velocity acquired by the stone is:
- A. 1.1 m/s forward
 - B. 1.1 m/s backward
 - C. 0.0017 m/s forward
 - D. 0.0017 m/s backward
 - E. none of these
- ans: B
33. A man is marooned at rest on level frictionless ice. In desperation, he hurls his shoe to the right at 15 m/s. If the man weighs 720 N and the shoe weighs 4.0 N, the man moves to the left with a speed of:
- A. 0
 - B. 2.1×10^{-2} m/s
 - C. 8.3×10^{-2} m/s
 - D. 15 m/s
 - E. 2.7×10^3 m/s
- ans: C

34. Two spacemen are floating together with zero speed in a gravity-free region of space. The mass of spaceman A is 120 kg and that of spaceman B is 90 kg. Spaceman A pushes B away from him with B attaining a final speed of 0.5 m/s. The final recoil speed of A is:
- A. zero
 - B. 0.38 m/s
 - C. 0.5 m/s
 - D. 0.67 m/s
 - E. 1.0 m/s

ans: B

35. A projectile in flight explodes into several fragments. The total momentum of the fragments immediately after this explosion:
- A. is the same as the momentum of the projectile immediately before the explosion
 - B. has been changed into kinetic energy of the fragments
 - C. is less than the momentum of the projectile immediately before the explosion
 - D. is more than the momentum of the projectile immediately before the explosion
 - E. has been changed into radiant energy

ans: A

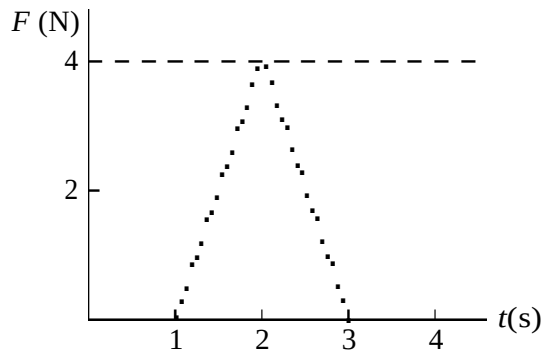
36. A rifle of mass M is initially at rest but free to recoil. It fires a bullet of mass m and velocity v (relative to the ground). After firing, the velocity of the rifle (relative to the ground) is:
- A. $-mv$
 - B. $-Mv/m$
 - C. $-mv/M$
 - D. $-v$
 - E. mv/M

ans: C

37. Bullets from two revolvers are fired with the same velocity. The bullet from gun #1 is twice as heavy as the bullet from gun #2. Gun #1 weighs three times as much as gun #2. The ratio of the momentum imparted to gun #1 to that imparted to gun #2 is:
- A. 2:3
 - B. 3:2
 - C. 2:1
 - D. 3:1
 - E. 6:1

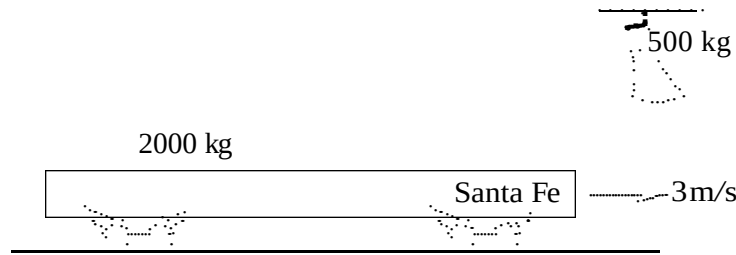
ans: C

38. A 5-kg object can move along the x axis. It is subjected to a force $\vec{F}$ in the positive x direction; a graph of F as a function of time t is shown below. Over the time the force is applied the change in the velocity of the object is:



- A. 0.8m/s
B. 1.1m/s
C. 1.6m/s
D. 2.3m/s
E. 4.0m/s
ans: A
39. Force:
- A. equals the negative integral (with respect to distance) of the potential energy function
B. is the ability to do work
C. is the rate of change of doing work
D. equals the time rate of change of momentum
E. has dimensions of momentum multiplied by time
ans: D
40. Cart A, with a mass of 0.20 kg, travels on a horizontal air track at 3.0 m/s and hits cart B, which has a mass of 0.40 kg and is initially traveling away from A at 2.0 m/s. After the collision the center of mass of the two cart system has a speed of:
- A. zero
B. 0.33 m/s
C. 2.3m/s
D. 2.5m/s
E. 5.0m/s
ans: B

41. A 500-kg sack of coal is dropped on a 2000-kg railroad flatcar which was initially moving at 3 m/s as shown. After the sack rests on the flatcar, the speed of the flatcar is:

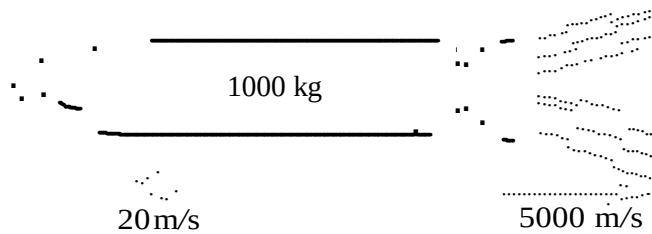


- A. 0.6m/s
 B. 1.2m/s
 C. 1.8m/s
 D. 2.4m/s
 E. 3.6m/s
 ans: D
42. A cart loaded with sand slides forward along a horizontal frictionless track. As the cart moves, sand trickles out at a constant rate through a hole in the back of the cart. The acceleration of the cart is:
- A. constant and in the forward direction
 B. constant and in the backward direction
 C. variable and in the forward direction
 D. variable and in the backward direction
 E. zero
 ans: E
43. The thrust of a rocket is:
- A. a gravitational force acting on the rocket
 B. the force of the exiting fuel gases on the rocket
 C. any force that is external to the rocket-fuel system
 D. a force that arises from the reduction in mass of the rocket-fuel system
 E. none of the above
 ans: B:
44. At one instant of time a rocket is traveling in outer space at 2500 m/s and is exhausting fuel at a rate of 100 kg/s. If the speed of the fuel as it leaves the rocket is 1500 m/s, relative to the rocket, the thrust is:
- A. 0
 B. 1.0×10^5 N
 C. 1.5×10^6 N
 D. 2.9×10^6 N
 E. 2.5×10^5 N
 ans: C

45. A rocket exhausts fuel with a velocity of 1500 m/s, relative to the rocket. It starts from rest in outer space with fuel comprising 80 per cent of the total mass. When all the fuel has been exhausted its speed is:
- A. 3600 m/s
 - B. 2400 m/s
 - C. 1200 m/s
 - D. 880 m/s
 - E. 400 m/s

ans: B

46. A 1000-kg space probe is motionless in space. To start moving, its main engine is fired for 5 s during which time it ejects exhaust gases at 5000 m/s. At the end of this process it is moving at 20 m/s. The approximate mass of the ejected gas is:



- A. 0.8 kg
- B. 4 kg
- C. 5 kg
- D. 20 kg
- E. 25 kg

ans: B

47. The physical quantity “impulse” has the same dimensions as that of:
- A. force
 - B. power
 - C. energy
 - D. momentum
 - E. work

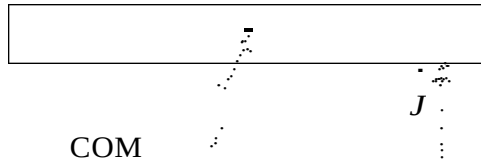
ans: D

48. The law of conservation of momentum applies to a system of colliding objects only if:
- A. there is no change in kinetic energy of the system
 - B. the coefficient of restitution is one
 - C. the coefficient of restitution is zero
 - D. the net external impulse is zero
 - E. the collisions are all elastic

ans: D

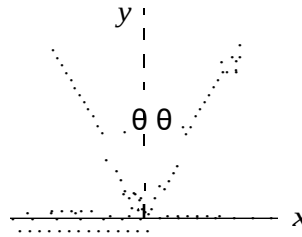
49. Sphere X, of mass 2 kg, is moving to the right at 10 m/s. Sphere Y, of mass 4 kg, is moving to the left at 10 m/s. The two spheres collide head-on. The magnitude of the impulse of X on Y is:
- A. twice the magnitude of the impulse of Y on X
 - B. half the magnitude of the impulse of Y on X
 - C. one-fourth the magnitude of the impulse of Y on X
 - D. four times the magnitude of the impulse of Y on X
 - E. the same as the magnitude of the impulse of Y on X
- ans: E
50. Two bodies of unequal mass, placed at rest on a frictionless surface, are acted on by equal horizontal forces for equal times. Just after these forces are removed, the body of greater mass will have:
- A. the greater speed
 - B. the greater acceleration
 - C. the smaller momentum
 - D. the greater momentum
 - E. the same momentum as the other body
- ans: E
51. A 0.2-kg rubber ball is dropped from the window of a building. It strikes the sidewalk below at 30 m/s and rebounds up at 20 m/s. The impulse on the ball during the collision is:
- A. 10 N · s upward
 - B. 10 N · s downward
 - C. 2.0 N · s upward
 - D. 2.0 N · s downward
 - E. 9.8 N · s upward
- ans: A
52. A 10-kg block of ice is at rest on a frictionless horizontal surface. A 1.0-N force is applied in an easterly direction for 1.0 s. During this time interval, the block:
- A. acquires a speed of 1 m/s
 - B. moves 10 cm
 - C. acquires a momentum of 1.0 kg · m/s
 - D. acquires a kinetic energy of 0.1 J
 - E. none of the above
- ans: C

53. A uniform narrow bar, resting on ice, is given a transverse horizontal impulse J at one end as shown. The center of mass of the bar COM will then:



- A. remain at rest
 - B. move in a circle
 - C. move in a straight line
 - D. move in a parabola
 - E. move along some other curve
- ans: C
54. What magnitude impulse will give a 2.0-kg object a momentum change of magnitude $+50 \text{ kg} \cdot \text{m/s}$?
- A. $+25 \text{ N} \cdot \text{s}$
 - B. $-25 \text{ N} \cdot \text{s}$
 - C. $+50 \text{ N} \cdot \text{s}$
 - D. $-50 \text{ N} \cdot \text{s}$
 - E. $+100 \text{ N} \cdot \text{s}$
- ans: C
55. A student's life was saved in an automobile accident because an airbag expanded in front of his head. If the car had not been equipped with an airbag, the windshield would have stopped the motion of his head in a much shorter time. Compared to the windshield, the airbag:
- A. causes a much smaller change in momentum
 - B. exerts a much smaller impulse
 - C. causes a much smaller change in kinetic energy
 - D. exerts a much smaller force
 - E. does much more work
- ans: D

56. A ball hits a wall and rebounds with the same speed, as diagramed below. The changes in the components of the momentum of the ball are:



- A. $\Delta p_x > 0, \Delta p_y > 0$
- B. $\Delta p_x < 0, \Delta p_y > 0$
- C. $\Delta p_x = 0, \Delta p_y > 0$
- D. $\Delta p_x = 0, \Delta p_y < 0$
- E. $\Delta p_x > 0, \Delta p_y < 0$

ans: C

57. A golf ball of mass m is hit by a golf club so that the ball leaves the tee with speed v . The club is in contact with the ball for time T . The magnitude of the average force on the club on the ball during the time T is:

- A. mvT
- B. mv/T
- C. $(1/2)mv^2T$
- D. $mv^2/(2T)$
- E. $mT^2/(2v)$

ans: B

58. A 640-N acrobat falls 5.0 m from rest into a net. The net tosses him back up with the same speed he had just before he hit the net. The magnitude of the average upward force exerted on him by the net during this collision is:

- A. 32 N
- B. 64 N
- C. 320 N
- D. 640 N
- E. impossible to determine from given data

ans: E

59. Whenever an object strikes a stationary object of equal mass:

- A. the two objects cannot stick together
- B. the collision must be elastic
- C. the first object must stop
- D. momentum is not necessarily conserved
- E. none of the above

ans: E

60. For a two-body collision involving objects with different masses, a frame of reference which has the same velocity relative to the laboratory as does the center of mass of the two objects is:
- A. a frame for which the momentum of the incident object is zero
 - B. a frame for which the momentum of the target object is zero
 - C. a frame for which the average momentum of the two objects is zero
 - D. a frame for which the total momentum of the two objects is zero
 - E. none of the above

ans: D

61. An inelastic collision is one in which:
- A. momentum is not conserved but kinetic energy is conserved
 - B. total mass is not conserved but momentum is conserved
 - C. neither kinetic energy nor momentum is conserved
 - D. momentum is conserved but kinetic energy is not conserved
 - E. the total impulse is equal to the change in kinetic energy

ans: D

62. A 4.0-N puck is traveling at 3.0 m/s. It strikes a 8.0-N puck, which is stationary. The two pucks stick together. Their common final speed is:
- A. 1.0m/s
 - B. 1.5m/s
 - C. 2.0m/s
 - D. 2.3m/s
 - E. 3.0m/s

ans: A

63. A 3.00-g bullet traveling horizontally at 400 m/s hits a 3.00-kg wooden block, which is initially at rest on a smooth horizontal table. The bullet buries itself in the block without passing through. The speed of the block after the collision is:
- A. 1.33m/s
 - B. 0.40m/s
 - C. 12.0m/s
 - D. 40.0m/s
 - E. 160 m/s

ans: B

64. A 2-kg cart, traveling on a horizontal air track with a speed of 3 m/s, collides with a stationary 4-kg cart. The carts stick together. The impulse exerted by one cart on the other has a magnitude of:
- A. 0
 - B. $4\text{ N} \cdot \text{s}$
 - C. $6\text{ N} \cdot \text{s}$
 - D. $9\text{ N} \cdot \text{s}$
 - E. $12\text{ N} \cdot \text{s}$

ans: B

65. A 3-g bullet is fired horizontally into a 10-kg block of wood suspended by a rope from the ceiling. The block swings in an arc, rising 3 mm above its lowest position. The velocity of the bullet was:
- A. unknown since the heat generated in the collision was not given
 - B. 8.0×10^2 m/s
 - C. 24.0m/s
 - D. 8.00m/s
 - E. 2.4×10^4 m/s
- ans: B
66. A 3.0-kg and a 2.0-kg cart approach each other on a horizontal air track. They collide and stick together. After the collision their total kinetic energy is 40 J. The speed of their center of mass is:
- A. zero
 - B. 2.8m/s
 - C. 4.0m/s
 - D. 5.2m/s
 - E. 6.3m/s
- ans: C
67. Blocks A and B are moving toward each other. A has a mass of 2.0 kg and a velocity of 50 m/s, while B has a mass of 4.0 kg and a velocity of ~~25~~ 25 m/s. They suffer a completely inelastic collision. The kinetic energy lost during the collision is:
- A. 0
 - B. 1250 J
 - C. 3750 J
 - D. 5000 J
 - E. 5600 J
- ans: C
68. For a completely inelastic two-body collision the kinetic energy retained by the objects is the same as:
- A. the total kinetic energy before the collision
 - B. the difference in the kinetic energies of the objects before the collision
 - C. $\frac{1}{2} M v_{\text{com}}^2$, where M is the total mass and v_{com} is the velocity of the center of mass
 - D. the kinetic energy of the more massive body before the collision
 - E. the kinetic energy of the less massive body before the collision
- ans: C
69. A 75-kg man is riding in a 30-kg cart at 2.0 m/s. He jumps off in such a way as to land on the ground with no horizontal velocity. The resulting change in speed of the cart is:
- A. zero
 - B. 2.0m/s
 - C. 3.0m/s
 - D. 5.0m/s
 - E. 7.0m/s
- ans: D

70. An elastic collision is one in which:
- A. momentum is not conserved but kinetic energy is conserved
 - B. total mass is not conserved but momentum is conserved
 - C. kinetic energy and momentum are both conserved
 - D. momentum is conserved but kinetic energy is not conserved
 - E. the total impulse is equal to the change in kinetic energy
- ans: C
71. Object A strikes the stationary object B head-on in an elastic collision. The mass of A is fixed, you may choose the mass of B appropriately. Then:
- A. for B to have the greatest recoil speed, choose $m_B = m_A$
 - B. for B to have the greatest recoil momentum, choose $m_B \gg m_A$
 - C. for B to have the greatest recoil kinetic energy, choose $m_B \gg m_A$
 - D. for B to have the least recoil speed, choose $m_B = m_A$
 - E. for B to have the greatest recoil kinetic energy, choose $m_B = m_A$
- ans: E
72. Block A, with a mass of 2.0 kg, moves along the x axis with a velocity of 5.0 m/s in the positive x direction. It suffers an elastic collision with block B, initially at rest, and the blocks leave the collision along the x axis. If B is much more massive than A, the speed of A after the collision is:
- A. 0
 - B. +5.0m/s
 - C. -5.0 m/s
 - D. +10m/s
 - E. -10 m/s
- ans: C
73. A very massive object traveling at 10 m/s strikes a very light object, initially at rest, and the light object moves off in the direction of travel of the heavy object. If the collision is elastic, the speed of the lighter object is:
- A. 5.0 m/s
 - B. 10m/s
 - C. 15m/s
 - D. 20m/s
 - E. Can't tell from the information given.
- ans: D
74. Sphere A has mass m and is moving with velocity v . It makes a head-on elastic collision with a stationary sphere B of mass $2m$. After the collision their speeds (v_A and v_B) are:
- A. 0, $v/2$
 - B. $-v/3$, $2v/3$
 - C. $-v$, v
 - D. $-2v/3$, $v/3$
 - E. none of these
- ans: B

75. Blocks A and B are moving toward each other along the x axis. A has a mass of 2.0 kg and a velocity of 50 m/s, while B has a mass of 4.0 kg and a velocity of 25 m/s. They suffer an elastic collision and move off along the x axis. The kinetic energy transferred from A to B during the collision is:

A. 0
B. 2500 J
C. 5000 J
D. 7500 J
E. 10000 J

ans: A

76. When a particle suffers a head-on elastic collision with another particle, initially at rest, the greatest fraction of kinetic energy is transferred if:

A. the incident particle is initially traveling very fast
B. the incident particle is traveling very slowly
C. the incident particle is much more massive than the target particle
D. the incident particle is much less massive than the target particle
E. the incident and target particle have the same mass

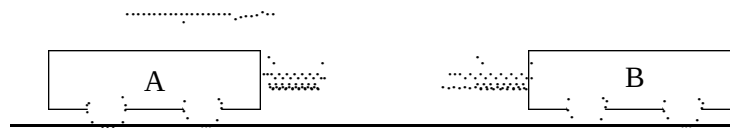
ans: E

77. Two objects, X and Y, are held at rest on a horizontal frictionless surface and a spring is compressed between them. The mass of X is $\frac{2}{5}$ times the mass of Y. Immediately after the spring is released, X has a kinetic energy of 50 J and Y has a kinetic energy of:

A. 20 J
B. 8 J
C. 310 J
D. 125 J
E. 50 J

ans: D

78. Two carts (A and B), having spring bumpers, collide as shown. Cart A has a mass of 2 kg and is initially moving to the right. Cart B has a mass of 3 kg and is initially stationary. When the separation between the carts is a minimum:



A. cart B is still at rest
B. cart A has come to rest
C. the carts have the same momentum
D. the carts have the same kinetic energy
E. the kinetic energy of the system is at a minimum

ans: E

79. Two identical carts travel at 1 m/s in opposite directions on a common horizontal surface. They collide head-on and are reported to rebound, each with a speed of 2 m/s. Then:
- momentum was not conserved; therefore, the report must be false
 - if some other form of energy were changed to kinetic during the collision, the report could be true
 - if the collision were elastic, the report could be true
 - if friction were present, the report could be true
 - if the duration of the collision were long enough, the report could be true

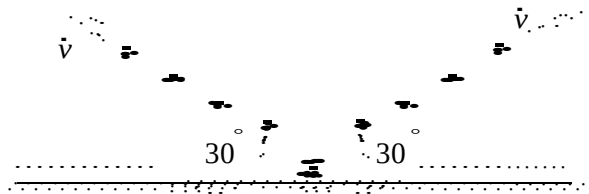
ans: B

80. A block moves at 5.0 m/s in the positive x direction and hits an identical block, initially at rest. A small amount of gunpowder had been placed on one of the blocks. The explosion does not harm the blocks but it doubles their total kinetic energy. After the explosion the blocks move along the x axis and the incident block has a speed in of:

- 1.8m/s
- 5.0m/s
- 6.8m/s
- 7.1m/s
- 11.8 m/s

ans: A

81. A stream of gas consists of n molecules. Each molecule has mass m and speed v . The stream is reflected elastically from a rigid surface as shown. The magnitude of the change in the total momentum of the stream is:



- $2mnv$
- $2mnv \sin 30^\circ$
- $mnv \sin 30^\circ$
- $mnv \cos 30^\circ$
- mnv

ans: B

Chapter 10: ROTATION

1. A radian is about:

- A. 25°
- B. 37°
- C. 45°
- D. 57°
- E. 90°

ans: D

2. One revolution is the same as:

- A. 1 rad
- B. 57 rad
- C. $\pi/2$ rad
- D. π rad
- E. 2π rad

ans: E

3. One revolution per minute is about:

- A. 0.0524 rad/s
- B. 0.105 rad/s
- C. 0.95 rad/s
- D. 1.57 rad/s
- E. 6.28 rad/s

ans: B

4. If a wheel turns with constant angular speed then:

- A. each point on its rim moves with constant velocity
- B. each point on its rim moves with constant acceleration
- C. the wheel turns through equal angles in equal times
- D. the angle through which the wheel turns in each second increases as time goes on
- E. the angle through which the wheel turns in each second decreases as time goes on

ans: C

5. If a wheel is turning at 3.0 rad/s, the time it takes to complete one revolution is about:

- A. 0.33 s
- B. 0.67 s
- C. 1.0 s
- D. 1.3 s
- E. 2.1 s

ans: E

6. If wheel turning at a constant rate completes 100 revolutions in 10 s its angular speed is:
- A. 0.31 rad/s
 - B. 0.63 rad/s
 - C. 10 rad/s
 - D. 31 rad/s
 - E. 63 rad/s

ans: E

7. The angular speed of the second hand of a watch is:
- A. $(\pi/1800)$ rad/s
 - B. $(\pi/60)$ m/s
 - C. $(\pi/30)$ m/s
 - D. (2π) m/s
 - E. (60) m/s

ans: C

8. The angular speed of the minute hand of a watch is:
- A. $(60/\pi)$ m/s
 - B. $(1800/\pi)$ m/s
 - C. (π) m/s
 - D. $(\pi/1800)$ m/s
 - E. $(\pi/60)$ m/s

ans: D

9. A flywheel is initially rotating at 20 rad/s and has a constant angular acceleration. After 9.0s it has rotated through 450 rad. Its angular acceleration is:
- A. 3.3 rad/s
 - B. 4.4 rad/s
 - C. 5.6 rad/s
 - D. 6.7 rad/s
 - E. 11 rad/s

ans: D

10. Ten seconds after an electric fan is turned on, the fan rotates at 300 rev/min. Its average angular acceleration is:
- A. 3.14 rad/s^2
 - B. 30 rad/s^2
 - C. 30 rev/s^2
 - D. 50 rev/min^2
 - E. 1800 rev/s^2

ans: A

11. A wheel rotates with a constant angular acceleration of $\pi \text{ rad/s}^2$. During a certain time interval its angular displacement is $\pi \text{ rad}$. At the end of the interval its angular velocity is $2\pi \text{ rad/s}$. Its angular velocity at the beginning of the interval is:
- A. zero
 - B. 1 rad/s
 - C. $\pi \text{ rad/s}$
 - D. $\pi \sqrt{2} \text{ rad/s}$
 - E. $2\pi \text{ rad/s}$
- ans: D
12. A flywheel rotating at 12 rev/s is brought to rest in 6 s . The magnitude of the average angular acceleration in rad/s^2 of the wheel during this process is:
- A. $1/\pi$
 - B. 2
 - C. 4
 - D. 4π
 - E. 72
- ans: D
13. A phonograph turntable, initially rotating at 0.75 rev/s , slows down and stops in 30 s . The magnitude of its average angular acceleration in rad/s^2 for this process is:
- A. 1.5
 - B. 1.5π
 - C. $\pi/40$
 - D. $\pi/20$
 - E. 0.75
- ans: D
14. The angular velocity of a rotating wheel increases by 2 rev/s every minute. The angular acceleration in rad/s^2 of this wheel is:
- A. $4\pi^2$
 - B. 2π
 - C. $1/30$
 - D. $\pi/15$
 - E. 4π
- ans: D
15. A wheel initially has an angular velocity of 18 rad/s . It has a constant angular acceleration of 2.0 rad/s^2 and is slowing at first. What time elapses before its angular velocity is 18 rad/s in the direction opposite to its initial angular velocity?
- A. 3.0 s
 - B. 6.0 s
 - C. 9.0 s
 - D. 18 s
 - E. 36 s
- ans: D

16. A wheel initially has an angular velocity of 36 rad/s but after 6.0 s its angular velocity is 24 rad/s . If its angular acceleration is constant its value is:
- A. 2.0 rad/s^2
 - B. -2.0 rad/s^2
 - C. 3.0 rad/s^2
 - D. -3.0 rad/s^2
 - E. 6.0 rad/s^2
- ans: B
17. A wheel initially has an angular velocity of -36 rad/s but after 6.0 s its angular velocity is -24 rad/s . If its angular acceleration is constant the value is:
- A. 2.0 rad/s^2
 - B. -2.0 rad/s^2
 - C. 3.0 rad/s^2
 - D. -3.0 rad/s^2
 - E. -6.0 rad/s^2
- ans: A
18. A wheel initially has an angular velocity of 18 rad/s but it is slowing at a rate of 2.0 rad/s^2 . By the time it stops it will have turned through:
- A. 81 rad
 - B. 160 rad
 - C. 245 rad
 - D. 330 rad
 - E. 410 rad
- ans: A
19. A wheel starts from rest and has an angular acceleration of 4.0 rad/s^2 . When it has made 10 rev its angular velocity is:
- A. 16 rad/s
 - B. 22 rad/s
 - C. 32 rad/s
 - D. 250 rad/s
 - E. 500 rad/s
- ans: B
20. A wheel starts from rest and has an angular acceleration of 4.0 rad/s^2 . The time it takes to make 10 rev is:
- A. 0.50 s
 - B. 0.71 s
 - C. 2.2 s
 - D. 2.8 s
 - E. 5.6 s
- ans: E

21. A wheel starts from rest and has an angular acceleration that is given by $\alpha(t) = (6 \text{ rad/s}^4)t^2$. The angle through which it turns in time t is given by:
- A. $[(1/8)t^4] \text{ rad}$
 - B. $[(1/4)t^4] \text{ rad}$
 - C. $[(1/2)t^4] \text{ rad}$
 - D. $(t^4) \text{ rad}$
 - E. 12 rad
- ans: C
22. A wheel starts from rest and has an angular acceleration that is given by $\alpha(t) = (6.0 \text{ rad/s}^4)t^2$. The time it takes to make 10 rev is:
- A. 2.8s
 - B. 3.3s
 - C. 4.0s
 - D. 4.7s
 - E. 5.3s
- ans: B
23. A wheel starts from rest and has an angular acceleration that is given by $\alpha(t) = (6.0 \text{ rad/s}^4)t^2$. After it has turned through 10 rev its angular velocity is:
- A. 63rad/s
 - B. 75rad/s
 - C. 89rad/s
 - D. 130 rad/s
 - E. 210 rad/s
- ans: B
24. A wheel is spinning at 27 rad/s but is slowing with an angular acceleration that has a magnitude given by $(3.0 \text{ rad/s}^4)t^2$. It stops in a time of:
- A. 1.7s
 - B. 2.6s
 - C. 3.0s
 - D. 4.4s
 - E. 7.3s
- ans: C
25. If the angular velocity vector of a spinning body points out of the page then, when viewed from above the page, the body is spinning:
- A. clockwise about an axis that is perpendicular to the page
 - B. counterclockwise about an axis that is perpendicular to the page
 - C. about an axis that is parallel to the page
 - D. about an axis that is changing orientation
 - E. about an axis that is getting longer
- ans: B

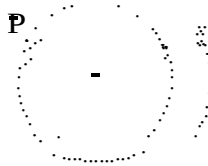
26. The angular velocity vector of a spinning body points out of the page. If the angular acceleration vector points into the page then:
- A. the body is slowing down
 - B. the body is speeding up
 - C. the body is starting to turn in the opposite direction
 - D. the axis of rotation is changing orientation
 - E. none of the above

ans: A

27. A child, riding on a large merry-go-round, travels a distance of 3000 m in a circle of diameter 40 m. The total angle through which she revolves is:
- A. 50rad
 - B. 75rad
 - C. 150rad
 - D. 314rad
 - E. none of these

ans: C

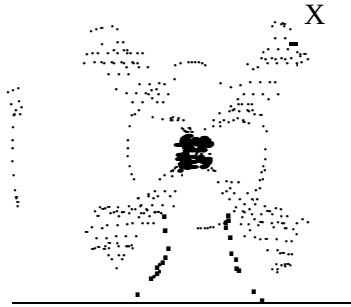
28. The figure shows a cylinder of radius 0.7 m rotating about its axis at 10 rad/s. The speed of the point P is:



- A. 7.0 m/s
- B. 14π rad/s
- C. 7.0π rad/s
- D. 0.70 m/s
- E. none of these

ans: A

29. The fan shown has been turned on and is now slowing as it rotates clockwise. The direction of the acceleration of the point X on the fan tip could be:



- A. $\curvearrowleft$
- B. ∇
- C. $\downarrow$
- D. $\leftarrow$
- E. $\rightarrow$

ans: D

30. A wheel of diameter 3.0 cm has a 4.0-m cord wrapped around its periphery. Starting from rest, the wheel is given a constant angular acceleration of 2.0 rad/s^2 . The cord will unwind in:

- A. 0.82 s
- B. 2.0s
- C. 8.0s
- D. 16 s
- E. 130 s

ans: D

31. A particle moves in a circular path of radius 0.10 m with a constant angular speed of 5 rev/s. The acceleration of the particle is:

- A. $0.10\pi \text{ m/s}^2$
- B. 0.50 m/s^2
- C. $500\pi \text{ m/s}^2$
- D. $1000\pi^2 \text{ m/s}^2$
- E. $10\pi^2 \text{ m/s}^2$

ans: E

32. A car travels north at constant velocity. It goes over a piece of mud, which sticks to the tire. The initial acceleration of the mud, as it leaves the ground, is:

- A. vertically upward
- B. horizontally to the north
- C. horizontally to the south
- D. zero
- E. upward and forward at 45° to the horizontal

ans: A

33. Wrapping paper is being from a 5.0-cm radius tube, free to rotate on its axis. If it is pulled at the constant rate of 10 cm/s and does not slip on the tube, the angular velocity of the tube is:
- A. 2.0 rad/s
 - B. 5.0 rad/s
 - C. 10rad/s
 - D. 25rad/s
 - E. 50rad/s

ans: A

34. String is wrapped around the periphery of a 5.0-cm radius cylinder, free to rotate on its axis. The string is pulled straight out at a constant rate of 10 cm/s and does not slip on the cylinder. As each small segment of string leaves the cylinder, its acceleration changes by:
- A. 0
 - B. 0.010 m/s^2
 - C. 0.020 m/s^2
 - D. 0.10 m/s^2
 - E. 0.20 m/s^2

ans: E

35. A flywheel of diameter 1.2 m has a constant angular acceleration of 5.0 rad/s^2 . The tangential acceleration of a point on its rim is:
- A. 5.0 rad/s^2
 - B. 3.0 m/s^2
 - C. 5.0 m/s^2
 - D. 6.0 m/s^2
 - E. 12 m/s^2

ans: B

36. For a wheel spinning with constant angular acceleration on an axis through its center, the ratio of the speed of a point on the rim to the speed of a point halfway between the center and the rim is:
- A. 1
 - B. 2
 - C. $1/2$
 - D. 4
 - E. $1/4$

ans: B

37. For a wheel spinning on an axis through its center, the ratio of the tangential acceleration of a point on the rim to the tangential acceleration of a point halfway between the center and the rim is:
- A. 1
 - B. 2
 - C. $1/2$
 - D. 4
 - E. $1/4$

ans: B

38. For a wheel spinning on an axis through its center, the ratio of the radial acceleration of a point on the rim to the radial acceleration of a point halfway between the center and the rim is:
- A. 1
 - B. 2
 - C. $1/2$
 - D. 4
 - E. $1/4$

ans: B

39. Two wheels are identical but wheel B is spinning with twice the angular speed of wheel A. The ratio of the magnitude of the radial acceleration of a point on the rim of B to the magnitude of the radial acceleration of a point on the rim of A is:
- A. 1
 - B. 2
 - C. $1/2$
 - D. 4
 - E. $1/4$

ans: D

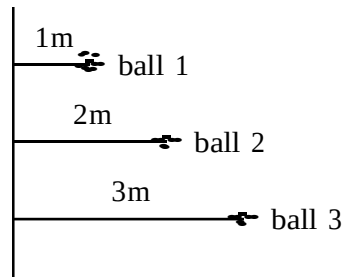
40. A wheel starts from rest and spins with a constant angular acceleration. As time goes on the acceleration vector for a point on the rim:
- A. decreases in magnitude and becomes more nearly tangent to the rim
 - B. decreases in magnitude and becomes more nearly radial
 - C. increases in magnitude and becomes more nearly tangent to the rim
 - D. increases in magnitude and becomes more nearly radial
 - E. increases in magnitude but retains the same angle with the tangent to the rim

ans: D

41. The magnitude of the acceleration of a point on a spinning wheel is increased by a factor of 4 if:
- A. the magnitudes of the angular velocity and the angular acceleration are each multiplied by a factor of 4
 - B. the magnitude of the angular velocity is multiplied by a factor of 4 and the angular acceleration is not changed
 - C. the magnitudes of the angular velocity and the angular acceleration are each multiplied by a factor of 2
 - D. the magnitude of the angular velocity is multiplied by a factor of 2 and the angular acceleration is not changed
 - E. the magnitude of the angular velocity is multiplied by a factor of 2 and the magnitude of the angular acceleration is multiplied by a factor of 4

ans: E

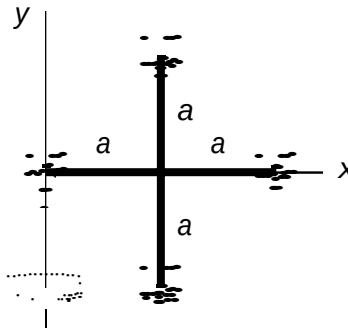
42. Three identical balls are tied by light strings to the same rod and rotate around it, as shown below. Rank the balls according to their rotational inertia, least to greatest.



- A. 1, 2, 3
- B. 3, 2, 1
- C. 3, then 1 and 2 tie
- D. 1, 3, 2
- E. All are the same

ans: A

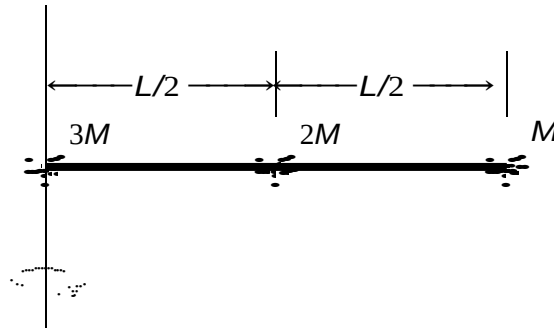
43. Four identical particles, each with mass m , are arranged in the x, y plane as shown. They are connected by light sticks to form a rigid body. If $m = 2.0$ kg and $a = 1.0$ m, the rotational inertia of this array about the y axis is:



- A. $4.0 \text{ kg} \cdot \text{m}^2$
- B. $12 \text{ kg} \cdot \text{m}^2$
- C. $9.6 \text{ kg} \cdot \text{m}^2$
- D. $4.8 \text{ kg} \cdot \text{m}^2$
- E. none of these

ans: B

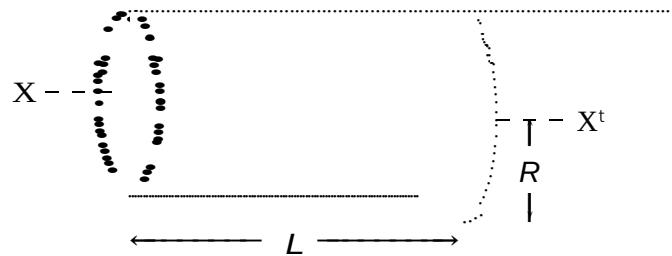
44. Three identical balls, with masses of M , $2M$, and $3M$, are fastened to a massless rod of length L as shown. The rotational inertia about the left end of the rod is:



- A. $ML^2/2$
- B. ML^2
- C. $3ML^2/2$
- D. $6ML^2$
- E. $3ML^2/4$

ans: E

45. The rotational inertia of a thin cylindrical shell of mass M , radius R , and length L about its central axis ($X-X'$) is:



- A. $MR^2/2$
- B. $ML^2/2$
- C. ML^2
- D. MR^2
- E. none of these

ans: D

46. The rotational inertia of a wheel about its axle does not depend upon its:

- A. diameter
- B. mass
- C. distribution of mass
- D. speed of rotation
- E. material composition

ans: D

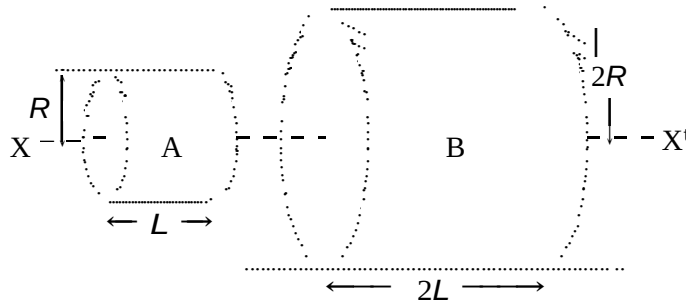
47. Consider four objects, each having the same mass and the same radius:
1. a solid sphere
 2. a hollow sphere
 3. a flat disk in the x, y plane
 4. a hoop in the x, y plane

The order of increasing rotational inertia about an axis through the center of mass and parallel to the z axis is:

- A. 1, 2, 3, 4
- B. 4, 3, 2, 1
- C. 1, 3, 2, 4
- D. 4, 2, 3, 1
- E. 3, 1, 2, 4

ans: C

48. A and B are two solid cylinders made of aluminum. Their dimensions are shown. The ratio of the rotational inertia of B to that of A about the common axis $X-X^t$ is:



- A. 2
- B. 4
- C. 8
- D. 16
- E. 32

ans: E

49. Two uniform circular disks having the same mass and the same thickness are made from different materials. The disk with the smaller rotational inertia is:
- A. the one made from the more dense material
 - B. the one made from the less dense material
 - C. neither — both rotational inertias are the same
 - D. the disk with the larger angular velocity
 - E. the disk with the larger torque

ans: A

50. A uniform solid cylinder made of lead has the same mass and the same length as a uniform solid cylinder made of wood. The rotational inertia of the lead cylinder compared to the wooden one is:
- A. greater
 - B. less
 - C. same
 - D. unknown unless the radii are given
 - E. unknown unless both the masses and the radii are given
- ans: B

51. To increase the rotational inertia of a solid disk about its axis without changing its mass:
- A. drill holes near the rim and put the material near the axis
 - B. drill holes near the axis and put the material near the rim
 - C. drill holes at points on a circle near the rim and put the material at points between the holes
 - D. drill holes at points on a circle near the axis and put the material at points between the holes
 - E. do none of the above (the rotational inertia cannot be changed without changing the mass)
- ans: B

52. The rotational inertia of a disk about its axis is $0.70 \text{ kg} \cdot \text{m}^2$. When a 2.0-kg weight is added to its rim, 0.40 m from the axis, the rotational inertia becomes:
- A. $0.38 \text{ kg} \cdot \text{m}^2$
 - B. $0.54 \text{ kg} \cdot \text{m}^2$
 - C. $0.70 \text{ kg} \cdot \text{m}^2$
 - D. $0.86 \text{ kg} \cdot \text{m}^2$
 - E. $1.0 \text{ kg} \cdot \text{m}^2$
- ans: E

53. When a thin uniform stick of mass M and length L is pivoted about its midpoint, its rotational inertia is $ML^2/12$. When pivoted about a parallel axis through one end, its rotational inertia is:
- A. $ML^2/12$
 - B. $ML^2/6$
 - C. $ML^2/3$
 - D. $7ML^2/12$
 - E. $13ML^2/12$
- ans: C

54. The rotational inertia of a solid uniform sphere about a diameter is $(2/5)MR^2$, where M is its mass and R is its radius. If the sphere is pivoted about an axis that is tangent to its surface, its rotational inertia is:
- A. MR^2
 - B. $(2/5)MR^2$
 - C. $(3/5)MR^2$
 - D. $(5/2)MR^2$
 - E. $(7/5)MR^2$
- ans: E

55. A solid uniform sphere of radius R and mass M has a rotational inertia about a diameter that is given by $(2/5)MR^2$. A light string of length $3R$ is attached to the surface and used to suspend the sphere from the ceiling. Its rotational inertia about the point of attachment at the ceiling is:

A. $(2/5)MR^2$
 B. $9MR^2$
 C. $16MR^2$
 D. $(47/5)MR^2$
 E. $(82/5)MR^2$

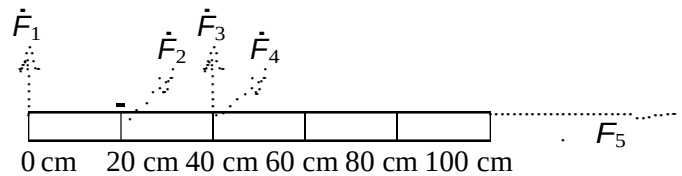
ans: E

56. A force with a given magnitude is to be applied to a wheel. The torque can be maximized by:

A. applying the force near the axle, radially outward from the axle
 B. applying the force near the rim, radially outward from the axle
 C. applying the force near the axle, parallel to a tangent to the wheel
 D. applying the force at the rim, tangent to the rim
 E. applying the force at the rim, at 45° to the tangent

ans: D

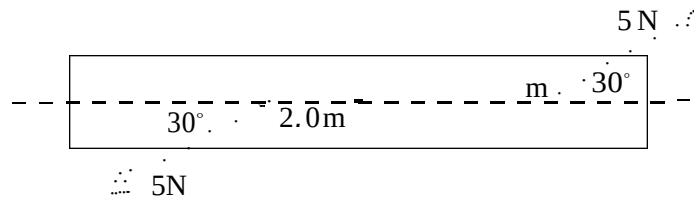
57. The meter stick shown below rotates about an axis through the point marked $\cdot$, 20 cm from one end. Five forces act on the stick: one at each end, one at the pivot point, and two 40 cm from one end, as shown. The magnitudes of the forces are all the same. Rank the forces according to the magnitudes of the torques they produce about the pivot point, least to greatest.



A. $\vec{F}_1, \vec{F}_2, \vec{F}_3, \vec{F}_4, \vec{F}_5$
 B. $\vec{F}_1$ and $\vec{F}_2$ tie, then $\vec{F}_3, \vec{F}_4, \vec{F}_5$
 C. $\vec{F}_2$ and $\vec{F}_5$ tie, then $\vec{F}_4, \vec{F}_1, \vec{F}_3$
 D. $\vec{F}_2, \vec{F}_5, \vec{F}_1$ and $\vec{F}_3$ tie, then $\vec{F}_4$
 E. $\vec{F}_2$ and $\vec{F}_5$ tie, then $\vec{F}_4$, then $\vec{F}_1$ and $\vec{F}_3$ tie

ans: E

58. A rod is pivoted about its center. A 5-N force is applied 4 m from the pivot and another 5-N force is applied 2 m from the pivot, as shown. The magnitude of the total torque about the pivot (in N·m) is:



- A. 0
- B. 5
- C. 8.7
- D. 15
- E. 26

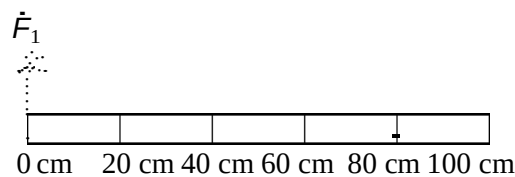
ans: D

59. $\tau = I\alpha$ for an object rotating about a fixed axis, where τ is the net torque acting on it, I is its rotational inertia, and α is its angular acceleration. This expression:

- A. is the definition of torque
- B. is the definition of rotational inertia
- C. is the definition of angular acceleration
- D. follows directly from Newton's second law
- E. depends on a principle of physics that is unrelated to Newton's second law

ans: D

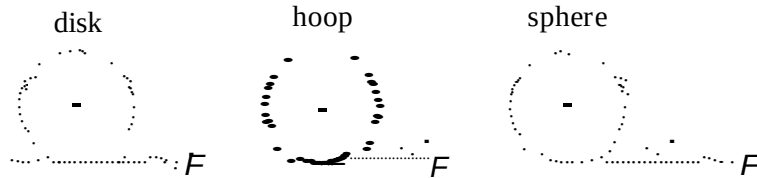
60. A meter stick on a horizontal frictionless table top is pivoted at the 80-cm mark. It is initially at rest. A horizontal force $\vec{F}_1$ is applied perpendicularly to the end of the stick at 0 cm, as shown. A second horizontal force $\vec{F}_2$ (not shown) is applied at the 100-cm end of the stick. If the stick does not rotate:



- A. $|\vec{F}_2| > |\vec{F}_1|$ for all orientations of $\vec{F}_2$
- B. $|\vec{F}_2| < |\vec{F}_1|$ for all orientations of $\vec{F}_2$
- C. $|\vec{F}_2| = |\vec{F}_1|$ for all orientations of $\vec{F}_2$
- D. $|\vec{F}_2| > |\vec{F}_1|$ for some orientations of $\vec{F}_2$ and $|\vec{F}_2| < |\vec{F}_1|$ for others
- E. $|\vec{F}_2| > |\vec{F}_1|$ for some orientations of $\vec{F}_2$ and $|\vec{F}_2| = |\vec{F}_1|$ for others

ans: A

61. A uniform disk, a thin hoop, and a uniform sphere, all with the same mass and same outer radius, are each free to rotate about a fixed axis through its center. Assume the hoop is connected to the rotation axis by light spokes. With the objects starting from rest, identical forces are simultaneously applied to the rims, as shown. Rank the objects according to their angular accelerations, least to greatest.



- A. disk, hoop, sphere
- B. hoop, disk, sphere
- C. hoop, sphere, disk
- D. hoop, disk, sphere
- E. sphere, disk, hoop

ans: D

62. A disk is free to rotate on a fixed axis. A force of given magnitude F , in the plane of the disk, is to be applied. Of the following alternatives the greatest angular acceleration is obtained if the force is:

- A. applied tangentially halfway between the axis and the rim
- B. applied tangentially at the rim
- C. applied radially halfway between the axis and the rim
- D. applied radially at the rim
- E. applied at the rim but neither radially nor tangentially

ans: B

63. A cylinder is 0.10 m in radius and 0.20 m in length. Its rotational inertia, about the cylinder axis on which it is mounted, is 0.020 kg m^2 . A string is wound around the cylinder and pulled with a force of 1.0 N. The angular acceleration of the cylinder is:

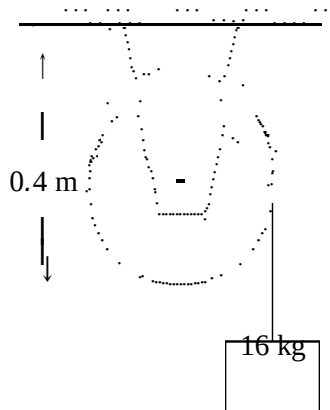
- A. 2.5 rad/s^2
- B. 5.0 rad/s^2
- C. 10 rad/s^2
- D. 15 rad/s^2
- E. 20 rad/s^2

ans: B

64. A disk with a rotational inertia of $2.0 \text{ kg} \cdot \text{m}^2$ and a radius of 0.40 m rotates on a frictionless fixed axis perpendicular to the disk faces and through its center. A force of 5.0 N is applied tangentially to the rim. The angular acceleration of the disk is:
- A. 0.40 rad/s^2
 - B. 0.60 rad/s^2
 - C. 1.0 rad/s^2
 - D. 2.5 rad/s^2
 - E. 10 rad/s^2
- ans: C
65. A disk with a rotational inertia of $5.0 \text{ kg} \cdot \text{m}^2$ and a radius of 0.25 m rotates on a frictionless fixed axis perpendicular to the disk and through its center. A force of 8.0 N is applied along the rotation axis. The angular acceleration of the disk is:
- A. 0
 - B. 0.40 rad/s^2
 - C. 0.60 rad/s^2
 - D. 1.0 rad/s^2
 - E. 2.5 rad/s^2
- ans: A
66. A disk with a rotational inertia of $5.0 \text{ kg} \cdot \text{m}^2$ and a radius of 0.25 m rotates on a frictionless fixed axis perpendicular to the disk and through its center. A force of 8.0 N is applied tangentially to the rim. If the disk starts at rest, then after it has turned through half a revolution its angular velocity is:
- A. 0.57 rad/s
 - B. 0.64 rad/s
 - C. 0.80 rad/s
 - D. 1.6 rad/s
 - E. 3.2 rad/s
- ans: D
67. A thin circular hoop of mass 1.0 kg and radius 2.0 m is rotating about an axis through its center and perpendicular to its plane. It is slowing down at the rate of 7.0 rad/s^2 . The net torque acting on it is:
- A. $7.0 \text{ N} \cdot \text{m}$
 - B. $14.0 \text{ N} \cdot \text{m}$
 - C. $28.0 \text{ N} \cdot \text{m}$
 - D. $44.0 \text{ N} \cdot \text{m}$
 - E. none of these
- ans: C

68. A certain wheel has a rotational inertia of $12 \text{ kg} \cdot \text{m}^2$. As it turns through 5.0 rev its angular velocity increases from 5.0 rad/s to 6.0 rad/s . If the net torque is constant its value is:
- A. $0.016 \text{ N} \cdot \text{m}$
 - B. $0.18 \text{ N} \cdot \text{m}$
 - C. $0.57 \text{ N} \cdot \text{m}$
 - D. $2.1 \text{ N} \cdot \text{m}$
 - E. $3.6 \text{ N} \cdot \text{m}$
- ans: D

69. A 16-kg block is attached to a cord that is wrapped around the rim of a flywheel of diameter 0.40 m and hangs vertically, as shown. The rotational inertia of the flywheel is $0.50 \text{ kg} \cdot \text{m}^2$. When the block is released and the cord unwinds, the acceleration of the block is:

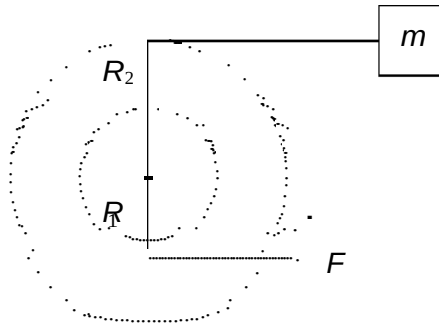


- A. $0.15g$
 - B. $0.56g$
 - C. $0.84g$
 - D. g
 - E. $1.3g$
- ans: B
70. A 8.0-cm radius disk with a rotational inertia of $0.12 \text{ kg} \cdot \text{m}^2$ is free to rotate on a horizontal axis. A string is fastened to the surface of the disk and a 10-kg mass hangs from the other end. The mass is raised by using a crank to apply a $9.0\text{-N} \cdot \text{m}$ torque to the disk. The acceleration of the mass is:
- A. 0.50 m/s^2
 - B. 1.7 m/s^2
 - C. 6.2 m/s^2
 - D. 12 m/s^2
 - E. 20 m/s^2
- ans: A

71. A 0.70-kg disk with a rotational inertia given by $MR^2/2$ is free to rotate on a fixed horizontal axis suspended from the ceiling. A string is wrapped around the disk and a 2.0-kg mass hangs from the free end. If the string does not slip, then as the mass falls and the cylinder rotates, the suspension holding the cylinder pulls up on the cylinder with a force of:
- A. 6.9 N
 - B. 9.8 N
 - C. 16 N
 - D. 26 N
 - E. 29 N

ans: B

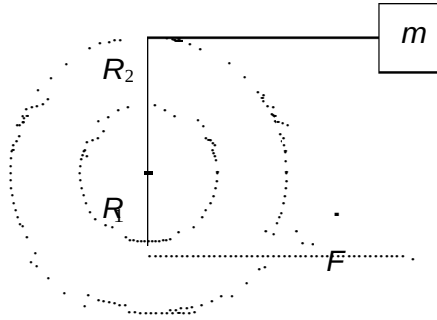
72. A small disk of radius R_1 is mounted coaxially with a larger disk of radius R_2 . The disks are securely fastened to each other and the combination is free to rotate on a fixed axle that is perpendicular to a horizontal frictionless table top, as shown in the overhead view below. The rotational inertia of the combination is I . A string is wrapped around the larger disk and attached to a block of mass m , on the table. Another string is wrapped around the smaller disk and is pulled with a force F as shown. The acceleration of the block is:



- A. $R F/mR$
- B. $R_1 R_2 F / (I - m R_2^2)$
- C. $R_1 R_2 F / (I + m R_2^2)$
- D. $R_1 R_2 F / (I - m R_1 R_2)$
- E. $R_1 R_2 F / (I + m R_1 R_2)$

ans: C

73. A small disk of radius R_1 is fastened coaxially to a larger disk of radius R_2 . The combination is free to rotate on a fixed axle, which is perpendicular to a horizontal frictionless table top, as shown in the overhead view below. The rotational inertia of the combination is I . A string is wrapped around the larger disk and attached to a block of mass m , on the table. Another string is wrapped around the smaller disk and is pulled with a force $\vec{F}$ as shown. The tension in the string pulling the block is:



- A. $R_1 F / R_2$
- B. $m R_1 R_2 F / (I - m R_2^2)$
- C. $m R_1 R_2 F / (I + m R_2^2)$
- D. $m R_1 R_2 F / (I - m R_1 R_2)$
- E. $m R_1 R_2 F / (I + m R_1 R_2)$

ans: C

74. A block is attached to each end of a rope that passes over a pulley suspended from the ceiling. The blocks do not have the same mass. If the rope does not slip on the pulley, then at any instant after the blocks start moving, the rope:
- A. pulls on both blocks, but exerts a greater force on the heavier block
 - B. pulls on both blocks, but exerts a greater force on the lighter block
 - C. pulls on both blocks and exerts the same magnitude force on both
 - D. does not pull on either block
 - E. pulls only on the lighter block

ans: A

75. A pulley with a radius of 3.0 cm and a rotational inertia of $4.5 \times 10^{-3} \text{ kg} \cdot \text{m}^2$ is suspended from the ceiling. A rope passes over it with a 2.0-kg block attached to one end and a 4.0-kg block attached to the other. The rope does not slip on the pulley. When the speed of the heavier block is 2.0 m/s the kinetic energy of the pulley is:

- A. 0.15 J
- B. 0.30 J
- C. 1.0 J
- D. 10 J
- E. 20 J

ans: D

76. A pulley with a radius of 3.0 cm and a rotational inertia of $4.5 \times 10^{-3} \text{ kg} \cdot \text{m}^2$ is suspended from the ceiling. A rope passes over it with a 2.0-kg block attached to one end and a 4.0-kg block attached to the other. The rope does not slip on the pulley. At any instant after the blocks start moving, the object with the greatest kinetic energy is:
- A. the heavier block
 - B. the lighter block
 - C. the pulley
 - D. either block (the two blocks have the same kinetic energy)
 - E. none (all three objects have the same kinetic energy)

ans: C

77. A disk with a rotational inertia of $5.0 \text{ kg} \cdot \text{m}^2$ and a radius of 0.25 m rotates on a fixed axis perpendicular to the disk and through its center. A force of 2.0 N is applied tangentially to the rim. As the disk turns through half a revolution the work done by the force is:
- A. 1.6J
 - B. 2.5J
 - C. 6.3J
 - D. 10 J
 - E. 40 J

ans: A

78. A circular saw is powered by a motor. When the saw is used to cut wood, the wood exerts a torque of $0.80 \text{ N} \cdot \text{m}$ on the saw blade. If the blade rotates with a constant angular velocity of 20 rad/s the work done on the blade by the motor in 1.0 min is:
- A. 0
 - B. 480J
 - C. 960J
 - D. 1400 J
 - E. 1800 J

ans: C

79. A disk has a rotational inertia of $6.0 \text{ kg} \cdot \text{m}^2$ and a constant angular acceleration of 2.0 rad/s^2 . If it starts from rest the work done during the first 5.0 s by the net torque acting on it is:
- A. 0
 - B. 30J
 - C. 60J
 - D. 300 J
 - E. 600 J

ans: D

80. A disk starts from rest and rotates around a fixed axis, subject to a constant net torque. The work done by the torque during the second 5 s is _____ as the work done during the first 5 s.
- A. the same
 - B. twice as much
 - C. half as much
 - D. four times as much
 - E. one-fourth as much

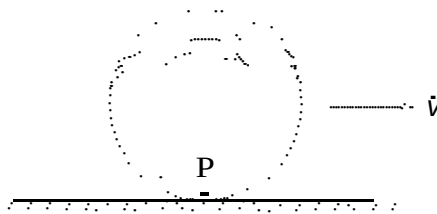
ans: D

81. A disk starts from rest and rotates about a fixed axis, subject to a constant net torque. The work done by the torque during the second revolution is ____ as the work done during the first revolution.
- A. the same
 - B. twice as much
 - C. half as much
 - D. four times as much
 - E. one-fourth as much

ans: A

Chapter 11: ROLLING, TORQUE, AND ANGULAR MOMENTUM

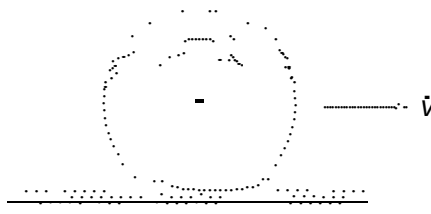
1. A wheel rolls without sliding along a horizontal road as shown. The velocity of the center of the wheel is represented by $\vec{v}$. Point P is painted on the rim of the wheel. The instantaneous velocity of point P is:



- A. $\rightarrow$
- B. $\leftarrow$
- C. $\uparrow$
- D. **3**
- E. zero

ans: E

2. A wheel of radius 0.5 m rolls without sliding on a horizontal surface as shown. Starting from rest, the wheel moves with constant angular acceleration 6 rad/s^2 . The distance traveled by the center of the wheel from $t = 0$ to $t = 3 \text{ s}$ is:



- A. zero
- B. 27 m
- C. 13.5m
- D. 18 m
- E. none of these

ans: C

3. Two wheels roll side-by-side without sliding, at the same speed. The radius of wheel 2 is twice the radius of wheel 1. The angular velocity of wheel 2 is:
- A. twice the angular velocity of wheel 1
 - B. the same as the angular velocity of wheel 1
 - C. half the angular velocity of wheel 1
 - D. more than twice the angular velocity of wheel 1
 - E. less than half the angular velocity of wheel 1

ans: C

4. A forward force on the axle accelerates a rolling wheel on a horizontal surface. If the wheel does not slide the frictional force of the surface on the wheel is:
- A. zero
 - B. in the forward direction
 - C. in the backward direction
 - D. in the upward direction
 - E. in the downward direction

ans: D

5. When the speed of a rear-drive car is increasing on a horizontal road the direction of the frictional force on the tires is:
- A. forward for all tires
 - B. backward for all tires
 - C. forward for the front tires and backward for the rear tires
 - D. backward for the front tires and forward for the rear tires
 - E. zero

ans: D

6. A solid wheel with mass M , radius R , and rotational inertia $\frac{1}{2}MR^2$, rolls without sliding on a horizontal surface. A horizontal force F is applied to the axle and the center of mass has an acceleration a . The magnitudes of the applied force F and the frictional force f of the surface, respectively, are:
- A. $F = Ma$, $f = 0$
 - B. $F = Ma$, $f = Ma/2$
 - C. $F = 2Ma$, $f = Ma$
 - D. $F = 2Ma$, $f = Ma/2$
 - E. $F = 3Ma/2$, $f = Ma/2$

ans: E

7. The coefficient of static friction between a certain cylinder and a horizontal floor is 0.40. If the rotational inertia of the cylinder about its symmetry axis is given by $I = \frac{1}{2}MR^2$, then the magnitude of the maximum acceleration the cylinder can have without sliding is:
- A. $0.1g$
 - B. $0.2g$
 - C. $0.4g$
 - D. $0.8g$
 - E. g

ans: D

8. A thin-walled hollow tube rolls without sliding along the floor. The ratio of its translational kinetic energy to its rotational kinetic energy (about an axis through its center of mass) is:
- A. 1
 - B. 2
 - C. 3
 - D. $1/2$
 - E. $1/3$

ans: A

9. A sphere and a cylinder of equal mass and radius are simultaneously released from rest on the same inclined plane and roll without sliding down the incline. Then:
- A. the sphere reaches the bottom first because it has the greater inertia
 - B. the cylinder reaches the bottom first because it picks up more rotational energy
 - C. the sphere reaches the bottom first because it picks up more rotational energy
 - D. they reach the bottom together
 - E. none of the above are true

ans: E

10. A hoop, a uniform disk, and a uniform sphere, all with the same mass and outer radius, start with the same speed and roll without sliding up identical inclines. Rank the objects according to how high they go, least to greatest.
- A. hoop, disk, sphere
 - B. disk, hoop, sphere
 - C. sphere, hoop, disk
 - D. sphere, disk, hoop
 - E. hoop, sphere, disk

ans: A

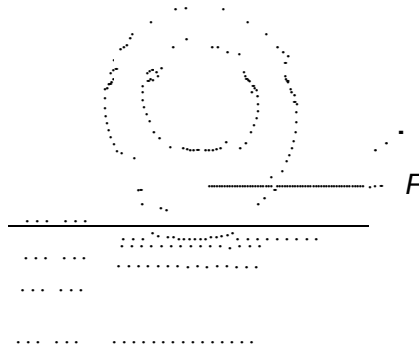
11. A hoop rolls with constant velocity and without sliding along level ground. Its rotational kinetic energy is:
- A. half its translational kinetic energy
 - B. the same as its translational kinetic energy
 - C. twice its translational kinetic energy
 - D. four times its translational kinetic energy
 - E. one-third its translational kinetic energy

ans: B

12. Two identical disks, with rotational inertia $I (= \frac{1}{2}MR^2)$, roll without sliding across a horizontal floor with the same speed and then up inclines. Disk A rolls up its incline without sliding. On the other hand, disk B rolls up a frictionless incline. Otherwise the inclines are identical. Disk A reaches a height 12 cm above the floor before rolling down again. Disk B reaches a height above the floor of:
- A. 24 cm
 - B. 18 cm
 - C. 12 cm
 - D. 8 cm
 - E. 6 cm

ans: D

13. A yo-yo, arranged as shown, rests on a frictionless surface. When a force $\vec{F}$ is applied to the string as shown, the yo-yo:



- A. moves to the left and rotates counterclockwise
 - B. moves to the right and rotates counterclockwise
 - C. moves to the left and rotates clockwise
 - D. moves to the right and rotates clockwise
 - E. moves to the right and does not rotate
- ans: B
14. When we apply the energy conservation principle to a cylinder rolling down an incline without sliding, we exclude the work done by friction because:
- A. there is no friction present
 - B. the angular velocity of the center of mass about the point of contact is zero
 - C. the coefficient of kinetic friction is zero
 - D. the linear velocity of the point of contact (relative to the inclined surface) is zero
 - E. the coefficient of static and kinetic friction are equal
- ans: D
15. Two uniform cylinders have different masses and different rotational inertias. They simultaneously start from rest at the top of an inclined plane and roll without sliding down the plane. The cylinder that gets to the bottom first is:
- A. the one with the larger mass
 - B. the one with the smaller mass
 - C. the one with the larger rotational inertia
 - D. the one with the smaller rotational inertia
 - E. neither (they arrive together)
- ans: E
16. A 5.0-kg ball rolls without sliding from rest down an inclined plane. A 4.0-kg block, mounted on roller bearings totaling 100 g, rolls from rest down the same plane. At the bottom, the block has:
- A. greater speed than the ball
 - B. less speed than the ball
 - C. the same speed as the ball
 - D. greater or less speed than the ball, depending on the angle of inclination
 - E. greater or less speed than the ball, depending on the radius of the ball
- ans: A

17. A cylinder of radius $R = 6.0$ cm is on a rough horizontal surface. The coefficient of kinetic friction between the cylinder and the surface is 0.30 and the rotational inertia for rotation about the axis is given by $MR^2/2$, where M is its mass. Initially it is not rotating but its center of mass has a speed of 7.0 m/s. After 2.0 s the speed of its center of mass and its angular velocity about its center of mass, respectively, are:
- A. 1.1 m/s, 0
 - B. 1.1 m/s, 19 rad/s
 - C. 1.1 m/s, 98 rad/s
 - D. 1.1 m/s, 200 rad/s
 - E. 5.9 m/s, 98 rad/s

ans: D

18. The fundamental dimensions of angular momentum are:
- A. mass·length·time⁻¹
 - B. mass·length⁻²·time⁻²
 - C. mass²·time⁻¹
 - D. mass·length²·time⁻²
 - E. none of these

ans: E

19. Possible units of angular momentum are:
- A. kg·m/s
 - B. kg·m²/s²
 - C. kg·m/s²
 - D. kg·m²/s
 - E. none of these

ans: D

20. The unit kg·m²/s can be used for:
- A. angular momentum
 - B. rotational kinetic energy
 - C. rotational inertia
 - D. torque
 - E. power

ans: A

21. The newton-second is a unit of:
- A. work
 - B. angular momentum
 - C. power
 - D. linear momentum
 - E. none of these

ans: D

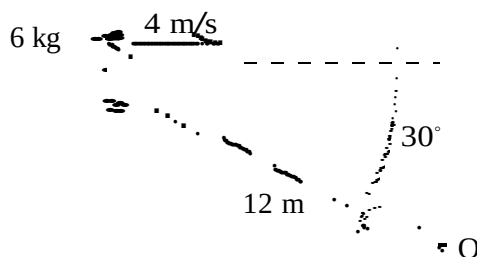
22. A 2.0-kg block travels around a 0.50-m radius circle with an angular velocity of 12 rad/s. The magnitude of its angular momentum about the center of the circle is:
- A. $6.0 \text{ kg} \cdot \text{m}^2/\text{s}$
 - B. $12 \text{ kg} \cdot \text{m}^2/\text{s}$
 - C. $48 \text{ kg/m}^2 \cdot \text{s}$
 - D. $72 \text{ kg} \cdot \text{m}^2/\text{s}^2$
 - E. $576 \text{ kg/m}^2 \cdot \text{s}^2$

ans: A

23. The angular momentum vector of Earth about its rotation axis, due to its daily rotation, is directed:
- A. tangent to the equator toward the east
 - B. tangent to the equator toward the west
 - C. north
 - D. south
 - E. toward the Sun

ans: C

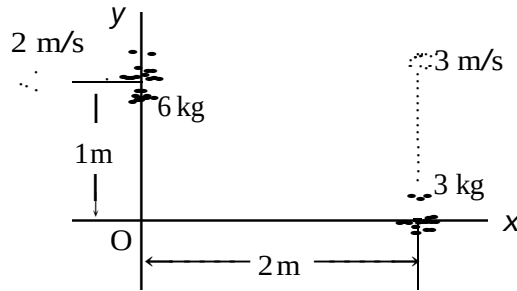
24. A 6.0-kg particle moves to the right at 4.0 m/s as shown. The magnitude of its angular momentum about the point O is:



- A. zero
- B. $288 \text{ kg} \cdot \text{m}^2/\text{s}$
- C. $144 \text{ kg} \cdot \text{m}^2/\text{s}$
- D. $24 \text{ kg} \cdot \text{m}^2/\text{s}$
- E. $249 \text{ kg} \cdot \text{m}^2/\text{s}$

ans: C

25. Two objects are moving in the x, y plane as shown. The magnitude of their total angular momentum (about the origin O) is:



- A. zero
- B. $6 \text{ kg} \cdot \text{m}^2/\text{s}$
- C. $12 \text{ kg} \cdot \text{m}^2/\text{s}$
- D. $30 \text{ kg} \cdot \text{m}^2/\text{s}$
- E. $78 \text{ kg} \cdot \text{m}^2/\text{s}$

ans: D

26. A 2.0-kg block starts from rest on the positive x axis 3.0 m from the origin and thereafter has an acceleration given by $\vec{a} = (4.0 \text{ m/s}^2)\hat{i} - (3.0 \text{ m/s}^2)\hat{j}$. At the end of 2.0 s its angular momentum about the origin is:

- A. 0
- B. $(-36 \text{ kg} \cdot \text{m}^2/\text{s})\hat{k}$
- C. $(+48 \text{ kg} \cdot \text{m}^2/\text{s})\hat{k}$
- D. $(-96 \text{ kg} \cdot \text{m}^2/\text{s})\hat{k}$
- E. $(+96 \text{ kg} \cdot \text{m}^2/\text{s})\hat{k}$

ans: B

27. A 15-g paper clip is attached to the rim of a phonograph record with a radius of 30 cm, spinning at 3.5 rad/s. The magnitude of its angular momentum is:

- A. $1.4 \times 10^{-3} \text{ kg} \cdot \text{m}^2/\text{s}$
- B. $4.7 \times 10^{-3} \text{ kg} \cdot \text{m}^2/\text{s}$
- C. $1.6 \times 10^{-2} \text{ kg} \cdot \text{m}^2/\text{s}$
- D. $3.2 \times 10^{-1} \text{ kg} \cdot \text{m}^2/\text{s}$
- E. $1.1 \text{ kg} \cdot \text{m}^2/\text{s}$

ans: B

28. As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, 0.75 m from the origin. The magnitude of its angular momentum around the origin is:

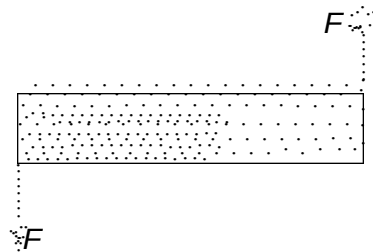
- A. $6.0 \text{ kg} \cdot \text{m}^2/\text{s}$
- B. $9.0 \text{ kg} \cdot \text{m}^2/\text{s}$
- C. $11 \text{ kg} \cdot \text{m}^2/\text{s}$
- D. $14 \text{ kg} \cdot \text{m}^2/\text{s}$
- E. $20 \text{ kg} \cdot \text{m}^2/\text{s}$

ans: C

29. As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, a distance of 0.75 m from the origin. The z component of the angular momentum around the origin is:
- A. $6.0 \text{ kg} \cdot \text{m}^2/\text{s}$
 - B. $9.0 \text{ kg} \cdot \text{m}^2/\text{s}$
 - C. $11 \text{ kg} \cdot \text{m}^2/\text{s}$
 - D. $14 \text{ kg} \cdot \text{m}^2/\text{s}$
 - E. $20 \text{ kg} \cdot \text{m}^2/\text{s}$
- ans: A
30. As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, 0.75 m from the origin. The component in the xy plane of the angular momentum around the origin has a magnitude of:
- A. 0
 - B. $6.0 \text{ kg} \cdot \text{m}^2/\text{s}$
 - C. $9.0 \text{ kg} \cdot \text{m}^2/\text{s}$
 - D. $11 \text{ kg} \cdot \text{m}^2/\text{s}$
 - E. $14 \text{ kg} \cdot \text{m}^2/\text{s}$
- ans: C
31. A uniform disk has radius R and mass M . When it is spinning with angular velocity ω about an axis through its center and perpendicular to its face its angular momentum is $I\omega$. When it is spinning with the same angular velocity about a parallel axis a distance h away its angular momentum is:
- A. $I\omega$
 - B. $(I + Mh^2)\omega$
 - C. $(I - Mh^2)\omega$
 - D. $(I + MR^2)\omega$
 - E. $(I - MR^2)\omega$
- ans: B
32. A pulley with radius R and rotational inertia I is free to rotate on a horizontal fixed axis through its center. A string passes over the pulley. A block of mass m_1 is attached to one end and a block of mass m_2 is attached to the other. At one time the block with mass m_1 is moving downward with speed v . If the string does not slip on the pulley, the magnitude of the total angular momentum, about the pulley center, of the blocks and pulley, considered as a system, is given by:
- A. $(m_1 - m_2)vR + Iv/R$
 - B. $(m_1 + m_2)vR + Iv/R$
 - C. $(m_1 - m_2)vR + Iv/R^2$
 - D. $(m_1 + m_2)vR + Iv/R^2$
 - E. none of the above
- ans: B

33. A single force acts on a particle situated on the positive x axis. The torque about the origin is in the negative z direction. The force might be:
- A. in the positive y direction
 - B. in the negative y direction
 - C. in the positive x direction
 - D. in the negative x direction
 - E. in the positive z direction
- ans: B

34. A rod rests on frictionless ice. Forces that are equal in magnitude and opposite in direction are then simultaneously applied to its ends as shown. The quantity that vanishes is its:



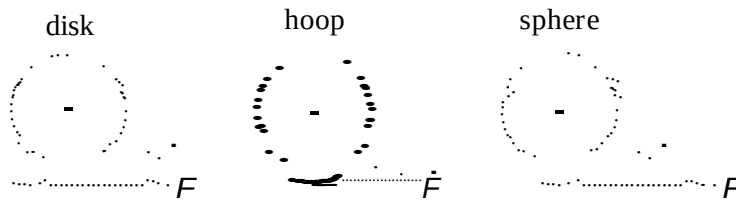
- A. angular momentum
 - B. angular acceleration
 - C. total linear momentum
 - D. kinetic energy
 - E. rotational inertia
- ans: C
35. A 2.0-kg stone is tied to a 0.50-m long string and swung around a circle at a constant angular velocity of 12 rad/s. The net torque on the stone about the center of the circle is:
- A. 0
 - B. $6.0 \text{ N} \cdot \text{m}$
 - C. $12 \text{ N} \cdot \text{m}$
 - D. $72 \text{ N} \cdot \text{m}$
 - E. $140 \text{ N} \cdot \text{m}$
- ans: A
36. A 2.0-kg stone is tied to a 0.50-m long string and swung around a circle at a constant angular velocity of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, 0.75 m from the origin. The magnitude of the torque about the origin is:
- A. 0
 - B. $6.0 \text{ N} \cdot \text{m}$
 - C. $14 \text{ N} \cdot \text{m}$
 - D. $72 \text{ N} \cdot \text{m}$
 - E. $108 \text{ N} \cdot \text{m}$
- ans: E

37. A 2.0-kg block starts from rest on the positive x axis 3.0 m from the origin and thereafter has an acceleration given by $\vec{a} = (4.0 \text{ m/s}^2)\hat{i} - (3.0 \text{ m/s}^2)\hat{j}$. The torque, relative to the origin, acting on it at the end of 2.0 s is:

A. 0
 B. $(-18 \text{ N} \cdot \text{m})\hat{k}$
 C. $(+24 \text{ N} \cdot \text{m})\hat{k}$
 D. $(-144 \text{ N} \cdot \text{m})\hat{k}$
 E. $(+144 \text{ N} \cdot \text{m})\hat{k}$

ans: B

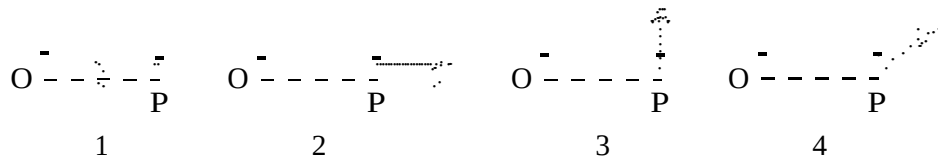
38. A uniform disk, a thin hoop, and a uniform sphere, all with the same mass and same outer radius, are each free to rotate about a fixed axis through its center. Assume the hoop is connected to the rotation axis by light spokes. With the objects starting from rest, identical forces are simultaneously applied to the rims, as shown. Rank the objects according to their angular momenta after a given time t , least to greatest.



A. all tie
 B. disk, hoop, sphere
 C. hoop, disk, sphere
 D. hoop, sphere, disk
 E. hoop, disk, sphere

ans: A

39. A single force acts on a particle P. Rank each of the orientations of the force shown below according to the magnitude of the time rate of change of the particle's angular momentum about the point O, least to greatest.

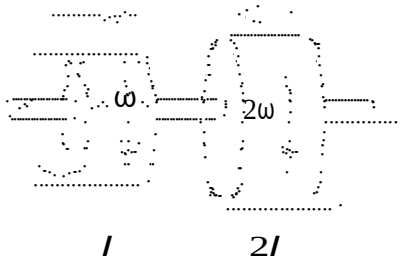


A. 1, 2, 3, 4
 B. 1 and 2 tie, then 3, 4
 C. 1 and 2 tie, then 4, 3
 D. 1 and 2 tie, then 3 and 4 tie
 E. All are the same

ans: C

40. A pulley with radius R is free to rotate on a horizontal fixed axis through its center. A string passes over the pulley. Mass m_1 is attached to one end and mass m_2 is attached to the other. The portion of the string attached to m_1 has tension T_1 and the portion attached to m_2 has tension T_2 . The magnitude of the total external torque, about the pulley center, acting on the masses and pulley, considered as a system, is given by:
- $|m_1 - m_2|gR$
 - $(m_1 + m_2)gR$
 - $|m_1 - m_2|gR + (T_1 + T_2)R$
 - $(m_1 + m_2)gR + (T_1 - T_2)R$
 - $|m_1 - m_2|gR + (T_2 - T_1)R$
- ans: A
41. An ice skater with rotational inertia I_0 is spinning with angular speed ω_0 . She pulls her arms in, thereby increasing her angular speed to $4\omega_0$. Her rotational inertia is then:
- I_0
 - $I_0/2$
 - $2I_0$
 - $I_0/4$
 - $4I_0$
- ans: D
42. A man, with his arms at his sides, is spinning on a light frictionless turntable. When he extends his arms:
- his angular velocity increases
 - his angular velocity remains the same
 - his rotational inertia decreases
 - his rotational kinetic energy increases
 - his angular momentum remains the same
- ans: E
43. A man, holding a weight in each hand, stands at the center of a horizontal frictionless rotating turntable. The effect of the weights is to double the rotational inertia of the system. As he is rotating, the man opens his hands and drops the two weights. They fall outside the turntable. Then:
- his angular velocity doubles
 - his angular velocity remains about the same
 - his angular velocity is halved
 - the direction of his angular momentum vector changes
 - his rotational kinetic energy increases
- ans: B

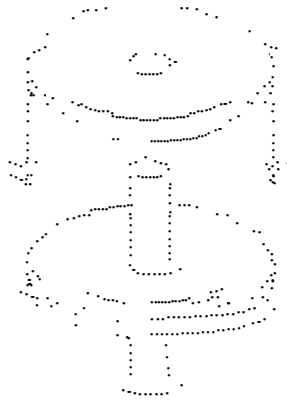
44. A uniform sphere of radius R rotates about a diameter with an angular momentum of magnitude L . Under the action of internal forces the sphere collapses to a uniform sphere of radius $R/2$. The magnitude of its new angular momentum is:
- $L/4$
 - $L/2$
 - L
 - $2L$
 - $4L$
- ans: C
45. When a man on a frictionless rotating stool extends his arms horizontally, his rotational kinetic energy:
- must increase
 - must decrease
 - must remain the same
 - may increase or decrease depending on his initial angular velocity
 - may increase or decrease depending on his angular acceleration
- ans: B
46. When a woman on a frictionless rotating turntable extends her arms out horizontally, her angular momentum:
- must increase
 - must decrease
 - must remain the same
 - may increase or decrease depending on her initial angular velocity
 - tilts away from the vertical
- ans: C
47. Two disks are mounted on low-friction bearings on a common shaft. The first disc has rotational inertia I and is spinning with angular velocity ω . The second disc has rotational inertia $2I$ and is spinning in the same direction as the first disc with angular velocity 2ω as shown. The two disks are slowly forced toward each other along the shaft until they couple and have a final common angular velocity of:



- $5\omega\sqrt{3}$
- $\omega\sqrt{3}$
- $\omega\sqrt{7/3}$
- ω
- 3ω

ans: A

48. A wheel with rotational inertia I , mounted on a vertical shaft with negligible rotational inertia, is rotating with angular speed ω_0 . A nonrotating wheel with rotational inertia $2I$ is suddenly dropped onto the same shaft as shown. The resultant combination of the two wheels and shaft will rotate at:



- A. $\omega_0/2$
- B. $2\omega_0$
- C. $\omega_0/3$
- D. $3\omega_0$
- E. $\omega_0/4$

ans: C

49. A phonograph record is dropped onto a freely spinning turntable. Then:
- A. neither angular momentum nor mechanical energy is conserved because of the frictional forces between record and turntable
 - B. the frictional force between record and turntable increases the total angular momentum
 - C. the frictional force between record and turntable decreases the total angular momentum
 - D. the total angular momentum remains constant
 - E. the sum of the angular momentum and rotational kinetic energy remains constant

ans: D

50. A playground merry-go-round has a radius R and a rotational inertia I . When the merry-go-round is at rest, a child with mass m runs with speed v along a line tangent to the rim and jumps on. The angular velocity of the merry-go-round is then:

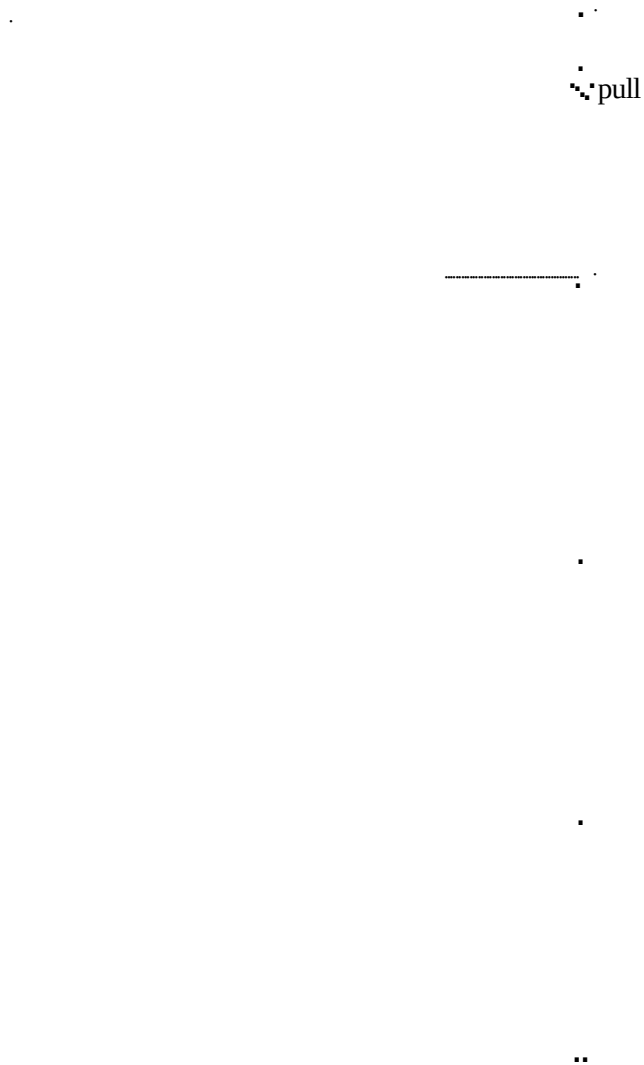
- A. mv/I
- B. v/R
- C. mRv/I
- D. $2mRv/I$
- E. $mRv/(mR^2 + I)$

ans: E

51. A playground merry-go-round has a radius of 3.0 m and a rotational inertia of $600 \text{ kg} \cdot \text{m}^2$. It is initially spinning at 0.80 rad/s when a 20-kg child crawls from the center to the rim. When the child reaches the rim the angular velocity of the merry-go-round is:
- A. 0.62 rad/s
 - B. 0.73 rad/s
 - C. 0.80 rad/s
 - D. 0.89 rad/s
 - E. 1.1 rad/s
- ans: A
52. Two pendulum bobs of unequal mass are suspended from the same fixed point by strings of equal length. The lighter bob is drawn aside and then released so that it collides with the other bob on reaching the vertical position. The collision is elastic. What quantities are conserved in the collision?
- A. Both kinetic energy and angular momentum of the system
 - B. Only kinetic energy
 - C. Only angular momentum
 - D. Angular speed of lighter bob
 - E. None of the above
- ans: A
53. A particle, held by a string whose other end is attached to a fixed point C, moves in a circle on a horizontal frictionless surface. If the string is cut, the angular momentum of the particle about the point C:
- A. increases
 - B. decreases
 - C. does not change
 - D. changes direction but not magnitude
 - E. none of these
- ans: C
54. A block with mass M , on the end of a string, moves in a circle on a horizontal frictionless table as shown. As the string is slowly pulled through a small hole in the table:



ans: A



- A. the angular momentum of the block remains constant
- B. the angular momentum of the block decreases
- C. the kinetic energy of the block remains constant
- D. the kinetic energy of the block decreases
- E. none of the above

Chapter 12: EQUILIBRIUM AND ELASTICITY

1. A net torque applied to a rigid object always tends to produce:
 - A. linear acceleration
 - B. rotational equilibrium
 - C. angular acceleration
 - D. rotational inertia
 - E. none of theseans: C
2. The conditions that the net force and the net torque both vanish:
 - A. hold for every rigid body in equilibrium
 - B. hold only for elastic solid bodies in equilibrium
 - C. hold for every solid body
 - D. are always sufficient to calculate the forces on a solid object in equilibrium
 - E. are sufficient to calculate the forces on a solid object in equilibrium only if the object is elasticans: A
3. For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:
 - A. the center of mass
 - B. the center of gravity
 - C. the geometrical center
 - D. the point of application of the force
 - E. the same pointans: E
4. For a body to be in equilibrium under the combined action of several forces:
 - A. all the forces must be applied at the same point
 - B. all of the forces form pairs of equal and opposite forces
 - C. the sum of the components of all the forces in any direction must equal zero
 - D. any two of these forces must be balanced by a third force
 - E. the lines of action of all the forces must pass through the center of gravity of the bodyans: C
5. For a body to be in equilibrium under the combined action of several forces:
 - A. all the forces must be applied at the same point
 - B. all of the forces form pairs of equal and opposite forces
 - C. any two of these forces must be balanced by a third force
 - D. the sum of the torques about any point must equal zero
 - E. the lines of action of all the forces must pass through the center of gravity of the bodyans: D

6. To determine if a rigid body is in equilibrium the vector sum of the gravitational forces acting on the particles of the body can be replaced by a single force acting at:
- A. the center of mass
 - B. the geometrical center
 - C. the center of gravity
 - D. a point on the boundary
 - E. none of the above

ans: C

7. The center of gravity coincides with the center of mass:
- A. always
 - B. never
 - C. if the center of mass is at the geometrical center of the body
 - D. if the acceleration due to gravity is uniform over the body
 - E. if the body has a uniform distribution of mass

ans: D

8. The location of which of the following points within an object might depend on the orientation of the object?
- A. Its center of mass
 - B. Its center of gravity
 - C. Its geometrical center
 - D. Its center of momentum
 - E. None of the above

ans: B

9. A cylinder placed so it can roll on a horizontal table top, with its center of gravity above its geometrical center, is:
- A. in stable equilibrium
 - B. in unstable equilibrium
 - C. in neutral equilibrium
 - D. not in equilibrium
 - E. none of the above

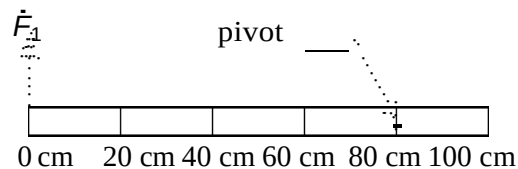
ans: B

10. A cylinder placed so it can roll on a horizontal table top, with its center of gravity below its geometrical center, is:
- A. in stable equilibrium
 - B. in unstable equilibrium
 - C. in neutral equilibrium
 - D. not in equilibrium
 - E. none of the above

ans: A

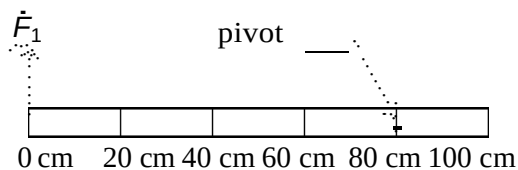
11. A cube balanced with one edge in contact with a table top and with its center of gravity directly above the edge is in equilibrium with respect to rotation about the edge and in equilibrium with respect to rotation about a horizontal axis that is perpendicular to the edge.
- stable, stable
 - stable, unstable
 - unstable, stable
 - unstable, unstable
 - unstable, neutral
- ans: C

12. A meter stick on a horizontal frictionless table top is pivoted at the 80-cm mark. A force $\vec{F}_1$ is applied perpendicularly to the end of the stick at 0 cm, as shown. A second force $\vec{F}_2$ (not shown) is applied perpendicularly at the 100-cm end of the stick. The forces are horizontal. If the stick does not move, the force exerted by the pivot on the stick:



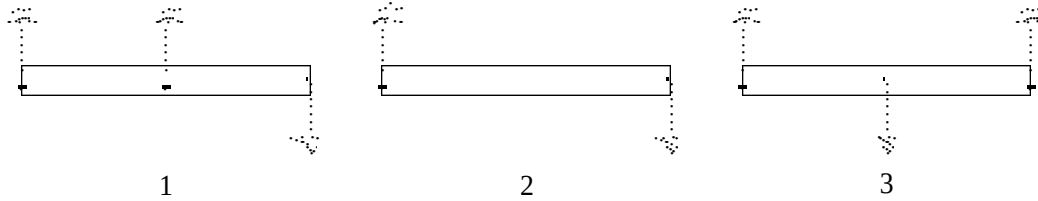
- must be zero
 - must be in the same direction as $\vec{F}_1$ and have magnitude $|\vec{F}_2| - |\vec{F}_1|$
 - must be directed opposite to $\vec{F}_1$ and have magnitude $|\vec{F}_2| - |\vec{F}_1|$
 - must be in the same direction as $\vec{F}_1$ and have magnitude $|\vec{F}_2| + |\vec{F}_1|$
 - must be directed opposite to $\vec{F}_1$ and have magnitude $|\vec{F}_2| + |\vec{F}_1|$
- ans: E

13. A meter stick on a horizontal frictionless table top is pivoted at the 80-cm mark. A force $\vec{F}_1$ is applied perpendicularly to the end of the stick at 0 cm, as shown. A second force $\vec{F}_2$ (not shown) is applied perpendicularly at the 60-cm mark. The forces are horizontal. If the stick does not move, the force exerted by the pivot on the stick:



- must be zero
 - must be in the same direction as $\vec{F}_1$ and have magnitude $|\vec{F}_2| - |\vec{F}_1|$
 - must be directed opposite to $\vec{F}_1$ and have magnitude $|\vec{F}_2| - |\vec{F}_1|$
 - must be in the same direction as $\vec{F}_1$ and have magnitude $|\vec{F}_2| + |\vec{F}_1|$
 - must be directed opposite to $\vec{F}_1$ and have magnitude $|\vec{F}_2| + |\vec{F}_1|$
- ans: B

14. Three identical uniform rods are each acted on by two or more forces, all perpendicular to the rods and all equal in magnitude. Which of the rods could be in static equilibrium if an additional force is applied at the center of mass of the rod?



- A. Only 1
 B. Only 2
 C. Only 3
 D. Only 1 and 2
 E. All three

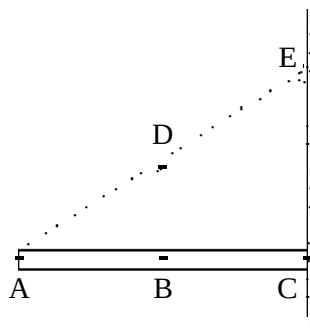
ans: C

15. A 160-N child sits on a light swing and is pulled back and held with a horizontal force of 100 N. The magnitude of the tension force of each of the two supporting ropes is:

- A. 60 N
 B. 94 N
 C. 120 N
 D. 190 N
 E. 260 N

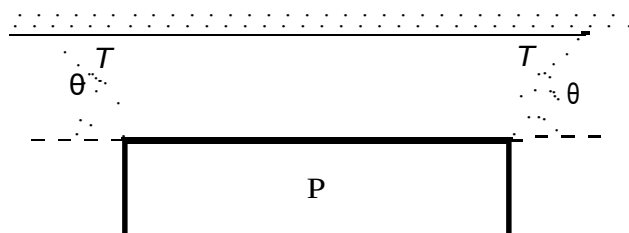
ans: B

16. The diagram shows a stationary 5-kg uniform rod (AC), 1 m long, held against a wall by a rope (AE) and friction between the rod and the wall. To use a single equation to find the force exerted on the rod by the rope at which point should you place the reference point for computing torque?



ans: C

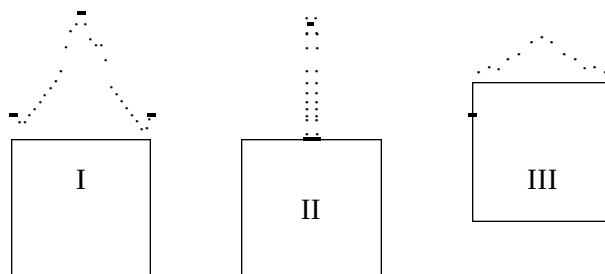
17. A picture P of weight W is hung by two strings as shown. The magnitude of the tension force of each string is T . The total upward pull of the strings on the picture is:



- A. $2W \cos \theta$
- B. $T \sin \theta$
- C. $T \cos \theta$
- D. $2T \sin \theta$
- E. $2T \cos \theta$

ans: D

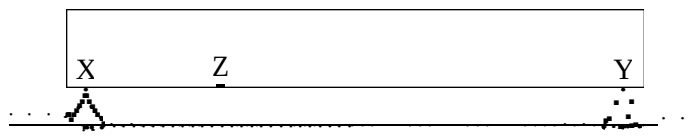
18. A picture can be hung on a wall with string in three different ways, as shown. The magnitude of the tension force of the string is:



- A. least in I
- B. greatest in I
- C. greatest in II
- D. least in III
- E. greatest in III

ans: E

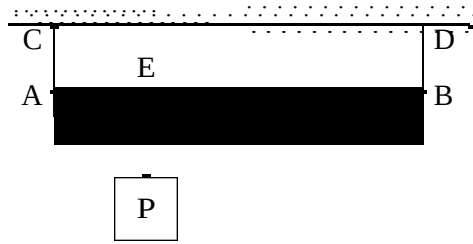
19. A uniform plank is supported by two equal 120-N forces at X and Y, as shown. The support at X is then moved to Z (half-way to the plank center). The supporting forces at Y and Z are then:



- A. $F_Y = 240 \text{ N}$, $F_Z = 120 \text{ N}$
- B. $F_Y = 200 \text{ N}$, $F_Z = 40 \text{ N}$
- C. $F_Y = 40 \text{ N}$, $F_Z = 200 \text{ N}$
- D. $F_Y = 80 \text{ N}$, $F_Z = 160 \text{ N}$
- E. $F_Y = 160 \text{ N}$, $F_Z = 80 \text{ N}$

ans: D

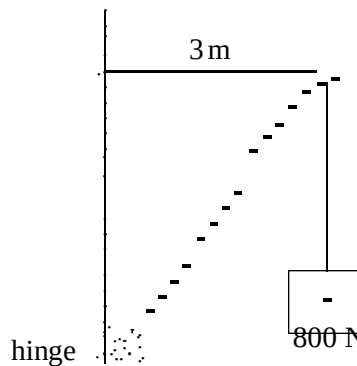
20. A uniform rod AB is 1.2 m long and weighs 16 N. It is suspended by strings AC and BD as shown. A block P weighing 96 N is attached at E, 0.30 m from A. The magnitude of the tension force of the string BD is:



- A. 8.0N
- B. 24 N
- C. 32 N
- D. 48 N
- E. 80 N

ans: C

21. A 5.0-m weightless strut, hinged to a wall, is used to support an 800-N block as shown. The horizontal and vertical components of the force of the hinge on the strut are:



- A. $F_H = 800 \text{ N}$, $F_V = 800 \text{ N}$
- B. $F_H = 600 \text{ N}$, $F_V = 800 \text{ N}$
- C. $F_H = 800 \text{ N}$, $F_V = 600 \text{ N}$
- D. $F_H = 1200 \text{ N}$, $F_V = 800 \text{ N}$
- E. $F_H = 0$, $F_V = 800 \text{ N}$

ans: B

22. A uniform plank is 6.0 m long and weighs 80 N. It is balanced on a sawhorse at its center. An additional 160 N weight is now placed on the left end of the plank. To keep the plank balanced, it must be moved what distance to the left?

- A. 6.0m
- B. 2.0m
- C. 1.5m
- D. 1.0m
- E. 0.50m

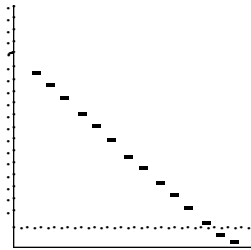
ans: B

23. A uniform 240-g meter stick can be balanced by a 240-g weight placed at the 100-cm mark if the fulcrum is placed at the point marked:

A. 75 cm
 B. 60 cm
 C. 50 cm
 D. 40 cm
 E. 80 cm

ans: A

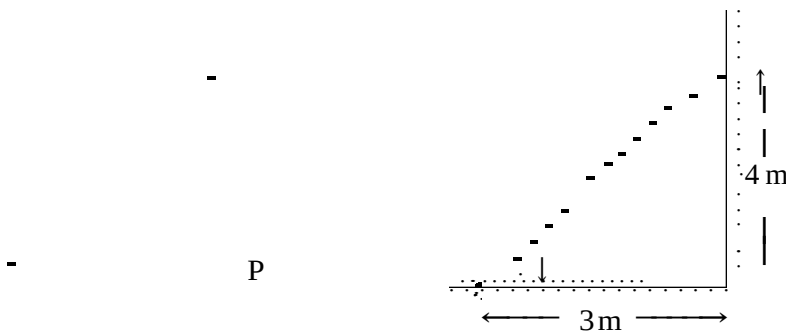
24. A ladder leans against a wall. If the ladder is not to slip, which one of the following must be true?



A. The coefficient of friction between the ladder and the wall must not be zero
 B. The coefficient of friction between the ladder and the floor must not be zero
 C. Both A and B
 D. Either A or B
 E. Neither A nor B

ans: B

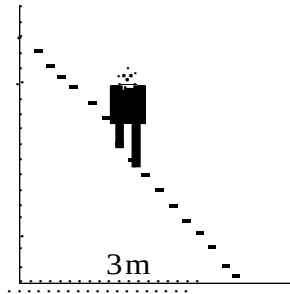
25. An 80-N uniform plank leans against a frictionless wall as shown. The magnitude of the torque (about point P) applied to the plank by the wall is:



A. $40 \text{ N} \cdot \text{m}$
 B. $60 \text{ N} \cdot \text{m}$
 C. $120 \text{ N} \cdot \text{m}$
 D. $160 \text{ N} \cdot \text{m}$
 E. $240 \text{ N} \cdot \text{m}$

ans: C

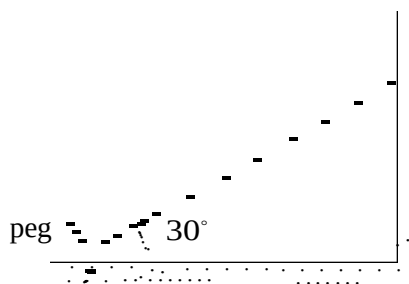
26. An 800-N man stands halfway up a 5.0-m long ladder of negligible weight. The base of the ladder is 3.0 m from the wall as shown. Assuming that the wall-ladder contact is frictionless, the wall pushes against the ladder with a force of magnitude:



- A. 150 N
- B. 300 N
- C. 400 N
- D. 600 N
- E. 800 N

ans: B

27. A uniform ladder is 10 m long and weighs 400 N. It rests with its upper end against a frictionless vertical wall. Its lower end rests on the ground and is prevented from slipping by a peg driven into the ground. The ladder makes a 30° angle with the horizontal. The magnitude of the force exerted on the peg by the ladder is:



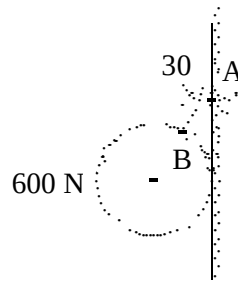
- A. zero
- B. 200 N
- C. 400 N
- D. 470 N
- E. 670 N

ans: D

28. A window washer attempts to lean a ladder against a frictionless wall. He finds that the ladder slips on the ground when it is placed at an angle of less than 75° to the ground but remains in place when the angle is greater than 75° . The coefficient of static friction between the ladder and the ground:
- A. is about 0.13
 - B. is about 0.27
 - C. is about 1.0
 - D. depends on the mass of the ladder
 - E. depends on the length of the ladder

ans: A

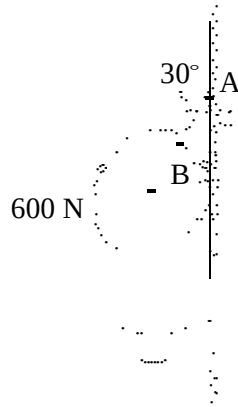
29. The 600-N ball shown is suspended on a string AB and rests against a frictionless vertical wall. The string makes an angle of 30° with the wall. The magnitude of the tension force of the string is:



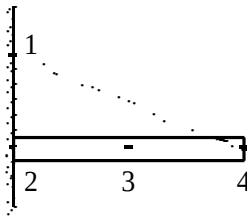
- A. 690 N
- B. 1200 N
- C. 2100 N
- D. 2400 N
- E. none of these

ans: A

30. The 600-N ball shown is suspended on a string AB and rests against a frictionless vertical wall. The string makes an angle of 30° with the wall. The ball presses against the wall with a force of magnitude:

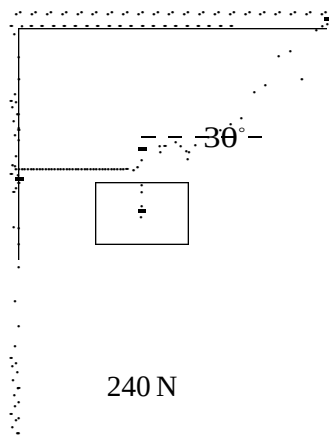


- A. 120 N
 B. 300 N
 C. 350 N
 D. 600 N
 E. 690 N
 ans: C
31. The uniform rod shown below is held in place by the rope and wall. Suppose you know the weight of the rod and all dimensions. Then you can solve a single equation for the force of the rope on the rod, provided you write expressions for the torques about the point:



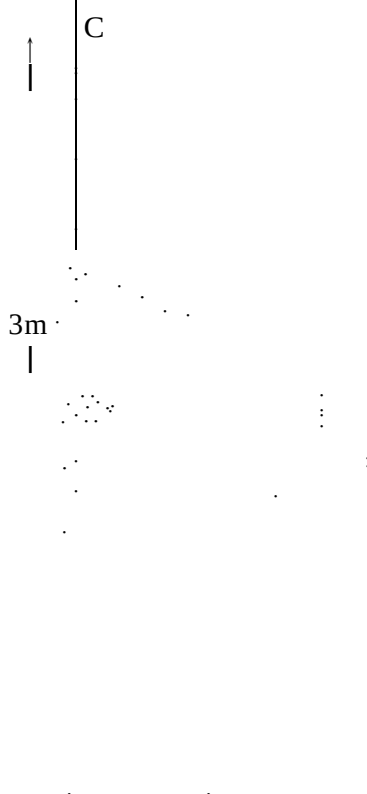
- A. 1
 B. 2
 C. 3
 D. 4
 E. 1, 2, or 3
 ans: B

32. A 240-N weight is hung from two ropes as shown. The tension force of the horizontal rope has magnitude:



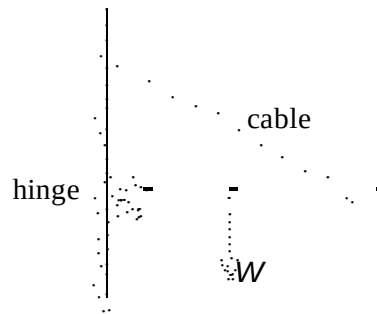
- A. 0
 B. 656 N
 C. 480 N
 D. 416 N
 E. 176 N
 ans: D

33. A 960-N block is suspended as shown. The beam AB is weightless and is hinged to the wall at A. The tension force of the cable BC has magnitude:



A $\xleftarrow{\quad} 4\text{ m} \xrightarrow{\quad}$ B

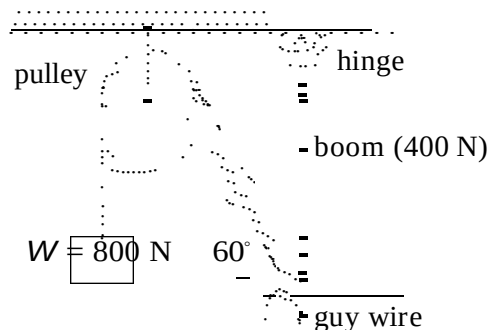
34. A horizontal beam of weight W is supported by a hinge and cable as shown. The force exerted on the beam by the hinge has a vertical component that must be:



- A. nonzero and up
- B. nonzero and down
- C. nonzero but not enough information given to know whether up or down
- D. zero
- E. equal to W

ans: A

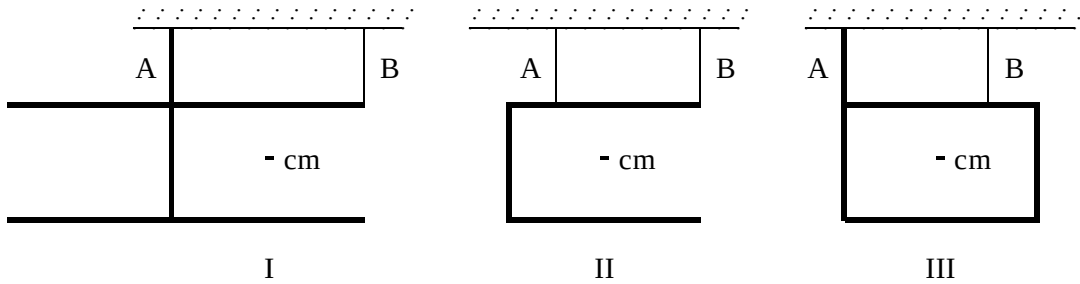
35. A 400-N uniform vertical boom is attached to the ceiling by a hinge, as shown. An 800-N weight W and a horizontal guy wire are attached to the lower end of the boom as indicated. The pulley is massless and frictionless. The tension force T of the horizontal guy wire has magnitude:



- A. 340 N
- B. 400 N
- C. 690 N
- D. 800 N
- E. 1200 N

ans: B

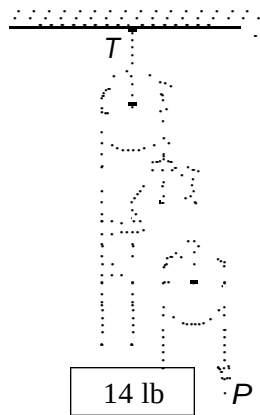
36. A picture is to be hung from the ceiling by means of two wires. Order the following arrangements of the wires according to the tension force of wire B, from least to greatest.



- A. I, II, III
 B. III, II, I
 C. I and II tie, then III
 D. II, I, III
 E. all tie

ans: D

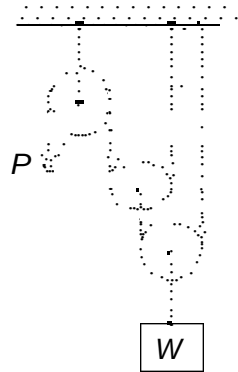
37. The pull P is just sufficient to keep the 14-N block and the weightless pulleys in equilibrium as shown. The magnitude T of the tension force of the upper cable is:



- A. 14 N
 B. 28 N
 C. 16 N
 D. 9.33 N
 E. 18.7 N

ans: C

38. The ideal mechanical advantage (i.e. the ratio of the weight W to the pull P for equilibrium) of the combination of pulleys shown is:



- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

ans: D

39. Stress can be measured in:

- A. N/m^2
- B. $\text{N}\cdot\text{m}^2$
- C. N/m
- D. $\text{N}\cdot\text{m}$
- E. none of these (it is unitless)

ans: A

40. Strain can be measured in:

- A. N/m^2
- B. $\text{N}\cdot\text{m}^2$
- C. N/m
- D. $\text{N}\cdot\text{m}$
- E. none of these (it is unitless)

ans: E

41. Young's modulus can be correctly given in:

- A. $\text{N}\cdot\text{m}$
- B. N/m^2
- C. $\text{N}\cdot\text{m/s}$
- D. N/m
- E. joules

ans: B

42. Young's modulus is a proportionality constant that relates the force per unit area applied perpendicularly at the surface of an object to:

- A. the shear
- B. the fractional change in volume
- C. the fractional change in length
- D. the pressure
- E. the spring constant

ans: C

43. Young's modulus can be used to calculate the strain for a stress that is:

- A. just below the ultimate strength
- B. just above the ultimate strength
- C. well below the yield strength
- D. well above the yield strength
- E. none of the above

ans: C

44. The ultimate strength of a sample is the stress at which the sample:

- A. returns to its original shape when the stress is removed
- B. remains underwater
- C. breaks
- D. bends 180°
- E. does none of these

ans: C

45. A certain wire stretches 0.90 cm when outward forces with magnitude F are applied to each end. The same forces are applied to a wire of the same material but with three times the diameter and three times the length. The second wire stretches:

- A. 0.10 cm
- B. 0.30 cm
- C. 0.90 cm
- D. 2.7 cm
- E. 8.1 cm

ans: B

46. A force of 5000 N is applied outwardly to each end of a 5.0-m long rod with a radius of 34.0 cm and a Young's modulus of $125 \times 10^8 \text{ N/m}^2$. The elongation of the rod is:

- A. 0.0020 mm
- B. 0.0040 mm
- C. 0.14 mm
- D. 0.55 mm
- E. 1.42 mm

ans: D

47. A 4.0-m long steel beam with a cross-sectional area of $1.0 \times 10^{-2} \text{ m}^2$ and a Young's modulus of $2.0 \times 10^{11} \text{ N/m}^2$ is wedged horizontally between two vertical walls. In order to wedge the beam, it is compressed by 0.020 mm. If the coefficient of static friction between the beam and the walls is 0.70 the maximum mass (including its own) it can bear without slipping is:
- A. 0
 - B. 3.6 kg
 - C. 36 kg
 - D. 71 kg
 - E. 710 kg
- ans: E

48. Two supports, made of the same material and initially of equal length, are 2.0 m apart. A stiff board with a length of 4.0 m and a mass of 10 kg is placed on the supports, with one support at the left end and the other at the midpoint. A block is placed on the board a distance of 0.50 m from the left end. As a result the board is horizontal. The mass of the block is:
- A. zero
 - B. 2.3 kg
 - C. 6.6 kg
 - D. 10 kg
 - E. 20 kg
- ans: E

49. The bulk modulus is a proportionality constant that relates the pressure acting on an object to:
- A. the shear
 - B. the fractional change in volume
 - C. the fractional change in length
 - D. Young's modulus
 - E. the spring constant
- ans: B

50. A cube with edges exactly 2 cm long is made of material with a bulk modulus of $3.5 \times 10^9 \text{ N/m}^2$. When it is subjected to a pressure of $3.0 \times 10^5 \text{ Pa}$ its volume is:
- A. 7.31 cm^3
 - B. 7.99931 cm^3
 - C. 8.00069 cm^3
 - D. 8.69 cm^3
 - E. none of these
- ans: B

51. A cube with 2.0-cm sides is made of material with a bulk modulus of $4.7 \times 10^5 \text{ N/m}^2$. When it is subjected to a pressure of $2.0 \times 10^5 \text{ Pa}$ the length of its any of its sides is:
- A. 0.85 cm
 - B. 1.15 cm
 - C. 1.66 cm
 - D. 2.0 cm
 - E. none of these
- ans: C

52. To shear a cube-shaped object, forces of equal magnitude and opposite directions might be applied:

- A. to opposite faces, perpendicular to the faces
- B. to opposite faces, parallel to the faces
- C. to adjacent faces, perpendicular to the faces
- D. to adjacent faces, neither parallel or perpendicular to the faces
- E. to a single face, in any direction

ans: B

53. A shearing force of 50 N is applied to an aluminum rod with a length of 10 m, a cross-sectional area of $1.0 \times 10^{-5} \text{ m}^2$, and a shear modulus of $2.5 \times 10^{10} \text{ N/m}^2$. As a result the rod is sheared through a distance of:

- A. zero
- B. 1.9 mm
- C. 1.9 cm
- D. 19 cm
- E. 1.9 m

ans: B

Chapter 13: GRAVITATION

1. In the formula $F = Gm_1m_2/r^2$, the quantity G :
 - A. depends on the local value of g
 - B. is used only when Earth is one of the two masses
 - C. is greatest at the surface of Earth
 - D. is a universal constant of nature
 - E. is related to the Sun in the same way that g is related to Earthans: D
2. The magnitude of the acceleration of a planet in orbit around the Sun is proportional to:
 - A. the mass of the planet
 - B. the mass of the Sun
 - C. the distance between the planet and the Sun
 - D. the reciprocal of the distance between the planet and the Sun
 - E. the product of the mass of the planet and the mass of the Sunans: B
3. Suitable units for the gravitational constant G are:
 - A. $\text{kg}\cdot\text{m}/\text{s}^2$
 - B. m/s^2
 - C. $\text{N}\cdot\text{s}/\text{m}$
 - D. $\text{kg}\cdot\text{m}/\text{s}$
 - E. $\text{m}^3/(\text{kg}\cdot\text{s}^2)$ans: E
4. The gravitational constant G has the derived units:
 - A. $\text{N}\cdot\text{m}$
 - B. $\text{N}\cdot\text{m}/\text{kg}$
 - C. $\text{N}\cdot\text{kg}/\text{m}$
 - D. $\text{N}\cdot\text{m}^2/\text{kg}^2$
 - E. $\text{N}\cdot\text{kg}^2/\text{m}^2$ans: D
5. Earth exerts a gravitational force on the Moon, keeping it in its orbit. The reaction to this force, in the sense of Newton's third law, is:
 - A. the centripetal force on the Moon
 - B. the nearly circular orbit of the Moon
 - C. the gravitational force on Earth by the Moon
 - D. the tides due to the Moon
 - E. the apple hitting Newton on the head.ans: C

6. A particle might be placed

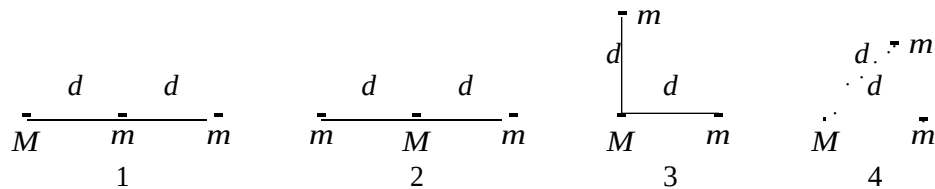
1. inside a uniform spherical shell of mass M , but not at the center
2. inside a uniform spherical shell of mass M , at the center
3. outside a uniform spherical shell of mass M , a distance r from the center
4. outside a uniform solid sphere of mass M , a distance $2r$ from the center

Rank these situations according to the magnitude of the gravitational force on the particle, least to greatest.

- A. All tie
- B. 1, 2, 3, 4
- C. 1 and 2 tie, then 3 and 4 tie
- D. 1 and 2 tie, then 3, then 4
- E. 1 and 2 tie, then 4, then 3

ans: D

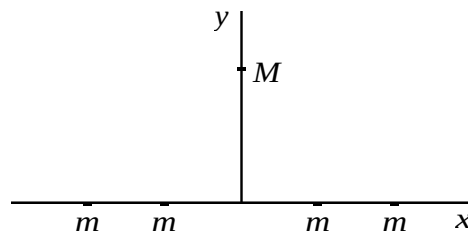
7. Three particles, two with mass m and one with mass M , might be arranged in any of the four configurations known below. Rank the configurations according to the magnitude of the gravitational force on M , least to greatest.



- A. 1, 2, 3, 4
- B. 2, 1, 3, 4
- C. 2, 1, 4, 3
- D. 2, 3, 4, 2
- E. 2, 3, 2, 4

ans: B

8. Four particles, each with mass m are arranged symmetrically about the origin on the x axis. A fifth particle, with mass M , is on the y axis. The direction of the gravitational force on M is:



- A. $\uparrow$
- B. $\downarrow$
- C. $\leftarrow$
- D. $\rightarrow$
- E. none of these directions

ans: B

9. Let F_1 be the magnitude of the gravitational force exerted on the Sun by Earth and F_2 be the magnitude of the force exerted on Earth by the Sun. Then:
- A. F_1 is much greater than F_2
 - B. F_1 is slightly greater than F_2
 - C. F_1 is equal to F_2
 - D. F_1 is slightly less than F_2
 - E. F_1 is much less than F_2

ans: C

10. Let M denote the mass of Earth and let R denote its radius. The ratio g/G at Earth's surface is:
- A. R^2/M
 - B. M/R^2
 - C. MR^2
 - D. M/R
 - E. R/M

ans: B

11. Venus has a mass of about 0.0558 times the mass of Earth and a diameter of about 0.381 times the diameter of Earth. The acceleration of a body falling near the surface of Venus is about:
- A. 0.21 m/s^2
 - B. 1.4 m/s^2
 - C. 2.8 m/s^2
 - D. 3.8 m/s^2
 - E. 25 m/s^2

ans: D

12. The approximate value of g at an altitude above Earth equal to one Earth diameter is:
- A. 9.8 m/s^2
 - B. 4.9 m/s^2
 - C. 2.5 m/s^2
 - D. 1.9 m/s^2
 - E. 1.1 m/s^2

ans: E

13. A rocket ship is coasting toward a planet. Its captain wishes to know the value of g at the surface of the planet. This may be inferred by:
- A. measuring the apparent weight of one of the crew
 - B. measuring the apparent weight of an object of known mass in the ship
 - C. measuring the diameter of the planet
 - D. measuring the density of the planet
 - E. observing the ship's acceleration and correcting for the distance from the center of the planet.

ans: E

14. To measure the mass of a planet with the same radius as Earth, an astronaut drops an object from rest (relative to the planet) from an altitude of one radius above the surface. When the object hits its speed is 4 times what it would be if the same experiment were carried out for Earth. In units of Earth masses, the mass of the planet is:
- A. 2
 - B. 4
 - C. 8
 - D. 16
 - E. 32

ans: D

15. Suppose you have a pendulum clock that keeps correct time on Earth (acceleration due to gravity = 9.8 m/s^2). Without changing the clock, you take it to the Moon (acceleration due to gravity = 1.6 m/s^2). For every hour interval (on Earth) the Moon clock will record:
- A. $(9.8/1.6) \text{ h}$
 - B. $1 \sqrt{\frac{9.8}{1.6}} \text{ h}$
 - C. $\frac{9.8}{1.6} \text{ h}$
 - D. $(1 \sqrt{\frac{1.6}{9.8}}) \text{ h}$
 - E. $\frac{1.6}{9.8} \text{ h}$

ans: E

16. The mass of an object:
- A. is slightly different at different locations on Earth
 - B. is a vector
 - C. is independent of the acceleration due to gravity
 - D. is the same for all objects of the same size and shape
 - E. can be measured directly and accurately on a spring scale

ans: C

17. An astronaut on the Moon simultaneously drops a feather and a hammer. The fact that they land together shows that:
- A. no gravity forces act on a body in a vacuum
 - B. the acceleration due to gravity on the Moon is less than on Earth
 - C. in the absence of air resistance all bodies at a given location fall with the same acceleration
 - D. the feather has a greater weight on the Moon than on Earth
 - E. $G = 0$ on the Moon

ans: C

18. The mass of a hypothetical planet is $1/100$ that of Earth and its radius is $1/4$ that of Earth. If a person weighs 600 N on Earth, what would he weigh on this planet?
- A. 24 N
 - B. 48 N
 - C. 96 N
 - D. 192 N
 - E. 600 N

ans: C

19. An object at the surface of Earth (at a distance R from the center of Earth) weighs 90 N. Its weight at a distance $3R$ from the center of Earth is:
- A. 10 N
 - B. 30 N
 - C. 90 N
 - D. 270 N
 - E. 810 N
- ans: A
20. An object is raised from the surface of Earth to a height of two Earth radii above Earth. Then:
- A. its mass increases and its weight remains constant
 - B. both its mass and weight remain constant
 - C. its mass remains constant and its weight decreases
 - D. both its mass and its weight decrease
 - E. its mass remains constant and its weight increases
- ans: C
21. A spring scale, calibrated in newtons, is used to weigh sugar. If it were possible to weigh sugar at the following locations, where will the buyer get the most sugar to a newton?
- A. At the north pole
 - B. At the equator
 - C. At the center of Earth
 - D. On the Moon
 - E. On Jupiter
- ans: C
22. Of the following where would the weight of an object be the least?
- A. 2000 miles above Earth's surface
 - B. At the north pole
 - C. At the equator
 - D. At the center of Earth
 - E. At the south pole
- ans: D
23. If Earth were to rotate only 100 times per year about its axis:
- A. airplanes flying west to east would make better time
 - B. we would fly off Earth's surface
 - C. our apparent weight would slightly increase
 - D. Earth's atmosphere would float into outer space
 - E. our apparent weight would slightly decrease
- ans: C

24. An astronaut in an orbiting spacecraft feels “weightless” because she:
- A. is beyond the range of gravity
 - B. is pulled outward by centrifugal force
 - C. has no acceleration
 - D. has the same acceleration as the spacecraft
 - E. is outside Earth’s atmosphere
- ans: D
25. Each of the four corners of a square with edge a is occupied by a point mass m . There is a fifth mass, also m , at the center of the square. To remove the mass from the center to a point far away the work that must be done by an external agent is given by:
- A. $4Gm^2/a$
 - B. $-\frac{4}{\sqrt{2}}Gm^2/a$
 - C. $4\frac{\sqrt{2}}{2}Gm^2/a$
 - D. $-4\frac{\sqrt{2}}{2}Gm^2/a$
 - E. $4Gm^2/a^2$
- ans: C
26. Two particles, each of mass m , are a distance d apart. To bring a third particle, with mass $2m$, from far away to a resting point midway between the two particles the work done by an external agent is given by:
- A. $4Gm^2/d$
 - B. $-4Gm^2/d$
 - C. $8Gm^2/d^2$
 - D. $-8Gm^2/d^2$
 - E. zero
- ans: D
27. The escape speed at the surface of Earth is approximately 8 km/s. What is the mass, in units of Earth’s mass, of a planet with twice the radius of Earth for which the escape speed is twice that for Earth?
- A. 2
 - B. 4
 - C. 8
 - D. 1/2
 - E. 1/4
- ans: C
28. Neglecting air resistance, a 1.0-kg projectile has an escape velocity of about 11 km/s at the surface of Earth. The corresponding escape velocity for a 2.0 kg projectile is:
- A. 3.5km/s
 - B. 5.5km/s
 - C. 7.1km/s
 - D. 10km/s
 - E. 11km/s
- ans: E

29. Neglecting air resistance, the escape speed from a certain planet for an empty space vehicle is 1.12×10^4 m/s. What is the corresponding escape speed for the fully loaded vehicle, which has triple the mass of the empty one?

A. 3.73×10^3 m/s
B. 1.12×10^4 m/s
C. 3.36×10^4 m/s
D. 9.98×10^4 m/s
E. 1.40×10^{12} m/s

ans: B

30. An object is dropped from an altitude of one Earth radius above Earth's surface. If M is the mass of Earth and R is its radius the speed of the object just before it hits Earth is given by:

A. $\sqrt{GM/R}$
B. $\sqrt{GM/2R}$
C. $\sqrt{2GM/R}$
D. $\sqrt{GM/R^2}$
E. $\sqrt{GM/2R^2}$

ans: A

31. A projectile is fired straight upward from Earth's surface with a speed that is half the escape speed. If R is the radius of Earth, the highest altitude reached, measured from the surface, is:

A. $R/4$
B. $R/3$
C. $R/2$
D. R
E. $2R$

ans: B

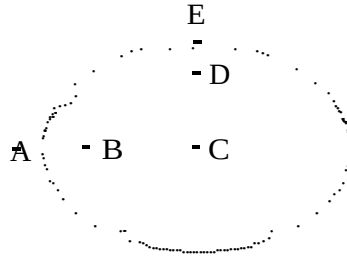
32. The mass density of a certain planet has spherical symmetry but varies in such a way that the mass inside every spherical surface with center at the center of the planet is proportional to the radius of the surface. If r is the distance from the center of the planet to a point mass inside the planet, the gravitational force on the mass is:

A. not dependent on r
B. proportional to r^2
C. proportional to r
D. proportional to $1/r$
E. proportional to $1/r^2$

ans: D

33. A spherical shell has inner radius R_1 , outer radius R_2 , and mass M , distributed uniformly throughout the shell. The magnitude of the gravitational force exerted on the shell by a point mass particle of m , located a distance d from the center, inside the inner radius, is:
- A. 0
 - B. GMm/R_1^2
 - C. GMm/d^2
 - D. $GMm/(R_2^2 - d^2)$
 - E. $GMm/(R_1 - d)^2$
- ans: A
34. A spherical shell has inner radius R_1 , outer radius R_2 , and mass M , distributed uniformly throughout the shell. The magnitude of the gravitational force exerted on the shell by a point mass m , located a distance d from the center, outside the outer radius, is:
- A. 0
 - B. GMm/R_1^2
 - C. GMm/d^2
 - D. $GMm/(R_2^2 - d^2)$
 - E. $GMm/(R_1 - d)^2$
- ans: C
35. A spherical shell has inner radius R_1 , outer radius R_2 , and mass M , distributed uniformly throughout the shell. The magnitude of the gravitational force exerted on the shell by a point particle of mass m located a distance d from the center, outside the inner radius and inside the outer radius, is:
- A. 0
 - B. GMm/d^2
 - C. $GMm/(R_2^3 - d^3)$
 - D. $GMm(d^3 - R_1^3)/d^2(R_2^3 - R_1^3)$
 - E. $GMm/(d^3 - R_1^3)$
- ans: D
36. An artificial satellite of Earth releases a bomb. Neglecting air resistance, the bomb will:
- A. strike Earth under the satellite at the instant of release
 - B. strike Earth under the satellite at the instant of impact
 - C. strike Earth ahead of the satellite at the instant of impact
 - D. strike Earth behind the satellite at the instant of impact
 - E. never strike Earth
- ans: E
37. An astronaut finishes some work on the outside of his satellite, which is in circular orbit around Earth. He leaves his wrench outside the satellite. The wrench will:
- A. fall directly down to Earth
 - B. continue in orbit at reduced speed
 - C. continue in orbit with the satellite
 - D. fly off tangentially into space
 - E. spiral down to Earth
- ans: C

38. The elliptical orbit of a planet around the Sun is shown on the diagram. Which of the following statements is true?



- A. the eccentricity of the orbit is less than zero
- B. the eccentricity of the orbit is greater than 1
- C. the sun might be at point C
- D. the sun might be at point D
- E. the sun might be at point B

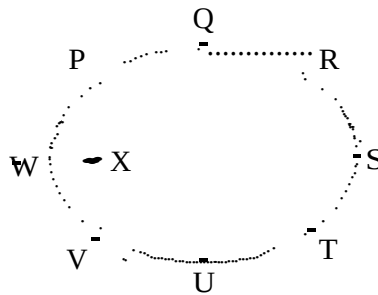
ans: E

39. Consider the statement: “Earth moves in a stable orbit around the Sun and is therefore in equilibrium”. The statement is:

- A. false, because no moving body can be in equilibrium
- B. true, because Earth does not fall into or fly away from the Sun
- C. false, because Earth is rotating on its axis and no rotating body can be in equilibrium
- D. false, because Earth has a considerable acceleration
- E. true, because if it were not in equilibrium then buildings and structures would not be stable

ans: D

40. A planet travels in an elliptical orbit about a star X as shown. The magnitude of the acceleration of the planet is:



- A. greatest at point Q
- B. greatest at point S
- C. greatest at point U
- D. greatest at point W
- E. the same at all points

ans: D

41. In planetary motion the line from the star to the planet sweeps out equal areas in equal times. This is a direct consequence of:

A. the conservation of energy
B. the conservation of momentum
C. the conservation of angular momentum
D. the conservation of mass
E. none of the above

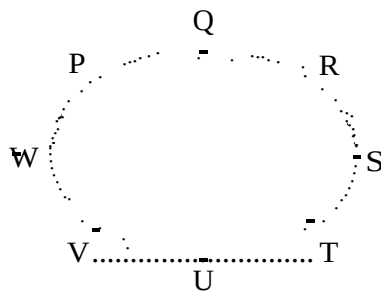
ans: C

42. The speed of a comet in an elliptical orbit about the Sun:

A. decreases while it is receding from the Sun
B. is constant
C. is greatest when farthest from the Sun
D. varies sinusoidally with time
E. equals $L/(mr)$, where L is its angular momentum, m is its mass, and r is its distance from the Sun

ans: A

43. A planet travels in an elliptical orbit about a star as shown. At what pair of points is the speed of the planet the same?

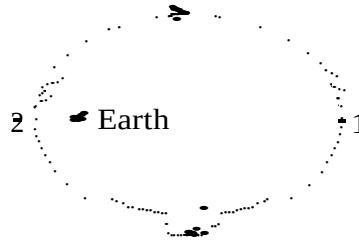


A. W and S
B. P and T
C. P and R
D. Q and U
E. V and R

ans: D

44. Planet 1 and planet 2 are both in circular orbits around the same central star. The orbit of planet 2 has a radius that is much larger than the radius of the orbit of planet 1. This means that:
- A. the period of planet 1 is greater than the period of planet 2 and the speed of planet 1 is greater than the speed of planet 2
 - B. the period of planet 1 is greater than the period of planet 2 and the speed of planet 1 is less than the speed of planet 2
 - C. the period of planet 1 is less than the period of planet 2 and the speed of planet 1 is less than the speed of planet 2
 - D. the period of planet 1 is less than the period of planet 2 and the speed of planet 1 is greater than the speed of planet 2
 - E. the planets have the same speed and the same period
- ans: D
45. For a planet in orbit around a star the perihelion distance is r_p and its speed at perihelion is v_p . The aphelion distance is r_a and its speed at aphelion is v_a . Which of the following is true?
- A. $v_a = v_p$
 - B. $v_a/r_a = v_p/r_p$
 - C. $v_a r_a^2 = v_p r_p^2$
 - D. $v_a/r_a^2 = v_p/r_p^2$
 - E. $v_a r_a^{2a} = v_p r_p^{2p}$
- ans: C
46. A planet is in circular orbit around the Sun. Its distance from the Sun is four times the average distance of Earth from the Sun. The period of this planet, in Earth years, is:
- A. 4
 - B. 8
 - C. 16
 - D. 64
 - E. 2.52
- ans: B
47. Two planets are orbiting a star in a distant galaxy. The first has a semimajor axis of 150×10^6 km, an eccentricity of 0.20, and a period of 1.0 Earth years. The second has a semimajor axis of 250×10^6 km, an eccentricity of 0.30, and a period of:
- A. 0.46 Earth years
 - B. 0.57 Earth years
 - C. 1.4 Earth years
 - D. 1.8 Earth years
 - E. 2.2 Earth years
- ans: E

48. A small satellite is in elliptical orbit around Earth as shown. If L denotes the magnitude of its angular momentum and K denotes kinetic energy:



- A. $L_2 > L_1$ and $K_2 > K_1$
- B. $L_2 > L_1$ and $K_2 = K_1$
- C. $L_2 = L_1$ and $K_2 = K_1$
- D. $L_2 < L_1$ and $K_2 = K_1$
- E. $L_2 = L_1$ and $K_2 > K_1$

ans: E

49. Assume that Earth is in circular orbit around the Sun with kinetic energy K and potential energy U , taken to be zero for infinite separation. Then, the relationship between K and U :

- A. is $K = U$
- B. is $K = -U$
- C. is $K = U/2$
- D. is $K = -U/2$
- E. depends on the radius of the orbit

ans: D

50. An artificial Earth satellite is moved from a circular orbit with radius R to a circular orbit with radius $2R$. During this move:

- A. the gravitational force does positive work, the kinetic energy of the satellite increases, and the potential energy of the Earth-satellite system increases
- B. the gravitational force does positive work, the kinetic energy of the satellite increases, and the potential energy of the Earth-satellite system decreases
- C. the gravitational force does positive work, the kinetic energy of the satellite decreases, and the potential energy of the Earth-satellite system increases
- D. the gravitational force does negative work, the kinetic energy of the satellite increases, and the potential energy of the Earth-satellite system decreases
- E. the gravitational force does negative work, the kinetic energy of the satellite decreases, and the potential energy of the Earth-satellite system increases

ans: E

51. An artificial satellite of Earth nears the end of its life due to air resistance. While still in orbit:

- A. it moves faster as the orbit lowers
- B. it moves slower as the orbit lowers
- C. it slowly spirals away from Earth
- D. it moves slower in the same orbit but with a decreasing period
- E. it moves faster in the same orbit but with an increasing period

ans: A

52. A spaceship is returning to Earth with its engine turned off. Consider only the gravitational field of Earth and let M be the mass of Earth, m be the mass of the spaceship, and R be the distance from the center of Earth. In moving from position 1 to position 2 the kinetic energy of the spaceship increases by:

A. $GMm \left(\frac{1}{R_2^2} - \frac{1}{R_1^2} \right)$

B. $GMm \left(\frac{1}{R_1^2} + \frac{1}{R_2^2} \right)$

C. $GMm \frac{R_1 - R_2}{R_1^2}$

D. $GMm \frac{R_1 - R_2}{R_1 R_2}$

E. $GMm \frac{R_1 R_2}{R_1^2 - R_2^2}$

ans: D

53. Given the perihelion distance, aphelion distance, and speed at perihelion of a planet, which of the following CANNOT be calculated?

- A. The mass of the star
- B. The mass of the planet
- C. The speed of the planet at aphelion
- D. The period of orbit
- E. The semimajor axis of the orbit

ans: B

54. The orbit of a certain satellite has a semimajor axis of 1.5×10^7 m and an eccentricity of 0.20. Its perigee (minimum distance) and apogee (maximum distance) are respectively:

- A. 1.2×10^7 m, 1.8×10^7 m
- B. 3.0×10^6 m, 1.2×10^7 m
- C. 9.6×10^6 m, 1.0×10^7 m
- D. 1.0×10^7 m, 1.2×10^7 m
- E. 9.6×10^6 m, 1.8×10^7 m

ans: A

55. A planet in another solar system orbits a star with a mass of 4.0×10^{30} kg. At one point in its orbit it is 250×10^6 km from the star and is moving at 35 km/s. Take the universal gravitational constant to be 6.67×10^{-11} m²/s² kg and calculate the semimajor axis of the planet's orbit. The result is:

- A. 79×10^6 km
- B. 160×10^6 km
- C. 290×10^6 km
- D. 320×10^6 km
- E. 590×10^6 km

ans: C

Chapter 14: FLUIDS

1. All fluids are:
 - A. gases
 - B. liquids
 - C. gases or liquids
 - D. non-metallic
 - E. transparentans: C

2. Gases may be distinguished from other forms of matter by their:
 - A. lack of color
 - B. small atomic weights
 - C. inability to form free surfaces
 - D. ability to flow
 - E. ability to exert a buoyant forceans: C

3. 1 Pa is:
 - A. 1 N/m
 - B. 1 m/N
 - C. 1 kg/m · s
 - D. 1 kg/m · s²
 - E. 1 N/m · sans: D

4. Mercury is a convenient liquid to use in a barometer because:
 - A. it is a metal
 - B. it has a high boiling point
 - C. it expands little with temperature
 - D. it has a high density
 - E. it looks silveryans: D

5. To obtain the absolute pressure from the gauge pressure:
 - A. subtract atmospheric pressure
 - B. add atmospheric pressure
 - C. subtract 273
 - D. add 273
 - E. convert to N/m²ans: B

6. Barometers and open-tube manometers are two instruments that are used to measure pressure.
- A. Both measure gauge pressure
 - B. Both measure absolute pressure
 - C. Barometers measure gauge pressure and manometers measure absolute pressure
 - D. Barometers measure absolute pressure and manometers measure gauge pressure
 - E. Both measure an average of the absolute and gauge pressures

ans: D

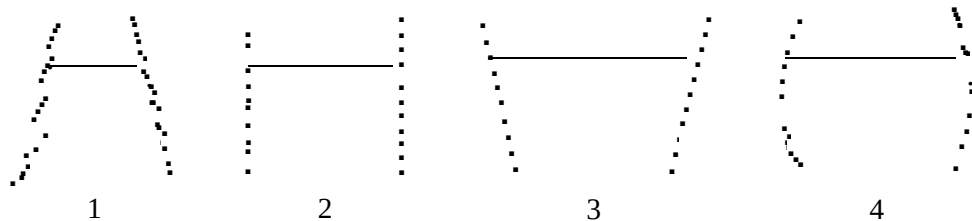
7. To measure moderately low pressures oil with a density of $8.5 \times 10^2 \text{ kg/m}^3$ is used in place of mercury in a barometer. A change in the height of the oil column of 1.0 mm indicates a change in pressure of about:
- A. $1.2 \times 10^{-7} \text{ Pa}$
 - B. $1.2 \times 10^{-5} \text{ Pa}$
 - C. 0.85 Pa
 - D. 1.2 Pa
 - E. 8.3 Pa

ans: E

8. The pressure exerted on the ground by a man is greatest when:
- A. he stands with both feet flat on the ground
 - B. he stands flat on one foot
 - C. he stands on the toes of one foot
 - D. he lies down on the ground
 - E. all of the above yield the same pressure

ans: C

9. The vessels shown below all contain water to the same height. Rank them according to the pressure exerted by the water on the vessel bottoms, least to greatest.

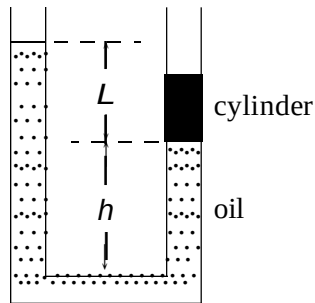


- A. 1, 2, 3, 4
- B. 3, 4, 2, 1
- C. 4, 3, 2, 1
- D. 2, 3, 4, 1
- E. All pressures are the same

ans: E

10. In a stationary homogeneous liquid:
- A. pressure is the same at all points
 - B. pressure depends on the direction
 - C. pressure is independent of any atmospheric pressure on the upper surface of the liquid
 - D. pressure is the same at all points at the same level
 - E. none of the above
- ans: D
11. Which of the following five statements, concerning the upper surface pressure of a liquid, is FALSE?
- A. It is independent of the surface area
 - B. It is the same for all points on that surface
 - C. It would not increase if the liquid depth were increased
 - D. It would increase if the liquid density were increased
 - E. It would increase if the atmospheric pressure increased
- ans: D
12. Several cans of different sizes and shapes are all filled with the same liquid to the same depth. Then:
- A. the weight of the liquid is the same for all cans
 - B. the force of the liquid on the bottom of each can is the same
 - C. the least pressure is at the bottom of the can with the largest bottom area
 - D. the greatest pressure is at the bottom of the can with the largest bottom area
 - E. the pressure on the bottom of each can is the same
- ans: E
13. An airtight box, having a lid of area 80 cm^2 , is partially evacuated. Atmospheric pressure is $1.01 \times 10^5\text{ Pa}$. A force of 600 N is required to pull the lid off the box. The pressure in the box was:
- A. $2.60 \times 10^4\text{ Pa}$
 - B. $6.35 \times 10^4\text{ Pa}$
 - C. $7.50 \times 10^4\text{ Pa}$
 - D. $1.38 \times 10^5\text{ Pa}$
 - E. $1.76 \times 10^5\text{ Pa}$
- ans: A
14. A closed hemispherical shell of radius R is filled with fluid at uniform pressure p . The net force of the fluid on the curved portion of the shell is given by:
- A. $2\pi R^2 p$
 - B. $\pi R^2 p$
 - C. $4\pi R^2 p$
 - D. $(4/3)\pi R^2 p$
 - E. $(4/3)\pi R^3 p$
- ans: B

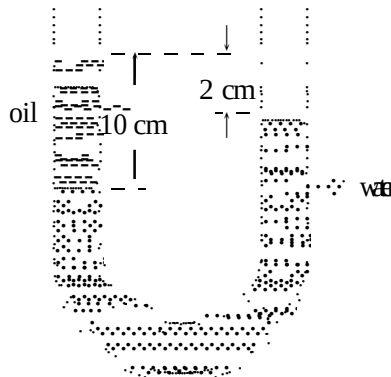
15. The diagram shows a U-tube with cross-sectional area A and partially filled with oil of density ρ . A solid cylinder, which fits the tube tightly but can slide without friction, is placed in the right arm. The system is in equilibrium. The weight of the cylinder is:



- A. $AL\rho g$
- B. $L^3\rho g$
- C. $A\rho(L + h)g$
- D. $A\rho(L - h)g$
- E. none of these

ans: A

16. The density of water is 1.0 g/cm^3 . The density of the oil in the left column of the U-tube shown below is:



- A. 0.20 g/cm^3
- B. 0.80 g/cm^3
- C. 1.0 g/cm^3
- D. 1.3 g/cm^3
- E. 5.0 g/cm^3

ans: B

17. A uniform U-tube is partially filled with water. Oil, of density 0.75 g/cm^3 , is poured into the right arm until the water level in the left arm rises 3 cm. The length of the oil column is then:
- A. 2.25 cm
 - B. 8 cm
 - C. 6 cm
 - D. 4 cm
 - E. need to know the cross-sectional area of the U-tube
- ans: B

18. A long U-tube contains mercury (density $= 14 \times 10^3 \text{ kg/m}^3$). When 10 cm of water (density $= 1.0 \times 10^3 \text{ kg/m}^3$) is poured into the left arm, the mercury in the right arm rises above its original level by:
- A. 0.36 cm
 - B. 0.72 cm
 - C. 14 cm
 - D. 35 cm
 - E. 70 cm
- ans: A

19. A bucket of water is pushed from left to right with increasing speed across a horizontal surface. Consider the pressure at two points at the same level in the water.
- A. It is the same
 - B. It is higher at the point on the left
 - C. It is higher at the point on the right
 - D. At first it is higher at the point on the left but as the bucket speeds up it is lower there
 - E. At first it is higher at the point on the right but as the bucket speeds up it is lower there
- ans: B

20. A bucket resting on the floor of an elevator contains an incompressible fluid of density ρ . When the elevator has an upward acceleration of magnitude a the pressure difference between two points in a fluid separated by a vertical distance Δh , is given by:
- A. $\rho a \Delta h$
 - B. $\rho g \Delta h$
 - C. $\rho(g + a) \Delta h$
 - D. $\rho(g - a) \Delta h$
 - E. $\rho g a \Delta h$
- ans: C

21. A bucket resting on the floor of an elevator contains an incompressible fluid of density ρ . When the elevator has a downward acceleration of magnitude a the pressure difference between two points in a fluid, separated by a vertical distance Δh , is given by:
- A. $\rho a \Delta h$
 - B. $\rho g \Delta h$
 - C. $\rho(g + a) \Delta h$
 - D. $\rho(g - a) \Delta h$
 - E. $\rho g a \Delta h$
- ans: D

22. “An object completely submerged in a fluid displaces its own volume of fluid”. This is:
- A. Pascal’s paradox
 - B. Archimedes’ principle
 - C. Pascal’s principle
 - D. true, but none of the above
 - E. false

ans: D

23. A certain object floats in fluids of density

- 1. $0.9\rho_0$
- 2. ρ_0
- 3. $1.1\rho_0$

Which of the following statements is true?

- A. the buoyant force of fluid 1 is greater than the buoyant forces of the other two fluids
- B. the buoyant force of fluid 3 is greater than the buoyant forces of the other two fluids
- C. the three fluids exert the same buoyant force
- D. the object displaces the same volume of all three fluids
- E. none of these are true

ans: C

24. A certain object floats in fluids of density

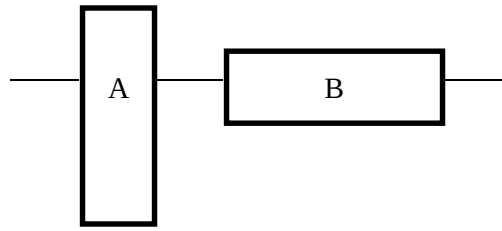
- 1. $0.9\rho_0$
- 2. ρ_0
- 3. $1.1\rho_0$

Rank these fluids according to the volume displaced by the object, least to greatest.

- A. 1, 2, 3
- B. 3, 2, 1
- C. 2, 3, 1
- D. 3, 1, 2
- E. All are the same

ans: B

25. Two identical blocks of ice float in water as shown. Then:



- A. block A displaces a greater volume of water since the pressure acts on a smaller bottom area
 - B. block B displaces a greater volume of water since the pressure is less on its bottom
 - C. the two blocks displace equal volumes of water since they have the same weight
 - D. block A displaces a greater volume of water since its submerged end is lower in the water
 - E. block B displaces a greater volume of water since its submerged end has a greater area
- ans: C
26. A block of ice at 0°C is floating on the surface of ice water in a beaker. The surface of the water just comes to the top of the beaker. When the ice melts the water level will:
- A. rise and overflow will occur
 - B. remain the same
 - C. fall
 - D. depend on the initial ratio of water to ice
 - E. depend on the shape of the block of ice
- ans: B
27. A block of ice at 0°C containing a piece of cork is floating on the surface of ice water in a beaker. When the ice has melted the water level:
- A. is higher
 - B. is lower
 - C. is the same
 - D. depends on the initial ratio of water to ice
 - E. depends on the shape of the ice block
- ans: C
28. A pirate chest rests at the bottom of an ocean. If the water is still, the net force it exerts on the chest:
- A. is upward
 - B. is downward
 - C. is zero
 - D. depends on the mass of the chest
 - E. depends on the contents of the chest
- ans: B

29. A small steel ball floats in a half-full container of mercury. When water is added:
- A. the ball will float on the water
 - B. the ball will rise slightly
 - C. the mercury will float on the water
 - D. the ball will sink to the bottom of the container
 - E. the ball will lower slightly more into the mercury
- ans: B
30. A cork floats on the surface of an incompressible liquid in a container exposed to atmospheric pressure. The container is then sealed and the air above the liquid is evacuated. The cork:
- A. sinks slightly
 - B. rises slightly
 - C. floats at the same height
 - D. bobs up and down about its old position
 - E. behaves erratically
- ans: C
31. An object hangs from a spring balance. The balance indicates 30 N in air and 20 N when the object is submerged in water. What does the balance indicate when the object is submerged in a liquid with a density that is half that of water?
- A. 20 N
 - B. 25 N
 - C. 30 N
 - D. 35 N
 - E. 40 N
- ans: B
32. A fir wood board floats in fresh water with 60% of its volume under water. The density of the wood in g/cm^3 is:
- A. 0.4
 - B. 0.5
 - C. 0.6
 - D. less than 0.4
 - E. more than 0.6
- ans: C
33. A boat floating in fresh water displaces 16, 000 N of water. How many newtons of saltwater would it displace if it floats in saltwater of specific gravity 1.17?
- A. 14, 500
 - B. 17, 600
 - C. 16, 000
 - D. 284
 - E. 234
- ans: C

34. A rock, which weighs 1400 N in air, has an apparent weight of 900 N when submerged in fresh water (998 kg/m^3). The volume of the rock is:
- A. 0.14 m^3
 - B. 0.60 m^3
 - C. 0.90 m^3
 - D. $5.1 \times 10^{-2} \text{ m}^3$
 - E. $9.2 \times 10^{-2} \text{ m}^3$
- ans: D
35. A loaded ship passes from a lake (fresh water) to the ocean (saltwater). Saltwater is more dense than fresh water and as a result the ship will:
- A. ride higher in the water
 - B. settle lower in the water
 - C. ride at the same level in the water
 - D. experience an increase in buoyant force
 - E. experience a decrease in buoyant force
- ans: A
36. The dimensions of a wooden raft (density = 150 kg/m^3) are $3.0 \text{ m} \times 3.0 \text{ m} \times 1.0 \text{ m}$. What maximum load can it carry in seawater (density = 1020 kg/m^3)?
- A. 1350 kg
 - B. 7800 kg
 - C. 9200 kg
 - D. 19, 500 kg
 - E. 24, 300 kg
- ans: B
37. A tin can has a volume of 1000 cm^3 and a mass of 100 g. Approximately what mass of lead shot can it carry without sinking in water?
- A. 900 g
 - B. 100 g
 - C. 1000 g
 - D. 1100 g
 - E. 980 g
- ans: A
38. A block of wood weighs 160 N and has a specific gravity of 0.60. To sink it in fresh water requires an additional downward force of:
- A. 54 N
 - B. 64 N
 - C. 96 N
 - D. 110 N
 - E. 240 N
- ans: D

39. A student standardizes the concentration of a saltwater solution by slowly adding salt until an egg will just float. The procedure is based on the assumption that:
- A. all eggs have the same volume
 - B. all eggs have the same weight
 - C. all eggs have the same density
 - D. all eggs have the same shape
 - E. the salt tends to neutralize the cholesterol in the egg
- ans: C
40. A solid has a volume of 8 cm^3 . When weighed on a spring scale calibrated in grams, the scale indicates 20 g. What does the scale indicate if the object is weighed while immersed in a liquid of density 2 g/cm^3 ?
- A. 4 g
 - B. 10 g
 - C. 12 g
 - D. 16 g
 - E. Zero, since the object will float
- ans: A
41. A 210-g object apparently loses 30 g when suspended in a liquid of density 2.0 g/cm^3 . The density of the object is:
- A. 7.0 g/cm^3
 - B. 3.5 g/cm^3
 - C. 1.4 g/cm^3
 - D. 14 g/cm^3
 - E. none of these
- ans: D
42. A steel ax and an aluminum piston have the same apparent weight in water. When they are weighed in air:
- A. they weigh the same
 - B. the ax is heavier
 - C. the piston is heavier
 - D. both weigh less than they did in water
 - E. depends on their shapes
- ans: C
43. The apparent weight of a steel sphere immersed in various liquids is measured using a spring scale. The greatest reading is obtained for that liquid:
- A. having the smallest density
 - B. having the largest density
 - C. subject to the greatest atmospheric pressure
 - D. having the greatest volume
 - E. in which the sphere was submerged deepest
- ans: A

44. A 0.50-N metal sinker appears (as measured using a spring scale) to have a weight of 0.45 N when submerged in water. The specific gravity of the metal is:
- A. 6
 - B. 8
 - C. 9
 - D. 10
 - E. 12
- ans: D
45. An object floats on the surface of a fluid. For purposes of calculating the torque on it, the buoyant force is taken to act at:
- A. the center of the bottom surface of the object
 - B. the center of gravity of the object
 - C. the center of gravity of the fluid that the object replaced
 - D. the geometric center of the object
 - E. none of the above
- ans: C
46. A blast of wind tips a sailboat in the clockwise direction when viewed from the stern. When the wind ceases the boat rotates back toward the upright position if, when it is tilted, the center of buoyancy:
- A. is above the center of gravity
 - B. is below the center of gravity
 - C. is to the right of the center of gravity
 - D. is to the left of the center of gravity
 - E. coincides with the center of gravity
- ans: C
47. A cork floats in water in a bucket resting on the floor of an elevator. The elevator then accelerates upward. During the acceleration:
- A. the cork is immersed more
 - B. the cork is immersed less
 - C. the cork is immersed the same amount
 - D. at first the cork is immersed less but as the elevator speeds up it is immersed more
 - E. at first the cork is immersed more but as the elevator speeds up it is immersed less
- ans: C
48. Two balls have the same shape and size but one is denser than the other. If frictional forces are negligible when they are dropped in air, which has the greater acceleration?
- A. The heavier ball
 - B. The lighter ball
 - C. They have the same acceleration
 - D. The heavier ball if atmospheric pressure is high, the lighter ball if it is low
 - E. The lighter ball if atmospheric pressure is high, the heavier ball if it is low
- ans: A

49. The principle of fluid pressure that is used in hydraulic brakes or lifts is that:
- A. pressure is the same at all levels in a fluid
 - B. increases of pressure are transmitted equally to all parts of a fluid
 - C. the pressure at a point in a fluid is due to the weight of the fluid above it
 - D. increases of pressure can only be transmitted through fluids
 - E. the pressure at a given depth is proportional to the depth in the fluid
- ans: B
50. Which of the following statements about Pascal's principle is true?
- A. It is valid only for incompressible fluids
 - B. It explains why light objects float
 - C. It explains why the pressure is greater at the bottom of a lake than at the surface
 - D. It is valid only for objects that are less dense than water
 - E. None of the above are true
- ans: E
51. The hydraulic automobile jack illustrates:
- A. Archimedes' principle
 - B. Pascal's principle
 - C. Hooke's law
 - D. Newton's third law
 - E. Newton's second law
- ans: B
52. One piston in a hydraulic lift has an area that is twice the area of the other. When the pressure at the smaller piston is increased by Δp the pressure at the larger piston:
- A. increases by $2\Delta p$
 - B. increases by $\Delta p/2$
 - C. increases by Δp
 - D. increases by $4\Delta p$
 - E. does not change
- ans: C
53. A hydraulic press has one piston of diameter 2.0 cm and the other piston of diameter 8.0 cm. What force must be applied to the smaller piston to obtain a force of 1600 N at the larger piston?
- A. 6.25 N
 - B. 25 N
 - C. 100 N
 - D. 400 N
 - E. 1600 N
- ans: C

54. The two arms of a U-tube are not identical, one having twice the diameter of the other. A cork in the narrow arm requires a force of 16 N to remove it. The tube is filled with water and the wide arm is fitted with a piston. The minimum force that must be applied to the piston to push the cork out is:
- A. 4 N
 - B. 8 N
 - C. 16 N
 - D. 32 N
 - E. 64 N

ans: E

55. A U-tube has dissimilar arms, one having twice the diameter of the other. It contains an incompressible fluid and is fitted with a sliding piston in each arm, with each piston in contact with the fluid. When the piston in the narrow arm is pushed down a distance d , the piston in the wide arm rises a distance:
- A. d
 - B. $2d$
 - C. $d/2$
 - D. $4d$
 - E. $d/4$

ans: E

56. A U-tube has dissimilar arms, one having twice the diameter of the other. It contains an incompressible fluid and is fitted with a sliding piston in each arm, with each piston in contact with the fluid. When an applied force does work W in pushing the piston in the narrow arm down, the fluid does work on the piston in the wide arm.
- A. W
 - B. $2W$
 - C. $W/2$
 - D. $4W$
 - E. $W/4$

ans: A

57. A fluid is undergoing “incompressible” flow. This means that:
- A. the pressure at a given point cannot change with time
 - B. the velocity at a given point cannot change with time
 - C. the velocity must be the same everywhere
 - D. the pressure must be the same everywhere
 - E. the density cannot change with time or location

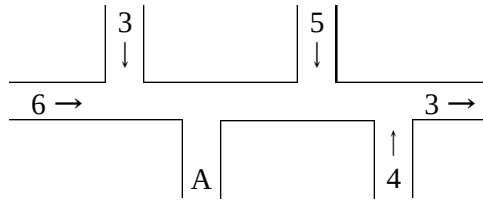
ans: E

58. A fluid is undergoing steady flow. Therefore:
- A. the velocity of any given molecule of fluid does not change
 - B. the pressure does not vary from point to point
 - C. the velocity at any given point does not vary with time
 - D. the density does not vary from point to point
 - E. the flow is not uphill or downhill

ans: C

59. If p is a pressure and ρ is a density then p/ρ has units of:
- A. m^2
 - B. m^2/s^2
 - C. N/m^2
 - D. kg/m^2
 - E. m^3/kg
- ans: B
60. One end of a cylindrical pipe has a radius of 1.5 cm. Water (density = $1.0 \times 10^3 \text{ kg}/\text{m}^3$) streams steadily out at 7.0 m/s. The rate at which mass is leaving the pipe is:
- A. 2.5 kg/s
 - B. 4.9 kg/s
 - C. 7.0 kg/s
 - D. 48 kg/s
 - E. $7.0 \times 10^3 \text{ kg/s}$
- ans: B
61. One end of a cylindrical pipe has a radius of 1.5 cm. Water (density = $1.0 \times 10^3 \text{ kg}/\text{m}^3$) streams steadily out at 7.0 m/s. The volume flow rate is:
- A. $4.9 \times 10^{-3} \text{ m}^3/\text{s}$
 - B. $2.5 \text{ m}^3/\text{s}$
 - C. $4.9 \text{ m}^3/\text{s}$
 - D. $7.0 \text{ m}^3/\text{s}$
 - E. $48 \text{ m}^3/\text{s}$
- ans: A
62. The equation of continuity for fluid flow can be derived from the conservation of:
- A. energy
 - B. mass
 - C. angular momentum
 - D. volume
 - E. pressure
- ans: B

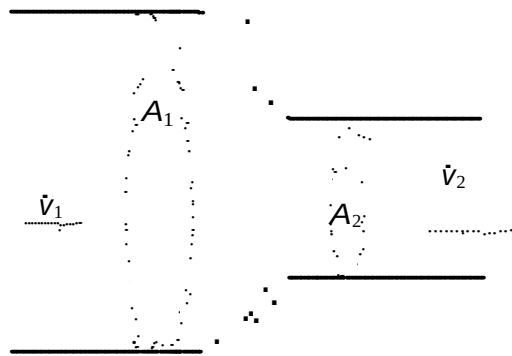
63. The diagram shows a pipe of uniform cross section in which water is flowing. The directions of flow and the volume flow rates (in cm^3/s) are shown for various portions of the pipe. The direction of flow and the volume flow rate in the portion marked A are:



- A. ↓ and $3 \text{ cm}^3/\text{s}$
- B. ↑ and $7 \text{ cm}^3/\text{s}$
- C. ↓ and $9 \text{ cm}^3/\text{s}$
- D. ↑ and $11 \text{ cm}^3/\text{s}$
- E. ↓ and $15 \text{ cm}^3/\text{s}$

ans: E

64. An incompressible liquid flows along the pipe as shown. The ratio of the speeds v_2/v_1 is:



- A. A_1/A_2
- B. A_2/A_1
- C. $\sqrt{A_1/A_2}$
- D. $\sqrt{A_2/A_1}$
- E. v_1/v_2

ans: A

65. Water flows through a cylindrical pipe of varying cross section. The velocity is 3.0 m/s at a point where the pipe diameter is 1.0 cm . At a point where the pipe diameter is 3.0 cm , the velocity is:

- A. 9 m/s
- B. 3 m/s
- C. 1 m/s
- D. 0.33 m/s
- E. 0.11 m/s

ans: D

66. A constriction in a pipe reduces its diameter from 4.0 cm to 2.0 cm. Where the pipe is narrow the water speed is 8.0 m/s. Where it is wide the water speed is:
- A. 2.0 m/s
 - B. 4.0 m/s
 - C. 8.0 m/s
 - D. 16m/s
 - E. 32m/s
- ans: C
67. Water flows from a 6.0-cm diameter pipe into an 8.0-cm diameter pipe. The speed in the 6.0-cm pipe is 5.0 m/s. The speed in the 8.0-cm pipe is:
- A. 2.8 m/s
 - B. 3.7 m/s
 - C. 6.6 m/s
 - D. 8.8 m/s
 - E. 9.9 m/s
- ans: A
68. A lawn sprinkler is made of a 1.0-cm diameter garden hose with one end closed and 25 holes, each with a diameter of 0.050 cm, cut near the closed end. If water flows at 2.0 m/s in the hose, the speed of the water leaving a hole is:
- A. 2.0 m/s
 - B. 32m/s
 - C. 40m/s
 - D. 600 m/s
 - E. 800 m/s
- ans: B
69. Bernoulli's equation can be derived from the conservation of:
- A. energy
 - B. mass
 - C. angular momentum
 - D. volume
 - E. pressure
- ans: A
70. Which of the following assumptions is NOT made in the derivation of Bernoulli's equation?
- A. Assume streamline flow
 - B. Neglect viscosity
 - C. Neglect friction
 - D. Neglect gravity
 - E. Neglect turbulence
- ans: D

71. The quantity y appearing in Bernoulli's equation MUST be measured:

- A. upward from the center of Earth
- B. upward from the surface of Earth
- C. upward from the lowest point in the flow
- D. downward from the highest point in the flow
- E. upward from any convenient level

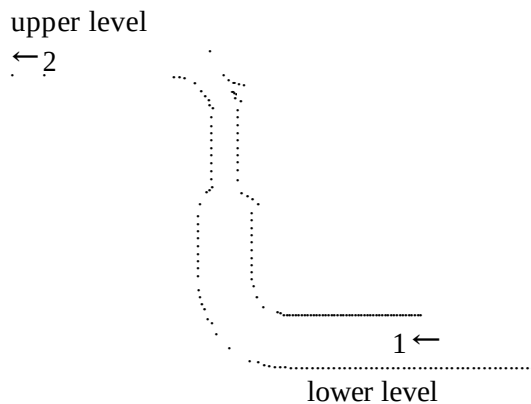
ans: E

72. Water flows through a constriction in a horizontal pipe. As it enters the constriction, the water's:

- A. speed increases and pressure decreases
- B. speed increases and pressure remains constant
- C. speed increases and pressure increases
- D. speed decreases and pressure increases
- E. speed decreases and pressure decreases

ans: A

73. Water is pumped through the hose shown below, from a lower level to an upper level. Compared to the water at point 1, the water at point 2:



- A. has greater speed and greater pressure
- B. has greater speed and less pressure
- C. has less speed and less pressure
- D. has less speed and greater pressure
- E. has greater speed and the same pressure

ans: B

74. A non-viscous incompressible liquid is flowing through a horizontal pipe of constant cross section. Bernoulli's equation and the equation of continuity predict that the drop in pressure along the pipe:
- A. is zero
 - B. depends on the length of the pipe
 - C. depends on the fluid velocity
 - D. depends on the cross-sectional area of the pipe
 - E. depends on the height of the pipe

ans: A

75. A non-viscous incompressible fluid is pumped steadily into the narrow end of a long tapered pipe and emerges from the wide end. The pressure at the input is greater than at the output. A possible explanation is:
- A. the fluid speed increases from input to output
 - B. the fluid speed is the same at the two ends
 - C. the fluid is flowing uphill
 - D. the fluid is flowing downhill
 - E. the fluid is flowing horizontally

ans: C

76. Water is pumped into one end of a long pipe at the rate of 40 L/min. It emerges at the other end at 24 L/min. A possible reason for this decrease in flow is:
- A. the water is being pumped uphill
 - B. the water is being pumped downhill
 - C. the diameter of the pipe is not the same at the two ends
 - D. friction in the pipe
 - E. a leak in the pipe

ans: E

77. Consider a pipe containing a fluid, with the fluid being at rest. To apply Bernoulli's equation to this situation:
- A. set v equal to zero because there is no motion
 - B. set g equal to zero because there is no acceleration
 - C. set v and g both equal to zero
 - D. set p equal to the atmospheric pressure
 - E. cannot be done, Bernoulli's equation applies only to fluids in motion

ans: A

78. Water (density = $1.0 \times 10^3 \text{ kg/m}^3$) flows through a horizontal tapered pipe. At the wide end its speed is 4.0 m/s. The difference in pressure between the two ends is $4.5 \times 10^3 \text{ Pa}$. The speed of the water at the narrow end is:
- A. 2.6 m/s
 - B. 3.4 m/s
 - C. 4.0 m/s
 - D. 4.5 m/s
 - E. 5.0 m/s

ans: E

79. Water is streaming downward from a faucet opening with an area of $3.0 \times 10^{-5} \text{ m}^2$. It leaves the faucet with a speed of 5.0 m/s. The cross-sectional area of the stream 0.50 m below the faucet is:
- A. $1.5 \times 10^{-5} \text{ m}^2$
 - B. $2.0 \times 10^{-5} \text{ m}^2$
 - C. $2.5 \times 10^{-5} \text{ m}^2$
 - D. $3.0 \times 10^{-5} \text{ m}^2$
 - E. $3.5 \times 10^{-5} \text{ m}^2$

ans: C

80. A large water tank, open at the top, has a small hole in the bottom. When the water level is 30 m above the bottom of the tank, the speed of the water leaking from the hole:
- A. is 2.5 m/s
 - B. is 24 m/s
 - C. is 44 m/s
 - D. cannot be calculated unless the area of the hole is given
 - E. cannot be calculated unless the areas of the hole and tank are given

ans: B

81. A large tank filled with water has two holes in the bottom, one with twice the radius of the other. In steady flow the speed of water leaving the larger hole is _ the speed of the water leaving the smaller.
- A. twice
 - B. four times
 - C. half
 - D. one-fourth
 - E. the same as

ans: E

82. A non-viscous incompressible fluid is pumped steadily up a vertical pipe with uniform cross section. The difference in pressure between points at the top and bottom:
- A. is the same as it would be if the fluid were motionless
 - B. is greater at higher flow rates than at lower flow rates
 - C. is less at higher flow rates than at lower flow rates
 - D. does not depend on the density of the fluid
 - E. is zero

ans: A

83. A water line enters a house 2.0 m below ground. A smaller diameter pipe carries water to a faucet 5.0 m above ground, on the second floor. Water flows at 2.0 m/s in the main line and at 7.0 m/s on the second floor. Take the density of water to be $1.0 \times 10^3 \text{ kg/m}^3$. The difference in pressure between the main line and the second floor is:
- A. $7.15 \times 10^4 \text{ Pa}$ with the main line at the higher pressure
 - B. $2.65 \times 10^4 \text{ Pa}$ with the main line at the higher pressure
 - C. $7.15 \times 10^4 \text{ Pa}$ with the main line at the lower pressure
 - D. $2.65 \times 10^4 \text{ Pa}$ with the main line at the lower pressure
 - E. $9.4 \times 10^4 \text{ Pa}$ with the main line at the higher pressure

ans: A

84. A person blows across the top of one arm of a U-tube partially filled with water. The water in that arm:
- A. rises slightly
 - B. drops slightly
 - C. remains at the same height
 - D. rises if the blowing is soft but drops if it is hard
 - E. rises if the blowing is hard but drops if it is soft

ans: A

Chapter 15: OSCILLATIONS

1. In simple harmonic motion, the restoring force must be proportional to the:
A. amplitude
B. frequency
C. velocity
D. displacement
E. displacement squared
ans: D
2. An oscillatory motion must be simple harmonic if:
A. the amplitude is small
B. the potential energy is equal to the kinetic energy
C. the motion is along the arc of a circle
D. the acceleration varies sinusoidally with time
E. the derivative, dU/dx , of the potential energy is negative
ans: D
3. In simple harmonic motion, the magnitude of the acceleration is:
A. constant
B. proportional to the displacement
C. inversely proportional to the displacement
D. greatest when the velocity is greatest
E. never greater than g
ans: B
4. A particle is in simple harmonic motion with period T . At time $t = 0$ it is at the equilibrium point. Of the following times, at which time is it furthest from the equilibrium point?
A. $0.5T$
B. $0.7T$
C. T
D. $1.4T$
E. $1.5T$
ans: B
5. A particle moves back and forth along the x axis from $x = -x_m$ to $x = +x_m$, in simple harmonic motion with period T . At time $t = 0$ it is at $x = +x_m$. When $t = 0.75T$:
A. it is at $x = 0$ and is traveling toward $x = +x_m$
B. it is at $x = 0$ and is traveling toward $x = -x_m$
C. it at $x = +x_m$ and is at rest
D. it is between $x = 0$ and $x = +x_m$ and is traveling toward $x = -x_m$
E. it is between $x = 0$ and $x = -x_m$ and is traveling toward $x = -x_m$
ans: A

6. A particle oscillating in simple harmonic motion is:

- A. never in equilibrium because it is in motion
- B. never in equilibrium because there is always a force
- C. in equilibrium at the ends of its path because its velocity is zero there
- D. in equilibrium at the center of its path because the acceleration is zero there
- E. in equilibrium at the ends of its path because the acceleration is zero there

ans: D

7. An object is undergoing simple harmonic motion. Throughout a complete cycle it:

- A. has constant speed
- B. has varying amplitude
- C. has varying period
- D. has varying acceleration
- E. has varying mass

ans: D

8. When a body executes simple harmonic motion, its acceleration at the ends of its path must be:

- A. zero
- B. less than g
- C. more than g
- D. suddenly changing in sign
- E. none of these

ans: E

9. A particle is in simple harmonic motion with period T . At time $t = 0$ it is halfway between the equilibrium point and an end point of its motion, traveling toward the end point. The next time it is at the same place is:

- A. $t = T$
- B. $t = T/2$
- C. $t = T/4$
- D. $t = T/8$
- E. none of the above

ans: E

10. An object attached to one end of a spring makes 20 complete oscillations in 10 s. Its period is:

- A. 2 Hz
- B. 10 s
- C. 0.5 Hz
- D. 2 s
- E. 0.50 s

ans: E

11. An object attached to one end of a spring makes 20 vibrations in 10 s. Its frequency is:
- A. 2 Hz
 - B. 10 s
 - C. 0.05 Hz
 - D. 2 s
 - E. 0.50 s
- ans: A
12. An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:
- A. 0.79 rad/s
 - B. 1.57 rad/s
 - C. 2.0 rad/s
 - D. 6.3 rad/s
 - E. 12.6 rad/s
- ans: E
13. Frequency f and angular frequency ω are related by
- A. $f = \pi\omega$
 - B. $f = 2\pi\omega$
 - C. $f = \omega/\pi$
 - D. $f = \omega/2\pi$
 - E. $f = 2\omega/\pi$
- ans: D
14. A block attached to a spring oscillates in simple harmonic motion along the x axis. The limits of its motion are $x = 10$ cm and $x = 50$ cm and it goes from one of these extremes to the other in 0.25 s. Its amplitude and frequency are:
- A. 40 cm, 2 Hz
 - B. 20 cm, 4 Hz
 - C. 40 cm, 2 Hz
 - D. 25 cm, 4 Hz
 - E. 20 cm, 2 Hz
- ans: B
15. A weight suspended from an ideal spring oscillates up and down with a period T . If the amplitude of the oscillation is doubled, the period will be:
- A. T
 - D. $1.5T$
 - B. $2T$
 - C. $T/2$
 - E. $4T$
- ans: A

16. In simple harmonic motion, the magnitude of the acceleration is greatest when:
- A. the displacement is zero
 - B. the displacement is maximum
 - C. the speed is maximum
 - D. the force is zero
 - E. the speed is between zero and its maximum

ans: B

17. In simple harmonic motion, the displacement is maximum when the:
- A. acceleration is zero
 - B. velocity is maximum
 - C. velocity is zero
 - D. kinetic energy is maximum
 - E. momentum is maximum

ans: C

18. In simple harmonic motion:
- A. the acceleration is greatest at the maximum displacement
 - B. the velocity is greatest at the maximum displacement
 - C. the period depends on the amplitude
 - D. the acceleration is constant
 - E. the acceleration is greatest at zero displacement

ans: A

19. The amplitude and phase constant of an oscillator are determined by:
- A. the frequency
 - B. the angular frequency
 - C. the initial displacement alone
 - D. the initial velocity alone
 - E. both the initial displacement and velocity

ans: E

20. Two identical undamped oscillators have the same amplitude of oscillation only if:
- A. they are started with the same displacement x_0
 - B. they are started with the same velocity v_0
 - C. they are started with the same phase
 - D. they are started so the combination $\omega^2 x_0^2 + v_0^2$ is the same
 - E. they are started so the combination $x_0^2 + \omega^2 v_0^2$ is the same

ans: D

21. The amplitude of any oscillator can be doubled by:
- A. doubling only the initial displacement
 - B. doubling only the initial speed
 - C. doubling the initial displacement and halving the initial speed
 - D. doubling the initial speed and halving the initial displacement
 - E. doubling both the initial displacement and the initial speed

ans: E

22. It is impossible for two particles, each executing simple harmonic motion, to remain in phase with each other if they have different:
- A. masses
 - B. periods
 - C. amplitudes
 - D. spring constants
 - E. kinetic energies
- ans: B
23. The acceleration of a body executing simple harmonic motion leads the velocity by what phase?
- A. 0
 - B. $\pi/8$ rad
 - C. $\pi/4$ rad
 - D. $\pi/2$ rad
 - E. π rad
- ans: D
24. The displacement of an object oscillating on a spring is given by $x(t) = x_m \cos(\omega t + \phi)$. If the initial displacement is zero and the initial velocity is in the negative x direction, then the phase constant ϕ is:
- A. 0
 - B. $\pi/2$ rad
 - C. π rad
 - D. $3\pi/2$ rad
 - E. 2π rad
- ans: B
25. The displacement of an object oscillating on a spring is given by $x(t) = x_m \cos(\omega t + \phi)$. If the object is initially displaced in the negative x direction and given a negative initial velocity, then the phase constant ϕ is between:
- A. 0 and $\pi/2$ rad
 - B. $\pi/2$ and π rad
 - C. π and $3\pi/2$ rad
 - D. $3\pi/2$ and 2π rad
 - E. none of the above (ϕ is exactly 0, $\pi/2$, π , or $3\pi/2$ rad)
- ans: B
26. A certain spring elongates 9.0 mm when it is suspended vertically and a block of mass M is hung on it. The natural angular frequency of this block-spring system:
- A. is 0.088 rad/s
 - B. is 33 rad/s
 - C. is 200 rad/s
 - D. is 1140 rad/s
 - E. cannot be computed unless the value of M is given
- ans: B

27. An object of mass m , oscillating on the end of a spring with spring constant k , has amplitude A . Its maximum speed is:

A. $A\sqrt{k/m}$
 B. $A^2\sqrt{k/m}$
 C. $A\sqrt{m/k}$
 D. Am/k
 E. A^2m/k

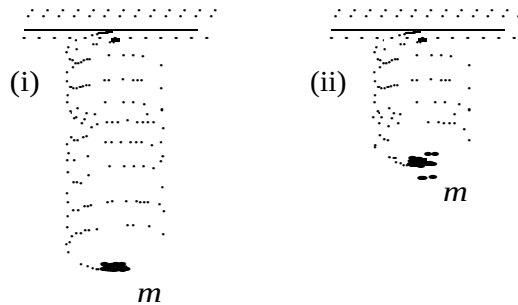
ans: A

28. A 0.20-kg object attached to a spring whose spring constant is 500 N/m executes simple harmonic motion. If its maximum speed is 5.0 m/s, the amplitude of its oscillation is:

A. 0.0020 m
 B. 0.10 m
 C. 0.20 m
 D. 25 m
 E. 250 m

ans: B

29. A simple harmonic oscillator consists of a particle of mass m and an ideal spring with spring constant k . Particle oscillates as shown in (i) with period T . If the spring is cut in half and used with the same particle, as shown in (ii), the period will be:



A. $2T$
 B. $\sqrt{2}T$
 C. $T/\sqrt{2}$
 D. T
 E. $T/2$

ans: C

30. A particle moves in simple harmonic motion according to $x = 2 \cos(50t)$, where x is in meters and t is in seconds. Its maximum velocity in m/s is:

A. $100 \sin(50t)$
 B. $100 \cos(50t)$
 C. 100
 D. 200
 E. none of these

ans: C

31. A 3-kg block, attached to a spring, executes simple harmonic motion according to $x = 2 \cos(50t)$ where x is in meters and t is in seconds. The spring constant of the spring is:
- A. 1 N/m
 - B. 100 N/m
 - C. 150 N/m
 - D. 7500 N/m
 - E. none of these
- ans: D
32. Let U be the potential energy (with the zero at zero displacement) and K be the kinetic energy of a simple harmonic oscillator. U_{avg} and K_{avg} are the average values over a cycle. Then:
- A. $K_{\text{avg}} > U_{\text{avg}}$
 - B. $K_{\text{avg}} < U_{\text{avg}}$
 - C. $K_{\text{avg}} = U_{\text{avg}}$
 - D. $K = 0$ when $U = 0$
 - E. $K + U = 0$
- ans: C
33. A particle is in simple harmonic motion along the x axis. The amplitude of the motion is x_m . At one point in its motion its kinetic energy is $K = 5$ J and its potential energy (measured with $U = 0$ at $x = 0$) is $U = 3$ J. When it is at $x = x_m$, the kinetic and potential energies are:
- A. $K = 5$ J and $U = 3$ J
 - B. $K = 5$ J and $U = -3$ J
 - C. $K = 8$ J and $U = 0$
 - D. $K = 0$ and $U = 8$ J
 - E. $K = 0$ and $U = -8$ J
- ans: D
34. A particle is in simple harmonic motion along the x axis. The amplitude of the motion is x_m . When it is at $x = x_1$, its kinetic energy is $K = 5$ J and its potential energy (measured with $U = 0$ at $x = 0$) is $U = 3$ J. When it is at $x = -x_1\frac{1}{2}$ the kinetic and potential energies are:
- A. $K = 5$ J and $U = 3$ J
 - B. $K = 5$ J and $U = -3$ J
 - C. $K = 8$ J and $U = 0$
 - D. $K = 0$ and $U = 8$ J
 - E. $K = 0$ and $U = -8$ J
- ans: A
35. A 0.25-kg block oscillates on the end of the spring with a spring constant of 200 N/m. If the system has an energy of 6.0 J, then the amplitude of the oscillation is:
- A. 0.06 m
 - B. 0.17 m
 - C. 0.24 m
 - D. 4.9 m
 - E. 6.9 m
- ans: C

36. A 0.25-kg block oscillates on the end of the spring with a spring constant of 200 N/m. If the system has an energy of 6.0 J, then the maximum speed of the block is:
- A. 0.06 m/s
 - B. 0.17 m/s
 - C. 0.24 m/s
 - D. 4.9 m/s
 - E. 6.9 m/s
- ans: E
37. A 0.25-kg block oscillates on the end of the spring with a spring constant of 200 N/m. If the oscillation is started by elongating the spring 0.15 m and giving the block a speed of 3.0 m/s, then the maximum speed of the block is:
- A. 0.13 m/s
 - B. 0.18 m/s
 - C. 3.7 m/s
 - D. 5.2 m/s
 - E. 13 m/s
- ans: D
38. A 0.25-kg block oscillates on the end of the spring with a spring constant of 200 N/m. If the oscillation is started by elongating the spring 0.15 m and giving the block a speed of 3.0 m/s, then the amplitude of the oscillation is:
- A. 0.13 m
 - B. 0.18 m
 - C. 3.7 m
 - D. 5.2 m
 - E. 13 m
- ans: B
39. An object on the end of a spring is set into oscillation by giving it an initial velocity while it is at its equilibrium position. In the first trial the initial velocity is v_0 and in the second it is $4v_0$. In the second trial:
- A. the amplitude is half as great and the maximum acceleration is twice as great
 - B. the amplitude is twice as great and the maximum acceleration is half as great
 - C. both the amplitude and the maximum acceleration are twice as great
 - D. both the amplitude and the maximum acceleration are four times as great
 - E. the amplitude is four times as great and the maximum acceleration is twice as great
- ans: C
40. A block attached to a spring undergoes simple harmonic motion on a horizontal frictionless surface. Its total energy is 50 J. When the displacement is half the amplitude, the kinetic energy is:
- A. zero
 - B. 12.5 J
 - C. 25 J
 - D. 37.5 J
 - E. 50 J
- ans: D

41. A mass-spring system is oscillating with amplitude A . The kinetic energy will equal the potential energy only when the displacement is:
- A. zero
 - B. $\pm A/\sqrt{2}$
 - C. $\pm A/2$
 - D. $\pm A/2$
 - E. anywhere between $-A$ and $+A$
- ans: C
42. If the length of a simple pendulum is doubled, its period will:
- A. halve
 - B. be greater by a factor of $\sqrt{2}$
 - C. be less by a factor of $\sqrt{2}$
 - D. double
 - E. remain the same
- ans: B
43. The period of a simple pendulum is 1 s on Earth. When brought to a planet where g is one-tenth that on Earth, its period becomes:
- A. $1\sqrt{10}$ s
 - B. $1/\sqrt{10}$ s
 - C. $\sqrt{10}$ s
 - D. 10 s
 - E. 10 s
- ans: D
44. The amplitude of oscillation of a simple pendulum is increased from 1° to 4° . Its maximum acceleration changes by a factor of:
- A. $1/4$
 - B. $1/2$
 - C. 2
 - D. 4
 - E. 16
- ans: D
45. A simple pendulum of length L and mass M has frequency f . To increase its frequency to $2f$:
- A. increase its length to $4L$
 - B. increase its length to $2L$
 - C. decrease its length to $L/2$
 - D. decrease its length to $L/4$
 - E. decrease its mass to $M/4$
- ans: D

46. A simple pendulum consists of a small ball tied to a string and set in oscillation. As the pendulum swings the tension force of the string is:
- A. constant
 - B. a sinusoidal function of time
 - C. the square of a sinusoidal function of time
 - D. the reciprocal of a sinusoidal function of time
 - E. none of the above
- ans: E
47. A simple pendulum has length L and period T . As it passes through its equilibrium position, the string is suddenly clamped at its midpoint. The period then becomes:
- A. $2T$
 - B. T
 - C. $T/2$
 - D. $T/4$
 - E. none of these
- ans: E
48. A simple pendulum is suspended from the ceiling of an elevator. The elevator is accelerating upwards with acceleration a . The period of this pendulum, in terms of its length L , g , and a is:
- A. $2\pi\sqrt{L/g}$
 - B. $2\pi\sqrt{L/(g+a)}$
 - C. $2\pi\sqrt{L/(g-a)}$
 - D. $2\pi\sqrt{L/g}$
 - E. $(1/2\pi)\sqrt{g/L}$
- ans: B
49. Three physical pendulums, with masses m_1 , $m_2 = 2m_1$, and $m_3 = 3m_1$, have the same shape and size and are suspended at the same point. Rank them according to their periods, from shortest to longest.
- A. 1, 2, 3
 - B. 3, 2, 1
 - C. 2, 3, 1
 - D. 2, 1, 3
 - E. All the same
- ans: E

50. Five hoops are each pivoted at a point on the rim and allowed to swing as physical pendulums. The masses and radii are

hoop 1: $M = 150\text{ g}$ and $R = 50\text{ cm}$
hoop 2: $M = 200\text{ g}$ and $R = 40\text{ cm}$
hoop 3: $M = 250\text{ g}$ and $R = 30\text{ cm}$
hoop 4: $M = 300\text{ g}$ and $R = 20\text{ cm}$
hoop 5: $M = 350\text{ g}$ and $R = 10\text{ cm}$

Order the hoops according to the periods of their motions, smallest to largest.

- A. 1, 2, 3, 4, 5
- B. 5, 4, 3, 2, 1
- C. 1, 2, 3, 5, 4
- D. 1, 2, 5, 4, 3
- E. 5, 4, 1, 2, 3

ans: B

51. A meter stick is pivoted at a point a distance a from its center and swings as a physical pendulum. Of the following values for a , which results in the shortest period of oscillation?

- A. $a = 0.1\text{ m}$
- B. $a = 0.2\text{ m}$
- C. $a = 0.3\text{ m}$
- D. $a = 0.4\text{ m}$
- E. $a = 0.5\text{ m}$

ans: C

52. The rotational inertia of a uniform thin rod about its end is $\frac{1}{3}ML^2$, where M is the mass and L is the length. Such a rod is hung vertically from one end and set into small amplitude oscillation. If $L = 1.0\text{ m}$ this rod will have the same period as a simple pendulum of length:

- A. 33 cm
- B. 50 cm
- C. 67 cm
- D. 100 cm
- E. 150 cm

ans: C

53. Two uniform spheres are pivoted on horizontal axes that are tangent to their surfaces. The one with the longer period of oscillation is the one with:

- A. the larger mass
- B. the smaller mass
- C. the larger rotational inertia
- D. the smaller rotational inertia
- E. the larger radius

ans: E

54. The x and y coordinates of a point each execute simple harmonic motion. The result might be a circular orbit if:
- A. the amplitudes are the same but the frequencies are different
 - B. the amplitudes and frequencies are both the same
 - C. the amplitudes and frequencies are both different
 - D. the phase constants are the same but the amplitudes are different
 - E. the amplitudes and the phase constants are both different
- ans: B

55. The x and y coordinates of a point each execute simple harmonic motion. The frequencies are the same but the amplitudes are different. The resulting orbit might be:
- A. an ellipse
 - B. a circle
 - C. a parabola
 - D. a hyperbola
 - E. a square
- ans: A

56. For an oscillator subjected to a damping force proportional to its velocity:
- A. the displacement is a sinusoidal function of time.
 - B. the velocity is a sinusoidal function of time.
 - C. the frequency is a decreasing function of time.
 - D. the mechanical energy is constant.
 - E. none of the above is true.
- ans: E

57. Five particles undergo damped harmonic motion. Values for the spring constant k , the damping constant b , and the mass m are given below. Which leads to the smallest rate of loss of mechanical energy?
- A. $k = 100 \text{ N/m}$, $m = 50 \text{ g}$, $b = 8 \text{ g/s}$
 - B. $k = 150 \text{ N/m}$, $m = 50 \text{ g}$, $b = 5 \text{ g/s}$
 - C. $k = 150 \text{ N/m}$, $m = 10 \text{ g}$, $b = 8 \text{ g/s}$
 - D. $k = 200 \text{ N/m}$, $m = 8 \text{ g}$, $b = 6 \text{ g/s}$
 - E. $k = 100 \text{ N/m}$, $m = 2 \text{ g}$, $b = 4 \text{ g/s}$
- ans: B

58. A sinusoidal force with a given amplitude is applied to an oscillator. To maintain the largest amplitude oscillation the frequency of the applied force should be:
- A. half the natural frequency of the oscillator
 - B. the same as the natural frequency of the oscillator
 - C. twice the natural frequency of the oscillator
 - D. unrelated to the natural frequency of the oscillator
 - E. determined from the maximum speed desired
- ans: B

59. A sinusoidal force with a given amplitude is applied to an oscillator. At resonance the amplitude of the oscillation is limited by:
- A. the damping force
 - B. the initial amplitude
 - C. the initial velocity
 - D. the force of gravity
 - E. none of the above

ans: A

60. An oscillator is subjected to a damping force that is proportional to its velocity. A sinusoidal force is applied to it. After a long time:
- A. its amplitude is an increasing function of time
 - B. its amplitude is a decreasing function of time
 - C. its amplitude is constant
 - D. its amplitude is a decreasing function of time only if the damping constant is large
 - E. its amplitude increases over some portions of a cycle and decreases over other portions

ans: C

61. A block on a spring is subjected to a damping force that is proportional to its velocity and to an applied sinusoidal force. The energy dissipated by damping is supplied by:
- A. the potential energy of the spring
 - B. the kinetic energy of the mass
 - C. gravity
 - D. friction
 - E. the applied force

ans: E

62. The table below gives the values of the spring constant k , damping constant b , and mass m for a particle in damped harmonic motion. Which of these takes the longest time for its mechanical energy to decrease to one-fourth of its initial value?

	k	b	m
A	k_0	b_0	m_0
B	$3k_0$	$2b_0$	m_0
C	$k_0/2$	$6b_0$	$2m_0$
D	$4k_0$	b_0	$2m_0$
E	k_0	b_0	$10m_0$

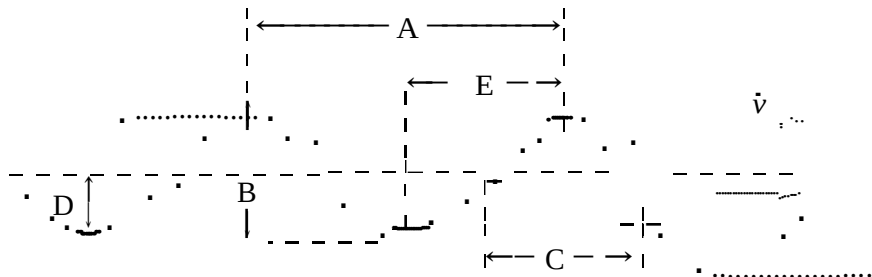
ans: E

Chapter 16: WAVES — I

- For a transverse wave on a string the string displacement is described by $y(x, t) = f(x - at)$, where f is a given function and a is a positive constant. Which of the following does NOT necessarily follow from this statement?
 - The shape of the string at time $t = 0$ is given by $f(x)$.
 - The shape of the waveform does not change as it moves along the string.
 - The waveform moves in the positive x direction.
 - The speed of the waveform is a .
 - The speed of the waveform is x/t .

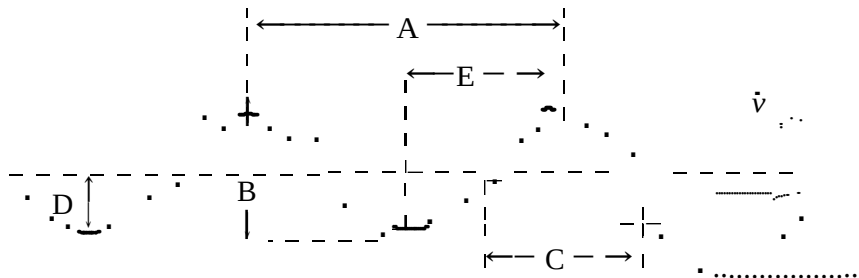
ans: E

- A sinusoidal wave is traveling toward the right as shown. Which letter correctly labels the amplitude of the wave?



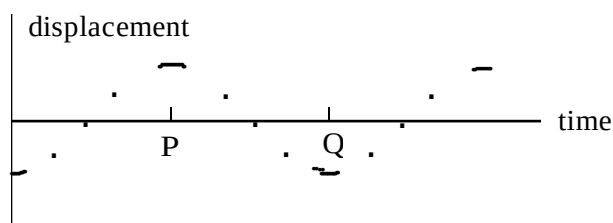
ans: D

- A sinusoidal wave is traveling toward the right as shown. Which letter correctly labels the wavelength of the wave?



ans: A

4. In the diagram below, the interval PQ represents:



- A. wavelength/2
- B. wavelength
- C. 2 x amplitude
- D. period/2
- E. period

ans: D

5. Let f be the frequency, v the speed, and T the period of a sinusoidal traveling wave. The correct relationship is:

- A. $f = 1/T$
- B. $f = v + T$
- C. $f = vT$
- D. $f = v/T$
- E. $f = T/v$

ans: A

6. Let f be the frequency, v the speed, and T the period of a sinusoidal traveling wave. The angular frequency is given by:

- A. $1/T$
- B. $2\pi/T$
- C. vT
- D. f/T
- E. T/f

ans: B

7. The displacement of a string is given by

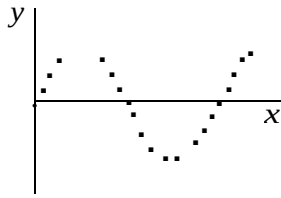
$$y(x, t) = y_m \sin(kx + \omega t) .$$

The wavelength of the wave is:

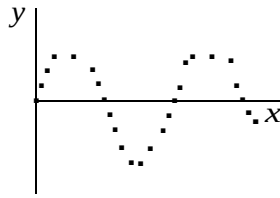
- A. $2\pi k/\omega$
- B. k/ω
- C. ωk
- D. $2\pi/k$
- E. $k/2\pi$

ans: D

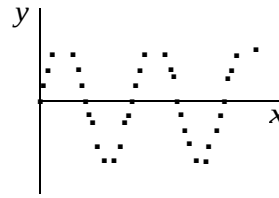
8. Three traveling sinusoidal waves are on identical strings, with the same tension. The mathematical forms of the waves are $y_1(x, t) = y_m \sin(3x - 6t)$, $y_2(x, t) = y_m \sin(4x - 8t)$, and $y_3(x, t) = y_m \sin(6x - 12t)$, where x is in meters and t is in seconds. Match each mathematical form to the appropriate graph below.



i



ii



iii

- A. y_1 : i, y_2 : ii, y_3 : iii
 B. y_1 : iii, y_2 : ii, y_3 : i
 C. y_1 : i, y_2 : iii, y_3 : ii
 D. y_1 : ii, y_2 : i, y_3 : iii
 E. y_1 : iii, y_2 : i, y_3 : ii

ans: A

9. The displacement of a string is given by

$$y(x, t) = y_m \sin(kx + \omega t).$$

The speed of the wave is:

- A. $2\pi k/\omega$
 B. ω/k
 C. ωk
 D. $2\pi/k$
 E. $k/2\pi$

ans: B

10. A wave is described by $y(x, t) = 0.1 \sin(3x + 10t)$, where x is in meters, y is in centimeters, and t is in seconds. The angular wave number is:

- A. 0.10 rad/m
 B. 3π rad/m
 C. 10 rad/m
 D. 10π rad/m
 E. 3.0 rad/cm

ans: E

11. A wave is described by $y(x, t) = 0.1 \sin(3x - 10t)$, where x is in meters, y is in centimeters, and t is in seconds. The angular frequency is:

- A. 0.10 rad/s
 B. 3.0π rad/s
 C. 10π rad/s
 D. 20π rad/s
 E. 10 rad/s

ans: E

12. Water waves in the sea are observed to have a wavelength of 300 m and a frequency of 0.07 Hz. The speed of these waves is:

A. 0.00021 m/s
 B. 2.1 m/s
 C. 21 m/s
 D. 210 m/s
 E. none of these

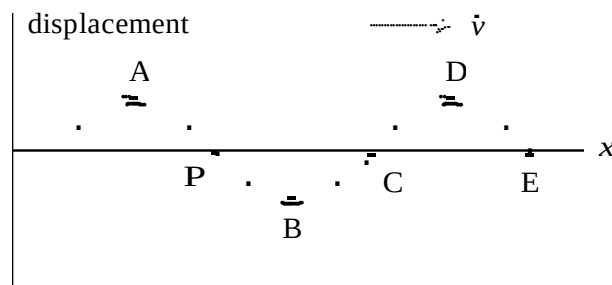
ans: C

13. Sinusoidal water waves are generated in a large ripple tank. The waves travel at 20 cm/s and their adjacent crests are 5.0 cm apart. The time required for each new whole cycle to be generated is:

A. 100 s
 B. 4.0 s
 C. 2.0 s
 D. 0.5 s
 E. 0.25 s

ans: E

14. A traveling sinusoidal wave is shown below. At which point is the motion 180° out of phase with the motion at point P?



ans: C

15. The displacement of a string carrying a traveling sinusoidal wave is given by

$$y(x, t) = y_m \sin(kx - \omega t - \phi).$$

At time $t = 0$ the point at $x = 0$ has a displacement of 0 and is moving in the positive y direction. The phase constant ϕ is:

A. 45°
 B. 90°
 C. 135°
 D. 180°
 E. 270°

ans: D

16. The displacement of a string carrying a traveling sinusoidal wave is given by

$$y(x, t) = y_m \sin(kx - \omega t - \phi) .$$

At time $t = 0$ the point at $x = 0$ has a velocity of 0 and a positive displacement. The phase constant ϕ is:

- A. 45°
- B. 90°
- C. 135°
- D. 180°
- E. 270°

ans: E

17. The displacement of a string carrying a traveling sinusoidal wave is given by

$$y(x, t) = y_m \sin(kx - \omega t - \phi) .$$

At time $t = 0$ the point at $x = 0$ has velocity v_0 and displacement y_0 . The phase constant ϕ is given by $\tan \phi =$:

- A. $v_0/\omega y_0$
- B. $\omega y_0/v_0$
- C. $\omega v_0/y_0$
- D. $y_0/\omega v_0$
- E. $\omega v_0 y_0$

ans: B

18. A sinusoidal transverse wave is traveling on a string. Any point on the string:

- A. moves in the same direction as the wave
- B. moves in simple harmonic motion with a different frequency than that of the wave
- C. moves in simple harmonic motion with the same angular frequency as the wave
- D. moves in uniform circular motion with a different angular speed than the wave
- E. moves in uniform circular motion with the same angular speed as the wave

ans: C

19. Here are the equations for three waves traveling on separate strings. Rank them according to the maximum transverse speed, least to greatest.

wave 1: $y(x, t) = (2.0 \text{ mm}) \sin[(4.0 \text{ m}^{-1})x - (3.0 \text{ s}^{-1})t]$

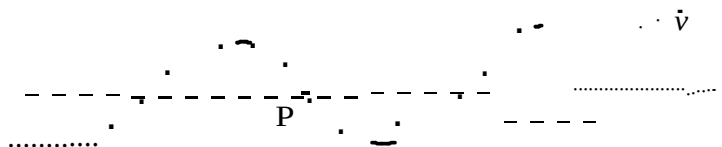
wave 2: $y(x, t) = (1.0 \text{ mm}) \sin[(8.0 \text{ m}^{-1})x - (4.0 \text{ s}^{-1})t]$

wave 3: $y(x, t) = (1.0 \text{ mm}) \sin[(4.0 \text{ m}^{-1})x - (8.0 \text{ s}^{-1})t]$

- A. 1, 2, 3
- B. 1, 3, 2
- C. 2, 1, 3
- D. 2, 3, 1
- E. 3, 1, 2

ans: C

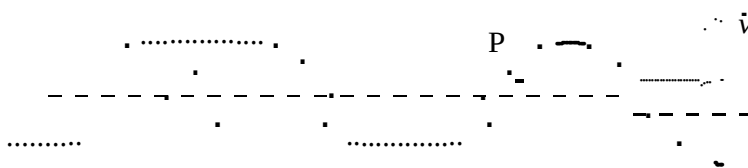
20. The transverse wave shown is traveling from left to right in a medium. The direction of the instantaneous velocity of the medium at point P is:



- A. $\uparrow$
- B. $\downarrow$
- C. $\rightarrow$
- D. **3**
- E. no direction since $v = 0$

ans: A

21. A wave traveling to the right on a stretched string is shown below. The direction of the instantaneous velocity of the point P on the string is:



- A. $\uparrow$
- B. $\downarrow$
- C. $\rightarrow$
- D. **3**
- E. no direction since $v = 0$

ans: B

22. Sinusoidal waves travel on five different strings, all with the same tension. Four of the strings have the same linear mass density, but the fifth has a different linear mass density. Use the mathematical forms of the waves, given below, to identify the string with the different linear mass density. In the expressions x and y are in centimeters and t is in seconds.

- A. $y(x, t) = (2 \text{ cm}) \sin(2x - 4t)$
- B. $y(x, t) = (2 \text{ cm}) \sin(4x - 10t)$
- C. $y(x, t) = (2 \text{ cm}) \sin(6x - 12t)$
- D. $y(x, t) = (2 \text{ cm}) \sin(8x - 16t)$
- E. $y(x, t) = (2 \text{ cm}) \sin(10x - 20t)$

ans: B

23. Any point on a string carrying a sinusoidal wave is moving with its maximum speed when:

- A. the magnitude of its acceleration is a maximum
- B. the magnitude of its displacement is a maximum
- C. the magnitude of its displacement is a minimum
- D. the magnitude of its displacement is half the amplitude
- E. the magnitude of its displacement is one-fourth the amplitude

ans: C

24. The mathematical forms for three sinusoidal traveling waves are given by

wave 1: $y(x, t) = (2 \text{ cm}) \sin(3x - 6t)$

wave 2: $y(x, t) = (3 \text{ cm}) \sin(4x - 12t)$

wave 3: $y(x, t) = (4 \text{ cm}) \sin(5x - 11t)$

where x is in meters and t is in seconds. Of these waves:

- A. wave 1 has the greatest wave speed and the greatest maximum transverse string speed
- B. wave 2 has the greatest wave speed and wave 1 has the greatest maximum transverse string speed
- C. wave 3 has the greatest wave speed and the greatest maximum transverse string speed
- D. wave 2 has the greatest wave speed and wave 3 has the greatest maximum transverse string speed
- E. wave 3 has the greatest wave speed and wave 2 has the greatest maximum transverse string speed

ans: D

25. Suppose the maximum speed of a string carrying a sinusoidal wave is v_s . When the displacement of a point on the string is half its maximum, the speed of the point is:

- A. $v_s/2$
- B. $2v_s$
- C. $v_s/4$
- D. $3v_s/4$
- E. $\sqrt{3}v_s/2$

ans: E

26. A string carries a sinusoidal wave with an amplitude of 2.0 cm and a frequency of 100 Hz. The maximum speed of any point on the string is:

- A. 2.0m/s
- B. 4.0m/s
- C. 6.3m/s
- D. 13 m/s
- E. unknown (not enough information is given)

ans: D

27. A transverse traveling sinusoidal wave on a string has a frequency of 100 Hz, a wavelength of 0.040 m, and an amplitude of 2.0 mm. The maximum velocity in m/s of any point on the string is:

- A. 0.2
- B. 1.3
- C. 4
- D. 15
- E. 25

ans: B

28. A transverse traveling sinusoidal wave on a string has a frequency of 100 Hz, a wavelength of 0.040 m, and an amplitude of 2.0 mm. The maximum acceleration in m/s^2 of any point on the string is:

A. 0
B. 130
C. 395
D. 790
E. 1600

ans: D

29. The speed of a sinusoidal wave on a string depends on:

A. the frequency of the wave
B. the wavelength of the wave
C. the length of the string
D. the tension in the string
E. the amplitude of the wave

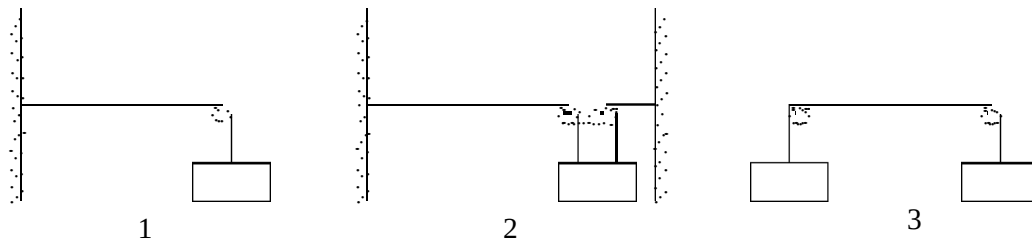
ans: D

30. The time required for a small pulse to travel from A to B on a stretched cord shown is NOT altered by changing:

A. the linear mass density of the cord
B. the length between A and B
C. the shape of the pulse
D. the tension in the cord
E. none of the above (changes in all alter the time)

ans: C

31. The diagrams show three identical strings that have been put under tension by suspending blocks of 5 kg each. For which is the wave speed the greatest?



A. 1
B. 2
C. 3
D. 1 and 3 tie
E. 2 and 3 tie

ans: D

32. For a given medium, the frequency of a wave is:
- A. independent of wavelength
 - B. proportional to wavelength
 - C. inversely proportional to wavelength
 - D. proportional to the amplitude
 - E. inversely proportional to the amplitude
- ans: C
33. The tension in a string with a linear mass density of 0.0010 kg/m is 0.40 N . A sinusoidal wave with a wavelength of 20 cm on this string has a frequency of:
- A. 0.0125 Hz
 - B. 0.25 Hz
 - C. 100 Hz
 - D. 630 Hz
 - E. 2000 Hz
- ans: C
34. When a 100-Hz oscillator is used to generate a sinusoidal wave on a certain string the wavelength is 10 cm . When the tension in the string is doubled the generator produces a wave with a frequency and wavelength of:
- A. 200 Hz and 20 cm
 - B. 141 Hz and 10 cm
 - C. 100 Hz and 20 cm
 - D. 100 Hz and 14 cm
 - E. 50 Hz and 14 cm
- ans: D
35. A source of frequency f sends waves of wavelength λ traveling with speed v in some medium. If the frequency is changed from f to $2f$, then the new wavelength and new speed are (respectively):
- A. $2\lambda, v$
 - B. $\lambda/2, v$
 - C. $\lambda, 2v$
 - D. $\lambda, v/2$
 - E. $\lambda/2, 2v$
- ans: B
36. A long string is constructed by joining the ends of two shorter strings. The tension in the strings is the same but string I has 4 times the linear mass density of string II. When a sinusoidal wave passes from string I to string II:
- A. the frequency decreases by a factor of 4
 - B. the frequency decreases by a factor of 2
 - C. the wavelength decreases by a factor of 4
 - D. the wavelength decreases by a factor of 2
 - E. the wavelength increases by a factor of 2
- ans: D

37. Three separate strings are made of the same material. String 1 has length L and tension τ , string 2 has length $2L$ and tension 2τ , and string 3 has length $3L$ and tension 3τ . A pulse is started at one end of each string. If the pulses start at the same time, the order in which they reach the other end is:
- A. 1, 2, 3
 - B. 3, 2, 1
 - C. 2, 3, 1
 - D. 3, 1, 2
 - E. they all take the same time

ans: A

38. A long string is constructed by joining the ends of two shorter strings. The tension in the strings is the same but string I has 4 times the linear mass density of string II. When a sinusoidal wave passes from string I to string II:
- A. the frequency decreases by a factor of 4
 - B. the frequency decreases by a factor of 2
 - C. the wave speed decreases by a factor of 4
 - D. the wave speed decreases by a factor of 2
 - E. the wave speed increases by a factor of 2

ans: E

39. Two identical but separate strings, with the same tension, carry sinusoidal waves with the same frequency. Wave A has an amplitude that is twice that of wave B and transmits energy at a rate that is that of wave B.
- A. half
 - B. twice
 - C. one-fourth
 - D. four times
 - E. eight times

ans: D

40. Two identical but separate strings, with the same tension, carry sinusoidal waves with the same frequency. Wave A has an amplitude that is twice that of wave B and transmits energy at a rate that is that of wave B.
- A. half
 - B. twice
 - C. one-fourth
 - D. four times
 - E. eight times

ans: D

41. A sinusoidal wave is generated by moving the end of a string up and down periodically. The generator must supply the greatest power when the end of the string
- A. has its greatest acceleration
 - B. has its greatest displacement
 - C. has half its greatest displacement
 - D. has one-fourth its greatest displacement
 - E. has its least displacement

ans: E

42. A sinusoidal wave is generated by moving the end of a string up and down periodically. The generator does not supply any power when the end of the string
- A. has its least acceleration
 - B. has its greatest displacement
 - C. has half its greatest displacement
 - D. has one-fourth its greatest displacement
 - E. has its least displacement

ans: B

43. The sum of two sinusoidal traveling waves is a sinusoidal traveling wave only if:
- A. their amplitudes are the same and they travel in the same direction.
 - B. their amplitudes are the same and they travel in opposite directions.
 - C. their frequencies are the same and they travel in the same direction.
 - D. their frequencies are the same and they travel in opposite directions.
 - E. their frequencies are the same and their amplitudes are the same.

ans: C

44. Two traveling sinusoidal waves interfere to produce a wave with the mathematical form

$$y(x, t) = y_m \sin(kx + \omega t + \alpha) .$$

If the value of ϕ is appropriately chosen, the two waves might be:

- A. $y_1(x, t) = (y_m/3) \sin(kx + \omega t)$ and $y_2(x, t) = (y_m/3) \sin(kx + \omega t + \phi)$
- B. $y_1(x, t) = 0.7y_m \sin(kx - \omega t)$ and $y_2(x, t) = 0.7y_m \sin(kx - \omega t + \phi)$
- C. $y_1(x, t) = 0.7y_m \sin(kx - \omega t)$ and $y_2(x, t) = 0.7y_m \sin(kx + \omega t + \phi)$
- D. $y_1(x, t) = 0.7y_m \sin[(kx/2) - (\omega t/2)]$ and $y_2(x, t) = 0.7y_m \sin[(kx/2) - (\omega t/2) + \phi]$
- E. $y_1(x, t) = 0.7y_m \sin(kx + \omega t)$ and $y_2(x, t) = 0.7y_m \sin(kx + \omega t + \phi)$

ans: E

45. Fully constructive interference between two sinusoidal waves of the same frequency occurs only if they:
- A. travel in opposite directions and are in phase
 - B. travel in opposite directions and are 180° out of phase
 - C. travel in the same direction and are in phase
 - D. travel in the same direction and are 180° out of phase
 - E. travel in the same direction and are 90° out of phase

ans: C

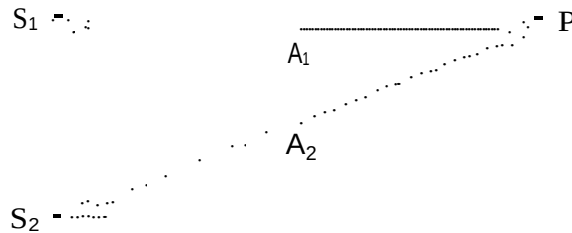
46. Fully destructive interference between two sinusoidal waves of the same frequency and amplitude occurs only if they:
- A. travel in opposite directions and are in phase
 - B. travel in opposite directions and are 180° out of phase
 - C. travel in the same direction and are in phase
 - D. travel in the same direction and are 180° out of phase
 - E. travel in the same direction and are 90° out of phase
- ans: D
47. Two sinusoidal waves travel in the same direction and have the same frequency. Their amplitudes are y_{1m} and y_{2m} . The smallest possible amplitude of the resultant wave is:
- A. $y_{1m} + y_{2m}$ and occurs if they are 180° out of phase
 - B. $|y_{1m} - y_{2m}|$ and occurs if they are 180° out of phase
 - C. $y_{1m} + y_{2m}$ and occurs if they are in phase
 - D. $|y_{1m} - y_{2m}|$ and occurs if they are in phase
 - E. $|y_{1m} - y_{2m}|$ and occurs if they are 90° out of phase
- ans: B
48. Two sinusoidal waves have the same angular frequency, the same amplitude y_m , and travel in the same direction in the same medium. If they differ in phase by 50° , the amplitude of the resultant wave is given by:
- A. $0.64y_m$
 - B. $1.3y_m$
 - C. $0.91y_m$
 - D. $1.8y_m$
 - E. $0.35y_m$
- ans: D
49. Two separated sources emit sinusoidal traveling waves that have the same wavelength λ and are in phase at their respective sources. One travels a distance A_1 to get to the observation point while the other travels a distance A_2 . The amplitude is a minimum at the observation point if $A_1 - A_2$ is:
- A. an odd multiple of $\lambda/2$
 - B. an odd multiple of $\lambda/4$
 - C. a multiple of λ
 - D. an odd multiple of $\pi/2$
 - E. a multiple of π
- ans: A

50. Two separated sources emit sinusoidal traveling waves that have the same wavelength λ and are in phase at their respective sources. One travels a distance A_1 to get to the observation point while the other travels a distance A_2 . The amplitude is a maximum at the observation point if $A_1 - A_2$ is:

A. an odd multiple of $\lambda/2$
 B. an odd multiple of $\lambda/4$
 C. a multiple of λ
 D. an odd multiple of $\pi/2$
 E. a multiple of π

ans: C

51. Two sources, S_1 and S_2 , each emit waves of wavelength λ in the same medium. The phase difference between the two waves, at the point P shown, is $(2\pi/\lambda)(A_2 - A_1) + s$. The quantity s is:



A. the distance S_1S_2
 B. the angle S_1PS_2
 C. $\pi/2$
 D. the phase difference between the two sources
 E. zero for transverse waves, π for longitudinal waves

ans: D

52. A wave on a stretched string is reflected from a fixed end P of the string. The phase difference, at P, between the incident and reflected waves is:

A. zero
 B. π rad
 C. $\pi/2$ rad
 D. depends on the velocity of the wave
 E. depends on the frequency of the wave

ans: B

53. The sinusoidal wave

$$y(x, t) = y_m \sin(kx - \omega t)$$

is incident on the fixed end of a string at $x = L$. The reflected wave is given by:

A. $y_m \sin(kx + \omega t)$
 B. $-y_m \sin(kx + \omega t)$
 C. $y_m \sin(kx + \omega t - kL)$
 D. $y_m \sin(kx + \omega t - 2kL)$
 E. $-y_m \sin(kx + \omega t + 2kL)$

ans: D

54. A wave on a string is reflected from a fixed end. The reflected wave:
- A. is in phase with the original wave at the end
 - B. is 180° out of phase with the original wave at the end
 - C. has a larger amplitude than the original wave
 - D. has a larger speed than the original wave
 - E. cannot be transverse
- ans: B
55. A standing wave:
- A. can be constructed from two similar waves traveling in opposite directions
 - B. must be transverse
 - C. must be longitudinal
 - D. has motionless points that are closer than half a wavelength
 - E. has a wave velocity that differs by a factor of two from what it would be for a traveling wave
- ans: A
56. Which of the following represents a standing wave?
- A. $y = (6.0 \text{ mm}) \sin[(3.0 \text{ m}^{-1})x + (2.0 \text{ s}^{-1})t] - (6.0 \text{ mm}) \cos[(3.0 \text{ m}^{-1})x + 2.0]$
 - B. $y = (6.0 \text{ mm}) \cos[(3.0 \text{ m}^{-1})x - (2.0 \text{ s}^{-1})t] + (6.0 \text{ mm}) \cos[(2.0 \text{ s}^{-1})t + 3.0 \text{ m}^{-1}x]$
 - C. $y = (6.0 \text{ mm}) \cos[(3.0 \text{ m}^{-1})x - (2.0 \text{ s}^{-1})t] - (6.0 \text{ mm}) \sin[(2.0 \text{ s}^{-1})t - 3.0]$
 - D. $y = (6.0 \text{ mm}) \sin[(3.0 \text{ m}^{-1})x - (2.0 \text{ s}^{-1})t] - (6.0 \text{ mm}) \cos[(2.0 \text{ s}^{-1})t + 3.0 \text{ m}^{-1}x]$
 - E. $y = (6.0 \text{ mm}) \sin[(3.0 \text{ m}^{-1})x] + (6.0 \text{ mm}) \cos[(2.0 \text{ s}^{-1})t]$
- ans: B
57. When a certain string is clamped at both ends, the lowest four resonant frequencies are 50, 100, 150, and 200 Hz. When the string is also clamped at its midpoint, the lowest four resonant frequencies are:
- A. 50, 100, 150, and 200 Hz
 - B. 50, 150, 250, and 300 Hz
 - C. 100, 200, 300, and 400 Hz
 - D. 25, 50, 75, and 100 Hz
 - E. 75, 150, 225, and 300 Hz
- ans: C
58. When a certain string is clamped at both ends, the lowest four resonant frequencies are measured to be 100, 150, 200, and 250 Hz. One of the resonant frequencies (below 200 Hz) is missing. What is it?
- A. 25 Hz
 - B. 50 Hz
 - C. 75 Hz
 - D. 125 Hz
 - E. 225 Hz
- ans: B

59. Two traveling waves $y_1 = A \sin[k(x - vt)]$ and $y_2 = A \sin[k(x + vt)]$ are superposed on the same string. The distance between the adjacent nodes is:

A. vt/π
 B. $vt/2\pi$
 C. $\pi/2k$
 D. π/k
 E. $2\pi/k$

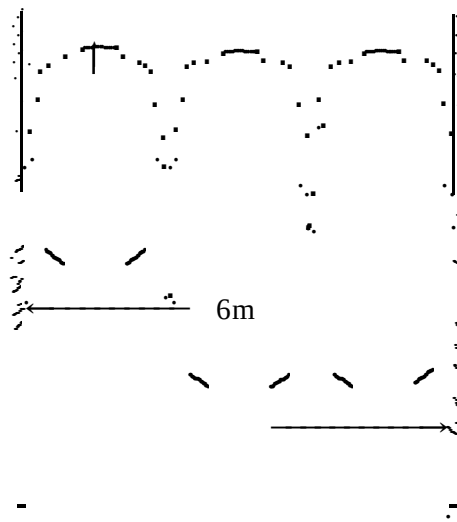
ans: D

60. If λ is the wavelength of each of the component sinusoidal traveling waves that form a standing wave, the distance between adjacent nodes in the standing wave is:

A. $\lambda/4$
 B. $\lambda/2$
 C. $3\lambda/4$
 D. λ
 E. 2λ

ans: B

61. A standing wave pattern is established in a string as shown. The wavelength of one of the component traveling waves is:



$\lambda = 0.5\text{m}$

$\lambda = 0.5\text{m}$

.. .. .

$$E_{\text{avg}} = \frac{1}{2} \epsilon_0 E_0^2$$

